

SMITHIAN FOLD THEORY - INFORMATION SCIENCE BRANCH PAPER 001

From Distinction to Information

An Exact, Parameter-Free and Machine-Closed Derivation of Information
Science from Smithian Fold Theory

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Third clean-room reconstruction - complete dated Information Science field

262 admitted derivations - 75,776 generated candidate structures

1,048 adverse controls - 262 independent reconstructions

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Abstract

This paper reports the complete dated Information Science reconstruction in Smithian Fold Theory V3: 262 of 262 frozen obligations admitted through the unchanged engine, with no free or fitted parameter, no imported axiom, no stochastic cause, no numerical-zero ontology, no negative proof quantity, no irrational or imaginary proof value, no floating proof arithmetic and no application-selected law. The branch executes 75,776 exact candidate structures and decisions, retains 262 unique survivors, passes 1,048 adverse controls and supplies 262 implementation-distinct reconstructions. The 250 complete-field successor laws additionally carry post-registry observation packages; the twelve foundational laws retain their original independently reconstructed receipts and are included in the later complete validation vector.

The reconstruction covers symbols, distinguishability and representation; records and provenance; source, process, sequence, spatial and network information; exact measures; signals, sampling and quantization; lossless and bounded-loss compression; channel structure and capacity; noise and error; block, convolutional, network, erasure and adversarial coding; conditional, mutual, directed and shared-information relations; sufficient records and coarse-graining; retrieval and knowledge organization; inference, filtering and estimation; privacy and leakage; information thermodynamics; classical, deterministic-probabilistic and quantum-support correspondence; semantic/reference boundaries; empirical/formal validation; and one-owner cross-branch handoffs.

The central scientific object is not a borrowed scalar. Information is complete generated alternative support together with exact records of what an observation retains, what it closes, how it transforms and what record is required to reverse a merge. Entropy is the full observation-class and unresolved-distinction ledger. Probability is an exact part of deterministic support rather than an uncaused transition. Compression, capacity and correction are exact finite separation laws. Semantic records preserve symbol, reference and context without replacing biological function, cognition or lived qualia. The branch is complete to its frozen dated census and explicitly open to lawful extension.

Keywords: Smithian Fold Theory; information science; distinguishability; representation; entropy; uncertainty; compression; channel capacity; noise; coding theory; inference; privacy; information thermodynamics; deterministic probability; quantum information; semantic information; computational proof; open science.

Results first: the complete information sciences from one exact distinction ledger

Headline finding	Exact result	Scientific meaning
Complete dated reconstruction	262/262 obligations, 20/20 families, 0 open (display 0 denotes absence)	Every frozen question has its own current engine receipt; future discoveries append by version rather than rewriting the record.
Exact exhaustive forcing	75, 776 candidates and decisions; 262 sole survivors	Each law survives a complete declared grammar, not fitting, authority, familiarity or target feedback.
Falsification and replication	1, 048 passing adverse controls; 262 implementation-distinct reconstructions	False premise, changed source, tampered artifact and boundary import are executable halts, not rhetorical caveats.
Information and entropy	Complete support plus retained/closed distinctions; entropy retains every exact observation class and unresolved pair	Uncertainty remains inspectable without importing a logarithm, stochastic cause or floating distribution.
Compression, channels and codes	Reconstructible descriptions, complete transport relations, exact confusability classes and disjoint error-image correction	Capacity and recovery are forced by finite support structure; all ties, collisions and adverse images remain visible.
Relations, inference and retrieval	Joint-incidence, conditional restriction, coarse-graining, query, ranking, relevance and estimator records retain full provenance	A result cannot hide the observations, contexts or exclusions that produced it.
Privacy and thermodynamics	Leakage is retained distinction access; irreversible merging requires predecessor custody for reversal	Maxwell-demon and Landauer/Bennett boundaries become exact information accounting rather than untracked gain or erasure.

Headline finding	Exact result	Scientific meaning
Classical-probabilistic-quantum correspondence	One held word, an unresolved deterministic class and complete quantum basis support share one exact carrier at their declared boundary	Operational phase, gates, entanglement and fault tolerance remain correctly owned by Quantum Computation.
Semantic and lived-experience boundary	Symbol-reference-context records are exact; cognition, consciousness, qualia and red-of-red remain named downstream handoffs	Information quantity does not erase or pretend to replace subjective experience.
Empirical/formal Grand Lock	All 244 pre-lock receipts and 976 controls are bound before the twelve validation receipts; six terminal handoffs complete the 262-row ledger	Favorable, adverse, absent, unresolved and scoped-boundary records are preserved under unchanged authority.

Complete dependency-ordered family census

Family	Frozen obligations	Current receipts
BASE	12	12/12
SYMREP	14	14/14
RECORD	12	12/12
SOURCE	14	14/14
MEASURE	16	16/16
SIGNAL	14	14/14
COMP	14	14/14
CHAN	18	18/18
NOISE	12	12/12
CODE	18	18/18
REL	14	14/14
COARSE	12	12/12
RETR	12	12/12
INFER	14	14/14
PRIV	10	10/10
THERM	10	10/10
CORR	16	16/16
SEM	12	12/12
VALID	12	12/12
HAND	6	6/6

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorize. This work therefore treats Maria Smith's authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific

contribution lost when financial gatekeeping is presented as rigor.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For Information Science, opacity is itself a technical loss. Paywalls, inaccessible datasets, undisclosed training corpora and terminal oracle scores close precisely the source distinctions needed to test premises, alternatives, data custody and failure conditions. A predictor may be an instrument, but an answer without that retained route is not an admitted scientific law.

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Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through <https://discord.gg/ucwGryVxGr>, or inspect the public project at <https://github.com/MettaMazza>.

1. Central scientific claim

The exact claim of this paper is:

Within the frozen SFT V3 Information Science current-knowledge census, all 262 registered obligations have depth-independent, model-admitted and independently reconstructed engine receipts; all 77 Information Science-owned V1/V2 atomic obligations remain closed at equal strength; no obligation in the dated census remains open, while lawful versioned extension remains permitted.

The complete-field census identity is

sha256:cf6d90095ada5fca5ca39579764ffbd79fa65194719260ba6a30261efa288965 and the terminal reconciliation identity is

sha256:48056bb5684a97ca5e9eba6bda72afe536a11dd1b9f970a0d220f82dc1ee3a54. They fix scope, ownership and dependency order before prose evaluation. Dated completeness applies to the exact generated-finite laws named in the census and remains open to lawful versioned discovery; operational quantum dynamics remains with its owning branch. Fold Protein, Fold Chess, Fold Go, Unison AI and all application experiments were excluded from law selection.

2. Standalone dependency foundation

This version is a same-paper successor patch to Information Science Branch Paper 001 in its existing DOI chain; it does not create a second competing paper. It is standalone in exposition while preserving scientific dependency identities. The Foundation supplies operational occurrence, structural One, complete positive finite count, exact held/whole parts, the minimal Fold, part equivalence, Fold assembly, finite form grammar, canonical form enforcement and the one-way measurement boundary.

The Mathematics branch supplies exact generated arithmetic, finite collections and relations, complete combinatorial families, graphs, algebra, order, finite geometry/topology, deterministic-support probability, optimization, dynamics, proof and composition. These are model-admitted dependencies with their own receipts. They are not silently replaced by conventional axioms. Every dependency path terminates at the sole root theorem, *there is no nothing*, and every Information Science registration has empty axiom and free-parameter tuples.

3. Exact information constitution

Admitted information objects are canonical symbols, held labels, generated words, complete support traces, exact observation classes, unordered source pairs, retained and closed distinction ledgers, exact positive support parts, relations, predecessor fibres, action traces and proof traces. Empty or identity cases use structural empty One. Semantic numerical zero, negative quantity, irrational or imaginary proof values, floating proof arithmetic, completed infinity and an ungenerated continuum remain outside the proof domain.

Host-language integers, Boolean verdicts, empty containers and process status values are implementation mechanics. They count or check artifacts and are not admitted SFT mathematical values. Exact parts use rational held/whole relations; absent cells are empty One rather than a numerical-zero probability. No logarithm is required to force support, uncertainty, dependence, channel separation or correction.

4. What forced, closed and validated mean

Each claim registers a finite product grammar of structural questions before execution. Every axis contains one explicitly rejected coordinate and one all-preserving coordinate. The engine generates their complete Cartesian product in canonical order, hashes every candidate trace, decides every candidate, records every elimination reason and requires exactly one survivor. A familiar name or successful example cannot bypass this census.

Minimality requires that replacing any admitted coordinate loses a registered preservation, completeness, exactness or provenance condition. Named-shape uniqueness requires exactly one all-preserving product coordinate. Generality requires a claim-specific structural-One base and successor certificate, yielding depth-independent closure over generated finite depth without positing a completed infinite object.

Four adverse controls are mandatory: a false premise, changed official source identity, missing/duplicated/additional survivor and excluded boundary import. A distinct Python process then regenerates the literal coordinate domains, candidate order, decision vector, unique survivor, closure flags and controls without importing the scientific law module. Only the shared admission engine may project an independently validated receipt into the model census.

5. Foundational BASE dependency order and executed census

Order	Claim	Candidate structures	Dimensions	Closure
1	SFT-INFO-SYMBOL-DISTINCTION-001	512	9	depth-independent
2	SFT-INFO-ENCODING-DECODING-001	1,024	10	depth-independent
3	SFT-INFO-QUANTITY-001	512	9	depth-independent
4	SFT-INFO-ENTROPY-UNCERTAINTY-001	512	9	depth-independent
5	SFT-INFO-COMPRESSION-REDUNDANCY-001	1,024	10	depth-independent
6	SFT-INFO-CHANNEL-CAPACITY-001	1,024	10	depth-independent
7	SFT-INFO-NOISE-ERROR-001	1,024	10	depth-independent
8	SFT-INFO-CODING-001	2,048	11	depth-independent
9	SFT-INFO-MUTUAL-CONDITIONAL-001	1,024	10	depth-independent
10	SFT-INFO-CONSERVATION-LOSS-001	1,024	10	depth-independent
11	SFT-INFO-CLASSICAL-PROBABILISTIC-001	1,024	10	depth-independent
12	SFT-INFO-QUANTUM-CORRESPONDENCE-001	1,024	10	depth-independent

The foundational BASE kernel contains **11,776** generated candidates and twelve survivors. The complete-field extension documented below adds 250 separately admitted laws. Across the dated branch the exact total is **75,776** candidates and decisions, 262 unique survivors and 1,048 passing adverse controls. Candidate counts describe complete grammar products, not every data object producible by the laws.

6. Derivation 1: Exact symbols, observation classes and distinguishability

Claim identity: SFT-INFO-SYMBOL-DISTINCTION-001

6.1 Question, necessity and theorem

Information cannot be quantified before the model derives what a symbol is and exactly when two possible forms remain distinguishable to an observation. Imported alphabets or meanings would select that boundary.

The exact theorem statement is:

An admitted symbol is a canonical generated form with a retained held label inside a complete canonical alphabet. Observation is a total single-valued source-bound classification. Two symbols are distinguishable exactly when their retained observation labels differ; symbols sharing one observation image remain distinct microforms but belong to one closed observation class. Every retained and closed distinction is recorded.

Admitted dependencies: SFT-FOUNDATION-FORM-ENFORCEMENT-001,
SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001, SFT-MATH-DISCRETE-001,
SFT-MATH-LOGIC-PROOF-001.

6.2 Derivation chain, grammar and boundary

Canonical form enforcement supplies exact identity; discrete relations supply total classification; logic supplies distinguished held orientations; the measurement boundary fixes one-way observation. Exhausting nine structural questions forces the symbol/observation-class kernel.

Generation rule: Generate the complete product of alphabet coverage, symbol identity, held label, observation coverage, observation single-valuedness, distinction rule, closed-class retention, finite generality and extra-symbol status.

Exact grammar boundary: All finite symbol systems generated from canonical Fold forms, held labels, complete alphabets, total observation relations and exact retained/closed pair ledgers.

The complete product contains 512 candidates. Its completeness-certificate identity is sha256:ca1f59aecaef95c9d31d3303d1a4d92673ad4b068e5ef58df58915e37e04becc. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-SYMBOL-DISTINCTION-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
alphabet	partial-or-duplicated-alphabet	Omission or duplication changes the possible symbol family.	complete-canonical-alphabet - Every generated symbol form occurs once.
identity	presentation-token	A presentation token may alias distinct generated forms.	canonical-form-identity - The complete construction trace fixes exact symbol identity.
label	unlabelled-form	Without a retained label the selected fibre identity is unavailable.	retained-held-label - The symbol carries its exact held fibre label.
observation	partial-observation	A missing symbol has no declared observation class.	total-source-bound-observation - Every alphabet member has one retained observation row.
image	multivalued-image	Multiple unheld images do not identify the observation made.	single-retained-image - Each source symbol has exactly one held observation label.
distinction	assumed-or-visual-difference	Presentation difference does not establish retained observational separation.	different-observation-labels - The observation relation retains distinct images for the two source forms.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
closure	source-identity-erased	Erasing sources prevents an exact account of closed distinctions.	closed-class-with-microforms - Merged symbols remain as exact source members of one observation class.
generality	sampled-alphabets	Examples do not establish extension to a fresh symbol.	alphabet-successor - A fresh canonical symbol adds one row and all new distinction pairs.
addition	extra-symbol-semantics	Imported meaning or frequency is not supplied by form and observation.	no-extra-symbol-semantics - Identity and distinction use only generated forms, labels and observation.

6.3 Exhaustion, unique survivor and laws

The symbol kernel is a complete canonical held-labelled alphabet, total single-valued observation, exact different-image distinguishability, retained closed classes, successor generality and no imported semantics.

The engine recorded 512 decisions, 511 eliminations and exactly one survivor, `complete-canonical-alphabet__canonical-form-identity__retained-held-label__total-source-bound-observation__single-retained-image__different-observation-labels__closed-class-with-microforms__alphabet-successor__no-extra-symbol-semantics`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- symbol identity is canonical construction identity plus retained held label.
- observation classes partition the complete alphabet without changing microform identity.
- distinguishability is symmetric and irreflexive on exact symbol pairs.
- coarsening closes distinctions only by merging observation images.
- the retained/closed pair ledgers exhaust every generated unordered symbol pair.

6.4 Operational demonstrations and adverse controls

- `total-observation` - The fine observation classifies every symbol exactly once. Result: PASS.
- `fine-distinction` - Fine observation retains every nonidentity pair distinction. Result: PASS.
- `coarse-closure` - Coarse observation closes only the a/b distinction while retaining both microforms. Result: PASS.
- `symmetry` - Swapping a distinguished pair does not change its status. Result: PASS.
- `irreflexivity` - A canonical symbol is not distinguished from itself. Result: PASS.
- `class-retention` - The coarse class retains both exact left microforms. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission or duplication changes the possible symbol family. Result: PASS; receipt
sha256:0fc110df76dc26a6061187a62ca6f4ab63284190dc52235042c8a4baf1aa6440.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported semantic meaning; no frequency or probability prior; no presentation alias replacing canonical identity; no ungenerated infinite alphabet Result: PASS; receipt
sha256:00999f5bc8b92e3bfaab8395e8e8ddf539ba74e61fe85e7e70092b74d98dcf46.

6.5 Depth-independent closure

Base: One canonical symbol has one observation row and no nonidentity pair distinction.

Successor: Adding one fresh canonical symbol adds exactly one source-bound observation row and one pair with every prior symbol. Each new pair is classified once as retained or closed by equality of its two observation labels.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

6.6 Meaning and scientific consequence

The first information law makes distinguishability operational rather than intuitive. A symbol is not a typographic mark or imported semantic token; it is a canonical generated form with a held label. An observation is a complete source-bound classification. Two source forms are distinguishable exactly when that classification retains different images. When images coincide, the source forms are not destroyed: they remain the complete microform membership of one closed observation class. This distinction ledger is the substrate from which every later quantity, uncertainty and loss statement is derived.

6.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: symbol, alphabet, observation, equivalence class, distinguishability. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

6.8 Exact exclusions and limitation

- no imported semantic meaning.
- no frequency or probability prior.
- no presentation alias replacing canonical identity.
- no ungenerated infinite alphabet.

This theorem closes finite structural symbols and observation-relative distinction. It does not assign linguistic meaning, occurrence frequency or information quantity; those require downstream laws.

6.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
sha256:9ab9dbeff82654d86ada02cd716d04a4d2a7d5a31358c418f8d5591142804311.
- Derivation seal: sha256:78cd2b2e6d03245ba4b96c7c6df45c3b2a2845ab3b1df64b90409384ed195abd.
- Independent implementation:
sha256:42ac391738828a48baa3b886e4017c8a99a037ff5fe39fce67dd387970cf0f34.
- Independent certificate:
sha256:75b1bf19a3e78fceca2ea74b37b86f17b74699f90b21d1b5e6b47262f923ae6a.
- External validation:
sha256:7cec4cb5666884fb6a81239c62e2e2259ee29654bf0143a467aeb8fd4ac85dc1.
- Engine receipt: sha256:e387d64c93f5241d90ecf2a447eb6a3518c1e9f473e7ec627d38b7f03ac9d941.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-SYMBOL-DISTINCTION-001-e387d64c93f5241d.json`.

7. Derivation 2: Exact encoding, decoding and declared representational loss

Claim identity: SFT - INFO - ENCODING - DECODING - 001

7.1 Question, necessity and theorem

Information requires lawful representation before quantity, compression or channels can be discussed. Encoding must expose exactly which distinctions it preserves and which it closes.

The exact theorem statement is:

An admitted encoding is a total single-valued source-bound map from a complete canonical source alphabet into generated codeword support, retaining every source-to-code relation. Exact decoding exists exactly when every used codeword class contains one source and the decoder is the resulting left inverse. A many-to-one encoding is lawful only as declared lossy representation whose complete closed-source classes remain recorded.

Admitted dependencies: SFT - FOUNDATION - FOLD - ASSEMBLY - 001, SFT - MATH - DISCRETE - 001, SFT - MATH - CATEGORY - TYPE - COMPOSITION - 001, SFT - INFO - SYMBOL - DISTINCTION - 001.

7.2 Derivation chain, grammar and boundary

Fold assembly supplies generated codewords; symbol law supplies canonical sources and distinctions; discrete maps supply totality; compositional mathematics supplies exact interfaces. Ten structural axes force the total relation, inverse and declared-loss kernel.

Generation rule: Generate the complete product of source coverage, code support, encoding totality, single-valuedness, provenance, decoder law, exact/lossy status, composition, finite generality and extra-code status.

Exact grammar boundary: All finite representations generated from complete canonical source alphabets, generated Fold codewords, total maps, inverse relations, retained source classes and exact interface composition.

The complete product contains 1,024 candidates. Its completeness-certificate identity is sha256:04b9b004111bd895088946afec50c717e90be78d7193e1d61e5c4c8a4d6c9117. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT - INFO - ENCODING - DECODING - 001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
source	partial-source	An omitted source symbol has no representation row.	complete-canonical-source - Every source symbol occurs once in the registered domain.
support	free-code-token	A free token has no generated Fold-word provenance.	generated-code-support - Every codeword is an exact generated held-label word.
coverage	partial-map	A partial map leaves at least one source without a codeword.	total-source-map - Every source has one encoding row.
image	unheld-multiple-images	Multiple images do not identify the codeword actually retained.	single-retained-codeword - Each source has exactly one held codeword image.
provenance	source-code-link-erased	Erasing links prevents reconstruction of source classes.	complete-source-code-relation - Every exact source-to-code pair is retained.
decoder	guessed-preimage	A used codeword can have several source predecessors.	singleton-class-left-inverse - Every used code class is singleton and the inverse map returns its source.
loss	silent-loss	Silent merging falsely claims unavailable distinctions remain.	declared-closed-source-classes - Every merged source family is retained as an explicit loss class.
composition	presentation-concatenation	Concatenation can join unequal code interfaces.	complete-interface-map - Every first-stage codeword has exactly one second-stage image.
generality	sampled-codebooks	Examples do not establish a fresh source row.	source-successor - A fresh source adds one exact encoding row and updates one code class.
addition	extra-code-convention	A borrowed numeral or lexical convention can select the code.	no-extra-code-convention - Only generated code support and exact relations are used.

7.3 Exhaustion, unique survivor and laws

The encoding kernel is complete source and generated code support with a total single-valued provenance relation, singleton-class left-inverse decoding, explicit loss classes, exact composition and no borrowed code.

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-source__generated-code-support__total-source-map__single-retained-codeword__complete-source__code-relation__singleton-class-left-inverse__declared-closed-source-classes__complete-interface-map__source-successor__no-extra-code-convention`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- every source symbol has exactly one retained codeword image.
- lossless exact decoding exists if and only if used codeword classes are singleton.
- many-to-one encoding closes exactly the distinctions inside its retained source classes.
- decoder-after-lossless-encoder returns the original canonical source.
- encoding composition preserves source provenance when every intermediate codeword is mapped.

7.4 Operational demonstrations and adverse controls

- `lossless-total` - The witness lossless relation maps every source once into generated support. Result: PASS.
- `left-inverse` - The exact decoder reconstructs every lossless source. Result: PASS.
- `loss-detected` - The lossy relation has no exact decoder and retains the merged a/b class. Result: PASS.
- `class-ledger` - Every used lossy codeword retains its exact source members. Result: PASS.
- `composition` - Complete code interface composition preserves each source relation. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An omitted source symbol has no representation row. Result: PASS; receipt
sha256: bf4e85ebd5f2d1856f83756999071a09416cb188f8f2b7575b387df85706d894.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no imported binary numeral system; no presentation-only token identity; no silent many-to-one loss; no infinite codebook as a completed object Result: PASS; receipt
sha256: 39ff4157377beb066b2a0fa22869a55e623bde5f0bddb7acd4e230983f12d04c.

7.5 Depth-independent closure

Base: One source symbol maps to one generated codeword and its singleton inverse decodes exactly.

Successor: Adding one fresh source adds exactly one encoding pair. A fresh codeword preserves singleton decoding; a used codeword extends its closed source class and changes the representation status to declared lossy.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

7.6 Meaning and scientific consequence

Encoding is reconstructed as an exact relation, not a convention selected in advance. Every canonical source has one retained generated codeword and every source/code link remains inspectable. Decoding is not guessed: it exists exactly when every

used codeword has one source predecessor. A many-to-one representation remains lawful only when it says what it closes and preserves the full merged source class. This establishes lossless and lossy representation before compression, channel or code claims can use those words.

7.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: alphabet, codeword, encoder, decoder, injective code, lossy representation. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

7.8 Exact exclusions and limitation

- no imported binary numeral system.
- no presentation-only token identity.
- no silent many-to-one loss.
- no infinite codebook as a completed object.

This theorem closes finite encoding and decoding structure. It does not yet assign code length value, compression optimality, channel reliability or correction power.

7.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
`sha256:d072e49df1f7903aa2987ccb29174b8261c9a32fa26eda6a856c65cee404b0f7`.
- Derivation seal: `sha256:ae856dbb457f937934a2fb5ea450d6902c4460ad478c9c596b8996540c23ebde`.
- Independent implementation:
`sha256:d93b45d2158dd50461fcf50bf1ef7ec0668dad57e2a498a1391e4a61fbf7531e`.
- Independent certificate:
`sha256:f03b8247debffe52a53757d246405cf4a44e98f22134494df12d6a9e665b2cab`.
- External validation:
`sha256:29719e28c425d45cd7a1c62e4a551776a08ff556c85a7bfe87587500c304dbaa`.
- Engine receipt: `sha256:dc3cbc708f5b88b3d9bbae5a646a644dc8feb7fe4a5d6f8367052f5062846d0f`.
- Engine receipt path:
`receipts/engine/model_admitted/SFT-INFO-ENCODING-DECODING-001-dc3cbc708f5b88b3.json`.

8. Derivation 3: Exact information quantity as retained distinction structure

Claim identity: SFT-INFO-QUANTITY-001

8.1 Question, necessity and theorem

Information quantity must be derived from exact alternatives and observation before entropy, capacity or compression. Importing logarithms would install a conventional numerical answer and forbidden proof values.

The exact theorem statement is:

Information quantity is the exact structure of distinguishable alternatives retained from a complete generated support: the support trace, observation-class trace, retained distinction ledger and closed distinction ledger. The first Fold supplies one native distinction unit. Complete Fold-word support across a canonical position trace supplies one independently witnessed unit per position; composition is complete pair support and concatenated position provenance. No logarithm, real scale or semantic zero is foundational.

Admitted dependencies: SFT-FOUNDATION-FOLD-ASSEMBLY-001, SFT-MATH-EXACT-ARITHMETIC-001, SFT-MATH-COMBINATORICS-001, SFT-INFO-SYMBOL-DISTINCTION-001, SFT-INFO-ENCODING-DECODING-001.

8.2 Derivation chain, grammar and boundary

Fold assembly supplies complete word support; symbol distinction supplies exact observation ledgers; encoding supplies representation; combinatorics supplies complete pairs. Nine structural questions force a ledger-valued quantity and independently witnessed Fold-position unit.

Generation rule: Generate the complete product of support coverage, distinction ledger, observation classes, native unit, position independence, quantity representation, composition, generality and extra-scale status.

Exact grammar boundary: All finite information profiles generated from complete canonical Fold-word support, total observation classes, exact pair distinctions, independently varied positions and pair-cell composition.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:40d0d2569464d104f73e01fed5c7bb9ef0cb4640c1d24709f6877e3674152e3c`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-QUANTITY-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
support	partial-support	Omitting a generated alternative changes every distinction count.	complete-canonical-support - Every registered alternative occurs once.
pairs	sampled-pairs	Sampling cannot prove a distinction ledger complete.	complete-unordered-pair-ledger - Every pair of distinct support forms occurs once.
observation	assumed-resolution	An assumed resolution has no source-to-image evidence.	exact-observation-classes - Image equality classifies every pair as retained or closed.
unit	imported-bit	A conventional binary unit would select the answer.	first-Fold-distinction - The minimal Fold's two held labels supply the first exact alternative distinction.
positions	word-length-only	Length alone does not show each position can vary independently.	independent-position-witness - Every position realizes the Fold distinction with all other positions held.
quantity	real-logarithm	A logarithm imports irrational and floating proof values.	exact-support-and-distinction-profile - The complete exact ledgers retain the quantity without a borrowed scale.
composition	count-product-only	A scalar product erases source-coordinate provenance.	complete-pair-support - Every left state pairs once with every right state and retains both word traces.
generality	depth-table	A table of small supports does not prove the next position.	position-successor - A fresh position appends every held label to every prior word.
addition	extra-log-base-or-unit	A chosen base or physical scale is a free parameter.	no-extra-information-scale - All quantity follows from Fold units, support and observation.

8.3 Exhaustion, unique survivor and laws

The information-quantity kernel is complete support and pair ledgers, exact retained/closed observation distinctions, first-Fold units, independently witnessed position traces, pair-support composition and no log scale.

The engine recorded 512 decisions, 511 eliminations and exactly one survivor, `complete-canonical-support__complete-unordered-pair-ledger__exact-observation-classes__first-Fold-distinction__independent-position-witness__exact-support-and-distinction-profile__complete-pair-support__position-successor__no-extra-information-scale`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- retained and closed distinction ledgers are disjoint and exhaust complete support pairs.
- fine observation retains every nonidentity support distinction.
- coarsening can only move pair distinctions from retained to closed.

- one complete Fold position supplies one independently realized distinction unit.
- composing supports retains both source coordinates and concatenates their unit traces.

8.4 Operational demonstrations and adverse controls

- `complete-support` - Three independently held Fold positions generate eight unique exact words. Result: PASS.
- `fine-ledger` - Fine observation retains every distinct support pair and closes none. Result: PASS.
- `coarse-ledger` - First-position observation retains cross-class pairs and closes within-class pairs. Result: PASS.
- `position-units` - Each of three positions independently realizes the Fold distinction. Result: PASS.
- `support-composition` - Two two-word supports compose into four coordinate-preserving words. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omitting a generated alternative changes every distinction count. Result: PASS; receipt
sha256: 66c8c2acb6c7e2805130d894dcbdcab51bd9665d1d065a1ca9860f3b23d14826.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no logarithm as a proof operation; no real-valued or irrational information quantity; no semantic numerical zero for identity information; no incomplete support presented as full quantity Result: PASS; receipt
sha256: 40cc647fb3b88cb6aca2ec0f86b8a4bc0e2a9c7e1810685d2ca3f5ce2df95355.

8.5 Depth-independent closure

Base: The first Fold position has complete two-label support and one independently realized distinction unit.

Successor: Appending one fresh position extends every prior word once with every held label. Prior position witnesses remain, the fresh position varies across its labels for each held prefix, and all new support pairs enter exactly once.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

8.6 Meaning and scientific consequence

Information quantity is carried by exact alternatives and their distinguishability, not by a borrowed real scale. The kernel retains complete word support, every unordered source pair, every observation class and the partition of pairs into retained and closed distinctions. The first Fold contributes the first exact two-label distinction. A longer word contributes another native unit only when that position can vary independently while the remaining positions are held. Logarithmic units may summarize a sealed finite support in correspondence, but they are not used to force the support or its distinctions.

8.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: information content, bit, alphabet size, logarithmic information, joint information. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

8.8 Exact exclusions and limitation

- no logarithm as a proof operation.

- no real-valued or irrational information quantity.
- no semantic numerical zero for identity information.
- no incomplete support presented as full quantity.

This theorem closes exact finite information quantity. Conventional logarithmic units may be post-seal correspondence summaries but are not SFT proof values.

8.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
`sha256:6a918de5f716d1371af1e475ce3a36d565c12e6d92f16ef430315c75dc2dad93.`
- Derivation seal: `sha256:268633b173cd9292188e15669163580ed1e76b99b66c861dd70181087b23bb43.`
- Independent implementation:
`sha256:67a80bd1df98d114b370eeca171e57f012f3a5c86e3056fb2f24c2f9161e479.`
- Independent certificate:
`sha256:5dddce624d62ebceb275a5a87beda0fe6886bcd348a59919c5145ac5af4b56ac.`
- External validation:
`sha256:325c8dcdbfc5b20ce21c7549baac1fe63ca0eabb138178d09b8f9b7eef013b9b.`
- Engine receipt: `sha256:a06381112b197ea1762d8dd06197c87f03055a4554429f100ead4a93bbb2362d.`
- Engine receipt path:
`receipts/engine/model_admitted/SFT-INFO-QUANTITY-001-a06381112b197ea1.json.`

9. Derivation 4: Exact entropy and uncertainty from deterministic observation classes

Claim identity: SFT-INFO-ENTROPY-UNCERTAINTY-001

9.1 Question, necessity and theorem

Entropy and uncertainty must be reconstructed without treating randomness or real logarithms as premises. Exact observation classes already contain the complete unresolved information.

The exact theorem statement is:

Uncertainty is the complete ledger of deterministic microstate distinctions closed inside each exact observation class. Entropy is the provenance-retaining family of those classes, their exact positive support parts, microstate traces and unresolved-pair ledgers; singleton certainty uses structural empty One. Refining observation can only remove closed pairs, while coarsening can only add them. No stochastic cause, logarithm, numerical zero or floating expectation enters the law.

Admitted dependencies: SFT-FOUNDATION-PART-EQUIVALENCE-001,
SFT-MATH-PROBABILITY-STATISTICS-001, SFT-INFO-SYMBOL-DISTINCTION-001,
SFT-INFO-QUANTITY-001.

9.2 Derivation chain, grammar and boundary

Mathematical probability supplies exact support parts; symbol distinction supplies observation classes; information quantity supplies retained/closed ledgers. Nine axes force an entropy object that preserves every class, part and unavailable distinction.

Generation rule: Generate the complete product of microstate support, observation partition, class weights, unresolved pairs, entropy representation, refinement order, deterministic status, generality and extra-entropy status.

Exact grammar boundary: All finite uncertainty structures generated from complete deterministic microstate support, total observation partitions, exact class parts, complete within-class pair ledgers and refinement relations.

The complete product contains 512 candidates. Its completeness-certificate identity is `sha256:77baae24e3e22695c0246fc61c48d8b34f73542f2bd557e1407861f24f21339d`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-ENTROPY-UNCERTAINTY-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
support	partial-microstate-support	Omission changes observation classes and uncertainty.	complete-deterministic-support - Every registered microstate occurs once.
partition	overlapping-or-partial-classes	Overlap or omission does not partition exact support.	total-disjoint-observation-classes - Every microstate belongs to exactly one observed class.
weights	floating-probability	A floating propensity imports precision and stochastic cause.	exact-class-whole-part - The class count is retained relative to complete support count.
unresolved	scalar-uncertainty-score	A scalar score erases which distinctions are unavailable.	complete-within-class-pairs - Every pair of distinct class microstates is retained once.
entropy	logarithmic-expectation	Logarithms and real expectations import forbidden proof values.	weighted-class-distinction-ledger - Every class retains exact part, members and unresolved distinctions.
refinement	assumed-more-information	A name does not prove one observation separates every fine class.	class-containment-witness - Every fine class lies inside one coarse class and its closed-pair set is contained.
cause	ontic-random-source	A stochastic source is neither generated nor required.	deterministic-support-observation - Uncertainty comes from distinctions closed by observation over deterministic support.
generality	sampled-distributions	Examples cannot establish a fresh microstate class update.	microstate-successor - A fresh state extends exactly one class and its new within-class pairs.
addition	extra-log-base-or-prior	A chosen log base or prior is a free parameter.	no-extra-entropy-convention - The exact ledger follows only from support and observation.

9.3 Exhaustion, unique survivor and laws

The entropy/uncertainty kernel is complete deterministic support, total observation classes, exact class parts, complete unresolved-pair ledgers, refinement containment, successor closure and no stochastic or log rule.

The engine recorded 512 decisions, 511 eliminations and exactly one survivor, `complete-deterministic-support__total-disjoint-observation-classes__exact-class-whole-part__complete-within-class-pairs__weighted-class-distinction-ledger__class-containment-witness__deterministic-support-observation__microstate-successor__no-extra-entropy-convention`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- observation classes form a disjoint complete partition of deterministic support.
- uncertainty is exactly the set of within-class nonidentity microstate pairs.
- singleton classes carry empty-One uncertainty rather than numerical zero.
- fine observation cannot have more closed pairs than a coarser observation it refines.
- entropy retains class provenance and exact part weights instead of collapsing to a floating scalar.

9.4 Operational demonstrations and adverse controls

- `fine-certainty` - Fine observation gives every state a singleton class with empty-One uncertainty. Result: PASS.
- `prefix-uncertainty` - Prefix observation has two exact half-support classes and one unresolved pair each. Result: PASS.
- `coarse-uncertainty` - Complete coarsening retains all six microstate pairs as unresolved. Result: PASS.
- `refinement-order` - Fine refines prefix, and prefix refines the one-class observation. Result: PASS.

- **closed-pair-monotonicity** - Refinement reduces closed distinctions exactly in the witness chain. Result: **PASS**.

The engine-level unfavorable controls are:

- **false_premise** - expected: reject a generated form missing the first required structural condition Observed: Omission changes observation classes and uncertainty. Result: **PASS**; receipt
sha256:b8cc7b2826b60bb835dce41750f0f9ed5e6ec9175437b2a571fbdb1981eea15a.
- **tampered_source** - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: **PASS**; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: **PASS**; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - expected: halt any attempt to import an excluded object or answer-producing model Observed: no stochastic source premise; no logarithmic entropy proof value; no numerical zero for singleton uncertainty; no fitted probability distribution Result: **PASS**; receipt
sha256:e2deb036c0db8c8405ccf1fc3515c5acfa97882e8e851aed41155d83d8769561.

9.5 Depth-independent closure

Base: One deterministic microstate forms one exact whole class with structural empty-One uncertainty.

Successor: Adding one fresh microstate assigns it to one retained observation image, updates that class's exact part and adds exactly one unresolved pair with every prior member of the same class; other classes remain unchanged.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

9.6 Meaning and scientific consequence

Uncertainty is the exact set of deterministic microstate pairs an observation cannot separate. Entropy is the provenance-retaining family of observation classes, exact positive support parts, class members and unresolved pair ledgers. A singleton class has structural empty-One uncertainty. Refining observation removes closed pairs; coarsening adds them. Nothing in this construction requires an ontic random cause, a logarithm, an expectation over reals or a fitted distribution. The result preserves more evidence than a scalar because every unresolved distinction remains named.

9.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: entropy, uncertainty, partition entropy, resolution, deterministic ensemble. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

9.8 Exact exclusions and limitation

- no stochastic source premise.
- no logarithmic entropy proof value.
- no numerical zero for singleton uncertainty.
- no fitted probability distribution.

This theorem closes exact finite entropy and uncertainty. Conventional scalar Shannon entropies are post-seal summaries only and require a separately declared correspondence map.

9.9 Machine evidence identities

- Registered status: **independently_replicated**.

- Source manifest:
sha256: 7154a737d0cc6c3db65e1dc129773dc38ede23ec9a78c5db34da4bbebedd72e1.
- Derivation seal: sha256: d7ea4b7d51ccbdba737ac0ac8118ef0fa25f6d93544a0e26bcd3284fd713682.
- Independent implementation:
sha256: b608e2eb5217eb81c46115f9952ae59f5d2c1366842cb508f2af62b8b9d856c1.
- Independent certificate:
sha256: b2f70d4486625da56092d8d2bcee3691720ffc312a91197f366adbd3d30520a9.
- External validation:
sha256: 2db327cc0f0f7585ef9ad15422f5f94e4d17686e69312321ab272b5754be99ae.
- Engine receipt: sha256: d604932214499ea1e11fc2df7b71f4ea0b67014e3001023720b325d6203d2b72.
- Engine receipt path: receipts/engine/model_admitted/SFT-INFO-ENTROPY-UNCERTAINTY-001-d604932214499ea1.json.

10. Derivation 5: Exact lossless compression and redundancy accounting

Claim identity: SFT-INFO-COMPRESSION-REDUNDANCY-001

10.1 Question, necessity and theorem

Compression claims can hide information in a dictionary, parser, model or decoder. The law must force exact reconstruction and count every retained resource before shortening has scientific meaning.

The exact theorem statement is:

Lossless compression is a complete description relation whose retained dictionary and token trace reconstruct the exact canonical message. Redundancy is a message distinction or occurrence reconstructible from retained description structure. Variable codewords are uniquely parseable only with a complete delimiter or prefix-free witness. Every dictionary and token resource is counted; no fixed complete word support can injectively map all its forms into the strictly shorter nonempty support.

Admitted dependencies: SFT-MATH-COMBINATORICS-001, SFT-MATH-OPTIMIZATION-001, SFT-INFO-ENCODING-DECODING-001, SFT-INFO-QUANTITY-001.

10.2 Derivation chain, grammar and boundary

Encoding supplies exact decoders; information quantity supplies resource traces; combinatorics supplies full fixed and shorter supports; optimization retains all minimal descriptions. Ten axes force lossless reconstruction, complete accounting and the universal-shortening boundary.

Generation rule: Generate the complete product of message coverage, description provenance, exact reconstruction, parsing, redundancy witness, resource accounting, shortening status, optimization, generality and extra-compressor status.

Exact grammar boundary: All finite lossless descriptions generated from canonical messages, retained dictionaries, token traces, exact decoders, unique parsing witnesses and complete positive resource traces.

The complete product contains 1,024 candidates. Its completeness-certificate identity is sha256: 428cb09aa9b7cb3be4e50612a4d3757c46b45e5fae06603e4780eb4ea7cd847a. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in claims/SFT-INFO-COMPRESSION-REDUNDANCY-001/candidate_census.json.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
message	partial-message	An omitted occurrence changes the reconstruction target.	complete-canonical-message - Every source occurrence remains in exact order.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
description	opaque-short-string	A short string without dictionary provenance cannot reconstruct the source.	dictionary-and-token-trace - Every reusable block and held token occurrence is retained.
reconstruction	similar-output	Similarity does not establish exact source identity.	exact-decoder-equality - Decoding reproduces the complete canonical message trace.
parsing	ambiguous-concatenation	One stream may admit several codeword parses.	prefix-or-delimiter-witness - A generated boundary makes the parse unique.
redundancy	frequency-assumption	Frequency alone does not prove reconstruction.	reconstructible-occurrence-trace - Omitted raw occurrences are generated exactly from retained description structure.
resources	payload-only	Ignoring a dictionary or decoder hides required information.	complete-description-resource - Dictionary, tokens, boundaries and decoder identity are all retained.
shortening	universal-shortening-claim	Strictly shorter support has fewer generated nonempty codewords than complete fixed support.	pigeonhole-boundary - At least one full-support message cannot receive a distinct strictly shorter codeword.
optimization	tuned-score	A floating objective or tie rule is unforced.	complete-undominated-descriptions - All exact descriptions are generated and every minimal resource trace tie is retained.
generality	sampled-messages	Examples do not establish a fresh repeated token.	token-successor - Appending one retained token reconstructs exactly its registered dictionary block.
addition	extra-statistical-model	A learned or fitted model can select the representation.	no-extra-compression-model - Only exact generated dictionaries, tokens and decoding are used.

10.3 Exhaustion, unique survivor and laws

The compression kernel is exact message reconstruction from a uniquely parseable dictionary/token description, complete resource accounting, reconstructible redundancy, pigeonhole-limited shortening and no fitted model.

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-message__dictionary-and-token-trace__exact-decoder-equality__prefix-or-delimiter-witness__reconstructible-occurrence-trace__complete-description-resource__pigeonhole-boundary__complete-undominated-descriptions__token-successor__no-extra-compression-model`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- losslessness is exact equality of reconstructed and source canonical traces.
- a reusable block is redundancy only when every replaced occurrence is reconstructible.
- dictionary, parser and decoder resources belong to the complete description.
- prefix freedom or an explicit delimiter is required for unique variable-word parsing.
- no injective code can assign every complete fixed-width message a strictly shorter nonempty word.

10.4 Operational demonstrations and adverse controls

- `exact-reconstruction` - Three retained block tokens reconstruct the six-cell message exactly. Result: PASS.
- `resource-accounting` - The description retains one dictionary entry and all three token occurrences. Result: PASS.
- `redundancy-provenance` - Every reconstructed raw cell retains the token and dictionary block that supplied it. Result: PASS.
- `prefix-control` - The witness prefix family parses uniquely while a prefix-containing family fails. Result: PASS.
- `universal-bound` - Complete two-label width-three support cannot inject into all shorter nonempty words. Result: PASS.

The engine-level unfavorable controls are:

- **false_premise** - expected: reject a generated form missing the first required structural condition Observed: An omitted occurrence changes the reconstruction target. Result: PASS; receipt
sha256: ba3756b2fcba103d3088afc0d25c904307b3e90947e5ff6158b0a2adf6b9c490.
- **tampered_source** - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - expected: halt any attempt to import an excluded object or answer-producing model Observed: no hidden learned model or trained weights; no payload-only size claim; no ambiguous variable-word parsing; no universal lossless shortening claim Result: PASS; receipt
sha256: ca800897b20ecc2e2616fa2a070b9de583c2e1943f29ea17d3036b4837bc734e.

10.5 Depth-independent closure

Base: One dictionary token reconstructs its one registered nonempty block and retains the complete description.

Successor: Appending one valid held token appends exactly its dictionary block to the reconstructed message and one token cell to the resource trace. Exact reconstruction and parsing remain local and replayable.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

10.6 Meaning and scientific consequence

Compression is lawful only when the retained dictionary, token trace, parser and decoder reconstruct the exact source. Resource comparison therefore counts the entire description system rather than reporting a shortened payload while hiding its dictionary. Redundancy is a structurally identified repeated or dependent position, not padding inferred from frequency. The complete-support control proves the finite pigeonhole boundary: a one-to-one code cannot assign every word a strictly shorter nonempty generated word. Compression gains are consequently source-family and representation dependent, never an unqualified universal shortening law.

10.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: lossless compression, redundancy, prefix code, dictionary coding, pigeonhole bound. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

10.8 Exact exclusions and limitation

- no hidden learned model or trained weights.
- no payload-only size claim.
- no ambiguous variable-word parsing.
- no universal lossless shortening claim.

This theorem closes finite structural compression. It does not assert a universal best compressor, a probability distribution or an asymptotic real-valued rate.

10.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
sha256: 72926e8f640dbad6cd14100e6e87e111f7b31aa9df24914e176632f7f35642fc.

- Derivation seal: sha256:075208fb299525387beb95e18f7ad83f1b6f26c24e32f0789b59188928e88bbe.
- Independent implementation:
sha256:483dc4a3279cd2a7deaa5358f916e771d8ca79dbfa9d75800d974f5c33891315.
- Independent certificate:
sha256:69db6a7db7c8c5c8edcf2ae986c4fc900de7b382cc89b190e73f5639e9072cda.
- External validation:
sha256:0ca99dbf97b53f602285051ef02a3449aee5b56396b921c8e1cd87d76da9f10f.
- Engine receipt: sha256:4d46ed451a5cd3c3a19fcb9d8b8cc3c32ffc8f92a9cc1d55cfb93f331023ee07.
- Engine receipt path: receipts/engine/model_admitted/SFT-INFO-COMPRESSION-REDUNDANCY-001-4d46ed451a5cd3c3.json.

11. Derivation 6: Exact finite channels, confusability and capacity

Claim identity: SFT-INFO-CHANNEL-CAPACITY-001

11.1 Question, necessity and theorem

Capacity must follow from which source distinctions a channel can preserve, not from a borrowed probabilistic rate formula. Complete finite transport and confusability make that boundary exact.

The exact theorem statement is:

An admitted channel is a complete canonical input carrier, complete output carrier and held source-bound transport relation giving every input at least one output. Inputs are confusable exactly when their output supports overlap. A zero-error code is a held input selection with pairwise disjoint output supports. Exact channel capacity is the complete family of greatest-cardinality distinguishable codes, with all ties retained; composition is relational path composition at an equal interface.

Admitted dependencies: SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-COMBINATORICS-001, SFT-MATH-OPTIMIZATION-001, SFT-INFO-ENCODING-DECODING-001, SFT-INFO-QUANTITY-001.

11.2 Derivation chain, grammar and boundary

Graphs supply relations and paths; encoding supplies messages and codewords; combinatorics exhausts selections; optimization retains all greatest codes; information quantity supplies exact distinction meaning. Ten axes force the finite zero-error channel and capacity family.

Generation rule: Generate the complete product of input coverage, output coverage, relation provenance, transport totality, confusability, code generation, capacity selection, composition, generality and extra-channel status.

Exact grammar boundary: All finite channels generated from canonical carriers, held transport relations, complete output supports, pairwise confusability checks, exhaustive input selections and exact relation composition.

The complete product contains 1,024 candidates. Its completeness-certificate identity is sha256:0f4bdf024d8eab5ad6d448ceb713d2eddad5e4fcac806ddb933fabebf43d7917. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in claims/SFT-INFO-CHANNEL-CAPACITY-001/candidate_census.json.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
inputs	partial-input-carrier	An omitted input changes possible messages and capacity.	complete-canonical-inputs - Every registered input occurs once.
outputs	partial-output-carrier	An omitted output hides a possible transport image.	complete-canonical-outputs - Every registered output occurs once.
relation	opaque-channel-box	An opaque box hides source-to-output alternatives.	held-transport-relation - Every possible source/output pair is retained.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
totality	missing-input-row	An input without an output is outside the claimed channel action.	nonempty-output-support-per-input - Every input has at least one retained image.
confusability	probability-threshold	A threshold imports a distribution and free scale.	overlapping-output-support - Confusability is exact shared output membership.
codes	sampled-input-set	Sampling cannot establish every pair separated.	pairwise-disjoint-output-selection - Every selected input pair has disjoint output support.
capacity	logarithmic-rate	A real logarithmic rate imports scale and asymptotic assumptions.	complete-greatest-code-family - Every input selection is generated and all maximum distinguishable ties survive.
composition	endpoint-guess	Endpoints alone omit the shared interface path.	exact-relational-paths - Every source-to-middle and middle-to-target path generates one composite cell.
generality	sampled-channels	Examples do not establish a fresh input relation.	carrier-successor - A fresh input adds its complete output support and all new confusability pairs.
addition	extra-transition-probability	A probability law is not supplied by the transport relation.	no-extra-channel-model - Capacity follows only from exact relation and distinguishability.

11.3 Exhaustion, unique survivor and laws

The channel kernel is complete input/output carriers and held total transport, exact overlap confusability, exhaustive zero-error codes, the complete greatest code family, relational composition and no stochastic rule.

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-inputs__complete-canonical-outputs__held-transport-relation__nonempty-output-support-per-input__overlapping-output-support__pairwise-disjoint-output-selection__complete-greatest-code-family__exact-relational-paths__carrier-successor__no-extra-channel-model`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- channel output support retains every possible output of one exact input.
- confusability is symmetric overlap of output supports.
- a zero-error input code has pairwise disjoint output supports.
- capacity retains every greatest distinguishable selection rather than applying an undeclared tie rule.
- composed output paths cannot restore an input distinction already merged at an intermediate interface.

11.4 Operational demonstrations and adverse controls

- `clean-capacity` - Disjoint clean outputs retain all three inputs as one greatest code. Result: PASS.
- `merged-confusability` - Inputs a and b are confusable exactly through shared output x. Result: PASS.
- `tied-capacity` - The merged channel retains both greatest two-input alternatives rather than tie-breaking. Result: PASS.
- `code-control` - The a/b selection fails zero-error distinction while a/c passes. Result: PASS.
- `composition` - Composing two clean identity-interface channels preserves exact source/output paths. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An omitted input changes possible messages and capacity. Result: PASS; receipt
sha256: d2e6c5c6e7916f77a1634ae43952a1a0fd1cee72f93edf62ebee79333ab10d18.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - expected: halt any attempt to import an excluded object or answer-producing model Observed: no transition probability or stochastic channel cause; no floating error threshold; no logarithmic or asymptotic capacity proof value; no ungenerated infinite code family Result: PASS; receipt
sha256:213dbc3640d5133d828b84453a6831298f15d164097b8958314613d00e7e33fc.

11.5 Depth-independent closure

Base: One input with one nonempty output support forms a one-member distinguishable code family.

Successor: Adding one fresh input adds its exact output support and one confusability decision with every prior input. Every prior code either extends by the fresh input when all supports are disjoint or remains unchanged.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

11.6 Meaning and scientific consequence

A channel is a complete relation from each input to every output reachable through a registered path. Inputs are confusable exactly when their output supports overlap. A zero-error code is a held input family with pairwise disjoint output supports. Capacity is not imported as a logarithm or fitted rate: it is the complete family of greatest distinguishable input selections, with ties retained. Composition joins channels only across an exact interface and preserves path provenance, making every limitation traceable to concrete overlaps.

11.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: communication channel, confusability graph, zero-error code, channel capacity, channel composition. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

11.8 Exact exclusions and limitation

- no transition probability or stochastic channel cause.
- no floating error threshold.
- no logarithmic or asymptotic capacity proof value.
- no ungenerated infinite code family.

This theorem closes finite zero-error structural capacity. Probabilistic error rates and asymptotic coding theorems require separately generated finite evidence and cannot be imported here.

11.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
sha256:19a1d66b7f0222f90955bb27db339eb93e576cd935d33e9838d6b62b2a9acd00.
- Derivation seal: sha256:c98482b0294a53ba5676104d268e34c95322175800e0d6dd54553aefe8f9d37c.
- Independent implementation:
sha256:a14b1fa8e95876ffaf760fdca476e7870a63ed4b7135aa4e37e6fb33a845da75.
- Independent certificate:
sha256:c036dd4a381bdeb6aeb7d0802f54991db22bfa0febf3fd47ac834369923a017c.
- External validation:
sha256:fd90d9edab2dff3e7ed79b658fc48d9c66e285fe265229537de02f86718c89b5.

- Engine receipt: sha256:fb47c7f7be3e832cb67658042b2c7c3224113d9262cb4c45df7924a1f3a1cc01.
- Engine receipt path:
receipts/engine/model_admitted/SFT-INFO-CHANNEL-CAPACITY-001-fb47c7f7be3e832c.json.

12. Derivation 7: Exact noise, error, detection and correction boundaries

Claim identity: SFT-INFO-NOISE-ERROR-001

12.1 Question, necessity and theorem

Coding cannot be derived until the model states exactly what noise changes, what detection observes and why a correction is unique. Probabilities and metrics are not necessary for these structural boundaries.

The exact theorem statement is:

Noise is a registered source-to-received transformation relation over complete canonical word carriers; error is the exact changed-position trace between a held source and received form. Detection is forced when a nonidentity image lies outside valid code support. Correction is forced only when the received form has one registered source predecessor; multiple predecessors remain an explicit closed source class. No random cause, error probability, distance threshold or unrecorded predecessor inference is admitted.

Admitted dependencies: SFT-MATH-GRAPH-NETWORK-001, SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-INFO-ENTROPY-UNCERTAINTY-001.

12.2 Derivation chain, grammar and boundary

Channels supply source/output relations; dynamics supplies transitions and predecessor loss; entropy supplies closed classes; graphs supply paths. Ten structural axes force exact action provenance, error traces and singleton-predecessor correction.

Generation rule: Generate the complete product of source support, received support, transformation provenance, error trace, detection, predecessor classes, correction, deterministic status, generality and extra-noise status.

Exact grammar boundary: All finite error systems generated from canonical source and received words, held transformation actions, exact changed-position traces, valid code support and complete predecessor classes.

The complete product contains 1,024 candidates. Its completeness-certificate identity is sha256:65da8756ca47477887e29bed39c882022116b985ee68ad461d8e24e26a349b4f. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-NOISE-ERROR-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
source	partial-source-support	An omitted source hides a possible predecessor.	complete-canonical-sources - Every registered clean word occurs once.
received	sampled-received-forms	Sampling cannot establish the transformation boundary complete.	complete-registered-images - Every generated image under registered actions is retained.
action	opaque-perturbation	An opaque change hides which source produced which image.	held-action-relation - Each cell retains source, image and action identity.
error	scalar-error-size	A scalar count erases changed positions and labels.	complete-changed-position-trace - Every differing position retains old and new labels; identity is empty One.
detection	probability-threshold	A threshold imports a distribution and free parameter.	invalid-codeword-witness - A changed received form outside valid code support is detected structurally.
predecessors	chosen-likely-source	Choosing one predecessor imports a prior.	complete-source-class - Every registered source mapping to the received form is retained.
correction	nearest-form-assumption	Nearest requires an unforced metric and can tie.	singleton-predecessor - Correction occurs exactly when one registered source remains.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
cause	stochastic-noise-source	A random cause is neither generated nor needed.	deterministic-transformation-family - Noise names the complete registered action relation over deterministic forms.
generality	sampled-masks	Examples do not establish a fresh action image.	action-successor - A fresh registered action adds its exact source/image cells and updates predecessor classes.
addition	extra-error-rate-or-metric	A rate, distribution or metric is not supplied by exact transformations.	no-extra-error-model - Detection and correction use only code support and predecessor identity.

12.3 Exhaustion, unique survivor and laws

The noise/error kernel is complete source and received support, held action provenance, exact position traces, invalid-code detection, complete predecessor classes, singleton correction and no stochastic or metric rule.

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-sources__complete-registered-images__held-action-relation__complete-changed-position-trace__invalid-codeword-witness__complete-source-class__singleton-predecessor__deterministic-transformation-family__action-successor__no-extra-error-model`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- identity transport has structural empty-One error trace.
- nonidentity error retains every changed position and both labels.
- detection requires a received form outside the valid codebook.
- exact correction is equivalent to a singleton registered predecessor class.
- ambiguous received forms preserve all source predecessors rather than choosing a likely one.

12.4 Operational demonstrations and adverse controls

- `identity-error` - Identity transport returns structural empty-One error. Result: PASS.
- `position-trace` - The first-label change retains its exact position and labels. Result: PASS.
- `detection` - A one-label image outside the repetition codebook is structurally detectable. Result: PASS.
- `unique-correction` - The registered b/a/a image has exactly source a. Result: PASS.
- `ambiguous-control` - The shared b/a/b image retains both predecessors and refuses unique correction. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: An omitted source hides a possible predecessor. Result: PASS; receipt
sha256: aa0a31d6b2f952723af21b700fd42b6e3524208149527bf5e2a28a1ef1db1f4f.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no stochastic noise cause; no floating error rate or threshold; no assumed metric or nearest-neighbor rule; no unrecorded source inference Result: PASS; receipt
sha256: e9746338cf385cfd4197fa5fbf34f715cc9519738827c592e17a26421e0859f.

12.5 Depth-independent closure

Base: One source under its identity action yields itself, empty-One error and a singleton predecessor class.

Successor: Adding one registered action adds one exact image cell for each source on which it acts. Error traces compare those forms positionwise, and each affected received predecessor class extends by precisely the new source cell.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

12.6 Meaning and scientific consequence

Noise is not assumed to be random. It is the complete registered family of source-to-received transformations, each retaining the source, received form and action identity. An error is the exact changed-position trace, including the previous and received labels. Detection follows when a nonidentity image leaves valid code support. Correction follows only when the received form has one registered source predecessor. If several sources remain, the machine returns their whole class and refuses a likely, nearest or prior-weighted choice.

12.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: noise, error pattern, error detection, syndrome, decoding ambiguity. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

12.8 Exact exclusions and limitation

- no stochastic noise cause.
- no floating error rate or threshold.
- no assumed metric or nearest-neighbor rule.
- no unrecorded source inference.

This theorem closes finite structural noise and error. It does not assign a stochastic error frequency, physical noise mechanism or asymptotic reliability rate.

12.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
sha256:84c7f7977b3822e236f0b3710e5a7a01465c4024aaa58b6a847bca4c1adcc410.
- Derivation seal: sha256:d3852e47b6d6b9aaa1328367c912800ffca2d78396fb432d0e56b3c5dc66c744.
- Independent implementation:
sha256:b07e2fda2a8b23a887a84ace1c0cf08613a2802ccbb98fc7f20500ed962532b5.
- Independent certificate:
sha256:f5c174a125958108183e7b9ea19ad76cdf799784b6b4ca9a32aa9bd44def50d.
- External validation:
sha256:8dba26a7c7b76d11f393d114625f8fffd0486e1b8c3c9bda9f71c44d096cbb92e.
- Engine receipt: sha256:bc42a3ab0291a59b278654bed369bb1d64b6abe1f9278af9f89eac78211f48b9.
- Engine receipt path:
receipts/engine/model_admitted/SFT-INFO-NOISE-ERROR-001-bc42a3ab0291a59b.json.

13. Derivation 8: Exact finite coding, detection and correction

Claim identity: SFT-INFO-CODING-001

13.1 Question, necessity and theorem

Coding theory must establish correction from complete structural error alternatives. A metric or probability may summarize a sealed code later but cannot select which received forms decode uniquely.

The exact theorem statement is:

A finite code is a nonempty held selection of complete generated equal-position word support plus a total source/codeword relation. Codeword difference is the exact changed-position trace. A registered error family is detectable when every nonidentity image lies outside the codebook and correctable when distinct codewords have disjoint complete error-image supports. Decoding is then the unique source whose image family contains the received word. Redundancy is admitted only when those retained positions force separation.

Admitted dependencies: SFT - FOUNDATION - FOLD - ASSEMBLY - 001, SFT - MATH - COMBINATORICS - 001, SFT - INFO - COMPRESSION - REDUNDANCY - 001, SFT - INFO - NOISE - ERROR - 001.

13.2 Derivation chain, grammar and boundary

Fold assembly supplies word support; noise supplies registered transformations and predecessor classes; compression supplies lawful redundancy; combinatorics exhausts image families. Eleven axes force finite detection and correction through exact support separation.

Generation rule: Generate the complete product of word support, codebook selection, source encoding, position identity, difference trace, error-family coverage, detection, correction, redundancy witness, generality and extra-code status.

Exact grammar boundary: All finite codes generated from complete held-label word support, held codebooks, exact source maps, common position carriers, registered error actions and complete error-image families.

The complete product contains 2,048 candidates. Its completeness-certificate identity is sha256:b1ec186cbef105762fbf52baed16e644e0f2f0944e6aa739f3b4da39f160a368. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-CODING-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
support	partial-word-support	Omission hides possible code or error images.	complete-generated-word-support - Every held-label word over the registered positions occurs once.
codebook	free-valid-list	A free list lacks a complete source support boundary.	held-support-selection - The codebook is an exact nonempty selection of generated words.
encoding	implicit-assignment	An implicit assignment erases source provenance.	total-source-code-relation - Every source message has one retained codeword.
positions	unequal-or-unlabeled-positions	Position changes cannot be located on unequal or anonymous carriers.	common-canonical-position-trace - Every word uses the same retained ordered position carrier.
difference	scalar-distance	A scalar erases which positions and labels differ.	complete-difference-trace - Every changed position retains both code labels; equality is empty One.
errors	sampled-error-outputs	Sampling cannot prove image supports disjoint.	complete-registered-error-images - Every registered action image of every codeword is generated.
detection	likely-corruption	Likelihood imports a prior and threshold.	noncodeword-image - Every nonidentity registered image lies outside the valid codebook.
correction	nearest-codeword	A metric can tie and is not structurally supplied.	disjoint-error-image-support - Every received image belongs to at most one source codeword family.
redundancy	unaccounted-padding	Padding without a correction witness is an unforced addition.	separation-forcing-positions - Every added position participates in the exact disjoint-image certificate.
generality	fixed-mask-example	One mask does not establish all registered actions.	error-action-successor - A fresh action extends every codeword image family and disjointness is rechecked.
addition	extra-distance-or-rate-rule	A borrowed metric or asymptotic rate can select the code.	no-extra-coding-rule - Code status follows only from generated support and error images.

13.3 Exhaustion, unique survivor and laws

The coding kernel is complete word support, held codebook and source map, common positions, exact differences, complete registered error images, structural detection, disjoint-support correction and witnessed redundancy.

The engine recorded 2,048 decisions, 2,047 eliminations and exactly one survivor, `complete-generated-word-support__held-support-selection__total-source-code-relation__common-canonical-position-trace__complete-difference-trace__complete-registered-error-images__noncodeword-image__disjoint-error-image-support__separation-forcing-positions__error-action-successor__no-extra-coding-rule`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- valid codewords are a held selection of complete generated word support.
- difference retains every changed canonical position and both labels.
- detection requires all nonidentity registered images outside the codebook.
- correction is equivalent to pairwise disjoint complete error-image supports.
- a correctable received form decodes by exact unique source membership without a prior.

13.4 Operational demonstrations and adverse controls

- `support` - Two labels over three positions generate eight unique words. Result: PASS.
- `difference` - The repetition codewords differ at all three retained positions. Result: PASS.
- `detection` - Every single-label change of a width-three repetition codeword lies outside the codebook. Result: PASS.
- `correction` - The two complete single-label error-image supports are disjoint. Result: PASS.
- `decoding` - A received b/a/a word belongs uniquely to the all-a source family. Result: PASS.
- `unprotected-control` - The one-position uncoded family cannot correct a single label change. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission hides possible code or error images. Result: PASS; receipt
sha256: f7172e9412e2f058233eedc9292241ada463109f88a567019995e1f842fa7acb.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no stochastic error model; no floating distance or likelihood threshold; no unaccounted redundancy; no asymptotic rate imported as a finite proof Result: PASS; receipt
sha256: dbc8aa5bf22488e77f3b48616b7f6856c06cc2ed7350c94e878078caaed92030.

13.5 Depth-independent closure

Base: One registered error action generates one image per codeword; detection and support disjointness are exhaustively checked.

Successor: Adding one error action appends its exact image to every affected codeword family. Detection checks valid-word membership and correction checks every newly possible intersection with all other source families.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert

finite generation into a completed infinity.

13.6 Meaning and scientific consequence

Coding theory follows from complete error-image separation. A codebook is a held selection of generated equal-position words, and every registered error action is applied to every codeword. Detection requires nonidentity images outside the codebook; correction requires disjoint complete image supports between distinct codewords. Width-three repetition is demonstrated by exhausting all single-label images, while an unprotected one-position family supplies the adverse control. No distance metric or error probability selects the code; a conventional distance may summarize the sealed difference traces afterward.

13.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: block code, Hamming distance, error detection, error correction, repetition code. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

13.8 Exact exclusions and limitation

- no stochastic error model.
- no floating distance or likelihood threshold.
- no unaccounted redundancy.
- no asymptotic rate imported as a finite proof.

This theorem closes exact finite classical coding for any explicitly registered error family. Quantum codes, fault tolerance and unbounded error thresholds belong to later branches.

13.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
`sha256:6e8db3894be5bea7163c2a18dbffa6879fa56308a2b3d3fb49a56f0d98c920e5`.
- Derivation seal: `sha256:357b7cd32f117b55c0bad9606247bf12db72c82b999710d1c38698a5d895523d`.
- Independent implementation:
`sha256:a7b52a43add0bb3b0e84de51a71a55320eba8beae57793ef328833b89f6427c1`.
- Independent certificate:
`sha256:8502145761cfda22f034d5ca13deca41128060420454b3d93f689bd2d822e7f7`.
- External validation:
`sha256:d1be5779eb0de948cc041a05e504a3dc9bb66fba0985f16bf5ed8d015fdc6a66`.
- Engine receipt: `sha256:7c7648bdba686025c8b1f647b3e63a242da24234aa953cac184021f878b07711`.
- Engine receipt path:
`receipts/engine/model_admitted/SFT-INFO-CODING-001-7c7648bdba686025.json`.

14. Derivation 9: Exact mutual and conditional information from joint observation incidence

Claim identity: SFT-INFO-MUTUAL-CONDITIONAL-001

14.1 Question, necessity and theorem

Conditional and mutual information must preserve the observation cells that create the relation. Collapsing them into an imported logarithmic scalar would discard the exact structural proof.

The exact theorem statement is:

Conditional information is the complete exact restriction of one total observation to every class of another. Mutual dependence is the complete joint-incidence ledger comparing each observed joint support part with the product of its exact marginal parts; absent cells remain structural empty One and no signed subtraction is used. Factorization holds exactly when every joint cell is present and its part equals the marginal product. Thus conditioning and dependence are derived from deterministic support and observation, without priors, logarithms, stochastic causes or floating values.

Admitted dependencies: SFT-MATH-PROBABILITY-STATISTICS-001, SFT-MATH-COMBINATORICS-001, SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-QUANTITY-001.

14.2 Derivation chain, grammar and boundary

Probability supplies exact support parts, entropy supplies observation classes and quantity supplies retained distinctions. Ten axes force conditional restriction, joint incidence and exact factorization.

Generation rule: Generate the complete product of support, two observations, marginal classes, joint incidence, conditional restriction, dependence comparison, factorization, deterministic status, generality and extra-measure status.

Exact grammar boundary: All finite mutual and conditional information structures generated from complete deterministic support, two total observation maps, their exact marginal classes, complete joint cells and exact positive part relations.

The complete product contains 1,024 candidates. Its completeness-certificate identity is sha256:b11ba0d7ab94f6fa10d3651be8f6c6935d4c4a70787d3ff51648695c1c477ea3. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-MUTUAL-CONDITIONAL-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
support	partial-source-sup port	Omission changes marginal, joint and conditional parts.	complete-deterministic-support - Every registered source form occurs exactly once.
observations	partial-or-multiva lued-views	An incomplete or multivalued view cannot form exact classes.	two-total-observation-maps - Each source form has exactly one image in each view.
marginals	named-marginal-sco res	Scores erase which source forms produce each image.	exact-provenance-classes - Each image retains its complete source class and exact whole part.
joint	sampld-joint-cell s	Sampling can hide absent or repeated combinations.	complete-joint-incidence-ledger - Every pair of marginal labels has an explicit member class or empty One.
conditional	renormalized-prior	A prior or guessed normalization is not generated.	exact-class-restriction - Each target class is intersected with the complete held given class.
mutual	scalar-log-ratio	A log ratio imports real values and erases the cells responsible.	joint-versus-product-ledger - Every joint part is compared exactly with its marginal product while retaining provenance.
factorization	uncorrelated-asser tion	A label does not establish every joint cell and weight.	all-cell-exact-equality - Every joint cell exists and equals its exact marginal product.
cause	stochastic-source- premise	A random cause is neither forced nor required.	deterministic-observation-incidence - Dependence is a relation among exact observation classes over deterministic support.
generality	small-table-only	A table does not establish the next source update.	source-successor-update - A fresh source enters one marginal class and one joint cell, updating exact parts.
addition	extra-base-prior-o r-scalar	The added choice is a free parameter.	no-extra-information-measure - The ledgers follow only from complete support and total observations.

14.3 Exhaustion, unique survivor and laws

The mutual/conditional kernel is complete deterministic support, two total observations, exact marginal and joint classes, exact conditional restrictions, provenance-retaining joint/product comparison, all-cell factorization, successor closure and no stochastic or logarithmic measure.

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-deterministic-support__two-total-observation-maps__exact-provenance-classes__complete-joint-incidence-ledger__exact-class-restriction__joint-versus-product-ledger__all-cell-exact-equality__deterministic-observation-incidence__source-successor-update__no-extra-information-measure`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- conditional rows partition every nonempty given class by target observation.
- every conditional part is exact and positive; absent intersections are empty One.
- the joint ledger explicitly contains every marginal-label pair cell.
- factorization holds exactly when every joint cell equals its marginal-part product.
- dependence retains the responsible joint cells and comparison direction without signed subtraction.

14.4 Operational demonstrations and adverse controls

- `conditional-parts` - Conditioning the second coordinate on the first yields two exact half-parts in each given class. Result: PASS.
- `factorized-grid` - The complete two-by-two coordinate grid factorizes exactly. Result: PASS.
- `empty-dependence` - A factorized grid has structural empty-One dependence ledger. Result: PASS.
- `resolved-copy` - An observation is conditionally resolved by an exact copy of itself. Result: PASS.
- `dependent-copy` - Two identical nontrivial views do not factorize and retain their responsible cells. Result: PASS.

The engine-level unfavorable controls are:

- `false_premise` - expected: reject a generated form missing the first required structural condition Observed: Omission changes marginal, joint and conditional parts. Result: PASS; receipt
sha256: f564b6aa6b8ba61accc8d69530492f79dc2980f78cc08304bea6d57f9801d75d.
- `tampered_source` - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - expected: halt any attempt to import an excluded object or answer-producing model Observed: no logarithmic mutual-information proof value; no signed or floating dependence score; no stochastic cause or prior distribution; no incomplete joint census presented as factorization Result: PASS; receipt
sha256: 23dc6a258a994a86504934396406af14c8c3d49dbbe6eaf38d4864f3f45aca3b.

14.5 Depth-independent closure

Base: One source form gives one whole marginal cell, one whole joint cell and one whole conditional cell.

Successor: Adding one fresh source form places it in exactly one class of each observation and therefore one joint cell; all marginal, joint and conditional counts and exact parts are recomputed from retained membership.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

14.6 Meaning and scientific consequence

Conditional information is exact restriction: within every class of a given observation, the target observation partitions the retained microstates and supplies exact positive parts. Mutual dependence is kept as a complete joint-incidence ledger. Every joint cell records members, exact joint part, exact marginal parts, their product and the comparison relation; an absent cell is

empty One. Factorization holds only when every cell is present and exactly matches the marginal product. This avoids signed subtraction and logarithmic ratios while retaining all evidence needed to identify where dependence occurs.

14.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: conditional information, mutual information, joint distribution, marginal, independence. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

14.8 Exact exclusions and limitation

- no logarithmic mutual-information proof value.
- no signed or floating dependence score.
- no stochastic cause or prior distribution.
- no incomplete joint census presented as factorization.

This closes exact finite structural conditional information and dependence. Conventional scalar Shannon mutual information is a post-seal correspondence summary, not an SFT proof value.

14.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
sha256: f8ea0d5e7e10a01d084c85bd9cfff114e02014a63432f38483131101c8c0b2982.
- Derivation seal: sha256: 34061aefc9a9322a79eedb4a54ca2755ab8959dd0894e432e747f56be3f443f1.
- Independent implementation:
sha256: 9ab64f13d6feff5206edd252cece25cc802249f6d39064d3bfd9c39defa7138d.
- Independent certificate:
sha256: ebc36a627bb287d5108dd3714f18855e44ffd50b871fdb82e91235506ae8dfd9.
- External validation:
sha256: 390455380b90b13b9cca82426eefca22b65315665c69dceaa64c056e2089217.
- Engine receipt: sha256: c311f27de12d70ea2307acb2d5b23f83cb065b6b28fe5c08ac4742f6f1aad26.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-MUTUAL-CONDITIONAL-001-c311f27de12d70ea.json`.

15. Derivation 10: Exact information conservation, loss and transformation

Claim identity: SFT-INFO-CONSERVATION-LOSS-001

15.1 Question, necessity and theorem

Information conservation and loss require pair-level accounting: image counts alone cannot show which source alternatives remain recoverable or what record reversal requires.

The exact theorem statement is:

An information transformation is a total single-valued map over complete source support. Every source distinction is conserved as an image distinction or closed inside an exact predecessor fibre; these two ledgers are disjoint and exhaustive. Reversibility is forced exactly by singleton fibres. An irreversible transformation can be reversed only when a retained predecessor label resolves each non-singleton fibre. Under composition, downstream retained distinctions are a subset of those retained upstream, so a map creates no unrecorded source distinction.

Admitted dependencies: SFT-MATH-DYNAMICAL-SYSTEMS-001, SFT-MATH-LOGIC-PROOF-001, SFT-INFO-ENCODING-DECODING-001, SFT-INFO-CHANNEL-CAPACITY-001, SFT-INFO-MUTUAL-CONDITIONAL-001.

15.2 Derivation chain, grammar and boundary

Encoding supplies inverse classes, channels supply relations, mutual information supplies joint incidence and dynamics supplies composition. Ten axes force exhaustive pair accounting and the compositional loss law.

Generation rule: Generate the complete product of source support, totality, provenance, pair accounting, fibres, loss, reversibility, composition, generality and extra-source status.

Exact grammar boundary: All finite deterministic information transformations generated from complete canonical source support, total single-valued maps, exact image fibres, complete source-pair ledgers and transformation composition.

The complete product contains 1,024 candidates. Its completeness-certificate identity is sha256:ad0454f13ea0f144524fa1e94dc4d23a7cae305a1d392ba12f2bfe7dbff59923. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-CONSERVATION-LOSS-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
support	partial-input-support	Omission hides distinctions that may be lost.	complete-canonical-source - Every registered source form occurs once.
map	partial-or-multivalued-rule	An incomplete or multivalued rule cannot account for every source.	total-single-valued-map - Each source has exactly one held image.
provenance	image-only-output	Image-only output erases which sources entered each result.	exact-predecessor-fibres - Every image retains its complete source class.
accounting	output-cardinality-only	A count erases which distinctions survive.	complete-source-pair-partition - Every source pair is classified exactly as retained or closed.
loss	lower-score-assertion	A score does not identify unavailable predecessors.	non-singleton-image-fibre - Loss is the exact source distinctions closed inside an image fibre.
reversal	assumed-inverse	An inverse cannot choose among merged predecessors.	singleton-fibre-or-held-record - Singleton fibres reverse directly; merged fibres require retained predecessor labels.
composition	fresh-downstream-distinction	A deterministic image map has no source for a distinction already closed.	retained-pair-subset - A downstream map can preserve or close an upstream distinction but cannot reopen it.
conservation	unchanged-scalar	An unchanged scalar can hide exchanges among distinctions.	retained-plus-closed-exhaustion - The two exact ledgers disjointly exhaust all source distinctions.
generality	sampled-transformations	Examples do not prove a fresh source image accounted.	source-successor-map-cell - A fresh source adds one held map cell and all new pair comparisons.
addition	unrecorded-created-distinction	A new distinction without source or record violates provenance.	no-extra-information-source - All output distinctions trace to separated source images or declared external input.

15.3 Exhaustion, unique survivor and laws

The transformation kernel is complete source support, a total held map, exact predecessor fibres, exhaustive retained/closed source distinctions, singleton-fibre reversibility, held reverse records, compositional retention monotonicity and no unrecorded information source.

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-source__total-single-valued-map__exact-predecessor-fibres__complete-source-pair-partition__non-singleton-image-fibre__singleton-fibre-or-held-record__retained-pair-subset__retained-plus-closed-exhaustion__source-successor-map-cell__no-extra-information-source`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- retained and closed ledgers are disjoint and exhaust every source distinction.
- information loss is exactly the source distinctions inside non-singleton image fibres.
- a transformation is reversible exactly when every predecessor fibre is singleton.
- a non-singleton fibre requires a retained predecessor label for exact reconstruction.
- composition cannot restore a distinction already closed by an upstream deterministic map.

15.4 Operational demonstrations and adverse controls

- **identity-conservation** - Identity retains all six source distinctions and closes none. Result: PASS.
- **prefix-loss** - Prefix projection retains four cross-prefix distinctions and closes two exact predecessor pairs. Result: PASS.
- **reversibility-boundary** - Identity is reversible while prefix projection is not. Result: PASS.
- **reverse-record** - Each prefix image retains its two possible source labels. Result: PASS.
- **data-processing** - Erasing the retained prefix cannot reopen either previously closed distinction. Result: PASS.

The engine-level unfavorable controls are:

- **false_premise** - expected: reject a generated form missing the first required structural condition Observed: Omission hides distinctions that may be lost. Result: PASS; receipt
sha256:e49483a81fecb2b86c78c6d275a16553a1d404e42699dcfd48c68e3cc368ce89.
- **tampered_source** - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - expected: halt any attempt to import an excluded object or answer-producing model Observed: no image without source provenance; no assumed inverse across merged predecessors; no scalar conservation law replacing distinction ledgers; no downstream reopening without a declared external record Result: PASS; receipt
sha256:985dfefb5e8a4ab2440265fb290f9431a0a8e901052f776a68ff919c0b2fd65f6.

15.5 Depth-independent closure

Base: One source maps to one held image, has one singleton fibre and structural empty-One pair ledgers.

Successor: Adding one fresh source adds one image membership and one pair against each prior source. Image equality places each new pair exactly in the retained or closed ledger and updates one predecessor fibre.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

15.6 Meaning and scientific consequence

Information conservation is pair accounting. A total map classifies every source distinction as retained by unequal images or closed inside one predecessor fibre. Those ledgers are disjoint and exhaustive. Reversibility is equivalent to singleton fibres; a merged fibre requires a retained predecessor label. Under composition, the downstream retained pair set is a subset of the upstream set, so a deterministic map cannot reopen an unrecorded distinction. Conservation therefore does not mean an opaque scalar remained constant: it means every source alternative has an exact disposition.

15.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: data processing, information loss, reversible map, many-to-one map, sufficient record. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by

changing the sealed law.

15.8 Exact exclusions and limitation

- no image without source provenance.
- no assumed inverse across merged predecessors.
- no scalar conservation law replacing distinction ledgers.
- no downstream reopening without a declared external record.

This theorem closes exact finite deterministic information transformation. Thermodynamic energy costs and operational quantum measurement belong to later physical and quantum branches.

15.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
`sha256:60ac768f9abc68df9e028bcd206af56b72e1de369257164082dbf31d0bb92c61`.
- Derivation seal: `sha256:9c9e86f6923115194a041aea4ee7d8c49133e1653f5f60a17e809c782dbe2517`.
- Independent implementation:
`sha256:d13db48f1136b543655727389a65b1c1e0dacf8cdcfaae856163ef3ffb36d051`.
- Independent certificate:
`sha256:5b195602b942201b80b630c2745f56f5bb1998039039e358240583c72005c8c9`.
- External validation:
`sha256:11ccba3a43e1918ab7a42d9ce2437b65c11c8f37e12b82bfd420be0655c1755e`.
- Engine receipt: `sha256:4e2892e85c1624a2234163fdd51377a7086219ce0945cbf6acfaa41664303f4`.
- Engine receipt path:
`receipts/engine/model_admitted/SFT-INFO-CONSERVATION-LOSS-001-4e2892e85c1624a2.json`.

16. Derivation 11: Exact classical and probabilistic information correspondence

Claim identity: SFT-INFO-CLASSICAL-PROBABILISTIC-001

16.1 Question, necessity and theorem

The relation between classical and probabilistic information must be explicit in a deterministic theory. Observation classes derive exact probability while preserving every underlying state.

The exact theorem statement is:

Classical information is one held member of a complete deterministic support with its provenance retained. Probabilistic information is the exact observation-class ledger over that same support: every class retains its deterministic microstates and exact positive part of the whole. Observation resolves a classical state only for a singleton class; otherwise it returns the complete unresolved class. Deterministic transformations push the ledger forward by exact predecessor grouping. Probability therefore describes unavailable distinctions and does not introduce stochastic dynamics, floating propensities, numerical zero or priors.

Admitted dependencies: SFT-MATH-PROBABILITY-STATISTICS-001,
SFT-INFO-ENTROPY-UNCERTAINTY-001, SFT-INFO-MUTUAL-CONDITIONAL-001,
SFT-INFO-CONSERVATION-LOSS-001.

16.2 Derivation chain, grammar and boundary

Mathematical probability supplies exact parts, entropy supplies observation classes, mutual information supplies joint restriction and conservation supplies deterministic pushforward. Ten axes force one shared support account.

Generation rule: Generate the complete product of deterministic support, classical state, observation, exact class parts, probabilistic meaning, resolution, transformation, whole closure, generality and extra-random status.

Exact grammar boundary: All finite classical/probabilistic information descriptions generated from complete deterministic support, one held classical state, total observation classes, exact positive support parts and total deterministic maps.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:e03cac175631127fde7130ed62fb48d555e106539d14910987ceeece484ee25f`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-CLASSICAL-PROBABILISTIC-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
support	sampld-or-duplicatd-support	Sampling or duplication changes every exact part.	complete-canonical-support - Every deterministic microstate occurs once.
classical	unheld-symbol-name	A name alone does not locate a state in complete support.	one-held-supported-state - One exact support member is retained with its provenance.
observation	free-weight-list	Weights without microstates import a distribution.	total-observation-classes - Every microstate belongs to exactly one observed class.
parts	floating-propensities	Floating propensities import precision and causal assumptions.	exact-class-whole-parts - Each nonempty class count is retained relative to complete support count.
meaning	ontic-random-cause	A random cause is neither generated nor needed.	unresolved-deterministic-distinction - Each probabilistic class explicitly retains the deterministic alternatives observation closes.
resolution	likely-state-selection	Selecting the likely state imports a prior.	singleton-class-only - Only one-member classes resolve; larger classes remain complete.
transformation	stochastic-transition-matrix	A free transition matrix adds parameters and random dynamics.	deterministic-predecessor-grouping - A total map groups exact predecessor support into image classes.
normalization	fitted-normalization	A fitted factor can conceal omitted states.	partition-exact-whole - Disjoint complete class parts reconstruct the exact whole.
generality	finite-examples-only	Examples do not prove the next microstate update.	microstate-successor - A fresh microstate enters one class and every exact part is recomputed from membership.
addition	extra-seed-prior-or-kernel	The addition is a free parameter and may select results.	no-extra-stochastic-law - All probabilistic information follows from support, observation and deterministic maps.

16.3 Exhaustion, unique survivor and laws

The classical/probabilistic kernel is complete deterministic support, one held classical state, total observation classes, exact positive class parts, singleton-only resolution, deterministic pushforward, exact-whole closure, successor closure and no stochastic law.

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `complete-canonical-support__one-held-supported-state__total-observation-classes__exact-class-whole-parts__unresolved-deterministic-distinction__singleton-class-only__deterministic-predecessor-grouping__partition-exact-whole__microstate-successor__no-extra-stochastic-law`.

Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- a classical information state is one held member of complete support.
- a probabilistic row is an exact nonempty observation class and positive whole part.
- the complete class ledger reconstructs the exact whole.
- observation selects a state only when its retained class is singleton.
- deterministic pushforward groups predecessors without introducing random transitions.

16.4 Operational demonstrations and adverse controls

- **classical-held-state** - A held aa state retains its complete four-state provenance. Result: PASS.
- **exact-half-classes** - Prefix observation produces two exact one-over-two deterministic classes. Result: PASS.
- **exact-unequal-parts** - A three-member class and one-member class retain exact three-over-four and one-over-four parts. Result: PASS.
- **resolution-boundary** - Fine observation resolves aa while prefix observation retains the unresolved aa/ab class. Result: PASS.
- **pushforward-whole** - Deterministic prefix pushforward retains both predecessors and reconstructs the exact whole. Result: PASS.
- **deterministic-realization** - Every probabilistic prefix row exposes its deterministic member trace. Result: PASS.

The engine-level unfavorable controls are:

- **false_premise** - expected: reject a generated form missing the first required structural condition Observed: Sampling or duplication changes every exact part. Result: PASS; receipt
sha256:0132e22625c62b433b3964d5cbe1d88179a76521617716a231969f070854eee2.
- **tampered_source** - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - expected: halt any attempt to import an excluded object or answer-producing model Observed: no ontic stochastic transition premise; no floating or fitted probability; no numerical zero probability row; absence is empty One; no prior-based selection inside a non-singleton class Result: PASS; receipt
sha256:74818e0565fb579afa88110c99f2f9434067dfd932e1cf70649dd060b5e5128b.

16.5 Depth-independent closure

Base: One deterministic microstate is one held classical state and one exact whole observation class.

Successor: Adding one fresh microstate extends complete support and exactly one observation class. The class ledgers and their exact whole parts update from retained membership, while each individual state remains deterministic.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

16.6 Meaning and scientific consequence

Classical and probabilistic information are two resolutions of one deterministic carrier. A classical state is one held support member with its provenance. A probabilistic row is an exact nonempty observation class and its positive part of complete support. Observation identifies a state only when that class is singleton; otherwise all predecessors remain. A deterministic map pushes the ledger forward by grouping exact predecessor support. Thus probability records unavailable distinctions without adding a stochastic transition law, prior, seed or floating propensity.

16.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: classical state, ensemble, probability distribution, pushforward, epistemic uncertainty. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

16.8 Exact exclusions and limitation

- no ontic stochastic transition premise.
- no floating or fitted probability.
- no numerical zero probability row; absence is empty One.
- no prior-based selection inside a non-singleton class.

This theorem closes finite classical/probabilistic information correspondence. Empirical frequencies require separately registered blind data protocols and do not alter the formal law.

16.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
sha256: 30456ab6222e0f2cf74333305772e57515b2c1590739d8072d6393fed8364e5f.
- Derivation seal: sha256: 1c937455ec2a12940b3d1521f5a88e3c21a1b51bb5f4b6d432f6498765a499f0.
- Independent implementation:
sha256: e2c15ace1a24f33267a6d3d92e02b4cacd33c59853328561eff2869b0885c606.
- Independent certificate:
sha256: 185f98f5264a57295ac54685e839caf0f3bb9a4de766f1635d308d8563885763.
- External validation:
sha256: 11d9cf6b6aff03928993b4c343b0a01ec8a857acaf3f22c573a7fcda53d24ce3.
- Engine receipt: sha256: ce0515fdf69d368bddb313faad3a88043550e90e3df21c4049267e1a5e54fe7c.
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-CLASSICAL-PROBABILISTIC-001-ce0515fdf69d368b.json`.

17. Derivation 12: Exact quantum-information support correspondence

Claim identity: SFT-INFO-QUANTUM-CORRESPONDENCE-001

17.1 Question, necessity and theorem

Quantum information must first establish its exact alternative carrier and its relation to classical and probabilistic information. Operational quantum effects cannot be used to select that foundation.

The exact theorem statement is:

One Fold distinction supplies the native information unit. Across k canonical positions, the complete generated held-label word family supplies the exact branch support corresponding to a finite quantum basis support, and complete pair cells supply joint-state support. Calling this support superposition-equivalent asserts equality of enumerated alternatives and provenance only: amplitudes, phase and dynamics are not imported. Observation returns an exact retained branch class and a complete reconstruction record. Thus classical held states, probabilistic observation classes and quantum basis support share one exact information carrier at this boundary.

Admitted dependencies: SFT-FOUNDATION-FOLD-ASSEMBLY-001,
SFT-MATH-CATEGORY-TYPE-COMPOSITION-001, SFT-INFO-QUANTITY-001,
SFT-INFO-CONSERVATION-LOSS-001, SFT-INFO-CLASSICAL-PROBABILISTIC-001.

17.2 Derivation chain, grammar and boundary

Fold assembly supplies words, information quantity supplies position units, conservation supplies records, classical/probabilistic correspondence supplies held states and observation classes, and composition supplies pair cells.

Generation rule: Generate the complete product of native unit, word support, basis correspondence, branch provenance, superposition boundary, joint support, observation record, reconstruction, generality and extra-quantum status.

Exact grammar boundary: All finite quantum-information support correspondences generated from the two held Fold labels, canonical word positions, complete branch support, bijective basis names, pair-cell composition and total observation classes.

The complete product contains 1,024 candidates. Its completeness-certificate identity is `sha256:0530d875a4941751645201917bb60d46a2b0605d5fa734874b2c5f9e8fadb1`. The table states every generating axis, its unfavorable coordinate and its forced coordinate. Their full row-level Cartesian product is preserved in `claims/SFT-INFO-QUANTUM-CORRESPONDENCE-001/candidate_census.json`.

Structural axis	Rejected alternative	Exact rejection	Forced coordinate
unit	imported-qubit	A conventional qubit premise would select the carrier.	one-Fold-distinction - The minimal Fold's two held labels supply one exact distinction unit.
support	sampled-branches	Sampling cannot establish complete finite support.	complete-Fold-word-support - Every held-label word over every registered position occurs once.
basis	unnamed-state-count	A count erases which Fold word corresponds to which basis state.	bijective-word-basis-trace - Every word has one unique retained basis name and inverse.
provenance	amplitude-only-row	A coefficient without a generated word erases branch origin.	position-label-trace - Every branch retains every canonical position and held label.
superposition	complex-amplitude-premise	Complex amplitudes, phases and normalization are not derived here.	complete-alternative-support - Superposition-equivalence means the exact complete branch family only.
composition	tensor-symbol-premise	A borrowed symbol does not enumerate joint alternatives.	complete-pair-cell-support - Every left branch pairs exactly once with every right branch and retains both origins.
observation	stochastic-collapse	A random collapse rule is not generated at this boundary.	exact-observation-class - The observed image retains its complete branch predecessor class.
record	discarded-predecessors	Discarding the record prevents exact support reconstruction.	complete-branch-image-record - Every pre-observation branch and image remains available for reconstruction.
generality	small-depth-table	A table does not establish the next support layer.	position-successor - A fresh position appends every held label to every prior branch.
addition	extra-amplitude-phase-or-gate-law	Those structures belong to later forced derivations and would select results here.	no-extra-quantum-operation - Only the exact information-support correspondence is admitted.

17.3 Exhaustion, unique survivor and laws

The quantum-information correspondence kernel is one Fold distinction per position, complete Fold-word branch support, bijective basis and provenance traces, support-only superposition equivalence, complete pair-cell joint support, exact observation classes, reconstructing records, position induction and no imported quantum operation.

The engine recorded 1,024 decisions, 1,023 eliminations and exactly one survivor, `one-Fold-distinction__complete-Fold-word-support__bijective-word-basis-trace__position-label-trace__complete-alternative-support__complete-pair-cell-support__exact-observation-class__complete-branch-image-record__position-successor__no-extra-quantum-operation`. Replacing any survivor coordinate with the rejected coordinate triggers the table's failure condition; the extra-rule coordinate prevents an unforced addition. This proves minimality and named-shape uniqueness inside the frozen grammar. The survivor entails:

- one Fold position supplies one independently realized information distinction.
- k positions generate every held-label word exactly once.
- basis correspondence is bijective and retains each Fold word.
- joint support is the complete pair product with both source coordinates retained.
- observation plus the complete branch-image record reconstructs pre-observation support exactly.

17.4 Operational demonstrations and adverse controls

- **complete-branch-support** - Two held labels over three positions generate eight exact unique branches. Result: PASS.
- **unit-trace** - Each of three positions independently realizes one Fold information distinction. Result: PASS.
- **basis-bijection** - Every branch has one unique retained basis name. Result: PASS.
- **support-only-boundary** - Superposition-equivalent support explicitly refuses amplitudes at this boundary. Result: PASS.
- **joint-composition** - Two one-position supports generate four exact joint pair cells. Result: PASS.
- **record-reconstruction** - First-label observation retains four branches and its record reconstructs all eight. Result: PASS.

The engine-level unfavorable controls are:

- **false_premise** - expected: reject a generated form missing the first required structural condition Observed: A conventional qubit premise would select the carrier. Result: PASS; receipt
sha256: c421b75768ca85be4d8d1a3b5ed28f6aca315c61acf1d1cc47de8957a467774e.
- **tampered_source** - expected: reject a changed official source identity Observed: the changed identity differs from the registered source manifest Result: PASS; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - expected: reject a missing, duplicated or additional survivor Observed: the generated product contains exactly one registered survivor Result: PASS; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - expected: halt any attempt to import an excluded object or answer-producing model Observed: no complex, irrational or imaginary proof value; no amplitude or phase law; no stochastic collapse postulate; no interference, entanglement or gate semantics claimed at the support boundary Result: PASS; receipt
sha256: cd66bc50dc444891797124878cf737140b4cfffdea7096e0bbc3446294a36c4bf.

17.5 Depth-independent closure

Base: The first Fold position has the two held labels, one exact distinction and a two-branch basis correspondence.

Successor: Adding one fresh canonical position appends each held label to every prior word exactly once. Prior branch provenance remains intact, the new position realizes one distinction and joint-support laws remain pairwise.

The executed product closes the registered representation alternatives. The base/successor certificate extends that rule to every generated finite support size, word depth, source extension or action extension named by the claim. It does not convert finite generation into a completed infinity.

17.6 Meaning and scientific consequence

The quantum result is deliberately exact and bounded. One Fold distinction is the native information unit. Complete held-label words across canonical positions correspond bijectively to finite basis support, and complete pair cells supply joint support. 'Superposition-equivalent' means equality of enumerated branch alternatives and provenance only. Observation retains an exact branch class, while the complete branch/image record reconstructs pre-observation support. Amplitude, complex phase, interference, entanglement, gate dynamics and collapse are not claimed here; their absence prevents support correspondence from silently becoming an operational quantum theory.

17.7 Correspondence only after sealing

After the SFT survivor and receipt are fixed, the following established terms may identify translation points: qubit support, computational basis, superposition support, joint quantum state, measurement record. They are not dependencies and did not provide formulas, thresholds or answer tables. A mismatch remains visible as a boundary result rather than being repaired by changing the sealed law.

17.8 Exact exclusions and limitation

- no complex, irrational or imaginary proof value.

- no amplitude or phase law.
- no stochastic collapse postulate.
- no interference, entanglement or gate semantics claimed at the support boundary.

This theorem closes only finite quantum-information support correspondence. Phase, interference, entanglement, measurement dynamics, transformations, gates, circuits and quantum fault tolerance remain quantum-branch obligations.

17.9 Machine evidence identities

- Registered status: `independently_replicated`.
- Source manifest:
`sha256:19d64ff5e92da9623652423b85f4d069864f440697b2b6c62e125d987ef7d564.`
- Derivation seal: `sha256:936f0834e7c4b89ea1963b45071b23ab3e5a6aa4097b9f2564d0020746afdf0d.`
- Independent implementation:
`sha256:1ca65f3dd9401285c04ed7460d06560ce61479d12dc0d003e0411624e3f0d031.`
- Independent certificate:
`sha256:66aa35ac197317adc2f25d7dd83eba7b10710dd192e8f51e5b54270bc5aa8c1b.`
- External validation:
`sha256:70288b5837290911752ee6245cc2d6b7ff8c9169859b0adc6403c6e70e6d96da.`
- Engine receipt: `sha256:01279dbd48775d7f5fa92b0c9abe07dcafeb2ba670d10374d9f3bb355582cc81.`
- Engine receipt path: `receipts/engine/model_admitted/SFT-INFO-QUANTUM-CORRESPONDENCE-001-01279dbd48775d7f.json.`

18. Cross-derivation synthesis: one information kernel

The branch is one dependency chain. Canonical symbols and total observation first establish the source alternatives and their distinguishability. Encoding then maps those alternatives to generated words while preserving source classes. Quantity collects complete support and retained/closed pair ledgers. Entropy reorganizes the same evidence by observation class and exact support part. Compression asks whether a complete representation reconstructs the source and counts the entire representation system.

Channels generalize encoding to complete input/output relations and identify confusability by support overlap. Noise adds registered transformation identity and exact changed positions. Coding chooses generated word families whose complete registered error images are separated. Mutual and conditional information then compare two observation partitions through exact joint cells. Conservation classifies every source distinction under transformation, and the classical/probabilistic law shows that one held state and an unresolved observation class use the same carrier.

Quantum-information support correspondence is last because it needs all earlier results: Fold-word support, native distinction units, joint composition, observation classes and conservation records. It cannot retroactively import amplitudes or gates into the symbol law. Likewise, later channel or code correspondences cannot change the exact observation ledger that forced their prerequisites.

19. Entropy without logarithms and probability without randomness

The conventional scalarization of finite information support is not used as foundational evidence. A logarithm needs a chosen base and can produce irrational or floating values outside the admitted proof domain. SFT retains the exact object from which such summaries are ordinarily calculated: every observation class, every member, its exact positive part of the whole and every unresolved pair. A singleton class uses empty One; an absent joint cell uses empty One.

This richer object makes refinement transparent. If one observation refines another, every fine class lies within a coarse class and its closed-pair ledger is a subset of the coarse ledger. Conditional information is exact restriction to a given class. Dependence is the complete comparison between joint cell parts and exact marginal products. No negative deviation is

needed: the relation records equal, joint-below-product, joint-above-product or empty-One.

Superdeterminism presents no contradiction here because probability is not a transition cause. The generated microstate support and every registered transformation remain deterministic. Probability describes the part occupied by an observation class when source distinctions are unavailable. A prior, seed or stochastic kernel would be an additional parameter and is rejected unless a later empirical claim separately registers and validates it.

20. Compression, capacity and correction as exact finite support laws

Compression, channel capacity and error correction are often reported by scalar rates. This branch derives the finite objects those rates summarize. A lossless description includes its dictionary, token trace, parser and decoder. A channel includes every registered source/output/path cell. A code includes every registered error image of every codeword. The machine can therefore determine reconstructability, confusability, detection and correction without an error distribution or distance threshold.

The exact capacity object is the complete family of greatest pairwise distinguishable input selections. Retaining the whole family prevents a tie from being hidden by one arbitrarily selected code. The exact correction criterion is disjoint image support; decoding then has one source predecessor. Conventional length, rate, Hamming distance and capacity units may be computed as post-seal correspondence summaries for a declared finite carrier, but they cannot replace the support evidence or select the survivor.

21. Quantum-information boundary and later obligations

The branch closes a precise classical-probabilistic-quantum information boundary. A classical state is one held word. A probabilistic state is an exact observation class over complete deterministic word support. The quantum support correspondence is the complete generated Fold-word alternative family with a bijective basis trace. Pair-cell composition yields complete joint support. Observation and a retained branch/image record yield exact class reduction and support reconstruction.

Support equality does not establish operational equivalence. No complex amplitude, phase rule, interference operation, entangling law, measurement dynamics, unitary transformation, gate, circuit, quantum code or fault threshold is used or claimed. Those are separately enumerated obligations of Reversible and Quantum Computation. This separation is a positive result: it states exactly what information structure is already available and prevents downstream quantum phenomena from being smuggled into the carrier definition.

22. Formal computational evidence versus natural empirical evidence

The claims in this paper are formal structural theorems. Their empirical evidence is computational: complete candidate products are generated, every decision executes, adverse controls are actually perturbed, independent validator processes run and stored receipts are replayable. This validates the reported machine behavior of the declared grammar and certificates. It is not mislabeled as observation of an external natural system.

A later claim about natural measurements must register before target access, seal its formal predictor, enforce target custody, disclose all rows and preserve falsification results. Blind opaque prediction is insufficient because a score does not expose premises, candidate alternatives, eliminated structures, source identity, information leakage or the conditions under which the claim must halt. Application domains may test a sealed law but cannot select or repair it.

23. Complete V1/V2 ownership and same-strength reconstruction

Version 1.4 preserves the complete registered branch closure and places its exact findings and public scientific mission before the evidence inventory rather than treating filenames or hashes as scientific headlines. The audit read all 356 V1 theorem-manifest rows and all 407 V2 numbered results. Composite source statements were split into atomic obligations so

that an information theorem could not absorb an unperformed physical measurement, and a downstream physical interpretation could not make a closed formal information result disappear. Twelve V1 rows and twenty-five V2 steps contain Information Science-owned content. They decompose into 77 atomic obligations; all 77 map to model-admitted V3 receipts at equal strength and none remains open. The authoritative ledger is `census/information_science_prior_obligations.json`.

The reconstruction retains the strongest prior content: one native distinction closes per Fold observation; held labels reconstruct the source; exact equal shares partition the One; entropy is complete observation-class and unresolved-pair structure; predecessor merging is exact information loss; recurrent return retains its distinctions; support uncertainty, branch support, deterministic probability, channel provenance, reverse records, finite coding, conditional reconstruction and the classical-probabilistic-quantum support boundary all remain explicit. None is reduced to a historical note or accepted merely because V1/V2 stated it.

Prior source	Atomic obligations	Admitted V3 law labels	Result
V1 Q5	2	ENCODING-DECODING, QUANTITY	2/2 closed
V1 Q6	1	CLASSICAL-PROBABILISTIC	1/1 closed
V1 PH2	1	ENTROPY-UNCERTAINTY	1/1 closed
V1 D6	1	ENTROPY-UNCERTAINTY, QUANTUM-CORRESPONDENCE	1/1 closed
V1 N7	2	CONSERVATION-LOSS, ENTROPY-UNCERTAINTY	2/2 closed
V1 C2s	1	SYMBOL-DISTINCTION	1/1 closed
V1 C5s	1	SYMBOL-DISTINCTION	1/1 closed
V1 C8s	2	CONSERVATION-LOSS, SYMBOL-DISTINCTION	2/2 closed
V1 I-2	2	ENTROPY-UNCERTAINTY, QUANTITY	2/2 closed
V1 I-10	2	CONSERVATION-LOSS	2/2 closed
V1 G1	2	CLASSICAL-PROBABILISTIC, QUANTUM-CORRESPONDENCE	2/2 closed
V1 G3	2	CHANNEL-CAPACITY, QUANTUM-CORRESPONDENCE	2/2 closed
V2 Step 21	2	CONSERVATION-LOSS, ENTROPY-UNCERTAINTY	2/2 closed
V2 Step 22	1	ENTROPY-UNCERTAINTY, QUANTUM-CORRESPONDENCE	1/1 closed
V2 Step 28	1	CONSERVATION-LOSS	1/1 closed
V2 Step 38	2	CLASSICAL-PROBABILISTIC, QUANTUM-CORRESPONDENCE	2/2 closed
V2 Step 98	2	CONSERVATION-LOSS, ENTROPY-UNCERTAINTY	2/2 closed
V2 Step 123	2	CONSERVATION-LOSS	2/2 closed
V2 Step 159	2	CLASSICAL-PROBABILISTIC, QUANTUM-CORRESPONDENCE	2/2 closed
V2 Step 192	2	CONSERVATION-LOSS	2/2 closed
V2 Step 203	2	CLASSICAL-PROBABILISTIC, ENTROPY-UNCERTAINTY	2/2 closed
V2 Step 252	2	CHANNEL-CAPACITY, QUANTUM-CORRESPONDENCE	2/2 closed
V2 Step 257	1	SYMBOL-DISTINCTION	1/1 closed

Prior source	Atomic obligations	Admitted V3 law labels	Result
V2 Step 326	2	QUANTITY, SYMBOL - DISTINCTION	2/2 closed
V2 Step 328	3	ENCODING - DECODING, SYMBOL - DISTINCTION	3/3 closed
V2 Step 329	5	CLASSICAL - PROBABILISTIC, CONSERVATION - LOSS, ENTROPY - UNCERTAINTY, QUANTITY	5/5 closed
V2 Step 348	2	CLASSICAL - PROBABILISTIC, ENCODING - DECODING	2/2 closed
V2 Step 349	2	CONSERVATION - LOSS, MUTUAL - CONDITIONAL	2/2 closed
V2 Step 356	1	COMPRESSION - REDUNDANCY	1/1 closed
V2 Step 384	3	CLASSICAL - PROBABILISTIC, ENTROPY - UNCERTAINTY	3/3 closed
V2 Step 385	3	COMPRESSION - REDUNDANCY, ENCODING - DECODING	3/3 closed
V2 Step 386	3	CHANNEL - CAPACITY	3/3 closed
V2 Step 387	3	NOISE - ERROR	3/3 closed
V2 Step 388	3	CODING	3/3 closed
V2 Step 389	3	MUTUAL - CONDITIONAL	3/3 closed
V2 Step 390	4	CLASSICAL - PROBABILISTIC, QUANTUM - CORRESPONDENCE	4/4 closed
V2 Step 402	2	CHANNEL - CAPACITY, QUANTITY	2/2 closed

Categorical separation is part of the result. The formal information component of thermodynamic erasure closes here; the physical heat value remains a Physics obligation. Exact branch support and unresolved-part accounting close here; Born dynamics and natural measurement remain Physics and Quantum Computation obligations. Finite support uncertainty closes here; physical position-momentum measurements remain in Physics. Classical held-word copying and quantum-support correspondence close here; operational no-cloning, gates and fault tolerance remain in Quantum Computation. These are retained downstream requirements, not exclusions or deflations.

Complete-field family: SYMREP

13. Canonical alphabet generation

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 001 Claim:
SFT - INFO - SYMREP - ALPHABET - GENERATION - 001

Question, law and forced result. An alphabet is the complete duplicate-free generated family of nonempty canonical symbols retained at one declared representation boundary.

SYMREP-001 uniquely retains complete-canonical-alphabet, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001

The current reconciliation binds this obligation to the current model-admitted receipt sha256:e882bc27084f4e2a64e33620edc12d084a952447e6dc9f097b672e5e730644c4. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-001 product before observation access.

Boundary: Every positive finite SYMREP-001 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-001 uniquely retains complete-canonical-alphabet, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__complete-canonical-alphabet__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: f487af420025543280fba6f3578de61d0087965df87d0458847ccc478c4211c6; generality

sha256: 1fe555354ce4e76c9a6bfd95ccfb7adb8e92b4f0fefec1a1b4b50734e20a9.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt
sha256: 559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
sha256: cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256: 5ff6fa7a186fd3666a9475812a8f6705f5f600bdfefee065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate

sha256: 73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal

sha256: f92e6e07ecb4974e675506db8568f3778009e44811a7c4df02d40b046a5048ed. Source manifest

sha256: 8f26ce0fc34a41369573466a8be4763b76e3264b2b9907f2bb735924e26958d6. Engine receipt

sha256: e882bc27084f4e2a64e33620edc12d084a952447e6dc9f097b672e5e730644c4 at receipts/engine/model_admitted/SFT-INFO-SYMREP-ALPHABET-GENERATION-001-e882bc27084f4e2a.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER

- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMREP-001: canonical_alphabet
- exact observation: {"complete":true,"duplicate_free":true,"symbols":["a","b","c"]}
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 4ead16aee4a3b726524f128bc1b84c050ca26b2cb12e9d8f71120328663ce444; empirical validation sha256: 35840993911a02129a1e1c91aa3753899ecd827466fde4a099b453998dc58a73; external validation

sha256: c1f50ce1a5345b2c3ae98ca86e667b76b383aaf7711736b05a7ffe2658e2161b; isolation sha256: 888e4ed37bbe72d924e2fee3f5631f50f7676cf76ebed5f5665fe4ed452b87b9; custody sha256: c7da631811424f1f8856c2c118a1114bca9a5e0f9cd2f8f90de9a223e71ab9bb.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

14. Symbol identity and distinguishability

Family: SYMREP Obligation: SFT-INFO-OBL-SYMREP-002 Claim:

SFT-INFO-SYMREP-SYMBOL-IDENTITY-DISTINCTION-002

Question, law and forced result. Symbol identity is canonical construction identity; two symbols are observationally distinguishable exactly when a total source-bound observation retains different labels.

SYMREP-002 uniquely retains source-bound-observation-classes, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-MATH-HAND-CROSS-BRANCH-COMPLETENESS-006
- SFT-INFO-QUANTUM-CORRESPONDENCE-001
- SFT-INFO-SYMREP-ALPHABET-GENERATION-001

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: cdcbf2734f3510b2b9cef328d695c9cefff9467419b6bc2822a3d9f308112e18. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-002 product before observation access.

Boundary: Every positive finite SYMREP-002 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-002 uniquely retains source-bound-observation-classes, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__source-bound-observation-classes__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:6563e6e69596002e38313c29bce0141fe2a276044c7ba3c4aca45d2efb05a379; generality

sha256:5ba8e3783b45cfada5ad996920e83c394718aa32e37eb638966342a49ee535f1.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt
sha256:559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
sha256:cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256:5ff6fa7a186fd3666a9475812a8f6705f5f600bdfef065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate

sha256:73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal

sha256:bf657491aaa7f8a76a8eeafdf85b34ef8268880ca3664a663d526fb1c9bf388a. Source manifest

sha256:47fd2937e4c8f2efd564626498bff2609678e3eb6e42a5b8f8f44cb90cea04d1. Engine receipt

sha256:cdcbf2734f3510b2b9cef328d695c9cefff9467419b6bc2822a3d9f308112e18 at receipts/engine/model_admitted/SFT-INFO-SYMREP-SYMBOL-IDENTITY-DISTINCTION-002-cdcbf2734f3510

b2.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMREP-002: observation_classes
- exact observation: {"coarse_classes": [{"L"}, {"a"}, {"b"}], [{"R"}, {"c"}]}, {"fine_class_count": 3, "microforms_retained": true}
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 4536d01baefed750d40e1a69e0b253d469e49c23886fb0ddcb715f823148d8f9; empirical validation sha256: 4de4f023fd98e4b08852cfa78b4e5826749d9fe98bb989c86b024661267bdf3d; external validation

sha256: 7d2372264ecbbe8c7ddfb5f8096e9f6b71b7905b8ec63f728616b7e50ed56b0c; isolation sha256: ddf5ebb65966d3e18ba63758972409c396b429170472fa219946d681ee4d8384; custody sha256: 5c7a664873cfe11d9fcd2e1eaa9e12fef01ebfaad79222bf338066899e3d0011.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

15. Codeword formation and parsing

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 003 Claim:

SFT - INFO - SYMREP - CODEWORD - PARSING - 003

Question, law and forced result. A codeword representation is exact concatenation of registered symbol images, and decoding is the complete enumeration of source words whose images equal the retained stream.

SYMREP-003 uniquely retains exact-concatenation-and-parse, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - SYMBOL - IDENTITY - DISTINCTION - 002

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: a382c2a8a75540d864aef2b338e235014c97adb731402dd745a24f27a92daa45. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-003 product before observation access.

Boundary: Every positive finite SYMREP-003 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-003 uniquely retains exact-concatenation-and-parse, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__exact-conc
atenation-and-parse__complete-boundary-aware-parse__complete-declared-product__root
-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explic
it-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:c1253def76cf084be0a742bf617d43437dbd61509d7d57b7d99acc55ef9e8505; generality
sha256:600630bcfba85cbf2529d375fb314332f705619390b02bb2546d177cd379934c.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt
sha256:559c879478b1f23f23be5edd952222106ad01e8c272129bbcf21343db735b61c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
sha256:cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256:5ff6fa7a186fd3666a9475812a8f6705f5f600bdfefee065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate
sha256:73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal
sha256:be8bec10906cf83fa5c2e938de2d27effcf67e84af9cb6925062b8a2be05d019. Source manifest
sha256:3dcfc10cb32763620fb738c25292fc5afa5a2bf36a70513ccbe3e4d563049c54. Engine receipt
sha256:a382c2a8a75540d864aef2b338e235014c97adb731402dd745a24f27a92daa45 at receipts/e
ngine/model_admitted/SFT-INFO-SYMREP-CODEWORD-PARSING-003-a382c2a8a75540d8.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMREP-003: codeword_parsing
- exact observation: {"encoded_units":["L","R","L","R","R"],"parse_count":1,"source_word":["a","b","c"]}
- complete family observation custody: 14 records; all rows preserved

- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:79ead775c88ad4007c6895836fa4450fd7ee851ad91bee4ce6bce1ac477b6231; empirical validation sha256:e849026eaf78ba560b1a03776fd353b5afb34a7dc3d23dc70d28d02212752b8; external validation

sha256:3aaf7aacdd9d03d06890b305dc65b717426f4e4edd8aee86b9f79cf50f362262; isolation sha256:f99f1ec995ca4a2579d744f267245edc6924fe3c027b94cb02526673e6ae9416; custody sha256:d407665a7c82ea52de0520c289d260d9c226383057dfa69617fc69df8cea1271.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

16. Prefix, uniquely decodable and instantaneous structure

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 004 Claim:

SFT - INFO - SYMREP - PREFIX - UNIQUE - DECODING - 004

Question, law and forced result. A prefix representation has no codeword as a proper initial segment of another; complete finite-word enumeration then forces instantaneous unique decoding.

SYMREP-004 uniquely retains prefix-free-unique-decoding, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - CODEWORD - PARSING - 003

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:6a565d3313fe90878e104066b71a405fce310317b0346215bbc972eab7a6d4c3. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-004 product before observation access.

Boundary: Every positive finite SYMREP-004 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-004 uniquely retains prefix-free-unique-decoding, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__prefix-free-unique-decoding__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit

-boundary.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256:635888f979fd7b5f2726d1ed73f52d48939cee40a9ab7332eb2e3df16b2fed08; generality
 sha256:1a7c2f72339e52744e5de5dc60e2c49df23d13bd7af7dee00b425fc1a1db1c64.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt
 sha256:559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
 sha256:cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256:5ff6fa7a186fd3666a9475812a8f6705f5f600bdfefee065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate
 sha256:73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal
 sha256:dfbd894fba00720bf8c89444a6a5218cd51a52a3111532f1d3701c3fcf9466bd. Source manifest
 sha256:6fb1e239e0890c13d57987f1a50b22c07357ca61ea38b57cdd9b91baf2d528c3. Engine receipt
 sha256:6a565d3313fe90878e104066b71a405fce310317b0346215bbc972eab7a6d4c3 at receipts/engine/model_admitted/SFT-INFO-SYMPREP-PREFIX-UNIQUE-DECODING-004-6a565d3313fe9087.js
 on. Closure scope `depth_independent`. External status
`empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMPREP-004: `prefix_unique_decoding`
- exact observation:
 {"all_unique":true,"codebook":{"a":["L"],"b":["R","L"],"c":["R","R"]},"generated_words_checked":39}
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:973ca363de91096db1ad01c0cfdc715f3f2030fb2ac60a558562a31e4c40de7a; empirical validation sha256:f12304dc0a947ffcc26fee06b55a497b864aa8ecd33b56777ed3826dda2b2542; external validation sha256:c874dc5c6d5daa1b492e98322e4c946c0003e08f086bcad27afc090616b638eb; isolation sha256:4e57d98f7c325afc7a3ede4010daa48f485e2b7445513b19a58f1f527918ec59; custody sha256:e0745cba3f49c2d9c30e4464cc00337df8ea2b543333cee913fdac19be776c2f.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

17. Grammar-constrained representation

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 005 Claim: SFT - INFO - SYMREP - GRAMMAR - REPRESENTATION - 005

Question, law and forced result. A grammar-constrained representation is exactly the closure produced by declared source-bound rewrites through the registered depth or successor certificate, with no ungenerated string admitted.

SYMREP-005 uniquely retains generated-grammar-language, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - PREFIX - UNIQUE - DECODING - 004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:1873a6a732469f5a8999951879ddb6c7a4c27304f311c3d3cea4558fa62b64bf. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-005 product before observation access.

Boundary: Every positive finite SYMREP-005 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-005 uniquely retains generated-grammar-language, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__generated-grammar-language__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: 77fab3236afadd25c5d4dc51c8ee8cf6bc2585814348b151930b2d60048f0ebb; generality
 sha256: c40a8f2758a70e4b8d71f6170f3314369cf5016a7d99b38a1937a1787960d0e5.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt
 sha256: 559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
 sha256: cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256: 5ff6fa7a186fd3666a9475812a8f6705f5f600bdfef065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate
 sha256: 73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal
 sha256: e072491ccfa1885a9f4361f0d0c3c494930ab49e9c46462bfaad1572c7b58649. Source manifest
 sha256: 81fc102a9595b23d66b4faef75857acf28c12b49011c5b44745da00217dd8d0b. Engine receipt
 sha256: 1873a6a732469f5a8999951879ddb6c7a4c27304f311c3d3cea4558fa62b64bf at `receipts/engine/model_admitted/SFT-INFO-SYMREP-GRAMMAR-REPRESENTATION-005-1873a6a732469f5a.js`
 on. Closure scope `depth_independent`. External status
`empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMREP-005: `grammar_language`
- exact observation: `{"complete_form_count":7,"depth":3,"forms":["S","aS","aaS","aaaS","aab","ab","b"]}`
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or

opened out of order.

Measurement receipt

sha256:7e13a451507927e8b78c54a9fe9c22713c12364b4f3c9f316e71d4ad99f11f2e; empirical validation sha256:d0273462e7b09a428a41e511df96bc921f16aaa3f4ea877e06ad3f806f62bd94;

external validation

sha256:17d9c43ea406fa0043370eb0c57cc962316414ca4881ab9aeeeb06096155a531; isolation

sha256:7b2b028207154af6aec063a327b37fdd446d0110adbeab3b49bc52f53aa24051; custody

sha256:059d2f4922ad8d48e6ddb6c5427e776da20d7f5468d2d3979408f3cc21c9f3d2.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

18. Representation equivalence and isomorphism

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 006 Claim:

SFT - INFO - SYMREP - EQUIVALENCE - ISOMORPHISM - 006

Question, law and forced result. Two finite representations are isomorphic exactly when a bijective symbol translation preserves and reflects every registered relation and admits an exact inverse.

SYMREP-006 uniquely retains bijective-structure-preserving-renaming, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - GRAMMAR - REPRESENTATION - 005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:a3f427723ba87a3f3489fe87c4d8e17aa8e628a0425b69a2f5c27da04ac12ddd. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-006 product before observation access.

Boundary: Every positive finite SYMREP-006 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-006 uniquely retains bijective-structure-preserving-renaming, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__bijective-structure-preserving-renaming__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:033ab83b71e945709054db63a8da305315458eb9614fad67ee39613ad0692290; generality

sha256:7b92510b14127a7f7c4222651b1cbe4d095c674934e0a5be22805e94a0a82f23.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt
sha256: 559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
sha256: cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256: 5ff6fa7a186fd3666a9475812a8f6705f5f600bdf065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate

sha256: 73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal

sha256: 58500d28d54f4dccf509fa0d2b27a07477e2abcaa5838370d3a05af4f107c25c. Source manifest

sha256: 1b4b0a03e9527fc417891ebfba5517267aaa1c67f55fd75248729dd4e544f98c. Engine receipt

sha256: a3f427723ba87a3f3489fe87c4d8e17aa8e628a0425b69a2f5c27da04ac12ddd at receipts/engine/model_admitted/SFT-INFO-SYMREP-EQUIVALENCE-ISOMORPHISM-006-a3f427723ba87a3f. json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMREP-006: representation_isomorphism
- exact observation:
{"bijjective":true,"preserved_and_reflected":true,"relation_rows":2,"source_symbols":3,"target_symbols":3}
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:178041b5ad3ca2721aae187a02181e43df7872ae8a99c71c1f1de92ef4f13ce6; empirical validation sha256:7735244c085dcc2b25c93706ac8c29dbcc1af8d22c32edd8812ceef2f98f8445; external validation sha256:d6eaa454a6c68e0eb3a880bdc1b4202b6a632cf9baf784b4b3d1dfc466d16209; isolation sha256:cb46ebdf4fc764e5dc2dbfd7dcc9b040a29881f08f2c771b0cf8d125a7df699f; custody sha256:426329467e9543d9f7079e843509b0b5624974e2af767ba5b7819b7a7a03e03c.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

19. Canonicalization and normalization

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 007 Claim:
SFT - INFO - SYMREP - CANONICAL - NORMALIZATION - 007

Question, law and forced result. Canonicalization is a terminating provenance-retaining rewrite to one canonical representative and normalization is idempotent on every generated token.

SYMREP-007 uniquely retains terminating-idempotent-canonicalization, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - EQUIVALENCE - ISOMORPHISM - 006

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:2f2885b4d6a1e043e5985511eb510c58db514ad30c8c16042adb7c6883b75859. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-007 product before observation access.

Boundary: Every positive finite SYMREP-007 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-007 uniquely retains terminating-idempotent-canonicalization, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__terminating-idempotent-canonicalization__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:2a6adfa80d4882fdc9a1137c335664e345b8caccbe5a1b17be5efacc53cd32e5; generality sha256:4ffd8b83ced575469e231bc7b6e1751602d4b3c5fa98f41e55b7d7a07f1dbd16.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt
sha256: 559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
sha256: cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256: 5ff6fa7a186fd3666a9475812a8f6705f5f600bdfef065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate

sha256: 73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal
sha256: 35ff8a08e6e01539c069194ef07f1c687a49467eb140b4d432a8edeef23f1fe. Source manifest
sha256: 7032d4d7f186452472c0b64a0539b8188a46d164b1a1b03972113544adab0b9f. Engine receipt
sha256: 2f2885b4d6a1e043e5985511eb510c58db514ad30c8c16042adb7c6883b75859 at receipts/engine/model_admitted/SFT-INFO-SYMREP-CANONICAL-NORMALIZATION-007-2f2885b4d6a1e043.json. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMREP-007: canonical_normalization
- exact observation: {"canonical": "a", "idempotent": true, "rewrite_chain": ["A1", "A", "a"]}
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 46a51fc2baa11fe1a286ebd164b988d18a6510020b8ab314c02b3afc86dfe634; empirical

validation sha256:4fbbfa1b0dc8bc42dfdc55fe76e86f1ce1f42bad6ab679ea822bb8db075dcae7;
 external validation
 sha256:4f7674cfb966ffdf2dafbcc884931748858e60c6fabd969209af8486084b4f4; isolation
 sha256:222c31f5d8eb73b8434f9fd3533da7ae5d91b6ef5d6868e95dfc02c36a7256bb; custody
 sha256:e8aa487b415fe4ca42a98df79ad118b6d0d7c054829b28cc0d6a7127bfccb38f.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

20. Variable-length representation boundaries

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 008 Claim:
 SFT - INFO - SYMREP - VARIABLE - LENGTH - BOUNDARY - 008

Question, law and forced result. Variable-length representation is decodable only when prefix structure, an explicit delimiter or an exact length record retains every codeword boundary; otherwise every lawful parse remains open.

SYMREP-008 uniquely retains retained-self-delimiting-boundary, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - CANONICAL - NORMALIZATION - 007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:c35239fea7e66aeaf96bf2fbee3df1c4c35033b0e25534b298dbdc6369fd555a. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-008 product before observation access.

Boundary: Every positive finite SYMREP-008 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-008 uniquely retains retained-self-delimiting-boundary, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__retained-self-delimiting-boundary__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:8e66e07af5155a1451b29f5244819be5f20f289612ddfec16906b7caa104b9cd; generality
 sha256:c29b1686ad4fae41770fbf2c68c69ac925de4ec22477512694225ae3114243b1.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted

magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt
sha256: 559c879478b1f23f23be5edd952222106ad01e8c272129bbcf21343db735b61c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
sha256: cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256: 5ff6fa7a186fd3666a9475812a8f6705f5f600bdfefee065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate

sha256: 73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal

sha256: 3c706aa8a6cd9d79b4270eb8dd3552bdb0b3a926c7f44c1c9087247cf0fb5d77. Source manifest

sha256: 06c4475af5c1011c03b2d0aaf0f0522b186626d0b60f564b375e5ab793bf2652. Engine receipt

sha256: c35239fea7e66aeaf96bf2fbee3df1c4c35033b0e25534b298dbdc6369fd555a at receipts/engine/model_admitted/SFT-INFO-SYMREP-VARIABLE-LENGTH-BOUNDARY-008-c35239fea7e66aea.

json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMREP-008: variable_length_boundary
- exact observation:
{ "boundary_required": true, "prefix_code_unique": true, "unbounded_parse_count": 2, "unbounded_stream": ["L", "L"] }
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: fdf3230bcb0ba8b6bb57b5d2ff5b7a74a14e3da327a5c02ce602b06994643b32; empirical

validation sha256: b51cc10e3b25c1120aec93bad95bd0670d0380703ee680b2e70125a5265d2322;

external validation

sha256:d3727760e0761990681df8987377c259e1c9edcb2e0f356e3e021a9c79df3e8a; isolation
 sha256:faa82e5575f03f78f24a635e189dda48e083d4a742f958432722f7c7519b87cc; custody
 sha256:26aed815f18b6f2ef77542ad8fc88f204545a4b2b5325b552ea244a9c4ffe8d6.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

21. Composite and product alphabets

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 009 Claim:
 SFT - INFO - SYMREP - PRODUCT - ALPHABET - 009

Question, law and forced result. The product alphabet is the complete generated ordered pair support of its component alphabets, retaining both coordinates and every cross-pair exactly once.

SYMREP-009 uniquely retains complete-ordered-product-alphabet, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - VARIABLE - LENGTH - BOUNDARY - 008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:f79e99c7b376f9a44004bb229c7e159817e7a35c769d771cca253fb83202e170. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-009 product before observation access.

Boundary: Every positive finite SYMREP-009 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-009 uniquely retains complete-ordered-product-alphabet, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__complete-ordered-product-alphabet__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:2f8168805fcfa436447d55c3c5db2e06ee7853304ea9b828852b3c251cab6171; generality
 sha256:7c54874a5354a47b32f4da25455ad4a8af2b189be24e4b819130f002d9e57bfa.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt
sha256: 559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
sha256: cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256: 5ff6fa7a186fd3666a9475812a8f6705f5f600bdfef065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate

sha256: 73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal

sha256: bb331e3574d6dd595731632a3dc81b1efeb9f0312e6fbfd331ed5a90369e8d5a. Source manifest

sha256: 05a161552c6c6a67dd0221afe1007d482769ba01cabb3699ad4dfb5b5852d870. Engine receipt

sha256: f79e99c7b376f9a44004bb229c7e159817e7a35c769d771cca253fb83202e170 at receipts/engine/model_admitted/SFT-INFO-SYMPREP-PRODUCT-ALPHABET-009-f79e99c7b376f9a4.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMPREP-009: product_alphabet
- exact observation: {"all_cross_pairs_once":true,"left_size":2,"ordered_pair_count":6,"right_size":3}
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 6009ba8bee53c3de43df2b3e035b4111be84d68364a30f64fbd8e3f3d2ca6a6; empirical validation sha256: 33e77cd3a98b6a847e7f3987735b9043d126c7f6bd8eb6cad57581132463823; external validation

sha256: 3b8385b578205f777efb0b35ba2ea8c7fbda3897b9ece4d4f8c7840504381e01; isolation

sha256: 08a537db693c17e866e2a6d6014179d198f666484508bbadf9d369760a4d31a3; custody

sha256: c72a1804d58beb7073b85ad665ba19eec1e2276d4ef1d3601326b5e31c18fb92.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

22. Hierarchical and typed symbols

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 010 Claim:
SFT - INFO - SYMREP - HIERARCHICAL - TYPED - SYMBOL - 010

Question, law and forced result. A typed symbol retains its type label and a value from exactly that type's generated fibre; hierarchical symbols recursively retain every enclosing boundary.

SYMREP-010 uniquely retains source-bound-dependent-symbol, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - PRODUCT - ALPHABET - 009

The current reconciliation binds this obligation to the current model-admitted receipt sha256:5ed07dc03dc62e4cbebe78449e0b483fef0a1e530bbb6c912906f6b2964c565b. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-010 product before observation access.

Boundary: Every positive finite SYMREP-010 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-010 uniquely retains source-bound-dependent-symbol, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__source-bound-dependent-symbol__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:5e5cdc10d408a2657eddbc3d8bef84bf08dc5aea36387b222424cecab179f390; generality sha256:461750943f77c5a6616b5c847a2ca86081cbd0494b8853855b4ee43c0d10c1a7.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt

sha256:559c879478b1f23f23be5edd952222106ad01e8c272129bbcf21343db735b61c.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
sha256:cb61f696ce7f4e7c9db9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256:5ff6fa7a186fd3666a9475812a8f6705f5f600bdf6e065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate
sha256:73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal
sha256:7a34e2cc74bec964b75d44d2a9a537882658cf6095e388123c8142e56a81ecf3. Source manifest
sha256:73efa901646178dd0540485299a2b0b1285040fc892faed1820bffc6a9225b02. Engine receipt
sha256:5ed07dc03dc62e4cbebe78449e0b483fef0a1e530bbb6c912906f6b2964c565b at receipts/engine/model_admitted/SFT-INFO-SYMREP-HIERARCHICAL-TYPED-SYMBOL-010-5ed07dc03dc62e4c.json. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMREP-010: typed_symbol
- exact observation: {"cross_fibre_value_rejected":true,"fibre_sizes":[2,2],"type_count":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:c933165280f81b97ce131af6d3775a06319f6b40a3eb4a8b475330988968bba7; empirical validation sha256:cf45d77e370715b760ba7c68c7452cc0eb5e497041dcec1b281134adee67274f; external validation
sha256:c51bc67f263de4683f776fa6465cc6e23a58b9561aec9c343560e1fac49a4acd; isolation
sha256:06a05b50dd9f37bb06b5c5e19f4f087d58077f80b3f2c3ac4a3555ba33a72d9d; custody
sha256:3eaa28a1c9dcfd1e3ed05804294ad6fe11be194c021aea55cc5725570d64281a.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result

- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

23. Representation conversion and transduction

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 011 Claim:
SFT - INFO - SYMREP - CONVERSION - TRANSDUCTION - 011

Question, law and forced result. A deterministic representation conversion is a total single-valued source-bound relation, and composed conversions equal the exact relational composition of their retained maps.

SYMREP-011 uniquely retains total-single-valued-provenance-transduction, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - HIERARCHICAL - TYPED - SYMBOL - 010

The current reconciliation binds this obligation to the current model-admitted receipt sha256:89ac0f4f8a9bb94981cd6b6da8c73af5ccf54948541e82412455c0ab7c76e65d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-011 product before observation access.

Boundary: Every positive finite SYMREP-011 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-011 uniquely retains total-single-valued-provenance-transduction, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__total-single-valued-provenance-transduction__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:8a263401fd0c6fb57d7233a18d84b3bae5b51dc0b3b5936d4055738009e861fa; generality sha256:78c20cf04c3896f4a76729205ee2670a4dbfd58d6d7283d8cb739d3e185a4adf.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt sha256:559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt

sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
sha256:cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256:5ff6fa7a186fd3666a9475812a8f6705f5f600bdfef065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate

sha256:73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal

sha256:2ed5179ab0b762a07800c7c7b7a4aaa65c3e23273e81e0478519dd53964498a7. Source manifest

sha256:ed1fbcfeccfbfb61652a8cdfbb48e5ab92cecc8ab5b1f0642d9294fff6bbf295. Engine receipt

sha256:89ac0f4f8a9bb94981cd6b6da8c73af5ccf54948541e82412455c0ab7c76e65d at receipts/engine/model_admitted/SFT-INFO-SYMREP-CONVERSION-TRANSDUCTION-011-89ac0f4f8a9bb949.j

son. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMREP-011: transduction
- exact observation:
{ "composition_equal":true, "intermediate":["x","y","x"], "source_word":["a","b","a"], "terminal":["L","R","L"] }
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:29e31c363f06e46d55404fc3cce00852202568be2ed5dbfd44f55cc57a0a3e2d; empirical validation sha256:42aa7cc73688fe916a7cc302a76b0a4862bb74ced0576d377db023f5f289743e; external validation

sha256:78d66338e707656f0b7b6c25d032d4cf8a539b26aad494030e2fcd79179ea527; isolation

sha256:2e4dd5911bea33b33596f4c018cc03874df68401606d6c87bfb8718a52acb91a; custody

sha256:c3ac2d4b74bac1e0c2f861f5228833a262d66f900787ddffc36099eea3b3cf40.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar

- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

24. Ambiguity detection and retained alternatives

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 012 Claim:

SFT - INFO - SYMREP - AMBIGUITY - ALTERNATIVES - 012

Question, law and forced result. Ambiguity exists exactly when complete parsing returns more than one source word; every alternative is retained and no preferred parse is silently selected.

SYMREP-012 uniquely retains complete-alternative-parse-ledger, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - CONVERSION - TRANSDUCTION - 011

The current reconciliation binds this obligation to the current model-admitted receipt sha256:de6d10b1160ea411911b7d761c195d707003a588de3d0d675163d6e9e51c1844. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-012 product before observation access.

Boundary: Every positive finite SYMREP-012 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-012 uniquely retains complete-alternative-parse-ledger, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__complete-alternative-parse-ledger__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:f889e0ddbac7bdd864b3f620ac6af16a38af91bd1d3bbaec826878ada1705dbc; generality sha256:d8c7abf22030990bf7d621ed764a09393adee5ba40377021248abd3ef60f135a.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt sha256:559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt
sha256:cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256:5ff6fa7a186fd3666a9475812a8f6705f5f600bdfef065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate

sha256:73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal
sha256:faa262cd706843e7f62be75cc8a87b473de4bf09aa5818586a36df664bbf77be. Source manifest
sha256:9c4c7e613d0ba88387eb0ae702942e950abf8716ab5e7ffbf6bee0aa7d9f73f4. Engine receipt
sha256:de6d10b1160ea411911b7d761c195d707003a588de3d0d675163d6e9e51c1844 at receipts/engine/model_admitted/SFT-INF0-SYMPREP-AMBIGUITY-ALTERNATIVES-012-de6d10b1160ea411.js
on. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMPREP-012: ambiguity_ledger
- exact observation: {"all_alternatives_retained":true,"parse_count":2,"parses":["a","a"],["b"],"stream":["L","L"]}
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:4033880bb93ccdf05faf1dec0b219aad4debe1902ef1b90b4eb494ef2282d8a2; empirical
validation sha256:34979eae9997d3dd5c65f121fac599bf74f8bddec83aef92fde66d8ca0fec123;
external validation
sha256:af36fc7df6852001f153aeccd5e234c018c5fb6432ae034fc785a9d91d5a7469; isolation
sha256:0d189d36fc6539551ff6e1211dbb6e50f324d0891660f87f339a83b3fea1e24b; custody
sha256:c10831f13954b3bed86006e1ccda5d0fa70b3a8315f3c7b69cd084b98dd41919.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier

- no failed route retires an obligation or changes protected authority

25. Finite-alphabet successor law

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 013 Claim:

SFT - INFO - SYMREP - FINITE - SUCCESSOR - 013

Question, law and forced result. Adding one fresh canonical symbol preserves the prior alphabet and representation laws while generating exactly one new observation row and one new pair against each prior symbol.

SYMREP-013 uniquely retains fresh-symbol-successor-extension, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - AMBIGUITY - ALTERNATIVES - 012

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:2410fd01a379abcb7091074c413976ebb71e8d3321d5447a8de823f6dc9af7ae. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-013 product before observation access.

Boundary: Every positive finite SYMREP-013 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-013 uniquely retains fresh-symbol-successor-extension, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__fresh-symbol-successor-extension__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:2df713699698f39fc03bb81940d49382917d5abfe6e725a1bd759063e2defecd; generality
sha256:caca57e2de8d0c15fc2c5c373b093a464652b94e38627ba73c74be6454e9f285.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt
sha256:559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt sha256:cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256:5ff6fa7a186fd3666a9475812a8f6705f5f600bdf065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate
 sha256:73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal
 sha256:20473dc89799ee267b0c208850cfbe1dc9f7535ac133fd2864bd4af70f7bd0b2. Source manifest
 sha256:f1bb0a02e09266b07f99dcca3c243e9792fe0a330c37163297a25fec40b61a96. Engine receipt
 sha256:2410fd01a379abcb7091074c413976ebb71e8d3321d5447a8de823f6dc9af7ae at receipts/engine/model_admitted/SFT-INFO-SYMPREP-FINITE-SUCCESSOR-013-2410fd01a379abcb.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMPREP-013: alphabet_successor
- exact observation:
 {"new_pair_count":3,"new_symbol":"d","prior_rows_preserved":true,"prior_size":3,"successor_size":4}
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:95c775e4cb8a4ab64c22cea28e4fa2802f12ad77d159c742523f363e033906fb; empirical validation sha256:0140f4abe1e4ac42f3e3ced6565c3f359f6e69d9f5426b68231bb6c7fc0f331f; external validation
 sha256:73a4446c6185430b4d6c9146bc9acad8c2dcbe9040f216662d925a56f0ae469c; isolation
 sha256:14bc5b9bac16a7eae3346bca4d6833436c69b01a2c8adddb8f111d34b3f255f0; custody
 sha256:c11f6ce83e4d0d8d85ea156a3ee3d95acbaa74c09079137d9e048b0a1e0becb5.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

26. Representation completeness and no-omission certificate

Family: SYMREP Obligation: SFT - INFO - OBL - SYMREP - 014 Claim: SFT - INFO - SYMREP - COMPLETENESS - 014

Question, law and forced result. Representation-family completeness is the one-to-one reconciliation of all fourteen frozen obligations with admitted receipts, exact observations and no duplicate or omitted owner.

SYMREP-014 uniquely retains fourteen-obligation-no-omission-ledger, complete representation custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - MATH - HAND - CROSS - BRANCH - COMPLETENESS - 006
- SFT - INFO - QUANTUM - CORRESPONDENCE - 001
- SFT - INFO - SYMREP - FINITE - SUCCESSOR - 013

The current reconciliation binds this obligation to the current model-admitted receipt sha256:8808d673f54abe8890b5d37c2cf9501726d2953e082b6a3ce7ff14fedcee7899. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SYMREP-014 product before observation access.

Boundary: Every positive finite SYMREP-014 alphabet, code, grammar, relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SYMREP-014 uniquely retains fourteen-obligation-no-omission-ledger, complete representation custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-carrier__construction-and-label-identity__fourteen-obligation-no-omission-ledger__complete-boundary-aware-parse__complete-declared-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:d32e584056bfb48edda4341a8f8ab5cdca83264e3eb4d9dab40c6f1d8128e959; generality sha256:f54ea13893e4b0588d7d280a6e3c500772ddd68cf65798a1f772a7f7ca33f573.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial or duplicated carrier changes the representation support.; receipt sha256:559c879478b1f23f23be5edd95222106ad01e8c272129bbcf21343db735b61c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported information model, coding theorem answer or target outcome selects the result; host 0 denotes structural absence or counts artifacts only and is not an SFT number object; no negative, irrational, imaginary or floating proof

scalar; no imported semantic meaning, frequency prior or stochastic cause; no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier; no failed route retires an obligation or changes protected authority; receipt sha256:cb61f696ce7f4e7cdb9d508419f1464afe67ff815e32f7d8312793d94243a04a.

Independent reconstruction and engine admission. Independent implementation

sha256:5ff6fa7a186fd3666a9475812a8f6705f5f600bdf6e065ae6c6a2f90b5e41791 regenerated the declared census and sole survivor; independent certificate

sha256:73e7200824947799384aa30e67e5e7b890da6168933f789d17b320392a2d74dc. Derivation seal sha256:a65d02e9ada011c1a6e1ad3ad5afde4c494e7384655d933d918222ce6d2a5bbc. Source manifest sha256:04aaa9e7a4ca8d455a8b53f377a2180a9ae8a31d7d33ee21b02dcb19ab06799c. Engine receipt sha256:8808d673f54abe8890b5d37c2cf9501726d2953e082b6a3ce7ff14fedcee7899 at receipts/engine/model_admitted/SFT-INFO-SYMPREP-COMPLETENESS-014-8808d673f54abe88.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-REPRESENTATION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SYMPREP-014: representation_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":14,"omitted_owners":0,"registered_obligations":14}
- complete family observation custody: 14 records; all rows preserved
- complete finite alphabet, code, parse, grammar and relation enumeration supplies the observation; no imported coding axiom, semantic prior or hidden parse enters

Falsification condition: Reject if the value-free registry, complete representation census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e56a46f135c5b88c5c3a738aadceab87e977fe446a9b56fb9a659e4c9b116c3e; empirical validation sha256:2f970c070e027972dbb83e41c6587e097b6ebe4e9a2c95dd4677e4573c9b05e4; external validation

sha256:e02c0412df8f1760f830b725a1459a72ce42ca78ad089332d5c251cf4d37685e; isolation sha256:d4172600c59e6198424b8787e9e3a9926828c364ea06ceb115b0a067c4e6e317; custody sha256:0466bcc316cdeba62b9a60969177123080c56b43cd8d489d5a2836c22c371de2.

Prohibited inputs.

- no axiom, imported information model, coding theorem answer or target outcome selects the result
- host 0 denotes structural absence or counts artifacts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no imported semantic meaning, frequency prior or stochastic cause
- no sampled alphabet, hidden parse, omitted relation or unregistered infinite carrier
- no failed route retires an obligation or changes protected authority

Complete-field family: RECORD

27. Data item and record identity

Family: RECORD Obligation: SFT-INFO-OBL-RECORD-001 Claim: SFT-INFO-RECORD-DATA-IDENTITY-001

Question, law and forced result. A data item is an exact source-bound field/value record whose canonical identity includes its provenance-bearing source, declared field and complete exact value.

RECORD-001 uniquely retains source-field-value-identity, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SYMREP-COMPLETENESS-014

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:188f5c8571a552f7a2e11ed331c20c752c20230612f7cc29b0c3576415caddfc. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-001 product before observation access.

Boundary: Every positive finite RECORD-001 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-001 uniquely retains source-field-value-identity, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__source-field-value-identity__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-success-or-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:a6f4c168324d04287d6e5ff1bed6f57d3ebb02cdf2edaf941c68f032b9db2d13; generality
sha256:e5e0470fe3a36712950055d648d5ddd9d129f627747071890de5afc1d82e35ae.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt
sha256:765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt
sha256:09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation
sha256:6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the

declared census and sole survivor; independent certificate

sha256:16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal sha256:5bd5bcc51c4dc3b3ac3baadf66e8cabcdce4afc38873919f21a4869fe9f97f58. Source manifest sha256:fd2e7fab16ab3511f9de16eaab981703831dd2b2026626cd1ce72b5fdb37653c. Engine receipt sha256:188f5c8571a552f7a2e11ed331c20c752c20230612f7cc29b0c3576415caddfc at receipts/engine/model_admitted/SFT-INFO-RECORD-DATA-IDENTITY-001-188f5c8571a552f7.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-001: data_identity
- exact observation: {"exact_value_retained":true,"field_retained":true,"identity_coordinates":3,"source_retained":true}
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:965a82d775da71710274fb55aa6449c540f493a29f7d682a15608394ef1bc9f4; empirical validation sha256:0f8cf76bd7e46d7e8a6b4ef7dbefaf86e6ccabea892572e37250ad463a84002e; external validation

sha256:0925c9003d333d503c39d0170fe8af6ec09e5d3f4942029cf071cc57ab2c6bc9; isolation sha256:b63bb65174b4111315dab415ddf4d7716379dac56a16af4dedc5d96194cdc6b9; custody sha256:66d424efdb2ca724fb7eb3c1501d8768f6044eea08896e25ed436ce20b86e9f0.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

28. Field, tuple and relation organization

Family: RECORD Obligation: SFT-INFO-OBL-RECORD-002 Claim: SFT-INFO-RECORD-TUPLE-RELATION-002

Question, law and forced result. A tuple instantiates every field of one ordered duplicate-free schema; a relation is the complete retained family of such typed tuples.

RECORD-002 uniquely retains complete-schema-tuple-relation, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SYMREP-COMPLETENESS-014
- SFT-INFO-RECORD-DATA-IDENTITY-001

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:9c96a61da557fa61aafc050738815f5e903f444f2db6ac7537217a709bdf5a1c. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-002 product before observation access.

Boundary: Every positive finite RECORD-002 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-002 uniquely retains complete-schema-tuple-relation, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__complete-schema-tuple-relation__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:46bf14b96cf03f5d20c0b1403964e0626e2d91122cf6caa043f761ea3823a317; generality
sha256:53f8b22fc57d40824eb6fe957179f5aaa260b30fa4f87e6b4f1f448e9012c4bb.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt
sha256:765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt
sha256:09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation

sha256:6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the declared census and sole survivor; independent certificate
sha256:16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal
sha256:2e5b6899fc462bc1df027c3f300a252d624686d3429574828bf0a6d04bbdbdb7. Source manifest
sha256:6ca4a5ba13048e7d002a9272db6949a24aa211345bed4a11eb5d424ba4fb0b53. Engine receipt
sha256:9c96a61da557fa61aafc050738815f5e903f444f2db6ac7537217a709bdf5a1c at receipts/engine/model_admitted/SFT-INFO-RECORD-TUPLE-RELATION-002-9c96a61da557fa61.json. Closure

scope `depth_independent`. External status `empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-002: `tuple_relation`
- exact observation: `{"complete_field_cells":4,"partial_rows":0,"rows":2,"schema_fields":2}`
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:ee4e4bb9ec8eace153e0aea90fff0d701c025f4daaeed4c3b57b6b41e68f071c; empirical validation sha256:9257fcdc0459d243db1ff192a3701734b644884a908612196314239d93acec46; external validation

sha256:74bcbe638ff972d203ec86ad1e3c82c5f249fe7b3fac98e8c1926707fb773a65; isolation sha256:e99f2e004e7f7e3796cd9760ecc7f2503b903d7e648da7cfb82379fd2b5268c2; custody sha256:b90da6628b95d2b512f4a5c09f0cae85e062606641d072c0b30bf62ae9557299.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

29. Metadata as retained interpretive record

Family: RECORD Obligation: SFT-INFO-OBL-RECORD-003 Claim: SFT-INFO-RECORD-METADATA-003

Question, law and forced result. Metadata is a separate retained record bound to the canonical data identity and carrying the declared schema, unit and source needed to interpret the data without replacing it.

RECORD-003 uniquely retains data-bound-metadata-record, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SYMREP-COMPLETENESS-014
- SFT-INFO-RECORD-TUPLE-RELATION-002

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:25e38daf05a46551ff43aab2b9e61f9528960e9e3afd3b13bafd1e31ec212f81. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-003 product before observation access.

Boundary: Every positive finite RECORD-003 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-003 uniquely retains data-bound-metadata-record, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__ata-bound-metadata-record__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:02f1c638654059a66c809bc0480f3ccc9fc93f1c66b26d76bf49522173fcb841; generality sha256:71a13b113b46951496d8bd902877f98d91ab10723f36528cea6bae5258745b86.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt sha256:765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt sha256:09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation

sha256:6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the declared census and sole survivor; independent certificate sha256:16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal sha256:58374d25d667c33143118ce60b3286c5a06bb9d25eb4d97636e507d6831518b4. Source manifest sha256:b8653e7ed136c53519c41e22a08c2622483df2b2f170c3fb0c1d9878b43f8229. Engine receipt sha256:25e38daf05a46551ff43aab2b9e61f9528960e9e3afd3b13bafd1e31ec212f81 at receipts/engine/model_admitted/SFT-INFO-RECORD-METADATA-003-25e38daf05a46551.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT - V3 - INDEPENDENT - EXACT - RECORD - OBSERVER
- SFT - V1 - V2 - INFORMATION - OBSERVATION - CORPUS

Observed, adverse and boundary records:

- RECORD-003: metadata
- exact observation: {"data_replaced":false,"metadata_fields":["record_identity","schema","unit","source"]}
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:36e141e7929038320fe5b568eb06b21a14132417058274472a6d16d3afd6f870; empirical validation sha256:b56ed281e502e8a862ffabf3b61e1a2e69101da9a407e524a3ed361b4f6f4598; external validation

sha256:c0bb8876b29f27e3d461c0b34bc72c47350c1c21043acab25e90abf0ac85847b; isolation sha256:b2a0778c56c8cd245ce41a8f983ba73c870c1ccc73dfe847b71bf9282d47669c; custody sha256:00ae8d3ccd8f7cb520ab7e5ee714eb6fc4dcecbd45917dc1c0a640609715dd0c.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

30. Schema and type custody

Family: RECORD Obligation: SFT - INFO - OBL - RECORD - 004 Claim: SFT - INFO - RECORD - SCHEMA - TYPE - 004

Question, law and forced result. A record conforms to a schema exactly when every declared field occurs once, no undeclared field enters and every value belongs to its generated type fibre.

RECORD-004 uniquely retains complete-field-type-custody, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SYMREP - COMPLETENESS - 014
- SFT - INFO - RECORD - METADATA - 003

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:f362adb1701f98f3aa6146adc30d98464694e832fbc910ee863d4fc61e000951. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-004 product before observation access.

Boundary: Every positive finite RECORD-004 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-004 uniquely retains complete-field-type-custody, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__complete-field-type-custody__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-success-or-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:80bc1cb36faee435a0daff07861e27cc609b3057743aad04c914f1df3508f0cb; generality sha256:c310af754c16c3fc2e2f5e578365c489e0ed9d63ab3e4e3980bddff514f938c7.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt sha256:765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt sha256:09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation

sha256:6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the declared census and sole survivor; independent certificate

sha256:16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal sha256:0511b41985ec07435ffb197b62ce08841a4b5cac08e5e1521600f5ef261cfe8a. Source manifest sha256:8d8680cd6cc48f7ccf536e919a868db0fdcf175f8fe08a345fd7195a3acdf19e. Engine receipt sha256:f362adb1701f98f3aa6146adc30d98464694e832fbc910ee863d4fc61e000951 at receipts/engine/model_admitted/SFT-INFO-RECORD-SCHEMA-TYPE-004-f362adb1701f98f3.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-004: schema_type
- exact observation: {"complete_record_accepted":true,"declared_fields":2,"missing_field_rejected":true}
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e6c53a6265e4c6242306ec68e457e20a168e0b71fafbe324aa32c59acac3ccfb; empirical validation sha256:0b5ae10cf4a94071d643c76f2f766ee99c4d5c20b73beaecf116d4f375a7e10c; external validation

sha256:c389df89ce337832bde80fbeeef067e1adc452095ed341af1d065af1a614e50; isolation sha256:1422fc3d2dd59a4948b7abe4afda542cef26b3737081564b2bb4b5bac5d213a0; custody sha256:ea7e0e1017b5b15480344feb1fe7adda0a6107b05ff6ed15d4648aa1d8db975a.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

31. Provenance chain composition

Family: RECORD Obligation: SFT - INFO - OBL - RECORD - 005 Claim: SFT - INFO - RECORD - PROVENANCE - 005

Question, law and forced result. Provenance is the complete ordered parent-bound transformation chain from source capture to present record; gaps, duplicates and cycles halt custody.

RECORD-005 uniquely retains contiguous-acyclic-provenance-chain, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SYMREP - COMPLETENESS - 014
- SFT - INFO - RECORD - SCHEMA - TYPE - 004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:7cbaa15ee285db6986d62029723d73cf3e72fe9878a354a7de4f65737c0d42e9. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-005 product before observation access.

Boundary: Every positive finite RECORD-005 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-005 uniquely retains contiguous-acyclic-provenance-chain, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__contiguous-acyclic-provenance-chain__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:1ed4eee98d3cdad7727e11b813da13340f6af8d995fd945df9b911057b7fc1c4; generality sha256:3d477ecf3bc2c72a52cb9aeff690053b6f5401f0f695bdfb487eb28b16a6ab35.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt sha256:765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt sha256:09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation

sha256:6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the declared census and sole survivor; independent certificate sha256:16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal sha256:f2ad869fe6910e52756ddc7d0e57d848307aa3ebdb026a3acd6d8309064c725b. Source manifest sha256:2a80596f57b48604f30f433547289e964c9d8b2a23a48a7d4c9d4514808986e3. Engine receipt sha256:7cbaa15ee285db6986d62029723d73cf3e72fe9878a354a7de4f65737c0d42e9 at receipts/engine/model_admitted/SFT-INFO-RECORD-PROVENANCE-005-7cbaa15ee285db69.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-005: provenance
- exact observation: {"contiguous":true,"duplicate_free":true,"gap_rejected":true,"steps":["capture","normalize","publish"]}
- complete family observation custody: 12 records; all rows preserved

- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:51decd38f97bd538200cfe30152c56ddd7439ef4f3a25856b9ba02ca0fad3b0b; empirical validation sha256:d7410e0fee452d249e67f2149749481e6686812590a518cd4d9d3ad88399ad98;

external validation

sha256:abda8b30987e82c16c65d89ea03ced74e724abdca44d94331b4a283371a67034; isolation sha256:a37438bad6671b909f02f5022ccf1845ad08ab09b5113a25c596007d7e03efa1; custody sha256:925d72b4eaab4d00e681d4b8fc669b58d0b33f131831548d86dd1c47cb944a18.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

32. Integrity and tamper evidence

Family: RECORD Obligation: SFT - INFO - OBL - RECORD - 006 Claim: SFT - INFO - RECORD - INTEGRITY - 006

Question, law and forced result. Integrity is equality between a retained canonical payload identity and a freshly reconstructed identity; any changed distinction produces a different identity and fails verification.

RECORD-006 uniquely retains identity-bound-integrity-check, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SYMREP-COMPLETENESS-014
- SFT-INFO-RECORD-PROVENANCE-005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:69fbb5e96b439ffdb77c551ace0e0c6fe920db4fb5bd6f413bbbcca8e800b92a. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-006 product before observation access.

Boundary: Every positive finite RECORD-006 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-006 uniquely retains identity-bound-integrity-check, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__identity-bound-integrity-check__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: 7a92b4f8c7c62310d57e3c579c687d0d1257e2ab77d8d5c3b6042ffcc18b7c78; generality
 sha256: be6737e29e59da9b4405d2d22d37f8dfbbfc6acf2e22785a1d04b002d2e518f.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt
 sha256: 765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt
 sha256: 09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation
 sha256: 6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the
 declared census and sole survivor; independent certificate
 sha256: 16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal
 sha256: 6e9416c4bf2bafb69adb60139a70a7c2260becd0c291c0cc43974a6e8bef791e. Source manifest
 sha256: bc3e2324daec36b393f53fc226d5cb10b57eb3b4b875c6df56b9a29bb8d750e5. Engine receipt
 sha256: 69fbb5e96b439ffdb77c551ace0e0c6fe920db4fb5bd6f413bbbcca8e800b92a at
 receipts/engine/model_admitted/SFT-INFO-RECORD-INTEGRITY-006-69fbb5e96b439ffd.json.
 Closure scope `depth_independent`. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows
 preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-006: integrity
- exact observation: {"changed_payload_rejected":true,"identity_reconstructed":true,"unchanged_payload_accepted":true}
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of

order.

Measurement receipt

sha256:c96bdb5fa5ea4d80a50dac66059713d69719085eac15ba04bcc0b6cfb2169f86; empirical validation sha256:2c6ede553f1341de2296e67a7c391bc149e6877e949277fd1b7ff4666ffa135e; external validation

sha256:4c0ae24eebb3a5a295062eb3f892245a45908fd47c8fc5c5941855da2b136ca8; isolation sha256:a4ae85fd0fc9af0f04ecb96be48d2b7c803ef2fefdf83684722c609d21e8378b; custody sha256:27246626fb5ef6798d1923a77697f1908db0abfdb75ee601ebad89f77bbb53cd.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

33. Version and revision identity

Family: RECORD Obligation: SFT - INFO - OBL - RECORD - 007 Claim:

SFT - INFO - RECORD - VERSION - REVISION - 007

Question, law and forced result. A version retains its own identifier, exact payload identity and explicit parent; revision never silently overwrites or aliases the prior record.

RECORD-007 uniquely retains payload-and-parent-version-identity, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SYMREP - COMPLETENESS - 014
- SFT - INFO - RECORD - INTEGRITY - 006

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:775b2a7f578c366836cb191538788f8b585f798f44586f48aed3f14faca63f57. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-007 product before observation access.

Boundary: Every positive finite RECORD-007 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-007 uniquely retains payload-and-parent-version-identity, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__payload-and-parent-version-identity__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:f022c0bb49abf876f0d28c9725a31845ff86a108df82652fae35d8e1272cf174; generality sha256:fc3e8fa53dbba716b9390247bccef885fb8851d648c7ee27c62e8cc5382d990c.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt
sha256: 765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt
sha256: 09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation

sha256: 6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the declared census and sole survivor; independent certificate

sha256: 16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal
sha256: 310011eb650df7f96eb90bf9162c61afc68829b3f88c338f653ddf55946f58c5. Source manifest
sha256: 01bdb432aec6647bd50e516eb00854e46b5216ba95c3ad28ec2ba66770f670a6. Engine receipt
sha256: 775b2a7f578c366836cb191538788f8b585f798f44586f48aed3f14faca63f57 at receipts/engine/model_admitted/SFT-INFO-RECORD-VERSION-REVISION-007-775b2a7f578c3668.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-007: version_revision
- exact observation: {"parent_retained": "v1", "payload_identities_distinct": true, "silent_overwrite": false, "versions": 2}
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: a26ba498b5704ebde25a69afb9961768a71a1f8290940912bdab58ad957fb785; empirical

validation sha256:6a1c9f6d5053cd75578aad111f30d44430bf77cbfa2321405d91bbe200b47fe1;
 external validation
 sha256:4ac886b9ebb95530a770d057ab486a7359bf5b5d6af1a2d8e1d025ac150961dd; isolation
 sha256:f3fb8242b80e61565769a4138c2e627ecbdf5d804b95076319dfb504dbb5d0d1; custody
 sha256:d35823684112a7e1e5ffcc1bfc1ffa6af6380c20068ece7be15d150d6b90619e.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

34. Missing, absent and unknown distinctions

Family: RECORD Obligation: SFT - INFO - OBL - RECORD - 008 Claim:
 SFT - INFO - RECORD - ABSENT - MISSING - UNKNOWN - 008

Question, law and forced result. Structural absence displayed as 0, an expected-but-missing record and a retained record with unknown value are distinct states and may never be collapsed into a numerical zero.

RECORD-008 uniquely retains three-way-absence-missing-unknown, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SYMREP - COMPLETENESS - 014
- SFT - INFO - RECORD - VERSION - REVISION - 007

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:89e4d009dfb6d60b36f34052373f73d7c772582f8d82dd5c56f4017fb2b2af11. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-008 product before observation access.

Boundary: Every positive finite RECORD-008 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-008 uniquely retains three-way-absence-missing-unknown, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__three-way-absence-missing-unknown__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:29f91b40e39c3e0de63f546aa3274bc24fbd91fd6e1fe306db79c99ba6f7b734; generality
 sha256:18709b623b7ce141e2ff710f325562850f0e067b4a48df459edd83c0269a991d.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT

numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt
sha256: 765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt
sha256: 09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation

sha256: 6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the declared census and sole survivor; independent certificate

sha256: 16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal

sha256: 8c8795df21b38cccd3bbc1f47b748419cba650c25246ff2f0b82b6178e8803c2. Source manifest

sha256: 12bec012f5a34e2cf3c6343b6a2dd07176f96e6058fc30f2e80cba93f982f713. Engine receipt

sha256: 89e4d009dfb6d60b36f34052373f73d7c772582f8d82dd5c56f4017fb2b2af11 at receipts/engine/model_admitted/SFT-INFO-RECORD-ABSENT-MISSING-UNKNOWN-008-89e4d009dfb6d60b.json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-008: absence_missing_unknown
- exact observation: {"distinct_state_count":3,"numeric_zero_claimed":false,"states":["absence","0"],["missing","expected-row"],["unknown","retained-row"]}
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 291e2509b0d8e84e97ba2e5befed5c662eb941c06c294a234633ed892afbbbeb5; empirical

validation sha256: 3c9859f71164ea0a3ba19aea6b5e8d06709b062e9db3d6f391a202b4ed2effde; external validation

sha256: f6c77c98c6524261de7b3b88b72eeb795019b1d6599bf08bee0e89973db6e655; isolation

sha256:3e96c90c30d5bed85805bda785e145fa8ac114de1f5113f1278aa4c82ebf0a25; custody
sha256:188c1ca5da730a924bb9e7c639bef53d2d7c659afad705031e9ddd1b533cb04d.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

35. Duplicate and alias resolution

Family: RECORD Obligation: SFT - INFO - OBL - RECORD - 009 Claim: SFT - INFO - RECORD - DUPLICATE - ALIAS - 009

Question, law and forced result. Duplicates are records with identical canonical identities; aliases retain distinct source tokens plus an explicit many-to-one canonical-name relation rather than silent deletion.

RECORD-009 uniquely retains canonical-duplicate-alias-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SYMREP - COMPLETENESS - 014
- SFT - INFO - RECORD - ABSENT - MISSING - UNKNOWN - 008

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:8bd9429bb1f87358f9b7e1d76bf5f024432dd4f6b986b9615295445d17745cfb. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-009 product before observation access.

Boundary: Every positive finite RECORD-009 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-009 uniquely retains canonical-duplicate-alias-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__canonical-duplicate-alias-ledger__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:939fd0018d123b7cda33bdc5e098e0b2486730a9abc39c56cc0572bb000f1987; generality
sha256:1fe22e955d89916ec76da6185837595157cea91c52de60622a89e808e7faf456.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt
sha256: 765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt
sha256: 09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation

sha256: 6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the declared census and sole survivor; independent certificate
sha256: 16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal
sha256: 6627299348ed9f31c54ccfbf84206629bb536867d9bc07e0bdd967d18bb812d1. Source manifest
sha256: ee4db5c86923d8ef48d1dbf069115290e9f52fcec63ecca0e13d81a84476de63. Engine receipt
sha256: 8bd9429bb1f87358f9b7e1d76bf5f024432dd4f6b986b9615295445d17745cfb at receipts/engine/model_admitted/SFT-INFO-RECORD-DUPLICATE-ALIAS-009-8bd9429bb1f87358.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-009: duplicate_alias
- exact observation:
{ "alias_source_count":2, "canonical_target_count":1, "duplicate_identity_equal":true, "source_tokens_retained":true }
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: f9adb1a8e363525a8fee99976198576b17775fc02ee819ec16b4667c2b5b9a07; empirical validation sha256: 35818e83a70d487f562b997e6b8cdb7f8bb11ebbcbe59754fd86e4ff8432f2cc; external validation
sha256: a0db45a0a91fdfb81408a3f9561e35eccef8f7aadadb1960645686ffcef67ada; isolation
sha256: 242c2fe0f277c96dc4d1e580018b6b7e96fb113da5e5e57b77aa98690314a1ce; custody
sha256: a79d821a0ba37c62a2e56b3a24849945a138d5ea79a4362a44feedd8276af178.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

36. Record linkage and identity boundaries

Family: RECORD Obligation: SFT - INFO - OBL - RECORD - 010 Claim:
SFT - INFO - RECORD - LINKAGE - IDENTITY - 010

Question, law and forced result. Record linkage is an exact declared relation between source identities; unresolved candidates remain explicit alternatives and no fitted similarity threshold silently selects identity.

RECORD-010 uniquely retains complete-link-alternative-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SYMREP - COMPLETENESS - 014
- SFT - INFO - RECORD - DUPLICATE - ALIAS - 009

The current reconciliation binds this obligation to the current model-admitted receipt
sha256: 3af4d67f4d641bdec40b47a8f234af19a9844f29f1b21b05d42fe4b80cf1f957. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-010 product before observation access.

Boundary: Every positive finite RECORD-010 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-010 uniquely retains complete-link-alternative-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__complete-link-alternative-ledger__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256: 65040d8b8aeb6016b4f65cd4941549bf699388505447a71496864b7ead9f83; generality
sha256: 0e3b297fdd1e186fd317c60e459fd04fde99561f8d21df7ac7a88212d4d5693e.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt

sha256: 765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt
sha256: 09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation

sha256: 6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the declared census and sole survivor; independent certificate
sha256: 16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal
sha256: 265b0460bc32303e15463bd157f3a6780e8544c2da216534553f4cdf4511fe94. Source manifest
sha256: 0cbf015709a2b4a812ce84542e9ad988076fccc272f259f4c95f1089f25a1b15. Engine receipt
sha256: 3af4d67f4d641bdec40b47a8f234af19a9844f29f1b21b05d42fe4b80cf1f957 at receipts/engine/model_admitted/SFT-INFO-RECORD-LINKAGE-IDENTITY-010-3af4d67f4d641bde.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-010: record_linkage
- exact observation: {"exact_links":1,"silent_selection":false,"unresolved_alternatives":2}
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: c177d8da5502d4195f59bbbd79eecec0f28d40a9078e36068822c9192f4dc6ec; empirical validation sha256: 2e48710d966461424781b98f9dc9b9f0951403007a86681685adcae68aa2c6c89; external validation
sha256: 46d4b35cb3c26dc738f1c9c8c13d77a00ec5e848311f15bc572f7e75faa96fa7; isolation
sha256: 23fe6161af211e8956234f45ab61a8158df9572320dee6f8a20818d9c7a64e5a; custody
sha256: 4b49ee7909028bd001082b4f4102020a2b1637d0f174b51f0da5ddb732b94de.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result

- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

37. Dataset completeness and omission ledger

Family: RECORD Obligation: SFT - INFO - OBL - RECORD - 011 Claim:
SFT - INFO - RECORD - DATASET - COMPLETENESS - 011

Question, law and forced result. Dataset completeness compares the full registered identity support with the observed support and separately retains every present, absent and unexpected row.

RECORD-011 uniquely retains retained-absent-unexpected-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SYMREP - COMPLETENESS - 014
- SFT - INFO - RECORD - LINKAGE - IDENTITY - 010

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:7298a45fcb8497b3bdc6739512bc51f83ce6cbcc53dc1f7b99f4227d3550b74. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-011 product before observation access.

Boundary: Every positive finite RECORD-011 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-011 uniquely retains retained-absent-unexpected-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__retained-absent-unexpected-ledger__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:208217f4fa72f7ddd4207f130974437dfcf1e7aa708a8d3a9ae6c46f32deae3c; generality
sha256:4290ddfe582293f1ad71623bdbaa779d3078b65ff5b9ea2d2b6ba319e5d156e4.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt
sha256:765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt

sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt
sha256:09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation

sha256:6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the declared census and sole survivor; independent certificate

sha256:16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal

sha256:8224fc00eccc42867dac871161215214e2aad9bc1476aa0e1bdb64afc05377a9. Source manifest

sha256:37684f025d919aaa896a6fcf8dfbce2f87aff28a1de83a99f368b8c8081ca1ae. Engine receipt

sha256:7298a45fcb8497b3bdcb6739512bc51f83ce6cbcc53dc1f7b99f4227d3550b74 at receipts/engine/model_admitted/SFT-INFO-RECORD-DATASET-COMPLETENESS-011-7298a45fcb8497b3.json

. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-011: dataset_completeness
- exact observation: {"absent":1,"all_rows_accounted":true,"expected":3,"observed":3,"retained":2,"unexpected":1}
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:fb5de8181982649e46c5b440a059ad0d1caf8b921f4f985b0c8df114d8688755; empirical

validation sha256:40e975847a18b35c7b0c68ad9205e5aa29af270e7d3092ca1bd739cd303fc596; external validation

sha256:1c51291eec00d3a1d7d444017d350cf98e7eea02cc81ea918da63e6842605611; isolation

sha256:eff8d08855f1a0e82cf38282fff84f6d81ce55a1f81f695af93c348d65695acd; custody

sha256:342e4cde3fe45d2288e5b16755294150c77891343dccc2c2f78ef24dfdf14454.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step

- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

38. Record custody and reproducibility certificate

Family: RECORD Obligation: SFT - INFO - OBL - RECORD - 012 Claim:
SFT - INFO - RECORD - CUSTODY - REPRODUCIBILITY - 012

Question, law and forced result. A reproducibility certificate binds data, metadata, provenance, schema and integrity identities so an independent reconstruction yields the same complete record identity or halts.

RECORD-012 uniquely retains reconstructing-custody-package, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SYMREP - COMPLETENESS - 014
- SFT - INFO - RECORD - DATASET - COMPLETENESS - 011

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:939f47572f8806dc4409f625369f53c39b0ee387ebc34c046d82c3428491eeb9. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RECORD-012 product before observation access.

Boundary: Every positive finite RECORD-012 data item, schema, provenance chain, dataset and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RECORD-012 uniquely retains reconstructing-custody-package, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-bound-carrier__canonical-content-and-source-identity__reconstructing-custody-package__complete-provenance-custody__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:35c4d804c3b1f310dae24e81aed1d747851678b43b17f6c9a10e7a1f40c52bc4; generality
sha256:015449062d83039500a18058a61bf0bb717c10d2f6f56da77479bc552a0c785f.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial carrier can omit a registered record.; receipt
sha256:765d8b319cc42159f139fcec23d1775027c72ad995181ad888d3f77ed2adaaba.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported data model, database convention or target outcome selects the result; displayed 0 denotes structural absence only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no silent overwrite, hidden missing value, erased alias or unrecorded provenance step; no sampled row family or unregistered infinite dataset; no failed route retires an obligation or changes protected authority; receipt
sha256:09c038e6b848dd165272c81ee48b43a6735fc15b966ea9ac5011d9c0e61d65d3.

Independent reconstruction and engine admission. Independent implementation

sha256:6c5a3e1c62d1fd22be94aac2acfa5fa388fcaa7189c7fde4132e37da36ce2be4 regenerated the declared census and sole survivor; independent certificate

sha256:16c00ac7e6d680c099d11da53c9072af9137bf9dd6807628853d5dcb986b0273. Derivation seal

sha256:ebca8b9b09627fedbe491178c6a42b1dca06acb814d14ce8481adaeaf89d79df. Source manifest

sha256:aa530aeb01fa2bed9874e1c9ca56fa678eec0a707d34ac2e72c908115399f93d. Engine receipt

sha256:939f47572f8806dc4409f625369f53c39b0ee387ebc34c046d82c3428491eeb9 at receipts/engine/model_admitted/SFT-INFO-RECORD-CUSTODY-REPRODUCIBILITY-012-939f47572f8806dc.j

son. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RECORD-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- RECORD-012: custody_reproducibility
- exact observation:
{"bound_components":["data","metadata","schema","provenance","integrity"],"independent_identity_equal":true}
- complete family observation custody: 12 records; all rows preserved
- complete finite data, schema, provenance, version and omission-ledger enumeration supplies the observation; no imported data-model axiom, silent overwrite or hidden missing row enters

Falsification condition: Reject if the value-free registry, complete record census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:d1e18484782fe8a9bf4bcd094179b6afb6f8af154c9534255a9796cfb0e93519; empirical validation sha256:bd94ff97fddc342831832da4dd8ed9750782e9e4714c961289eeaf5076ec5d76;

external validation

sha256:42d5f9c30f9da397ef841e6cc4de9e4c54979d353b65ff3f111199dfae298291; isolation

sha256:b262d4d630f33d3763a9c0e7ea974d0886b3f33ba353fda27a991eb7eaf34eca; custody

sha256:6186979ed894b9a7004725ef0a2014530bbe0b0786bea788c0b362436a3463dc.

Prohibited inputs.

- no axiom, imported data model, database convention or target outcome selects the result
- displayed 0 denotes structural absence only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no silent overwrite, hidden missing value, erased alias or unrecorded provenance step
- no sampled row family or unregistered infinite dataset
- no failed route retires an obligation or changes protected authority

Complete-field family: SOURCE

39. Finite source-support generation

Family: SOURCE Obligation: SFT - INFO - OBL - SOURCE - 001 Claim: SFT - INFO - SOURCE - SUPPORT - 001

Question, law and forced result. A finite information source is the complete duplicate-free family of canonical forms available at one declared observation boundary; no frequency or ontic randomness is presumed.

SOURCE-001 uniquely retains complete-canonical-source-support, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RECORD - CUSTODY - REPRODUCIBILITY - 012

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:1d4cf50615da67505bdb9be8da733e118d30cb4087d6882633f5173b86f73f44. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-001 product before observation access.

Boundary: Every positive finite SOURCE-001 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-001 uniquely retains complete-canonical-source-support, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__complete-canonical-source-support__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:f54ae0b281a20b88b4e521d11a906d300cc54b8a5320b4bf12f673e8e5c34e03; generality
sha256:a7243e86de6b41fb609f4aec368b280bad3a87f8dd55f95bf80d19c090a1394a.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt
sha256:b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar;

no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
 sha256:b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256:820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate

sha256:1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal
 sha256:5140900d61c540cb465314b24176caa498235913e274a46fb5b79ad5a30ddfa4. Source manifest
 sha256:6447320b20422dd1ae3a814d2b5a34af2c168f2d8685301c272a1145e9776789. Engine receipt
 sha256:1d4cf50615da67505bdb9be8da733e118d30cb4087d6882633f5173b86f73f44 at
 receipts/engine/model_admitted/SFT-INFO-SOURCE-SUPPORT-001-1d4cf50615da6750.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-001: source_support
- exact observation: {"canonical_forms":["a","b","c"],"duplicates":0,"support_size":3}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:8e39b54ce13de4166049bf7053dee8f0d660406e1d2984b517ea8e6470f0df25; empirical
 validation sha256:ea1969fa482cafe9fac9633a0adbfa9c1657308487a5ac5433bbab036d264983;
 external validation
 sha256:39af414fd7b1b4d86a3ec9c3c7e06264ba466418b10949f063e3ce62f5acdcb; isolation
 sha256:c087575a9065c31e7a893e36aafec07e999fd83b5c1e2c98ae0b99e174cece4; custody
 sha256:ee375f4974407aa9db6ef58a869dd44320acbeac91508fc0d6abda40626581e1.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

40. Sequence information and retained order

Family: SOURCE Obligation: SFT-INFO-OBL-SOURCE-002 Claim: SFT-INFO-SOURCE-SEQUENCE-ORDER-002

Question, law and forced result. Sequence information is a complete successor-indexed source record; equal values at different positions remain distinct occurrences and permutation changes the ordered record.

SOURCE-002 uniquely retains complete-position-labelled-sequence, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RECORD-CUSTODY-REPRODUCIBILITY-012
- SFT-INFO-SOURCE-SUPPORT-001

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:96609683f424ffcc3875ff030c519a90547bf86d3830775f9d912a7de84d6a6b. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-002 product before observation access.

Boundary: Every positive finite SOURCE-002 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-002 uniquely retains complete-position-labelled-sequence, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__complete-position-labelled-sequence__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:abe6f875dbafaeb83ad50106dc551939be26e060fb707aae322efef3d694d7b2; generality
sha256:d70be692c64d0bd9e2af83b81123baf2c77e6390a041f9c4aca72a7448b52928.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt
sha256:b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256:b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256:820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate

sha256:1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal sha256:4b93fc04cd10660ea27d85c53cc5e176457939d5a47894dd9139f2956fc300d3. Source manifest sha256:3c361faaa7c00bec14026b59bfa5b9636873faa46015994bc6d8d7393e6368e0. Engine receipt sha256:96609683f424ffcc3875ff030c519a90547bf86d3830775f9d912a7de84d6a6b at receipts/engine/model_admitted/SFT-INFO-SOURCE-SEQUENCE-ORDER-002-96609683f424ffcc.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-002: sequence_order
- exact observation: {"permutation_distinguished":true,"positions":[1,2,3],"values":["a","b","a"]}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:a5f95dfb232d12c254d594240b0d0d027595b665fb4a7902c14f03986171b14b; empirical validation sha256:4e114019f0dc05e8a8d4599be0e95bd061529cb0f9a965408d20e0cf1461ea45; external validation

sha256:36e504a6856ef8182512328f7da2660cd97732788343b56818f1482a3fd511b8; isolation sha256:e0de0a85cda043095e6a496263bdabb72ab6020cb3496a6d27aecdbdb0a1cda4; custody sha256:338fc194c2b34c5b25e595bc282c83a9eb8319252f5ebd7bf51a2f97d917b162.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

41. Process information and transition custody

Family: SOURCE Obligation: SFT-INFO-OBL-SOURCE-003 Claim: SFT-INFO-SOURCE-PROCESS-TRANSITION-003

Question, law and forced result. Process information retains the initial state and every adjacent declared transition in order, so the terminal state and complete path can be independently reconstructed.

SOURCE-003 uniquely retains state-and-adjacent-transition-trace, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RECORD-CUSTODY-REPRODUCIBILITY-012
- SFT-INFO-SOURCE-SEQUENCE-ORDER-002

The current reconciliation binds this obligation to the current model-admitted receipt sha256:29aaaf0bd2c2447744f6d34374759ca64c232527960213b45d8b15b39b76f28d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-003 product before observation access.

Boundary: Every positive finite SOURCE-003 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-003 uniquely retains state-and-adjacent-transition-trace, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__state-and-adjacent-transition-trace__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:02eb4ebdf976792b93bf01f7e7ece4594358373990e903c08266b642319819ea; generality sha256:a1ada8bd50f57b1b74b8bd0ab476c99649321884a008eb4d50a9975b90685256.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt sha256:b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256:820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate

sha256:1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal sha256:a82488cc9681fbe9d89d35348e7fe6f6d28188e9d9e6292afe58a3e27485386c. Source manifest

sha256:badd19d12f1973eddacf114b4d578b08ae587d1d2324a7efa02b0b10f3d35a88. Engine receipt sha256:29aaaf0bd2c2447744f6d34374759ca64c232527960213b45d8b15b39b76f28d at receipts/engine/model_admitted/SFT-INFO-SOURCE-PROCESS-TRANSITION-003-29aaaf0bd2c24477.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-003: process_transition
- exact observation:
 - { "complete_trace_valid":true,"missing_transition_rejected":true,"states":["ready","read","emit"],"transitions":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:664a2417f99516d6870914a6f3a939d182f0dbbee567e4fcea70847c56d78345; empirical validation sha256:bf0571b56af837d01f8901b0c4f225fb114db2b9f2a7fbbc01ba66e679efffbf; external validation

sha256:8a65acb8a71d64fbd20f26840fc0be1dd67d98af018613e076be2889f01d45c4; isolation sha256:a70b09f7f8a75f4b3d795416176cc7b7112f8b92f9276d9462b8da00827f71a3; custody sha256:ae41e2dda9882fd8bc1e938e4fbf11aea3928acbad723b8be16fbceaaef7ea.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

42. Spatial information and adjacency custody

Family: SOURCE Obligation: SFT-INFO-OBL-SOURCE-004 Claim:

SFT-INFO-SOURCE-SPATIAL-ADJACENCY-004

Question, law and forced result. Spatial information is a complete canonical cell carrier plus a held adjacency relation; symmetry, orientation or direction is admitted only when exhaustively present in that relation.

SOURCE-004 uniquely retains complete-held-spatial-adjacency, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RECORD-CUSTODY-REPRODUCIBILITY-012
- SFT-INFO-SOURCE-PROCESS-TRANSITION-003

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:e77634cca7e04f0ec50c336b75dbde83045540816a1a34a79afcdea51590a3c3. A missing
 dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-004 product before observation access.

Boundary: Every positive finite SOURCE-004 support, sequence, transition, spatial or network relation and registered
 successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly
 one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-004 uniquely retains complete-held-spatial-adjacency, complete source
 custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__complete-h
 eld-spatial-adjacency__complete-transition-and-path-custody__complete-declared-sour
 ce-product__root-bound-forward-forcing__post-registry-exact-observation__finite-suc
 cessor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:c123e4b2a2c2bb04efb9e6c188f9796bdae7b33c87c1fb15a641da46e5e5c4c1; generality
 sha256:dd5c28e4d578c10216633c2c5d77c29b55f8718bc1fc05f014921c4cd9ab70c2.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained
 distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit
 downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted
 magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT
 numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed:
 Partial or duplicate support changes source possibilities.; receipt
 sha256:b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs
 from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated
 product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no
 axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural
 absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar;
 no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered
 completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
 sha256:b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256:820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the
 declared census and sole survivor; independent certificate
 sha256:1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal
 sha256:d26cf260afa08b6bd282fb21fb2752b95fd01673fd0161309cfcbeeb82f18ad66. Source manifest
 sha256:a88766709c0cf31e15ed9aa9d487915e84cd446c7f2e03a3c0b45225ec3f9e4b. Engine receipt
 sha256:e77634cca7e04f0ec50c336b75dbde83045540816a1a34a79afcdea51590a3c3 at receipts/
 engine/model_admitted/SFT-INFO-SOURCE-SPATIAL-ADJACENCY-004-e77634cca7e04f0e.json.

Closure scope `depth_independent`. External status `empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-004: `spatial_adjacency`
- exact observation: `{"cells":4,"directed_rows":8,"self_rows":0,"symmetric":true}`
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:0941e0054df86c5d946a5465b53efdbb1453c4a854cab2f209d315908d9d54d0; empirical validation sha256:38c1e33901d43623a273f8ae2dda826e75308f4fa9b3fae2a4a3c00a8d55c9f7; external validation

sha256:2e87a4c24357e9eaa97a0a17b1bbc0f423768b276021a7ca9db084d599c2f69b; isolation sha256:735bc64b21633f6a08c44ceb770518a05b73e938c34322e298a77da163b2c0cd; custody sha256:2af43113a9aa404e16e8aa9bf137f8b317d0872a80ed43a83974f4ea4fcb5e74.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

43. Network information and path custody

Family: SOURCE Obligation: SFT-INFO-OBL-SOURCE-005 Claim: SFT-INFO-SOURCE-NETWORK-PATH-005

Question, law and forced result. Network information retains nodes, directed links and every generated path at the registered finite boundary; alternative routes remain distinct rather than being collapsed to reachability alone.

SOURCE-005 uniquely retains complete-network-path-ledger, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RECORD-CUSTODY-REPRODUCIBILITY-012
- SFT-INFO-SOURCE-SPATIAL-ADJACENCY-004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:035f5f93b5209dc1a07b7b77fe3f93b94b493beca2ddf443a74934c569ab85ea. A missing

dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-005 product before observation access.

Boundary: Every positive finite SOURCE-005 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-005 uniquely retains complete-network-path-ledger, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__complete-network-path-ledger__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:0ccd18737084f566d820182e7d954c5c60a79d9ee171f887796ac4c0bdf76a11; generality

sha256:5e794373b585d6e4965a45900295afce0d28b365f557f242dfc752896f4dd466.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt
sha256:b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256:b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256:820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate

sha256:1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal sha256:39fd84c7beb301f1c6d21831ec31054beb8a08169cae603a02d754e2113f4361. Source manifest sha256:0519bad99552cf87b596d91a6fc1613c3bd663f18b455b9862f7df92a5f4bd14. Engine receipt sha256:035f5f93b5209dc1a07b7b77fe3f93b94b493beca2ddf443a74934c569ab85ea at receipts/engine/model_admitted/SFT-INFO-SOURCE-NETWORK-PATH-005-035f5f93b5209dc1.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT - V3 - INDEPENDENT - EXACT - SOURCE - OBSERVER
- SFT - V1 - V2 - INFORMATION - OBSERVATION - CORPUS

Observed, adverse and boundary records:

- SOURCE-005: network_paths
- exact observation: {"complete_path_count":2,"paths":[["a","b","d"],["a","c","d"]],"source":"a","target":"d"}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e42d17c8f349276cf0e47630e9fa39f90473cee181297b762f1b3f78ca01421e; empirical validation sha256:3712d5c5985063296959edb8d39296d47bfc4ffabb93e4be7fc3e3bdd6e78b14; external validation

sha256:3dd1b602084569d3db145c20bd414e76dd133081b4e441da5d45321f333b51f1; isolation sha256:40636e9f7a4dc3631c8ba3d70357981120b7876c503a66c81ebb9f4f03d5ff25; custody sha256:c112e0c5188ce90688c87fd52bab2b67e075e9da4b8603a8fe978a92e9d5897.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

44. Source refinement and coarsening

Family: SOURCE Obligation: SFT - INFO - OBL - SOURCE - 006 Claim:
SFT - INFO - SOURCE - REFINEMENT - COARSENING - 006

Question, law and forced result. One source observation refines another exactly when each fine class lies wholly inside one coarse class; coarsening closes only the between-microform distinctions explicitly merged.

SOURCE-006 uniquely retains partition-refinement-order, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RECORD - CUSTODY - REPRODUCIBILITY - 012
- SFT - INFO - SOURCE - NETWORK - PATH - 005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:80d59f7d08df01bde1b7283271c33458c7c17c4d3c1d91a4727dd1a9ae5b5d08. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-006 product before observation access.

Boundary: Every positive finite SOURCE-006 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-006 uniquely retains partition-refinement-order, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__partition-refinement-order__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:ad44777b2b5e69b959a09c1b212bc6c4ca4fc736952d15c2a2c46d0201fcb1fc; generality

sha256:f50be3a1734f22ba7919c731ba80214530342d5147afd373e1440a1c1ea2846d.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt
sha256:b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256:b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256:820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate

sha256:1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal

sha256:09566b4a3fee6cbaeec9e568a184f4bbb6b6db0a4c6fa3d30654d446b3e20874. Source manifest

sha256:883594e11dc220ba34a5a09ef3444335ca217b5050a204f9cf05a7dd1b97b279. Engine receipt

sha256:80d59f7d08df01bde1b7283271c33458c7c17c4d3c1d91a4727dd1a9ae5b5d08 at receipts/engine/model_admitted/SFT-INFO-SOURCE-REFINEMENT-COARSENING-006-80d59f7d08df01bd.json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-006: refinement_coarsening
- exact observation: {"all_fine_classes_contained":true,"coarse_classes":2,"fine_classes":3,"microforms_retained":3}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:fbee123e2ec1a445816f24a69f4a7b98e505776f917bcd76a13fa0649e70d046; empirical validation sha256:3f03c2f968be7a558719a35bdb46bef4fdde604c91a3172a6fc69441d3bf99dd; external validation sha256:2e6c41d2a5b76e9e96b9b5a2e2ec7edef91c5471051e0b9b037feba5bb88d67e; isolation sha256:d172ae1f2a09bce2c1d450fb1f1355a0189841b3a12dda5704b43ab1ed1731b5; custody sha256:740ac40f3ac26380b7e0b3d82a2aff30606a496865348833870720571528acd8.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

45. Stationary-support correspondence

Family: SOURCE Obligation: SFT - INFO - OBL - SOURCE - 007 Claim: SFT - INFO - SOURCE - STATIONARY - SUPPORT - 007

Question, law and forced result. A finite source is support-stationary exactly when the complete canonical possibility family is identical at every retained position; this is a support law, not an imported stochastic cause.

SOURCE-007 uniquely retains position-invariant-complete-support, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RECORD - CUSTODY - REPRODUCIBILITY - 012
- SFT - INFO - SOURCE - REFINEMENT - COARSENING - 006

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:34709bf730aeae360faaeaa589331b12f01444495cb1dd574fbf08684f2a4155. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-007 product before observation access.

Boundary: Every positive finite SOURCE-007 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-007 uniquely retains position-invariant-complete-support, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__position-invariant-complete-support__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256: 2e5448b238d75d20bfa42792cf189d3305c6be4154bd5a51f10b24d666ff9b4c; generality
 sha256: 8a058d07c75f48b8d9d13d12bce2ad7918bceeee2bba1428e501f7baf9da0278.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt
 sha256: b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
 sha256: b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256: 820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate

sha256: 1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal

sha256: 9aeb2a88607a67fd9c66121635da8754ab83a20819ad78449a7aee1180664d47. Source manifest

sha256: 8d6d4ae651dae7c9adf087e5d904485a51d18f18e2a7e326b3f9a1a9ffb70927. Engine receipt

sha256: 34709bf730aeae360faaeaa589331b12f01444495cb1dd574fbf08684f2a4155 at receipts/engine/model_admitted/SFT-INFO-SOURCE-STATIONARY-SUPPORT-007-34709bf730aeae36.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-007: stationary_support
- exact observation: {"all_supports_identical":true,"positions":3,"support_size_each":[2,2,2]}
- complete family observation custody: 14 records; all rows preserved

- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:1a68a2455fec90465638295176f56ce5420cc55878c8da3d7d46b253f6f749c6; empirical validation sha256:e19e9b8b24256d97669ba09ff7b04acdf6b1d22bfbdd3ee4a6e6454c72611990; external validation

sha256:48c9b62c64adc50334c170b7b53b56e45ace5dd2104b1240a9c4b83c8a2c498a; isolation sha256:ac31fcf56b529a7e949be365e4c9e5dc88cc6ec17cad4868c3157205edb5f8e0; custody sha256:aa3e1649b8a9a3e5160d0b7fcd449057f9e8e19246426d06c2b78100cddb6d66.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

46. Nonstationary-support correspondence

Family: SOURCE Obligation: SFT - INFO - OBL - SOURCE - 008 Claim:

SFT - INFO - SOURCE - NONSTATIONARY - SUPPORT - 008

Question, law and forced result. A source is support-nonstationary when at least one retained position has a different complete possibility family; every change position and support is preserved.

SOURCE-008 uniquely retains position-labelled-changing-support, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RECORD - CUSTODY - REPRODUCIBILITY - 012
- SFT - INFO - SOURCE - STATIONARY - SUPPORT - 007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:d9e4deda247cb25d96458cf6b0e02604b352c0a6acb7be6c75729923ba2b87e2. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-008 product before observation access.

Boundary: Every positive finite SOURCE-008 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-008 uniquely retains position-labelled-changing-support, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__position-labelled-changing-support__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: 5f898f93acb74ba0b346eca5b0fc1f1878750e923290022f6d9e6e58ed5ab0ce; generality
 sha256: cf8dbfa658c31fa5df5ec0154ea77366129c06f4c0bce06e8eff380724a2b961.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt
 sha256: b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
 sha256: b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256: 820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate
 sha256: 1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal
 sha256: a9afe9081fa537a67fbc14ca4c191388d1c4374c32af65ba91f7cbabcac8f46. Source manifest
 sha256: a59ea5921b1e4ee21fc5372c674588645687906280401bc5b9c109509611b005. Engine receipt
 sha256: d9e4deda247cb25d96458cf6b0e02604b352c0a6acb7be6c75729923ba2b87e2 at receipts/engine/model_admitted/SFT-INFO-SOURCE-NONSTATIONARY-SUPPORT-008-d9e4deda247cb25d.js
 n. Closure scope `depth_independent`. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-008: nonstationary_support
- exact observation: {"all_supports_retained":true,"change_positions":[2],"positions":3}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of

order.

Measurement receipt

sha256:4c186f0e81b649d2e7a1b6321b8e8056b2fc5bab10502061af70ea908d2825a0; empirical validation sha256:c494c70cba7c803ba91e062dc47c9a558af433fe6a140e5fc6b1fb0b2b66ad52;

external validation

sha256:591e853c75e5c35aa2d83926e110d489112c8dff9a99126a11aa1ee53ec9e3c4; isolation

sha256:dddfe34a94dd1364d8f5cf56518cd84019e18027ae81fa016d1d374e74ec5efc; custody

sha256:c8770dd6a8b869d132b1e097f1a74393c63fb69e866dfbcd7960b9aa02486750.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

47. Memoryless-source correspondence

Family: SOURCE Obligation: SFT - INFO - OBL - SOURCE - 009 Claim: SFT - INFO - SOURCE - MEMORYLESS - 009

Question, law and forced result. Memoryless support correspondence is forced when every position admits the same alphabet independently and the complete word support equals the full ordered product.

SOURCE-009 uniquely retains complete-position-product-support, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RECORD - CUSTODY - REPRODUCIBILITY - 012
- SFT - INFO - SOURCE - NONSTATIONARY - SUPPORT - 008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:77ba00dd7077d887cc2d386bb7bc287190642f89baf53b526676ba344410b904. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-009 product before observation access.

Boundary: Every positive finite SOURCE-009 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-009 uniquely retains complete-position-product-support, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__complete-position-product-support__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a17219420ce8f60642fa7268e933d4d16117d7eeaed58e546efec65dd067e013; generality

sha256:c19b3007cde492250936f5272a3f73ae7d19b6d7bcb3037bfff3a0166b453b598.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt
sha256: b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256: b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256: 820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate

sha256: 1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal
sha256: 488fafe6e42b09a9099d605d415a93b5e19a37f2ada803d9baafc8d51dbb41d3. Source manifest
sha256: f7ec3e02f92162f7f1ffffef334aed8ba1db2b458d8d11e4c77c55fa4c136ad84. Engine receipt
sha256: 77ba00dd7077d887cc2d386bb7bc287190642f89baf53b526676ba344410b904 at receipts/engine/model_admitted/SFT-INFO-SOURCE-MEMORYLESS-009-77ba00dd7077d887.json. Closure scope
depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INDEPENDENT- EXACT- SOURCE- OBSERVER
- SFT-V1-V2- INFORMATION- OBSERVATION- CORPUS

Observed, adverse and boundary records:

- SOURCE-009: memoryless_support
- exact observation: {"alphabet_size":2,"full_product":true,"positions":2,"word_count":4}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 65142aa3276ab0534467d9dcac1a5ddbc7be02ebf4c14b48b69dde32e709bd5; empirical validation sha256: 9a317b459f4b6cd9bfbe0cfef5966138c67cb55a6b604e4f2fa862327e8cf9ba;

external validation

sha256: a1567fa9436c0bbadfbfb928c8c1e691da7e7942466a9e384e2ec74642c72ae; isolation
sha256: 67065e5d7ddd7901d4237ab74e6af6f8de5f0944b08d504cddc55f2910ccc42f; custody
sha256: d0ab10a253bbd53c87d3c2038881c537885598b027e01cf9beb4ffca44173ece.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

48. Finite-memory source structure

Family: SOURCE Obligation: SFT - INFO - OBL - SOURCE - 010 Claim: SFT - INFO - SOURCE - FINITE - MEMORY - 010

Question, law and forced result. Finite-memory source structure retains a bounded predecessor context and admits exactly the words whose successive contexts occur in the complete declared transition relation.

SOURCE-010 uniquely retains bounded-context-transition-support, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RECORD - CUSTODY - REPRODUCIBILITY - 012
- SFT - INFO - SOURCE - MEMORYLESS - 009

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 47256b3ea98d459ace50ed78c82e267fa2ccd3536809e7d10f40724db5cc8b30. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-010 product before observation access.

Boundary: Every positive finite SOURCE-010 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-010 uniquely retains bounded-context-transition-support, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__bounded-context-transition-support__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: e6d0053bf2f74c9f602c7de7604d4e0988c426e92105ef2ee40179d5e3a5d614; generality
sha256: dd0d13501a46a14a75ba30c4198a4a332b78ede08f58e9e4d4f58dea67a7cb30.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt
sha256: b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256: b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256: 820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate

sha256: 1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal
sha256: 646043e1a0a0a018cb9f9995c2f2fdcaa8d195dba13614bb23e715ef3369d725. Source manifest
sha256: 24d95e33cb79c09c9a9c2d8ea0ff9077cc0c4651942a42200960d07cb0afc8c1. Engine receipt
sha256: 47256b3ea98d459ace50ed78c82e267fa2ccd3536809e7d10f40724db5cc8b30 at receipts/engine/model_admitted/SFT-INFO-SOURCE-FINITE-MEMORY-010-47256b3ea98d459a.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INDEPENDENT- EXACT- SOURCE- OBSERVER
- SFT-V1-V2- INFORMATION- OBSERVATION- CORPUS

Observed, adverse and boundary records:

- SOURCE-010: finite_memory
- exact observation: {"allowed_transitions":3,"context_width":1,"forbidden_word_rejected":true,"valid_words_checked":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: d574fc5dce06460539c5f913c725051b86751fd274a6f44c79b4176f54993168; empirical validation sha256: ece9001eb88db34f9bfd701c185db8826a2c21aa21773f934b114b56bebb6b95; external validation

sha256: 616c71d744a43399c81b65290cd0a1fe2c5135792c78cbc67f2562afc7e66dc5; isolation sha256: 02cb0cfa83b8b96f5ce7a088a7c63b6f41efcd506ee08c6f654fde1352d84ec8; custody sha256: d5e219cb66aa07ed131c3194fe337a873756853053297d70bcf840d766bfb4c2.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

49. Joint-source composition

Family: SOURCE Obligation: SFT - INFO - OBL - SOURCE - 011 Claim:

SFT - INFO - SOURCE - JOINT - COMPOSITION - 011

Question, law and forced result. Joint-source composition is the complete generated ordered-pair support with both coordinate projections retained and each cross-pair occurring exactly once before any dependence restriction.

SOURCE-011 uniquely retains complete-ordered-joint-support, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RECORD - CUSTODY - REPRODUCIBILITY - 012
- SFT - INFO - SOURCE - FINITE - MEMORY - 010

The current reconciliation binds this obligation to the current model-admitted receipt sha256:0342e263902de2bc7898782d759b278f18f9fcd9ae6dbb4a68eeee6d363bd514. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-011 product before observation access.

Boundary: Every positive finite SOURCE-011 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-011 uniquely retains complete-ordered-joint-support, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__complete-ordered-joint-support__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:d750d2ad91f72a25414d93770b700146d29bd6372918e616cd96646f3b50a0a5; generality sha256:e98fe2c7450b7dd8bb1b7f10121c64d1223e64bbfa9634e299224b101dc611a2.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt sha256:b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256: b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256: 820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate
sha256: 1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal
sha256: f6485ca125dca0d3a481578caca42db0862970be70bd53f2f24986ae2993b241. Source manifest
sha256: e9031ee840e548903be2b9b66b3688a2c2bb8706f4b5d16fa87361e4b365f402. Engine receipt
sha256: 0342e263902de2bc7898782d759b278f18f9fcd9ae6dbb4a68eeee6d363bd514 at receipts/engine/model_admitted/SFT-INFO-SOURCE-JOINT-COMPOSITION-011-0342e263902de2bc.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-011: joint_composition
- exact observation: {"both_projections_complete":true,"joint_cells":6,"left_support":2,"right_support":3}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 4d79b11e50be79e7a81016e1d405c53061fae3c0d650048bc88761cb3570cd66; empirical validation sha256: b35e6a285836e3064d45f1d77d18c354e4261dd823c68331979ecca6dc2ca649; external validation
sha256: 8e2da3fb3a7aec79a9d245ec7a0b54eca4369abeb726ccbfc015043c463d85ef; isolation
sha256: d9f410d83b2eb716694c288ad61319a3d7bd3d048399992a5a13971a5c2aaf6a; custody
sha256: 9d3529d48de6a286e6de43629993904ce852443f5752297871bdf26160922152.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object

- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

50. Source dependence and common support

Family: SOURCE Obligation: SFT - INFO - OBL - SOURCE - 012 Claim:
SFT - INFO - SOURCE - DEPENDENCE - COMMON - SUPPORT - 012

Question, law and forced result. Source dependence is a retained restriction of joint support that is not the full product while preserving the declared marginal projections; missing joint cells remain explicit closed possibilities.

SOURCE-012 uniquely retains proper-joint-support-with-projections, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RECORD - CUSTODY - REPRODUCIBILITY - 012
- SFT - INFO - SOURCE - JOINT - COMPOSITION - 011

The current reconciliation binds this obligation to the current model-admitted receipt sha256:d548d93e0d37da8ac037cac7406f0828cba867aa0b6379cd563ae3e1a6312064. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-012 product before observation access.

Boundary: Every positive finite SOURCE-012 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-012 uniquely retains proper-joint-support-with-projections, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__proper-joint-support-with-projections__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:fb66116ac9893d7cf3a5cfceb6d9f0642a18b1b31b45f92a97c1880748ddbe4d; generality sha256:9bc12e2912bb986c64bbc142f37dd99bc0933614799d5903ca6bd75f18d873ba.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt sha256:b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256:b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256:820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate

sha256:1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal

sha256:83bc52eb4c8187f719ede75b7d6c155bd18f5e83305848b14d1a87c03697b33c. Source manifest

sha256:a41874c61ff1d792cf3960d0d2490212734f7fa3d467c439aca16b6fffb85115b. Engine receipt

sha256:d548d93e0d37da8ac037cac7406f0828cba867aa0b6379cd563ae3e1a6312064 at receipts/engine/model_admitted/SFT-INFO-SOURCE-DEPENDENCE-COMMON-SUPPORT-012-d548d93e0d37da8a

.json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-012: dependent_support
- exact observation: {"marginals_complete":true,"product_cells":4,"proper_subset":true,"retained_joint_cells":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:fbf633a957fcae77c96f52121c34aa1c2b44435da63053df0bb20e98f22195d5; empirical validation sha256:04cbca89558e119f7a0a930ebb46e1796cde443042605a8c8b21ebbb00d8d01a; external validation

sha256:377353c4b1c5204546e061808f382c4e0479e4ebd75f01ae77f690a3b27748a6; isolation

sha256:bb1b78aebfdf204c1a6974cdabf89f9462209cbb8de66b2e4a4e12c857b85b16; custody

sha256:2bf95f4b43a30280b0b97a99bb877c929477eb1780a559335eb31d2dd515b905.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source

- no failed route retires an obligation or changes protected authority

51. Source extension by exact successor

Family: SOURCE Obligation: SFT - INFO - OBL - SOURCE - 013 Claim: SFT - INFO - SOURCE - SUCCESSOR - 013

Question, law and forced result. Adding one fresh canonical source form preserves the prior support, generates each new joint and observation relation exactly once and introduces no frequency parameter.

SOURCE-013 uniquely retains fresh-source-form-successor, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RECORD-CUSTODY-REPRODUCIBILITY-012
- SFT-INFO-SOURCE-DEPENDENCE-COMMON-SUPPORT-012

The current reconciliation binds this obligation to the current model-admitted receipt sha256:20ce59552d421058110dc981a1ba4783d236c0c264b8dd5fd34dfcdb8aa80449. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-013 product before observation access.

Boundary: Every positive finite SOURCE-013 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-013 uniquely retains fresh-source-form-successor, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__fresh-source-form-successor__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:dd46b2ed73904610f9de6ac5c1b4dd6da58871417244c979a21b676f8cd0c566; generality sha256:35056759adf7e89742b2ddc20a48eff7a0253bae21f10d96190183d64191973f.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt sha256:b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural

absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
 sha256: b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256: 820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate
 sha256: 1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal
 sha256: 7386c1b63d11b69aec43d0c4149c557aa12853d3a2927e03313616adb754e777. Source manifest
 sha256: 4fb56f5c233271ba654af10623b9c047a885f76e033fa4f1b2e1424746df2377. Engine receipt
 sha256: 20ce59552d421058110dc981a1ba4783d236c0c264b8dd5fd34dfcdb8aa80449 at
 receipts/engine/model_admitted/SFT-INFO-SOURCE-SUCCESSOR-013-20ce59552d421058.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-013: source_successor
- exact observation:
 {"frequency_parameter_added":false,"prior_forms_preserved":true,"prior_support":2,"successor_support":3}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: db25d52db728b36489df77c803bb91e2f64d93c1fb1d3685d98317feaf05c335; empirical validation sha256: b0c8dc61aabcdf3c5236b42ef6bed4c8b693c9db1e62c2ae68b59a399b955f19; external validation
 sha256: 7d634bce6a9b72a0938b55fb041c65bcf31bcf3e0c5718820687e09320d1ef37; isolation
 sha256: 181ae51c356b6b1883285e96c0d935233f4e81c865408aa58b525ae6642277a6; custody
 sha256: 0a0568251c4631dc9a869e6d2e1101f946340b40ec6d06c881b453eb538b64b1.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

52. Source completeness and boundary certificate

Family: SOURCE Obligation: SFT-INFO-OBL-SOURCE-014 Claim: SFT-INFO-SOURCE-COMPLETENESS-014

Question, law and forced result. Source-family completeness is the one-to-one reconciliation of all fourteen frozen obligations with exact receipts, observations and ownership boundaries.

SOURCE-014 uniquely retains fourteen-source-obligation-ledger, complete source custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RECORD-CUSTODY-REPRODUCIBILITY-012
- SFT-INFO-SOURCE-SUCCESSOR-013

The current reconciliation binds this obligation to the current model-admitted receipt sha256:ecfa32db51e739c58d4dd241a4406a1fe1d09a1fb689c3a4e98230d3f03b4c05. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SOURCE-014 product before observation access.

Boundary: Every positive finite SOURCE-014 support, sequence, transition, spatial or network relation and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SOURCE-014 uniquely retains fourteen-source-obligation-ledger, complete source custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-position-or-node-label__fourteen-source-obligation-ledger__complete-transition-and-path-custody__complete-declared-source-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:d307e6ae745de1b437c94d823de65cd8f081bde75d560a50058096e07d91090; generality sha256:852e52fc062d39426bc4cef91af62031575fbcaad40a0efc1f4036f25f8e0979.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial or duplicate support changes source possibilities.; receipt sha256:b8ccca085e25652a7b63fd86afa8f1f3b02c5c7f7f32086fc8bbaf495e3f048a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic source, probability distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no ontic randomness, hidden state path, erased route or omitted source possibility; no sampled support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:b5c50b8a55825d10a90728c0162f274ccf028a016ca2fe2776bbf18f5335a390.

Independent reconstruction and engine admission. Independent implementation

sha256:820989e83c85c81aef3c274ff26d6fba8b2417aa4e7e1ed4dae2eec164364126 regenerated the declared census and sole survivor; independent certificate

sha256:1379063baaf558956faa71045fd1278be5407240e4b45d98870b7fae1b9657ac. Derivation seal sha256:85b6b00480207b67747cb09fd6293b6dcf42080e34b37d4040c612d610f5a979. Source manifest sha256:58701c9e7c74aecdf5def91d6930bdb2c63c1545b19d35b7eb242862ffc8ba7c. Engine receipt sha256:ecfa32db51e739c58d4dd241a4406a1fe1d09a1fb689c3a4e98230d3f03b4c05 at receipts/engine/model_admitted/SFT-INFO-SOURCE-COMPLETENESS-014-ecfa32db51e739c5.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SOURCE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SOURCE-014: source_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":14,"omitted_owners":0,"registered_obligations":14}
- complete family observation custody: 14 records; all rows preserved
- complete finite support, sequence, transition, spatial, network and joint-source enumeration supplies the observation; no imported stochastic cause, hidden path or omitted possibility enters

Falsification condition: Reject if the value-free registry, complete source census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:b9838aca2f23e065a5c4de35167ca03a364a585e4eff8291a931d62c84114bfe; empirical validation sha256:f9b1cbb86e515029af20e756d40cff0040fb1a8caf8f70f7c0ae0e60905ad42b; external validation

sha256:702f2b450ec5e8b87e8e45dbbf1f153ab5ce9b98e6cff139810b3e1eb816604f; isolation sha256:2cffacf02a3919276dcb95cef869e9dde1f9277b15a2a4f3f3341369d93dab52; custody sha256:80c80adc2fa48b59f01187dd924511f45dc7585c67a5c86933f8f01944a7e63d.

Prohibited inputs.

- no axiom, imported stochastic source, probability distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no ontic randomness, hidden state path, erased route or omitted source possibility
- no sampled support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

Complete-field family: MEASURE

53. Distinction-count information measure

Family: MEASURE Obligation: SFT-INFO-OBL-MEASURE-001 Claim: SFT-INFO-MEASURE-DISTINCTION-COUNT-001

Question, law and forced result. Distinction-count information is the exact count of unordered source pairs separated by a complete observation partition, with every closed pair retained in the complementary ledger.

MEASURE-001 uniquely retains complete-retained-pair-count, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SOURCE-COMPLETENESS-014

The current reconciliation binds this obligation to the current model-admitted receipt sha256:9439d2704d1942362e1ab678342db62738c3d78f6de66baf8d10f20c9f6e0784. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-001 product before observation access.

Boundary: Every positive finite MEASURE-001 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-001 uniquely retains complete-retained-pair-count, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__complete-retained-pair-count__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:8cd36d287be832a376689ff98a057bc56cb185ceef0afb72086340c46a67a2b4; generality sha256:80ba92fddc09741bf401107181e33d82c5153572ab84c30ecd8889e0fe84ae66.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt sha256:534af192501c93e31482880b8440c07057aeaa05e17d359d3237d5b3ed67ff30.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal

sha256:e84b09d49d94f3d7204fcb152ac2fd180e36a22b980b40f96d4a97ad69330364. Source manifest sha256:4bb3338bc299544c3bee9720c33f50bda01d71eb735673eb59332700ab00fae7. Engine receipt sha256:9439d2704d1942362e1ab678342db62738c3d78f6de66baf8d10f20c9f6e0784 at receipts/engine/model_admitted/SFT-INFO-MEASURE-DISTINCTION-COUNT-001-9439d2704d194236.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT- INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-001: distinction_count
- exact observation: {"fine_retained_pairs":6,"possible_pairs":6,"source_forms":4}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:cf34c6c39a402c229946c43046fd042411ddfdd618efda8de25fb748d4115c42; empirical validation sha256:e1d9b45be14458963c3021d47036180b4ce01682b1b004f0ac19e610e1dede6b; external validation sha256:1d552dc1ab76f3a3d56aa9611e5525d9e33b8baf7bba4d2ebe76b476cf9393be; isolation sha256:e72c0aba36860cb53c9b8c8ea9ca1ab910d48a48048d6df9737bf0ffd873fb79; custody sha256:41f3a0f88f3265d12817c8cc148aa3bc39142b36e8969b21f1b02a9bf5c296b2.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

54. Combinatorial information quantity

Family: MEASURE Obligation: SFT- INFO- OBL- MEASURE- 002 Claim: SFT- INFO- MEASURE- COMBINATORIAL- QUANTITY- 002

Question, law and forced result. Combinatorial information quantity is the exact cardinality and generated-position structure of the complete canonical possibility support; no analytic logarithm is required as a premise.

MEASURE-002 uniquely retains complete-generated-support-cardinality, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT- ROOT- THERE- IS- NO- NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SOURCE-COMPLETENESS-014
- SFT-INFO-MEASURE-DISTINCTION-COUNT-001

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:9dc794f87f65d1f7e6d6b4aa989b0d36aa45ba851af5f466a5b41e99d6efd451. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-002 product before observation access.

Boundary: Every positive finite MEASURE-002 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-002 uniquely retains complete-generated-support-cardinality, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__complete-generated-support-cardinality__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:6170fed3266c1a59a8dfea47cc9e08842d6c1e820f7aa55ae1cd9ecee02505e1; generality
sha256:976a8b73e72b9ae6d8592db5ed5018d9a93af8cf959abf8d29cf2f146b6f3c7a.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt
sha256:534af192501c93e31482880b8440c07057aeaa05e17d359d3237d5b3ed67ff30.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate
sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal
sha256:318f6272b98066683bfff1fb47b37f2693790f5893a6211c6f390db4ec08e75ab. Source manifest
sha256:506a96415855c530eeb8a2a9594fa81bd061994bc23af40dbe1b45f5e51c01cf. Engine receipt
sha256:9dc794f87f65d1f7e6d6b4aa989b0d36aa45ba851af5f466a5b41e99d6efd451 at receipts/engine/model_admitted/SFT-INFO-MEASURE-COMBINATORIAL-QUANTITY-002-9dc794f87f65d1f7.j

son. Closure scope `depth_independent`. External status `empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-002: `combinatorial_quantity`
- exact observation: `{"alphabet_size":2,"complete_word_support":8,"positions":3}`
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:67513adb76945f77f61ced26e4bb57ba0d7f32c48ba90bf9f02be6bfd14ef5cd; empirical validation sha256:7f218ff535bb4aa40435a99d791e561c8bbf7b32ab903277ae4a4767157c2653; external validation

sha256:16f864c901f217b707f7b35ad171d57e8fea33d375efe4634359028a93e80c93; isolation sha256:beb949263950673fd05d1dd53606e2d4555ba727ff16d47f2154be7eec9e3d91; custody sha256:06f445b75a3002ee2c21c67ae7315ef4f056bfe47270711f705069e3b0067021.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

55. Operational discrimination cost

Family: MEASURE Obligation: SFT-INFO-OBL-MEASURE-003 Claim: SFT-INFO-MEASURE-OPERATIONAL-COST-003

Question, law and forced result. Operational discrimination cost is the least generated family of allowed observations whose joint signatures distinguish every source form, proven by complete smaller-family elimination.

MEASURE-003 uniquely retains least-complete-observation-family, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SOURCE-COMPLETENESS-014
- SFT-INFO-MEASURE-COMBINATORIAL-QUANTITY-002

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:8f164e185a31b3f03f1910a4955707e00fa261b5f5d39adaa5fde94581e4b483. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-003 product before observation access.

Boundary: Every positive finite MEASURE-003 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-003 uniquely retains least-complete-observation-family, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__least-complete-observation-family__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:b3d781d0b9899e997758caa1cf0e11068172d075644384d98a1f8d06f6d817ea; generality
 sha256:a8550c70cb0f8a9410dd301990ff901021399a43f73eee2dd61725d8fc4e69f3.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt
 sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate
 sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal
 sha256:2f18c725f97d8fa35e79721adb2e5d168846da5a14c16e59c80082553dbc2d66. Source manifest
 sha256:08212b63b09f60138cc928fb1ca8269fc3aee5738ac983d92ee63ab387214e2b. Engine receipt
 sha256:8f164e185a31b3f03f1910a4955707e00fa261b5f5d39adaa5fde94581e4b483 at receipts/engine/model_admitted/SFT-INFO-MEASURE-OPERATIONAL-COST-003-8f164e185a31b3f0.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-003: `operational_cost`
- exact observation: `{"all_three_distinguish":true,"coordinate_observations":3,"every_two_fail":true,"forms":8}`
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:58dfd641a24ff205420f25193f7b03028c5f3d1e368b151ccc120f24430e1911; empirical validation sha256:127b7081ecec6b63d676d1c4820312b45bd49bf0437229fd3e32b3dc65d7e91ec; external validation sha256:65fcc06f9d56276304b8dd12dd967c5c3dadfeb10da2f3a1c3b8730631c7525d; isolation sha256:6144ccb1ef49a881787e97c79af850464a3f7ca37c2f3f2617f1ba97fd7f4b8d; custody sha256:9882667f039dd6038d4c225aafd5c5f2bdf401969c3291ec0316dd66d56e919c.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

56. Description length on a fixed grammar

Family: MEASURE Obligation: SFT-INFO-OBL-MEASURE-004 Claim: SFT-INFO-MEASURE-DESCRIPTION-LENGTH-004

Question, law and forced result. Description length is the exact retained token count of a valid expression in one frozen grammar, including every delimiter and instruction needed for reconstruction.

MEASURE-004 uniquely retains registered-token-trace-length, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SOURCE-COMPLETENESS-014
- SFT-INFO-MEASURE-OPERATIONAL-COST-003

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:4a5376dd6e2283698cbfbd0d6c84ba93afee70339f93f274ced62ae37a6a47d9. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-004 product before observation access.

Boundary: Every positive finite MEASURE-004 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-004 uniquely retains registered-token-trace-length, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__registered-token-trace-length__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:038a6fca17564ae20b9dc88b3c10442fe0525773c36584c5ebae456ad7ea1dda; generality sha256:8ed90ff06ea64871e3075c1956732a968052527257ddf2bb9f06e0e1b0683feb.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal sha256:8e8fe2dc6dbca9a5542a5c1ce623074c6a7a8cac3145b93c3b1e56bd6913a70e. Source manifest sha256:42c0ff18bd383d69aecd41d5e1502a9fe6593de9362be70e82ff4a2a88376d90. Engine receipt sha256:4a5376dd6e2283698cbfbd0d6c84ba93afee70339f93f274ced62ae37a6a47d9 at receipts/engine/model_admitted/SFT-INFO-MEASURE-DESCRIPTION-LENGTH-004-4a5376dd6e228369.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT - V3 - INDEPENDENT - EXACT - INFORMATION - MEASURE - OBSERVER
- SFT - V1 - V2 - INFORMATION - OBSERVATION - CORPUS

Observed, adverse and boundary records:

- MEASURE-004: description_length
- exact observation: {"grammar_tokens":4,"retained_description_tokens":4,"unregistered_tokens":0}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: ae74d214126d501a2b13dd30c17adb947a8963bc391194ed2ac2699cb590bdf7; empirical validation sha256: c70db9c2ed37b8bb8364acafa4d4c23831f031c68a5ed18a9f2f253b0ddb38db; external validation sha256: ec785cc6655e7a3b58fc05382423fed5b72511f91c6b63e47afbc296425a64c7; isolation sha256: 5d93c0ae5ffac419d43c9046c68e1b380d3b5940b9e2f68b149660f1574cdd1d; custody sha256: 877434509e09df372d1b26cf1d33ba039e4bce1b7f044a6f9fee28343b43d0b1.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

57. Algorithmic-description correspondence boundary

Family: MEASURE Obligation: SFT - INFO - OBL - MEASURE - 005 Claim:
SFT - INFO - MEASURE - ALGORITHMIC - BOUNDARY - 005

Question, law and forced result. Algorithmic-description correspondence is admissible only for a completely generated fixed program grammar and resource boundary; the least target-producing description is found by exhaustive comparison, while unrestricted universal complexity remains a computability boundary.

MEASURE-005 uniquely retains fixed-grammar-least-program, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SOURCE - COMPLETENESS - 014
- SFT - INFO - MEASURE - DESCRIPTION - LENGTH - 004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: fa512d0c822f7f1a3fe46b6868ce46f84765f015a78e8beeffabe15a999d1388. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-005 product before observation access.

Boundary: Every positive finite MEASURE-005 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-005 uniquely retains fixed-grammar-least-program, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__fixed-grammar-least-program__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:3a40990ff3e12b1b402320787c4fb39ec15b042161f2d27c21a16bce05a6f076; generality sha256:23de7e632ed64e1f305ac9839b7d484bcc71e0c665254eba507c17248e21b317.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal sha256:269fde79047df1a945fe3b77680fb5739dfbac7174b3a556f42d87963738c07. Source manifest sha256:67f7313ae4643bdcec7dd5d3f11a51f09b3bd95e16f4cf3fa6f78b88b3ea25b4. Engine receipt sha256:fa512d0c822f7f1a3fe46b6868ce46f84765f015a78e8beeffabe15a999d1388 at receipts/engine/model_admitted/SFT-INFO-MEASURE-ALGORITHMIC-BOUNDARY-005-fa512d0c822f7f1a.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT- INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-005: algorithmic_boundary
- exact observation: {"least_cost":3,"least_target_program":"alternate","registered_programs":3,"target_programs":2,"unrestricted_complexity_claimed":false}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:6ff97fc0fc09a46c08fc953ba13500b5bfb56f9eb2ff0f4916e2b0eb81ddecbb4; empirical validation sha256:f55f7949e743f24e483afdbdae60cf8c4145bdf7c10b161cbc091a7fc3b7add5; external validation

sha256:51f06dfb156383c0b317e4b952fc7a1e6e90f967c0141d3b14b1d6c45c839ef3; isolation sha256:32cc426ad5d7f03ba60f4001c802e848471caa7aa428d06b69f3b23cb5844fd0; custody sha256:82c4e2723d9207e4bdbd2f3b2ca80b641bb4395845f9f84c39e807368d502554.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

58. Partition refinement information order

Family: MEASURE Obligation: SFT - INFO - OBL - MEASURE - 006 Claim: SFT - INFO - MEASURE - PARTITION - REFINEMENT - 006

Question, law and forced result. Observation information is ordered by partition refinement: every fine class lies within one coarse class, so refinement can retain distinctions but cannot silently close a previously retained pair.

MEASURE-006 uniquely retains class-containment-refinement-order, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SOURCE - COMPLETENESS - 014
- SFT - INFO - MEASURE - ALGORITHMIC - BOUNDARY - 005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:e09105740ebd9d57f1e50a4d25f63ccd056a322e0563420790978b5bab16b577. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-006 product before observation access.

Boundary: Every positive finite MEASURE-006 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-006 uniquely retains class-containment-refinement-order, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__class-containment-refinement-order__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-success-or-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:01ce0bacc882b7b5796f2a881e4f1fa9192a01e9c2066e7216087e53b86dbf93; generality sha256:9bc17b7a0905ec0e737d0dc3426b58402a1de2cf4c5359db95ac81eaa5cf1d7c.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal sha256:5f48e41ff849c2d08d711787c58a80452b1dd57c8c8a226890c993e3f8d26974. Source manifest sha256:5fa09a824f9c0f6051f00fd14d3467bcf453bafef9191cc9d86641551267ea90. Engine receipt sha256:e09105740ebd9d57f1e50a4d25f63ccd056a322e0563420790978b5bab16b577 at receipts/engine/model_admitted/SFT-INFO-MEASURE-PARTITION-REFINEMENT-006-e09105740ebd9d57.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT- INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-006: partition_order

- exact observation: {"coarse_classes":2,"containment_chain":true,"fine_classes":4,"middle_classes":3}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:47502514ffc7814444ff912f0c6c46379106e3d4e6fd68ed200207c4937bc0b4; empirical validation sha256:d1bf03e46b4d4503fba5a031ee1e0f3f4721b67129dd3d657bdf3a28605ba641; external validation

sha256:9ad0dfc78aad1335efec8a2399f8e7dd47adf701f08c25d54731b833a348bdba; isolation sha256:2d165a610924171ea7394a91ce48395ec44eabdeccc747dde629c0945aa624f3; custody sha256:aaef9e9a9e5956064ef524a6cffb727bb1caf7d72e2be6e0628454470c7de68b.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

59. Additivity on independent product support

Family: MEASURE Obligation: SFT - INFO - OBL - MEASURE - 007 Claim: SFT - INFO - MEASURE - PRODUCT - ADDITIVITY - 007

Question, law and forced result. For independent complete word supports, composition concatenates retained positions, multiplies support cardinalities and adds the exact position-based information units.

MEASURE-007 uniquely retains position-additive-product-support, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SOURCE - COMPLETENESS - 014
- SFT - INFO - MEASURE - PARTITION - REFINEMENT - 006

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:fe66cd4675b04dd5a7d1a55443e011b097509ad9ddd03fcf97a9cd1e31178131. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-007 product before observation access.

Boundary: Every positive finite MEASURE-007 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-007 uniquely retains position-additive-product-support, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__position-additive-product-support__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:381c4ce2ccd1e7847ce5dcf5b6c4f33909cf067e5f1bd1402f325723e7793f11; generality sha256:15594af3c579ede0067fdc4112d6cbd8a267d6373518a60e4ccb7c7107c73965.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate

sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal

sha256:db9bf03a1ad88efb13968a2ed6a86b63283897f54ab6c83f00dbf8cb96ff53ee. Source manifest

sha256:8bacd46792c0f8a5d44f5de96f37ad64cc41705995f64ed3abf5da75b78c4136. Engine receipt

sha256:fe66cd4675b04dd5a7d1a55443e011b097509ad9ddd03fcf97a9cd1e31178131 at receipts/engine/model_admitted/SFT-INFO-MEASURE-PRODUCT-ADDITIVITY-007-fe66cd4675b04dd5.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-007: product_additivity
- exact observation: {"joint_positions":5,"joint_support":32,"left_positions":2,"left_support":4,"right_positions":3,"right_support":8}

- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e21d79cd0964999e20d91ded5ad128a459470f22a59f25ffe5f2a8679da400ea; empirical validation sha256:e3f377430013670279216d0e79bec0445ddb89b62ba7a8527020af29ca037123; external validation

sha256:f574c298557d043cbee16be17056f9d7d21676ef881dca91d72d89bd0dec29b2; isolation sha256:c77de63a239c0435235bc524264eb3eac35b6dfffbb15ed96cc8439eacb8cf640; custody sha256:a5583142a250625d2d8478cf283fa0d8af361b2d4a6f80810b378c28cd2bcfb1.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

60. Subadditivity under shared support

Family: MEASURE Obligation: SFT - INFO - OBL - MEASURE - 008 Claim: SFT - INFO - MEASURE - SHARED - SUBADDITIVITY - 008

Question, law and forced result. For restricted joint support, a complete pair of marginal descriptions reconstructs each joint form, so the least joint discrimination cost cannot exceed their combined retained costs.

MEASURE-008 uniquely retains joint-cost-bounded-by-marginal-costs, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SOURCE - COMPLETENESS - 014
- SFT - INFO - MEASURE - PRODUCT - ADDITIVITY - 007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:e094503f68b90836697b64cd3c27e061d4be8248322a4e18429a44e161d5dadd. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-008 product before observation access.

Boundary: Every positive finite MEASURE-008 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-008 uniquely retains joint-cost-bounded-by-marginal-costs, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__joint-cost-bounded-by-marginal-costs__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure

-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:1ab36a52bfac63a61c4e35366ae56a7a0ffec7dea4a04dc7a2e26abdf19d91d; generality sha256:3da2cf1a5094e4556e858d081c309e9e7e49da06aeceb6e2935a2a4225245b9c.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate

sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal sha256:57956f02d9a7080770bdfd802e3219f06cafd2e2fa2c6a3935a346cfee5cd4c8. Source manifest sha256:b2596360e70444f82968120c8ea2872dfa739bdf7b6b24278f23b453dbff8175. Engine receipt sha256:e094503f68b90836697b64cd3c27e061d4be8248322a4e18429a44e161d5dadd at receipts/engine/model_admitted/SFT-INFO-MEASURE-SHARED-SUBADDITIVITY-008-e094503f68b90836.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-008: shared_subadditivity
- exact observation: {"joint_cost":1,"joint_not_greater":true,"marginal_costs":[1,1]}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:18e0366c95a95ae8929c257de693b9c9407f77970791eaab718b60de598841b2; empirical validation sha256:9461c6cd428d26a2b8997ed657594a4b1779222313523016f078009527ea14cc; external validation sha256:6434132547b7c2a6d5fe058f5e7cc469860b6c0a5a5936395bbbcef0c6bc6148; isolation sha256:d62df12d1aa4ee4c5b3f49798d0c544fb1df517254f6e54eebdc39389c2eee11; custody sha256:718bb4752d0500f3e5dff05059eac3adf0712bebdea50e49e506a257beba827b.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

61. Monotonicity under observation coarsening

Family: MEASURE Obligation: SFT - INFO - OBL - MEASURE - 009 Claim: SFT - INFO - MEASURE - COARSENING - MONOTONICITY - 009

Question, law and forced result. Observation coarsening merges partition classes and therefore cannot increase the complete retained-pair ledger; every lost distinction appears in the closed ledger.

MEASURE-009 uniquely retains nonincreasing-retained-distinction-count, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SOURCE - COMPLETENESS - 014
- SFT - INFO - MEASURE - SHARED - SUBADDITIVITY - 008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:3c02f0a645253b849b2b9bb524388b03628b57fa58d335c709fb091dc5a26b54. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-009 product before observation access.

Boundary: Every positive finite MEASURE-009 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-009 uniquely retains nonincreasing-retained-distinction-count, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__nonincreasing-retained-distinction-count__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:060a3e6950a0dd8d667b2612c200d2dba3b311bb70b148624ce965c5ba68d; generality

sha256:ea55b3213e07cbe5736b5e2a92b9ced71c61b5a678df05beff9ab792a401d9d5.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt
sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate

sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal

sha256:334dd81af5430a74c757896deb53586b3369a32d9f086ee506f048c2d8bfe2b8. Source manifest

sha256:69da2e3b9123a030f96151452e9275771da65fbdf7fab41df5045265515070a6. Engine receipt

sha256:3c02f0a645253b849b2b9bb524388b03628b57fa58d335c709fb091dc5a26b54 at receipts/engine/model_admitted/SFT-INFO-MEASURE-COARSENING-MONOTONICITY-009-3c02f0a645253b84.

json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-009: coarsening_monotonicity
- exact observation: {"nonincreasing":true,"retained_pair_counts":[6,5,4,0],"zero_is_structural_absence":true}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:8a9b7b1ed1448c2b0288611d2e1e52918eec1701340ba4c4c6c2f4cc520ad978; empirical validation sha256:d3ce2abe1e323679c5379782f6f31ce48bcf525cf0ede3c12047be994cea6600; external validation sha256:59e37a78bd26a29163ce07f079d7f28072fb0220ac95997726135aa0533d2d6a; isolation sha256:7f767bdfdd2366b81c182720f2d85a041f89ea24fbaa1a003412e3fa86dc35aa; custody sha256:eed02d8a89e99830934bb807749219388a3f01018e1c216b1fdd64908f6ff42c.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

62. Exact information balance ledger

Family: MEASURE Obligation: SFT - INFO - OBL - MEASURE - 010 Claim: SFT - INFO - MEASURE - BALANCE - LEDGER - 010

Question, law and forced result. At one fixed source support, retained and closed distinction ledgers are disjoint and together exhaust the complete pair support, forcing an exact transformation balance.

MEASURE-010 uniquely retains retained-plus-closed-exhaustion, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SOURCE - COMPLETENESS - 014
- SFT - INFO - MEASURE - COARSENING - MONOTONICITY - 009

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:f9e9e4248e852e3851ccccdcacf11f02f49b5da05df2e4d85c2b7a1a3dc5fc30d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-010 product before observation access.

Boundary: Every positive finite MEASURE-010 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-010 uniquely retains retained-plus-closed-exhaustion, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__retained-plus-closed-exhaustion__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:c9c7c0a58767d5e56ac7c5e8235014ce398e265f13185dc2341b7215aeaf0c8a; generality sha256:79276d5cf0a0107f11f9b370a33b1d58970fc5898a17cca601fca3ab086866b3.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit

downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt
sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate

sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal

sha256:a7a65b15ede2b456467f1f3b333f15e502d16fbe6ce86b5a3a10ab08ca145f97. Source manifest

sha256:871c80576f9f568084e516b20c483fb0e4fc241e149e4f154f7b06c171fb3290. Engine receipt

sha256:f9e9e4248e852e3851cccdcaf11f02f49b5da05df2e4d85c2b7a1a3dc5fc30d at receipts/engine/model_admitted/SFT-INFO-MEASURE-BALANCE-LEDGER-010-f9e9e4248e852e38.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT- INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-010: balance_ledger
- exact observation: {"closed_pairs":2,"disjoint_and_exhaustive":true,"retained_pairs":4,"source_pairs":6}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:7b5adf44e43f97fe720cce2e9333823e85f5bce1266a83abc4efb193d2970821; empirical

validation sha256:2f3a0c706c40ae5ef862b7a5465fa728b3ef209d8135e726db1471e0d75cff47;

external validation

sha256:7af725f916cff67a65202356ee73994e0443d2c2d01705b54ca4eec398c0e8c1; isolation
 sha256:375273c2bf8fa64780a07b951f3697f5ce6d7c55417559e67b99a148dfe8ea35; custody
 sha256:da57d1ad5c72d3a2a9a82099da268d8971d9e53dd484b81f31ecc595c0ca42e0.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

63. Relative information between observations

Family: MEASURE Obligation: SFT - INFO - OBL - MEASURE - 011 Claim: SFT - INFO - MEASURE - RELATIVE - 011

Question, law and forced result. Relative information from one observation to its refinement is exactly the source-pair distinctions retained by the refinement but closed by the coarser observation.

MEASURE-011 uniquely retains refinement-restored-distinction-ledger, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SOURCE - COMPLETENESS - 014
- SFT - INFO - MEASURE - BALANCE - LEDGER - 010

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:796a221d4c72c016b019feedbb84f9da2af8de73664bde26f22538613f04d686. A missing
 dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-011 product before observation access.

Boundary: Every positive finite MEASURE-011 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-011 uniquely retains refinement-restored-distinction-ledger, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__refinement-restored-distinction-ledger__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:a15247e02fd5d44eb73a2f11be1e4250abc7b0ba0f00a239326fa7f4e928d1c8; generality
 sha256:267342f1ed13d66507e48b770e91b6b477319b58de8ed69a5219df327d0dce83.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt
sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate

sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal

sha256:b698e7a55f1c85a7de90bdc64f1875e3e2e3c4995ed1f6e1189d3c117ca530db. Source manifest

sha256:47f288f8842f70ed509ad9450f909797a0e6962cecd8be8f44104b6822a454db. Engine receipt

sha256:796a221d4c72c016b019feedbb84f9da2af8de73664bde26f22538613f04d686 at

receipts/engine/model_admitted/SFT-INFO-MEASURE-RELATIVE-011-796a221d4c72c016.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT- INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-011: relative_information
- exact observation: {"coarse_retained":4,"fine_retained":6,"restored_distinctions":2}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:2756668e765b1e6034ed13db117074b8832b2a9f7b241ce8de75ef3c11f834d4; empirical

validation sha256:3797a7013919912199462f7d43edfaea28af2ebb9d3ecf80e68d0a723e302d28;

external validation

sha256:891c1af0892794eca3144d0800024d802b40ec238feb74c23e333821251d8cde; isolation

sha256:25d86591ea59d77eab5cddc7c0d91f1b78dd67d1e9b7800067014dbdab275e70; custody

sha256:8da96dfa17bcf9d3727393ab0a9e2b51e174d70ae5f4c9fed86e9929c0ab317a.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

64. Information divergence correspondence

Family: MEASURE Obligation: SFT - INFO - OBL - MEASURE - 012 Claim: SFT - INFO - MEASURE - DIVERGENCE - 012

Question, law and forced result. Finite information divergence correspondence is the exact symmetric difference between two complete retained-distinction ledgers; structural absence denotes identical observations and a positive count denotes disagreement.

MEASURE-012 uniquely retains partition-disagreement-count, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SOURCE - COMPLETENESS - 014
- SFT - INFO - MEASURE - RELATIVE - 011

The current reconciliation binds this obligation to the current model-admitted receipt sha256:6a0996db7794d1ee473d4b42c50c096ff1dd60e88a43c924ea7d33d739db39db. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-012 product before observation access.

Boundary: Every positive finite MEASURE-012 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-012 uniquely retains partition-disagreement-count, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__partition-disagreement-count__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:f9568630ddcdf5e6323491211149f3a0ec46d77e993c8371bacd7c041c3286e1; generality sha256:5df0104a3c449aac2e58eff5ae29b129c7fa2822e71edbd5ff1e7cda91d0dc0b.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256: 5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256: c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate
sha256: 47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal
sha256: f1c424c3fadba4df2054a92c96e5fb508d771955307537e277a58d05d94a1a9f. Source manifest
sha256: f047b64b792607c355a1f1c1fe169b82fcbec722172ef26a8cb03c1b12c79559. Engine receipt
sha256: 6a0996db7794d1ee473d4b42c50c096ff1dd60e88a43c924ea7d33d739db39db at receipts/engine/model_admitted/SFT-INFO-MEASURE-DIVERGENCE-012-6a0996db7794d1ee.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-012: divergence
- exact observation:
{ "changed_partition_disagreement": 1, "identical_partition_disagreement": 0, "zero_is_structural_absence": true }
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 8a89fc686138affa9aea7b02baffff125e547e1f5028d0899e6f313e1f7c81b52; empirical validation sha256: 1b1fc6236085629a53e41ff72995caf6be4a8b0cb7e2906805749125d6d2610e; external validation
sha256: 8a3b5ac2b3d7fb63767a41df8e9034f7de094d8ba21d20f9d9dbbf0adf7b93ae9; isolation
sha256: 016bbf8bbd657497ffdc9e340bc7dfdba14f1abfa887c4825d2ccce5a2d2e941; custody
sha256: 162bbf1d1afe0125738c3e2be8142a980093dae5ce4ad9af9cf666e27289c9c2.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar

- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

65. Information geometry on exact parts

Family: MEASURE Obligation: SFT - INFO - OBL - MEASURE - 013 Claim: SFT - INFO - MEASURE - GEOMETRY - 013

Question, law and forced result. Complete observation partitions form an exact finite information geometry under retained-pair symmetric-difference distance, which is symmetric and satisfies the triangle relation without irrational coordinates.

MEASURE-013 uniquely retains pair-ledger-symmetric-difference-metric, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SOURCE - COMPLETENESS - 014
- SFT - INFO - MEASURE - DIVERGENCE - 012

The current reconciliation binds this obligation to the current model-admitted receipt sha256:9b99c3d97aab0b792f05c5b22ad69b3dc00068fde9d921553d0171e481fc0741. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-013 product before observation access.

Boundary: Every positive finite MEASURE-013 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-013 uniquely retains pair-ledger-symmetric-difference-metric, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__pair-ledger-symmetric-difference-metric__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:0cda6a381684ff02fc9fb77ad6805b88211db46764eada9b6d1234e706deae45; generality sha256:146ad2cfa44b38835220092b27b2b2733f9f019bbe740241d501feb17de9e6d7.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate

sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal sha256:88f25ec55fd93adea453a486bb74571efafa44fd9e9742c35dcebb8a6017c669. Source manifest sha256:3d55ba33b6eb2981d615201c7f0e8459b2e392188129d007379de6ea28cd5f2a. Engine receipt sha256:9b99c3d97aab0b792f05c5b22ad69b3dc00068fde9d921553d0171e481fc0741 at receipts/engine/model_admitted/SFT-INFO-MEASURE-GEOMETRY-013-9b99c3d97aab0b79.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT- INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-013: information_geometry
- exact observation: {"irrational_coordinates_used":false,"symmetric":true,"triangle_relation":true}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:655969727b12ed4c759b4cfb4827d7212e13c63ccf60878cb99c6b2be6d4999a; empirical validation sha256:889ff3d49519fd8ecf9455b858ef2f0ef7fa28c057ab8aab481cd06542393dbe; external validation

sha256:062c22626ebf9bc0f4c49ec1efaff2b488152a971c507fa63d393d3f30b71e51; isolation sha256:537ec3ebcd43838fbc15f914a42ba72d7931144e46541509f0f3e56c1927423e; custody sha256:03f86d400f217419d413542c67a8eab93856c4b986b1f72b13ab54c168ce8144.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source

- no failed route retires an obligation or changes protected authority

66. Multi-scale information decomposition

Family: MEASURE Obligation: SFT - INFO - OBL - MEASURE - 014 Claim: SFT - INFO - MEASURE - MULTISCALE - 014

Question, law and forced result. A nested observation chain decomposes information into exact disjoint distinction increments whose accumulated ledger telescopes to the fine-versus-coarse difference.

MEASURE-014 uniquely retains telescoping-refinement-increments, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SOURCE - COMPLETENESS - 014
- SFT - INFO - MEASURE - GEOMETRY - 013

The current reconciliation binds this obligation to the current model-admitted receipt sha256:67be345678ea8083fcd99c189de049c6e12e9f14e42a7ace547a5c397ff2294. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-014 product before observation access.

Boundary: Every positive finite MEASURE-014 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-014 uniquely retains telescoping-refinement-increments, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__telescoping-refinement-increments__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:d9cf51ac44311dba49cc4ee2f64da3fc6a7877e6f689102f2bc8d4ffa7d40f00; generality sha256:1566581ffb8607cbb31ee9bed9b6b5c70e2a770a11d456cbb6b6219b751cee41.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural

absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate

sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal

sha256:ca6ae613ed62b9d6c57194e548491e95ef67acd54af0eaa0a1a935d11495a3b0. Source manifest

sha256:48b636c26bc166b993d1f6eed173e2c19f9f8b975e4f93bc0d6656deea3f9e78. Engine receipt

sha256:67be345678ea8083fcfd99c189de049c6e12e9f14e42a7ace547a5c397ff2294 at receipts/engine/model_admitted/SFT-INFO-MEASURE-MULTISCALE-014-67be345678ea8083.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT- INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-014: multiscale
- exact observation: {"coarse_to_middle_increment":4,"middle_to_fine_increment":2,"telescopes":true,"total_increment":6}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:660e2c1c654b0e6c3b78d11d386fd44920b5b3d2f425097f846e5e30c5d48ac2; empirical validation sha256:ec0c30237a376062e26cec0ee4dd9bcc9000e171950be2cc9c30028fac9848a8; external validation

sha256:6b4e7f41fda2bf0aeeb5801ff98c1866ed8f77c4861b97fff306685531e29e95; isolation

sha256:f2a312a9f7b0d1ba473a5f12abd8276d71329ecb67674b2fde995e49acd9282c; custody

sha256:086d6076eed1a63a2c200c0d5aa935c438380cb247707607f21146dd33c4c554.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

67. Measure conversion and unit custody

Family: MEASURE Obligation: SFT-INFO-OBL-MEASURE-015 Claim: SFT-INFO-MEASURE-UNIT-CUSTODY-015

Question, law and forced result. Information-unit conversion is admitted only by exact equality of complete supports; the alphabet base and position width remain attached so numerically equal support counts do not erase units.

MEASURE-015 uniquely retains equal-support-unit-labelled-conversion, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SOURCE-COMPLETENESS-014
- SFT-INFO-MEASURE-MULTISCALE-014

The current reconciliation binds this obligation to the current model-admitted receipt sha256:61431200deb07c07eb2dd1ea2c53fcc9e7ed4eba34cf20a763ab8f74214723eb. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-015 product before observation access.

Boundary: Every positive finite MEASURE-015 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-015 uniquely retains equal-support-unit-labelled-conversion, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__equal-support-unit-labelled-conversion__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:787352be6680d42cc5f43bbf58839603d524a277c960e275ce246e1dc7d7ddfd; generality sha256:d4e07febf3575495ec6a8a6e43e71d94e5950f6878f4b3935d14d2f3747ab8e5.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation

sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate

sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal sha256:409b852f90f5e524e49017bcad5193530cef6b2679c7eb61fdcc67f6f72c8dc1. Source manifest sha256:11a89ecb467ee8250931289a3e4ca53aef3819e6380f2cb6731210d737badefa. Engine receipt sha256:61431200deb07c07eb2dd1ea2c53fcc9e7ed4eba34cf20a763ab8f74214723eb at receipts/engine/model_admitted/SFT-INFO-MEASURE-UNIT-CUSTODY-015-61431200deb07c07.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-015: unit_custody
- exact observation: {"base_eight_positions":1,"support_each":8,"two_label_positions":3,"unit_tags_retained":true}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:d8c1cae073a0c32bb122e2ca927ea27920c046b330b40d6938c670efe7011db4; empirical validation sha256:3504e5ab3ec29f680327f41e39811ce59988ae74463573b1b7302e78a6acb5f6; external validation

sha256:5dea8b54c8c4820d22a1765660b72055abc330e3c3255440bba12d3553d73235; isolation sha256:5ce71d57a7d73f078c362117d8ee16c5dbd9865557463cdc4364a3b9cb74a5f5; custody sha256:226faf0a6d2c660ad950a9f35e00f5498d908707467039e37c1602b6213fe8b3.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

68. Information-measure completeness certificate

Family: MEASURE Obligation: SFT-INFO-OBL-MEASURE-016 Claim: SFT-INFO-MEASURE-COMPLETENESS-016

Question, law and forced result. Measure-family completeness is the one-to-one reconciliation of all sixteen frozen obligations with exact receipts, observations, units and declared computability boundaries.

MEASURE-016 uniquely retains sixteen-measure-obligation-ledger, complete measure custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SOURCE-COMPLETENESS-014
- SFT-INFO-MEASURE-UNIT-CUSTODY-015

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:f267f72a1b16ab7d14d45c1e7004001a33e721f13ee3f289fad4acea7a5e41b3. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis MEASURE-016 product before observation access.

Boundary: Every positive finite MEASURE-016 support, partition, description grammar, unit record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- MEASURE-016 uniquely retains sixteen-measure-obligation-ledger, complete measure custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-support__sixteen-measure-obligation-ledger__complete-retained-and-closed-ledger__support-and-unit-custody__complete-declared-measure-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:d62a3dee2a549942afe5795703b29514140bd032aa307c3c44f466fc89685d20; generality
sha256:f6d5e8fab5ead7ccfe44f60841285b47d394b3ef9eb82b25a587ac7c8571f4a9.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the possible distinctions.; receipt
sha256:534af192501c93e31482880b8440c07057aeea05e17d359d3237d5b3ed67ff30.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold; no sampled partition support or unregistered completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:5a1a9c58f6097cf2a89e8c8303484e965316bfec4c3a47d7f52d6407561e1aa6.

Independent reconstruction and engine admission. Independent implementation
sha256:c56fec9da954374b54abcefc3ee6fde86e0e78cf5a304e24a58a18b3e06af67 regenerated the declared census and sole survivor; independent certificate
sha256:47bc7f3d6c3cc4a22f09e15e3629c5fbf0367620965ec9f50421bcbf74192580. Derivation seal
sha256:7836f24b1fce64d6fb4a0f28fc8463d3929f844f4694d768ce525bbb9b9c8b5e. Source manifest

sha256:ae340df6eef14da955621f8d6a5dee16a2814c86206a7f1ea0b940dbfd77e6ab. Engine receipt sha256:f267f72a1b16ab7d14d45c1e7004001a33e721f13ee3f289fad4acea7a5e41b3 at receipts/engine/model_admitted/SFT-INFO-MEASURE-COMPLETENESS-016-f267f72a1b16ab7d.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT- INFORMATION-MEASURE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- MEASURE-016: measure_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":16,"omitted_owners":0,"registered_obligations":16}
- complete family observation custody: 16 records; all rows preserved
- complete finite support, partition, description-program, unit and retained/closed-ledger enumeration supplies the observation; no imported entropy formula, continuum logarithm, universal oracle or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete measure census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:877ac191cd45eeb2fee43fc445e543b9aaf452990c951d47ae0bcde0803ac6e0; empirical validation sha256:a096f48af40022ca363bc72899057b586e3cc732b47cf47d299ca171cea7f3e1; external validation sha256:68873a0d7f27d5f3c37949afa67d14e5d75e37e03e10388b3064a5cbe1d1d1fc; isolation sha256:bc9a72b5d79d18412539c0d07b3e05ed3efcea7340f778035fbf8fe3d660c0a6; custody sha256:3aa734d7f4709ae2083bad7ae79405d10ec1d57da058f35dd69b7392b0acb2bc.

Prohibited inputs.

- no axiom, imported entropy formula, logarithmic continuum or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no unregistered grammar, universal-complexity oracle, hidden unit conversion or fitted threshold
- no sampled partition support or unregistered completed-infinite source
- no failed route retires an obligation or changes protected authority

Complete-field family: SIGNAL

69. Signal as an ordered observation record

Family: SIGNAL Obligation: SFT-INFO-OBL-SIGNAL-001 Claim: SFT-INFO-SIGNAL-ORDERED-RECORD-001

Question, law and forced result. A signal is a complete successor-indexed observation record whose exact position labels and exact part-valued amplitude labels remain jointly reconstructible.

SIGNAL-001 uniquely retains complete-position-amplitude-record, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-MEASURE-COMPLETENESS-016

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:e1480fb2ec65d8ff917eb9fb8d628251f72c320eb84708049a8e4e34651f4bec. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-001 product before observation access.

Boundary: Every positive finite SIGNAL-001 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-001 uniquely retains complete-position-amplitude-record, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__complete-position-amplitude-record__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:f1119238c614b215f799e8ca8e06eeb4bfd5d32a8b907b810e9f8a1663e78587; generality
 sha256:299b3fe67b047a16e5fa3efd850caff18f0c9725ba15873670d7b81e9014ae1a.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt
 sha256:2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt
 sha256:83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256:de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate
 sha256:0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal
 sha256:21b225562338532471927a11bc971c3789841da1f26f08447f35613829302d95. Source manifest
 sha256:b00ac37d81f178eb17339be9e24ca3d6e36a2251daaa6f7af37ca3e1e36cc941. Engine receipt
 sha256:e1480fb2ec65d8ff917eb9fb8d628251f72c320eb84708049a8e4e34651f4bec at receipts/engine/model_admitted/SFT-INFO-SIGNAL-ORDERED-RECORD-001-e1480fb2ec65d8ff.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-001: `ordered_signal`
- exact observation: `{"complete":true,"exact_amplitudes":["1/4","1/2","3/4","1/1"],"positions":4}`
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: `b5d5ca2930bf1558a4ca5ac7562a9e4c476b2fab06c6a026e8c644fc845c85ad`; empirical validation sha256: `d7f6ab17832ccd83317cef2f599f5a9215712171a41d49113a6f5fd02be26a51`; external validation

sha256: `d6535b62fb7a52e594279d74f538f410b2ceea318c3226254db6c6b5c9415aa1`; isolation sha256: `6707498561c580ac3eb86f5632c4021e05afc9bcbe67de31639113b51c530d1d`; custody sha256: `5db4ba3f93a8cf398eeb9d19ff7fa56ba6f12b91f4cc276af5b5ab912983747b`.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

70. Amplitude-label and support separation

Family: SIGNAL Obligation: SFT-INFO-OBL-SIGNAL-002 Claim:

SFT-INFO-SIGNAL-AMPLITUDE-SUPPORT-002

Question, law and forced result. Signal support identifies where observations occur; amplitude labels identify what exact part is retained there. Neither carrier may silently replace the other.

SIGNAL-002 uniquely retains separate-support-and-amplitude-custody, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-MEASURE-COMPLETENESS-016
- SFT-INFO-SIGNAL-ORDERED-RECORD-001

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: `aa715eb732002cf90cef0de1f03151147633926b6434c5dcba15efbe43bd4c2`. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-002 product before observation access.

Boundary: Every positive finite SIGNAL-002 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-002 uniquely retains separate-support-and-amplitude-custody, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__separate-support-and-amplitude-custody__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:5ec0b60110511af6e118dbc6f014adc78f66c8ce0dd6b1d363fdd8bc94e2d786; generality sha256:8ede0844e9a6b80f155a8cbdd1df7ecb5a335a9b5c4e8f840777f440d5856e77.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt sha256:2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt sha256:83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256:de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate sha256:0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal sha256:9f768c04b2e43b8f8f390b0fb5bbe75b36eaa1e4b0f889a778e8a2886930a667. Source manifest sha256:6d890765def092ab2138bc250e7582a47380d3b413b6b02a30bf75d8bb5b685c. Engine receipt sha256:aa715eb732002cf90cef0de1f03151147633926b6434c5dcba15efbe43bd4c2 at receipts/engine/model_admitted/SFT-INFO-SIGNAL-AMPLITUDE-SUPPORT-002-aa715eb732002cf9.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-002: support_amplitude
- exact observation: {"amplitude_rows":4,"separate_and_linked":true,"support_positions":[1,2,3,4]}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: ae77f30e0ce7b3f073cfe951b361854bac831da869680d6083f838bc5cb63ce3; empirical validation sha256: 33ddf7b46953220f38bf4cee100d05e3f2594e4324b1e883534c057fe2779e4d; external validation

sha256: 06164bcf871f5401455c89f09da109dd1b1c045a597851d25ecfb8d6a3d8ac0a; isolation sha256: 1b57297018c71f18fb6612aff636b72d6750102be600b8d8b3789bdae6e5e88c; custody sha256: 04fdb772f31d63e84636707b934db5a1f2aaeaac23d0f9ad16264cab5409f4e5.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

71. Sampling as retained position selection

Family: SIGNAL Obligation: SFT-INFO-OBL-SIGNAL-003 Claim: SFT-INFO-SIGNAL-SAMPLING-SELECTION-003

Question, law and forced result. Sampling is a held selection of declared source positions that retains each chosen position, amplitude and provenance link to the complete signal.

SIGNAL-003 uniquely retains source-bound-position-selection, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-MEASURE-COMPLETENESS-016
- SFT-INFO-SIGNAL-AMPLITUDE-SUPPORT-002

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 94cdf76f9079431270f155fbadf45b3ef98b44993cee48fac81977947a728c2d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-003 product before observation access.

Boundary: Every positive finite SIGNAL-003 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- **SIGNAL-003** uniquely retains source-bound-position-selection, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__source-bound-position-selection__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:e265cb1d706e4ed04d569c04071d5b39e5db0435c4355d2dae1f74209a9b81ed; generality

sha256:1a5d7d9908ea375606e8b2ce884a55c73dc1b441d94cc17748e7570e9af097f8.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt
sha256:2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt
sha256:83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256:de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate

sha256:0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal

sha256:19d87312f1a6e14f04630b57578afaa5880295200e34deb5064ac625662b4651. Source manifest

sha256:906f88ee9114193f960c044fb4e3986dd39ddef72dd4daa9f16ce437fea609a4. Engine receipt

sha256:94cdf76f9079431270f155fbadf45b3ef98b44993cee48fac81977947a728c2d at receipts/engine/model_admitted/SFT-INFO-SIGNAL-SAMPLING-SELECTION-003-94cdf76f90794312.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-003: sampling
- exact observation: {"provenance_retained":true,"retained_amplitudes":["1/4","3/4"],"selected_positions":[1,3]}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:bf182bc2b30295153c77ca3a5fc2c31eca7c8a128f3387ddf9414f05810b665a; empirical validation sha256:d96c3ebc8b24f3055d183c69c147d33d31b59e382480a76bf788524ffe66dd49; external validation sha256:21b8b0112a230e7d681cb9fd2774cf1ac1aa4ed73d3bd952ead58e9efdebcca5; isolation sha256:135b9e3e997d6ef568fe98f2a3b866bc48cabfa82dbba61e297af9e320315b1c; custody sha256:8bb0af32320f7767aff02ff5ad55d25cf76f1f23ae18456d0f77ae2f62b25c07.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

72. Exact finite sampling sufficiency

Family: SIGNAL Obligation: SFT - INFO - OBL - SIGNAL - 004 Claim:
SFT - INFO - SIGNAL - FINITE - SUFFICIENCY - 004

Question, law and forced result. A sample is sufficient for a frozen finite signal grammar exactly when every generated signal consistent with that sample reconstructs to one source form; complete candidate enumeration proves the boundary.

SIGNAL-004 uniquely retains grammar-exhaustive-reconstruction-sufficiency, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - MEASURE - COMPLETENESS - 016
- SFT - INFO - SIGNAL - SAMPLING - SELECTION - 003

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:3b89ab82ee1792ce20128af5fba7ad5affa5cbe2bd071f54521e544bdb73e3e6. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-004 product before observation access.

Boundary: Every positive finite SIGNAL-004 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-004 uniquely retains grammar-exhaustive-reconstruction-sufficiency, complete signal custody, root forcing, post-registry observation and no extra

rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__grammar-exhaustive-reconstruction-sufficiency__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256: fe7be7bab3806b77ad37be784dcd58c2e21fcff44431a6a25d2a478c10a46724; generality
sha256: 658266b5f7b20c8e32dec7e404e8e3bdd7dd4003cea868224df95c58a15b0175.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt
sha256: 2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt
sha256: 83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256: de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate
sha256: 0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal
sha256: ce80e9d7b435fbeb00d939387631dfb75d8c11058a05a0fc9ccc362c763de6ba. Source manifest
sha256: 3af058d6c5b8cf06477107e1a4d9e80c0219c417b522ea89b0da63c963c5a114. Engine receipt
sha256: 3b89ab82ee1792ce20128af5fba7ad5affa5cbe2bd071f54521e544bdb73e3e6 at receipts/engine/model_admitted/SFT-INFO-SIGNAL-FINITE-SUFFICIENCY-004-3b89ab82ee1792ce.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-004: finite_sufficiency

- exact observation: {"all_reconstructed":true,"generated_signals":4,"grammar":"two-label alternating length six","samples_required":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:64d51d79ee5240ebe42935dfdbbf9ae3d5ada5c979e35b696c8c9630165700a8; empirical validation sha256:625d0a333c18020df70b5522fe45637f4feb42b9b83fa0ab8c6aef914bec00e; external validation sha256:ea872237e7007c8d6175585f1de92e35d060c693697a59fdf40c9075f262ecc7; isolation sha256:ce3513fe955a0dd1d1f1b2e64e4db459b61e07daee5d137239bbdf53a50c261; custody sha256:e6cfceb401f0c84bbf2ef5e9ba238621ad4ce9f0450fb3b516054aa2fb3b7372.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

73. Aliasing as closed distinction

Family: SIGNAL Obligation: SFT - INFO - OBL - SIGNAL - 005 Claim: SFT - INFO - SIGNAL - ALIASING - 005

Question, law and forced result. Aliasing occurs exactly when distinct source signals map to one sampled record; the observation closes their difference while the source ledger retains every aliased predecessor.

SIGNAL-005 uniquely retains equal-sample-distinct-source-alias, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - MEASURE - COMPLETENESS - 016
- SFT - INFO - SIGNAL - FINITE - SUFFICIENCY - 004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:57d7937c2c41f4cc234024e13df5472eb9621dfe199cc8bb861f1abe3a4a8d70. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-005 product before observation access.

Boundary: Every positive finite SIGNAL-005 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-005 uniquely retains equal-sample-distinct-source-alias, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__equal-sample-distinct-source-alias__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:445a0cc9bca0bbd8cf354474fb02e9adcc1b5565d4851512613326a1ec95bbae; generality
 sha256:d1502fa8bc215744fa383785f91eae8815f2e4c1ed544b53f4289b59b8320ad6.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt
 sha256:2ecd5eae281c7bddfb914fd62a69d5938847291094905d4dc1812294b522adc.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt
 sha256:83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256:de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate

sha256:0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal

sha256:44bc7a90924f7adbbcb62a3a3f2b9673b3398b7c0abfc045af76f8c43f2d63f96. Source manifest

sha256:8863aba7f1a3c67b64563d8c663869b6dd6867df4f342d0d0b3b9f3e09222c1e. Engine receipt

sha256:57d7937c2c41f4cc234024e13df5472eb9621dfe199cc8bb861f1abe3a4a8d70 at

receipts/engine/model_admitted/SFT-INFO-SIGNAL-ALIASING-005-57d7937c2c41f4cc.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-005: aliasing
- exact observation:
 {"distinct_sources":2,"predecessors_retained":2,"sample_records_equal":true,"selected_positions":[1,3]}

- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:8fc606bdcd92ac00c97848ee8d05f17f075ce8ee52488383de58dc9ff5274859; empirical validation sha256:153341d2495abaaef1a4e1cae4f33391a1ff5f738127103c6be5d097b8bc7567; external validation

sha256:d51c6ca4da1c1e305059708cee168dd4476b6ebd7d24bb4bb305113e4a6cdcbb; isolation sha256:e53d1b18d42626233fba8ffdc4c40f00cd164f73262f3803b8de3b180f1c6e13; custody sha256:d8a59ac0379b00db1f3cb3ab2b880f57aade4512ad9c39446462c31b08592321.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

74. Quantization as observation partition

Family: SIGNAL Obligation: SFT - INFO - OBL - SIGNAL - 006 Claim: SFT - INFO - SIGNAL - QUANTIZATION - PARTITION - 006

Question, law and forced result. Quantization is a complete nonoverlapping partition of the declared exact amplitude support with one retained class label for every source amplitude.

SIGNAL-006 uniquely retains complete-exact-amplitude-partition, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - MEASURE - COMPLETENESS - 016
- SFT - INFO - SIGNAL - ALIASING - 005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:5fd34246d405251f93db300ed1bf0ca5e3736f577a5611affcb85d7b99970739. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-006 product before observation access.

Boundary: Every positive finite SIGNAL-006 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-006 uniquely retains complete-exact-amplitude-partition, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__complete-exact-amplitude-partition__source-sample-reconstruction-ledger__complete-declared-

signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:b58810632459baf7b6543644c2430af0f6ccd880adbad8facd578829e65b8811; generality
 sha256:34f7204d5fc340626327668b3aae058ab512b571ad4412d70f2b60a5301686b6.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt
 sha256:2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt
 sha256:83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256:de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate
 sha256:0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal
 sha256:c37094da325fda667097f68bdcff3220fbfb5c4b4215bb9973e757d81d7715a5. Source manifest
 sha256:abd10c6ebd7a8d1cec6b52409a42801026eb547e9f11e0101a34fec52fe65a5d. Engine receipt
 sha256:5fd34246d405251f93db300ed1bf0ca5e3736f577a5611affcb85d7b99970739 at receipts/engine/model_admitted/SFT-INFO-SIGNAL-QUANTIZATION-PARTITION-006-5fd34246d405251f.js
 on. Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-006: quantization_partition
- exact observation: {"amplitude_support":4,"classes":2,"omissions":0,"overlap":0}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:ce99e9dbdaa23a3f7579064d010e67561fdf1adc3990a03e4eec040232c145da; empirical validation sha256:82dfd920887b29a5feb6e9a4ff9b58412b1f31e48112faa2cbf7e412b8b8e74e; external validation sha256:1dc3f844e743c6303220cc67bd366b3760f87a23e564d43e53f6eb2365e4ad21; isolation sha256:c1d1eb26c60bb6b7a1186e3c9248a941bffa5b2770a748aa74c47ec28a05f896; custody sha256:b9e0ee1d8305ec485a69b8347ea867b72835bac3afadd297c22ad7d9ec4e5611.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

75. Quantization error custody

Family: SIGNAL Obligation: SFT - INFO - OBL - SIGNAL - 007 Claim:
SFT - INFO - SIGNAL - QUANTIZATION - ERROR - 007

Question, law and forced result. Quantization error is the exact part between a source amplitude and its retained representative together with held/opposed orientation; no negative or floating scalar enters.

SIGNAL-007 uniquely retains held-oriented-exact-error-part, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - MEASURE - COMPLETENESS - 016
- SFT - INFO - SIGNAL - QUANTIZATION - PARTITION - 006

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:c1241d38a332b2e4173d249e367f3e847686d1cbc7eb2cedce6e7c7285a07541. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-007 product before observation access.

Boundary: Every positive finite SIGNAL-007 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-007 uniquely retains held-oriented-exact-error-part, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__held-oriented-exact-error-part__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:b50b95a02d93352539a2093be97d56ceffd7440cca72425003378089e65cd4e0; generality

sha256:c67d8833cca953e7d27fb682aef3d4d31ee8eb7c317333ce8bab5ffaa1afb4a2.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt
sha256:2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt
sha256:83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256:de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate

sha256:0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal

sha256:7af73f50e8cee60080648b0367164e3266984eefec1cb7fffeb3f34c49da9e7f. Source manifest

sha256:4a3a5cc6493c8e8b579c68444e0c4f1627f4bafabbeca77fc86b6297a4eda377. Engine receipt

sha256:c1241d38a332b2e4173d249e367f3e847686d1cbc7eb2cedce6e7c7285a07541 at receipts/engine/model_admitted/SFT-INFO-SIGNAL-QUANTIZATION-ERROR-007-c1241d38a332b2e4.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-007: quantization_error
- exact observation: {"examples":2,"magnitude_each":"1/8","negative_scalar_used":false,"orientations":["opposed","held"]}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:97dae12bfc9da50e1916d062e19e842ef028063189fcfc29d09f092d59cef0c3; empirical validation sha256:5a5def0057f9a992db5b976af5412ea21a80607d2befd311dc6b21d90531c3fe; external validation sha256:48a8ae6a95a1a987ef3e3a7b720c37b14a05278e668011c013fe5cc7d186091b; isolation sha256:c4e89b9b2f1fb77863ebb027965d7f7704e8fada06ecb4ed9e022fa0e3747fae; custody sha256:d781f7a88444f944d0521ea031aa32564d9370b97bb81fa3bf8fa3834701fd05.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

76. Reconstruction from complete retained samples

Family: SIGNAL Obligation: SFT - INFO - OBL - SIGNAL - 008 Claim: SFT - INFO - SIGNAL - RECONSTRUCTION - 008

Question, law and forced result. Signal reconstruction is valid exactly when the retained sample record and declared grammar select one source signal and reproduce every position-amplitude row.

SIGNAL-008 uniquely retains unique-source-bound-reconstruction, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - MEASURE - COMPLETENESS - 016
- SFT - INFO - SIGNAL - QUANTIZATION - ERROR - 007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:772cb633ffc72b3fcfd3a2f72e66422e37f7fcc76152f849db9dfdb27c888ed9. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-008 product before observation access.

Boundary: Every positive finite SIGNAL-008 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-008 uniquely retains unique-source-bound-reconstruction, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__unique-source-bound-reconstruction__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:39ee4e5488fc6f6e80a4f12afa0b35409afb4cdda01c2a6b07abddc3f4e4c554; generality sha256:bccb6a54fae2e73f3390c8f6a6795d770a7116eadce782d5066c110e09a3bcd6.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted

magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt
sha256: 2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt
sha256: 83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256: de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate

sha256: 0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal
sha256: 3e69b9ec631539b6cf3cd72530c1add653a247dd154010fbb4f10a7c23e7cd97. Source manifest
sha256: d5e807cc14330f49707d4dc17dad215e9e975664e898aab360083f21ea644efe. Engine receipt
sha256: 772cb633ffc72b3fcfd3a2f72e66422e37f7fcc76152f849db9dfdb27c888ed9 at receipts/engine/model_admitted/SFT-INFO-SIGNAL-RECONSTRUCTION-008-772cb633ffc72b3f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-008: reconstruction
- exact observation: {"samples":4,"source_rows":4,"unique_reconstruction":true}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 83cc17a457ca5927aad6643a4b2cd17370274956a0ddd40ec170a753d60eb5140; empirical validation
sha256: 52a79c9e9427efc9a4fa1b35c9bb3f0996b643e40a6a0391fd46dfb606caf325; external validation

sha256: 309a74bac425b077fe7ef9207452061c847bf972d49c436f335278b1624e0641; isolation
sha256: adbee7f9bf047e761e6a0337bad13d4db11ec10f6104149d4801f981930f9559; custody

sha256:5533a18ec78994e75066e895f3db49ef68fd9fc5d549c3f87a5009b6d848d7.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

77. Interpolation correspondence boundary

Family: SIGNAL Obligation: SFT - INFO - OBL - SIGNAL - 009 Claim:

SFT - INFO - SIGNAL - INTERPOLATION - BOUNDARY - 009

Question, law and forced result. Interpolation is admitted only over a frozen exact finite candidate grammar whose consistent forms are exhaustively enumerated; an ungenerated continuum is not inferred from finite samples.

SIGNAL-009 uniquely retains finite-generated-interpolation-candidates, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - MEASURE - COMPLETENESS - 016
- SFT - INFO - SIGNAL - RECONSTRUCTION - 008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:c94acd30ab94af39f242d95d247db33eab69c577ba3715432874ae6ce70b885a. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-009 product before observation access.

Boundary: Every positive finite SIGNAL-009 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-009 uniquely retains finite-generated-interpolation-candidates, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__finite-generated-interpolation-candidates__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:8d1b8ae2bf108484645604cb8488fc66dc5db19936dc64f7bc71b6033f9847a4; generality

sha256:07b1a7a2eab648f5dbae4e5a158dd5ebdaefee066b05954db212d1d94053ca2.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt
sha256: 2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt
sha256: 83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256: de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate
sha256: 0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal
sha256: c62e22110b4736649e77e8a20361ac10961c08af95141f748913cb30a155a04b. Source manifest
sha256: ef68d7be73dac817ebcfaf02c2608efede72bb3e7bb5942af6d7c966f12449066. Engine receipt
sha256: c94acd30ab94af39f242d95d247db33eab69c577ba3715432874ae6ce70b885a at receipts/engine/model_admitted/SFT-INFO-SIGNAL-INTERPOLATION-BOUNDARY-009-c94acd30ab94af39.js
on. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-009: interpolation_boundary
- exact observation: {"candidate_count":4,"candidate_values":["1/4","1/2","3/4","1/1"],"continuum_claimed":false}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 2aa9c6b4bbe2ab8aacf07c83d1124a524ab134f7403bc1406fe4071191641548; empirical validation sha256: d918ee728ec6f692018d172fa84dd154907fbee323e78f7729c2fa2f17b446f2; external validation

sha256: 70390d33e1eb1d4445210a8d904717fcb4cd1402c0fbba5ad60e68af469c3fba; isolation
sha256: cbebd136632d145b1863902f003fc60a1203273f7489a9b817fa0714f6f4fe3f; custody
sha256: a83a77b8e53838fd9feafcf88523bc7d5194c40c25b1f55c76ce0762e701463.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

78. Transform representation of finite signals

Family: SIGNAL Obligation: SFT - INFO - OBL - SIGNAL - 010 Claim:
SFT - INFO - SIGNAL - TRANSFORM - REPRESENTATION - 010

Question, law and forced result. A lossless finite signal transform is a bijection on the complete generated support with a separately implemented inverse; transformation changes representation without closing distinctions.

SIGNAL-010 uniquely retains bijective-finite-support-transform, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - MEASURE - COMPLETENESS - 016
- SFT - INFO - SIGNAL - INTERPOLATION - BOUNDARY - 009

The current reconciliation binds this obligation to the current model-admitted receipt sha256:b01be7af191f4e0d72fddb44b51ed469b7f604f9d955961e5fb9e122dbfe8a6a. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-010 product before observation access.

Boundary: Every positive finite SIGNAL-010 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-010 uniquely retains bijective-finite-support-transform, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__bijective-finite-support-transform__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:e087dbf1eb939d3cdaebc1f775cad1d15c135c6b0a001464bea0bcdf3dec439e; generality sha256:8646786fafb9973523c412a27abd0fbca7a6983134ebe543c87ee9dcf0a32788.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt sha256:2ecd5eae281c7bddfb914fd62a69d5938847291094905d4dc1812294b522adc.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt
sha256:83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256:de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate
sha256:0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal
sha256:df7d7b3bc8e843a7bfd7c687c0ea9b5295e02def718f7e1e56bc598490923054. Source manifest
sha256:bcf94c9042bb05fa0ac3153106dd2ee23ebc94361ecb7fd467dc0a2e3772759f. Engine receipt
sha256:b01be7af191f4e0d72fddb44b51ed469b7f604f9d955961e5fb9e122dbfe8a6a at receipts/engine/model_admitted/SFT-INFO-SIGNAL-TRANSFORM-REPRESENTATION-010-b01be7af191f4e0d.json. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-010: finite_transform
- exact observation: {"bijective":true,"image_forms":4,"independent_inverse":true,"support_forms":4}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e565263d1391c65df095d6a1b0adc51cee887ecc1d726ac2b771822e3408d450; empirical validation sha256:b8a0bad117c1606887ca8787afe6a64ee31ff0c39c53836a09ba4aeefd5a8f2c; external validation
sha256:4b150013e6d2886fbb6295b500ba7bdb8aff2b8ad6e26d78da904ea02232f2d9; isolation
sha256:ca375df7478a83c51a31fb3831f65d77754833765dc50edb16c48153baf642fa; custody
sha256:9ee3ae6624a9eee5625587616606f8ec9d743cc822e886325967b22dfb39ef52.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object

- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

79. Time-frequency support correspondence

Family: SIGNAL Obligation: SFT-INFO-OBL-SIGNAL-011 Claim: SFT-INFO-SIGNAL-TIME-FREQUENCY-011

Question, law and forced result. Finite time-frequency correspondence retains ordered positions together with exact recurrence/change labels, forming a joint record without importing continuum frequency.

SIGNAL-011 uniquely retains position-and-change-joint-record, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-MEASURE-COMPLETENESS-016
- SFT-INFO-SIGNAL-TRANSFORM-REPRESENTATION-010

The current reconciliation binds this obligation to the current model-admitted receipt sha256:2b959f97bc880c0763c1843f4b90b8d30b5c35ceb366b57151df9f0f9eb9666e. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-011 product before observation access.

Boundary: Every positive finite SIGNAL-011 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-011 uniquely retains position-and-change-joint-record, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__position-and-change-joint-record__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:08973c5f540224f647c183fc973682aa7a4850272685f4a2966c18939067bd81; generality sha256:ec9f987a896eab250136b652cf5927f23179a5683354a34456e86b8127a09d00.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt sha256:2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt
sha256:83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbfb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256:de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate

sha256:0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal
sha256:20f2b5248887c21a503ee8b6d5e1f91c621e4d79104cedcbbf3d3c0106141362. Source manifest
sha256:69d4a24bcc973cd039e00d2e23082594e2a52b9cf82ea741d1363c9fe6c31f5d. Engine receipt
sha256:2b959f97bc880c0763c1843f4b90b8d30b5c35ceb366b57151df9f0f9eb9666e at receipts/engine/model_admitted/SFT-INF0-SIGNAL-TIME-FREQUENCY-011-2b959f97bc880c07.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-011: time_frequency
- exact observation: {"change_rows":2,"joint_record_complete":true,"positions":4,"same_rows":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:228c5002445645b631a79cbfe62db899d8558a1725d0377647a86fd9d22be1fb; empirical validation sha256:43bd7b13a5699c32b86152bf674f663b39bed0580ec7bafc8e6c948026954872; external validation

sha256:8b50c340e275c5b12c7c548c109b848f131f2f691b8abc4295fa0afd1c59e41e; isolation sha256:0f1670665093a33fd841a25f79a4aa678ec98dc4ddbee173abc53cd5b7f9c2; custody sha256:ef083b1866750c09239d0e9625dfc39675804108682b7dba6208881c727e04b7.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

80. Spatial and multidimensional sampling

Family: SIGNAL Obligation: SFT - INFO - OBL - SIGNAL - 012 Claim:
SFT - INFO - SIGNAL - SPATIAL - MULTIDIMENSIONAL - 012

Question, law and forced result. Multidimensional signal support is the complete ordered coordinate product; sampling retains selected coordinate tuples and their source-bound amplitudes.

SIGNAL-012 uniquely retains complete-coordinate-product-support, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - MEASURE - COMPLETENESS - 016
- SFT - INFO - SIGNAL - TIME - FREQUENCY - 011

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:bb24e7bb1f2f3480a436c9387b1b143aabad5f8e4d9e2d0227c6bd24f8d96eab. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-012 product before observation access.

Boundary: Every positive finite SIGNAL-012 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-012 uniquely retains complete-coordinate-product-support, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__complete-coordinate-product-support__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:dce5b57351a67c823d9e3d74798225cdf6f3b1ec0d609e3f9ee71e44e334524d; generality
sha256:1213fcb229e948011b863f8859a92b7377678cdad59f3c903fa3901dc2afd17f.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt
sha256:2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar;

no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt sha256: 83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256: de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate

sha256: 0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal sha256: a3fcdb905587d041c5109c98de594bda18b5d48609194c2f176c3bb215d912e4. Source manifest sha256: d311656757b4699769acb1dcc66b5511768cdb4f8e432dacf1b04c2b6e6a7fac. Engine receipt sha256: bb24e7bb1f2f3480a436c9387b1b143aabad5f8e4d9e2d0227c6bd24f8d96eab at receipts/engine/model_admitted/SFT-INFO-SIGNAL-SPATIAL-MULTIDIMENSIONAL-012-bb24e7bb1f2f3480.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-012: spatial_support
- exact observation: {"all_coordinates_once":true,"axis_sizes":[2,3],"coordinate_count":6}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: b5e4cec15812d29fd11c0a83336ed2c0941cab4ad4267450f42871e26fb48616; empirical validation sha256: fd83850973db2272ca960694506258c37cefbe7f681a53eb2a16dd3cad203d0c; external validation

sha256: 99b30b5133cf54d50632df88dbb58d7e8bb61b13dcc5163e71b67d565e15ecb0; isolation sha256: eda4b28a525c926f004b5757f251ff2a98e257a163c457d3f83c473d26ae5e00; custody sha256: b38d43ab55272c678f0d1cbbf9cfbec9198829fe659ef0c47e148e89262fa164.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

81. Signal-to-record provenance

Family: SIGNAL Obligation: SFT-INFO-OBL-SIGNAL-013 Claim: SFT-INFO-SIGNAL-PROVENANCE-013

Question, law and forced result. Signal provenance is the complete ordered custody chain from source capture through sampling and representation transformation to the retained record.

SIGNAL-013 uniquely retains capture-sample-transform-provenance-chain, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-MEASURE-COMPLETENESS-016
- SFT-INFO-SIGNAL-SPATIAL-MULTIDIMENSIONAL-012

The current reconciliation binds this obligation to the current model-admitted receipt sha256:08248d2357f25c3ddb67623827f0f99f6d779402477bb72288090459bc56c0be. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-013 product before observation access.

Boundary: Every positive finite SIGNAL-013 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-013 uniquely retains capture-sample-transform-provenance-chain, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__capture-sample-transform-provenance-chain__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:c18e75635e8c6fe2ddf71f2a10a2ee7e72756d168a3bd0100478f2ac6cee56e4; generality sha256:6f35aed5be9fe1d2d43288b3afbef739099cccca65a179ee9424a92e90884ff6.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt sha256:2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt sha256:83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfbb7efa431b4.

Independent reconstruction and engine admission. Independent implementation

sha256: de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate

sha256: 0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal sha256: f7484ed1365f4964fa92b3ca80c3e4fa2a8db958487b5e162662841c75fefa31. Source manifest sha256: b77805f6bba7c9999a1fc0ed316896f5a02f58fdd6041fbed4eb6c6a5e5141a. Engine receipt sha256: 08248d2357f25c3ddb67623827f0f99f6d779402477bb72288090459bc56c0be at receipts/engine/model_admitted/SFT-INFO-SIGNAL-PROVENANCE-013-08248d2357f25c3d.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-013: signal_provenance
- exact observation: {"ordered":true,"step_count":3,"steps":["capture","sample","quantize"]}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 58e5a40d24a4d3aa8097e4183241938c26ce14e795eba59a8cfe7c0cd2cedf13; empirical validation sha256: b0fffd29d56ffe08b9dbdaf5bf3ff67ffa8c40900dcf0cc1b749c9f40d69778b7; external validation

sha256: fd0d578925c037ac425b3616c58ca62bbe8cfc2b4609612f4a55ca488331c313; isolation sha256: bd374be0d806935e5a5d600336cbf440ec92f2915070bf38c4cc9883dc6fd1a6; custody sha256: 1990d7696344eb4713c850d958b1bb8ba3c2a7d1e5105e51303382447931b1ae.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

82. Sampling and reconstruction completeness certificate

Family: SIGNAL Obligation: SFT-INFO-OBL-SIGNAL-014 Claim: SFT-INFO-SIGNAL-COMPLETENESS-014

Question, law and forced result. Signal-family completeness is the one-to-one reconciliation of all fourteen frozen obligations with exact receipts, observations, reconstruction checks and explicit continuum boundaries.

SIGNAL-014 uniquely retains fourteen-signal-obligation-ledger, complete signal custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-MEASURE-COMPLETENESS-016
- SFT-INFO-SIGNAL-PROVENANCE-013

The current reconciliation binds this obligation to the current model-admitted receipt sha256:078dcfb462d56aa57863f88a97a9a5f588170274c668c6302f17f82cff0ffff2. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SIGNAL-014 product before observation access.

Boundary: Every positive finite SIGNAL-014 support, sample, quantizer, transform, provenance chain and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SIGNAL-014 uniquely retains fourteen-signal-obligation-ledger, complete signal custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-position-labelled-support__exact-part-and-held-label__fourteen-signal-obligation-ledger__source-sample-reconstruction-ledger__complete-declared-signal-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:d138f28a1ca6d4922dd14e7a4fcd2465f06295ee32a3d1445b3bd26c167d11ee; generality sha256:c9ae22eb914acb3fb412c99b596dc1baccc6ff0280dbdf16e13518ec07081100.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the signal.; receipt sha256:2ecd5eae281c7bddfba914fd62a69d5938847291094905d4dc1812294b522adc.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported continuum signal model, Fourier answer or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant; no sampled candidate family or completed-infinite signal support; no failed route retires an obligation or changes protected authority; receipt sha256:83c5b308992c671e0f121892663c0c9eb3681224b2548fefafbfb7ef431b4.

Independent reconstruction and engine admission. Independent implementation sha256:de5f55ae4b65f83f53fd08852fc347a8de7922359abbe17a8ab980cb7d58bd78 regenerated the declared census and sole survivor; independent certificate sha256:0c2e73cc0e07417ca58041320bedecc8b5e3325f340e931bce10398560dfcaa4. Derivation seal sha256:61042e2f37008ebd274c626f5ca5e51378083ecf1429a5a8e27513cf072ef71a. Source manifest

sha256:36ff7891a7cd18d6306faad66db0708dfcecf8f8be165ffe5a56668364805165a. Engine receipt sha256:078dcfb462d56aa57863f88a97a9a5f588170274c668c6302f17f82cff0ffff2 at receipts/engine/model_admitted/SFT-INFO-SIGNAL-COMPLETENESS-014-078dcfb462d56aa5.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-SIGNAL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- SIGNAL-014: signal_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":14,"omitted_owners":0,"registered_obligations":14}
- complete family observation custody: 14 records; all rows preserved
- complete finite position, amplitude, sample, alias, quantization, reconstruction, transform and provenance enumeration supplies the observation; no imported continuum theorem, hidden predecessor or fitted threshold enters

Falsification condition: Reject if the value-free registry, complete signal census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:eb2073f542db8d59f6c476a7bbdefaf108175c066a81c995c1fa90d6a8d2e558; empirical validation sha256:c678f87593f3272eb91b9b8ea1febf54ca3fc6860b8f5a22428bed185d3821eb; external validation sha256:2b477048b99892ffd4f9d18fc695dcacc9a52dc6a4588ad3c7b9d17f2acba5d9; isolation sha256:9f6b923e9f196807e5a4716a8e30cf03b4d427be2486056ce7df01d9bec36729; custody sha256:9907146d4f0a031dfe8c7ff4ca7707696487ec538f2f5f4bd477d5c71a3ee670.

Prohibited inputs.

- no axiom, imported continuum signal model, Fourier answer or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden sample position, erased alias predecessor, fitted threshold or unregistered interpolant
- no sampled candidate family or completed-infinite signal support
- no failed route retires an obligation or changes protected authority

Complete-field family: COMP

83. Lossless code and exact reconstruction

Family: COMP Obligation: SFT-INFO-OBL-COMP-001 Claim: SFT-INFO-COMP-LOSSLESS-RECONSTRUCTION-001

Question, law and forced result. A lossless code is an injective source-to-code relation with a total inverse on its image; decoding reproduces every source symbol and position exactly.

COMP-001 uniquely retains bijective-source-code-reconstruction, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

• SFT-INFO-SIGNAL-COMPLETENESS-014

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:75c800388923e6076a00a11487aed2e59bb3d2f30ad730e0eab0d1d9b9c8f542. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-001 product before observation access.

Boundary: Every positive finite COMP-001 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-001 uniquely retains bijective-source-code-reconstruction, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__bijective-source-code-reconstruction__retain ed-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-co de-product__root-bound-forward-forcing__post-registry-exact-observation__finite-suc cessor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:ec94105c36e5929429fb85af8539c41b5dae392961c1a513a5a989d4bee9c906; generality
sha256:591b158a7752a36e482b3cc047ab965bc6e17d9931342071b6b50b198269bcad.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt
sha256:d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256:e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256:3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate
sha256:b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal
sha256:d719036a1b593cd46daf2a421c65e000d3ed7879540b37682c0756781885e803. Source manifest
sha256:2cba1764f5e39358c804324a689cb09107b02807adc04ad1a7e7ce3e4a810ff4. Engine receipt
sha256:75c800388923e6076a00a11487aed2e59bb3d2f30ad730e0eab0d1d9b9c8f542 at receipts/e ngine/model_admitted/SFT-INFO-COMP-LOSSLESS-RECONSTRUCTION-001-75c800388923e607.json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-001: lossless_reconstruction
- exact observation: {"encoded_units":6,"exact_reconstruction":true,"parse_count":1,"source_symbols":4}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:51d019580ef74a16818106011b2a361ffa5228fa31e861d9fdf51566c4ddc6b; empirical validation sha256:3d016cfd0906ec9324c8ca55e165c76705d3ffd8a66fbad3e8c5bc0029f9d2a; external validation

sha256:579b807906dc7869a1c3c23077244e1018d2537272a7b17bc3fa68827775e698; isolation sha256:e2e18bf50ce74057e11a281b999490800e2d354a6162fa4a6b0aa91322551f20; custody sha256:75097531c2d694f699a1724de7d2b2d395c6b6974141f4ed74d528604fecc703.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

84. Prefix-tree compression structure

Family: COMP Obligation: SFT-INFO-0BL-COMP-002 Claim: SFT-INFO-COMP-PREFIX-TREE-002

Question, law and forced result. A prefix-tree code places each source symbol at one generated leaf and forbids any codeword leaf from lying on the path to another, forcing boundary-free unique parsing.

COMP-002 uniquely retains prefix-free-leaf-code, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SIGNAL-COMPLETENESS-014
- SFT-INFO-COMP-LOSSLESS-RECONSTRUCTION-001

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:b5cacf1e07d7c25c5d9ff67b23f16edebc86a9ae88f28a9ebbf824e35338fa9. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-002 product before observation access.

Boundary: Every positive finite COMP-002 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-002 uniquely retains prefix-free-leaf-code, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__prefix-free-leaf-code__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:535fa21879dcb1f6c5d137436045a0a2a682f3ef23a0adc9929ffe59278002de; generality
sha256:e6a0663a7325c93d437a6c66abed7af6ecf40a32bb199f4a4407b4be6dddaf6b.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt
sha256:d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256:e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256:3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate
sha256:b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal
sha256:13a29f14abe7ccfcb13d075e1691a69d29753558a10e31277cb5f0a67f708c6e. Source manifest
sha256:c298639224141af1e138304c2db0e4f6d7cca75d6534d6a8abe82d6f03ac887a. Engine receipt
sha256:b5cacf1e07d7c25c5d9ff67b23f16edebc86a9ae88f28a9ebbf824e35338fa9 at
receipts/engine/model_admitted/SFT-INFO-COMP-PREFIX-TREE-002-b5cacf1e07d7c25c.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER

- SFT - V1 - V2 - INFORMATION - OBSERVATION - CORPUS

Observed, adverse and boundary records:

- COMP-002: prefix_tree
- exact observation: {"instantaneous":true,"leaves":3,"prefix_collisions":0}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:26b509c25eee89333ee2d7a9cc53e12c5b79d076a9ec5e31cf52ca476258c7bd; empirical validation sha256:b9b5df43f62fc73c93bc1e6489abb7ac69d1cf69d7f727d868d09a75327fc46c; external validation

sha256:ac8023103dcb25bf98248a5d2eef191aa866e569414736a71e6c6f48fac93768; isolation sha256:f257139b60b599efaa42d06b9d73a1c3fd0e767def82eaae8fb85a14fcf7dcb8; custody sha256:d9d11245fa61497e181c9dffef8f3cf3c7e2382f481153dd36c8f8a613a53c2cd.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

85. Dictionary compression structure

Family: COMP Obligation: SFT - INFO - OBL - COMP - 003 Claim: SFT - INFO - COMP - DICTIONARY - 003

Question, law and forced result. Dictionary compression replaces registered source phrases with retained tokens; the complete token dictionary and ordered token stream reconstruct the source exactly.

COMP-003 uniquely retains source-bound-dictionary-expansion, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SIGNAL - COMPLETENESS - 014
- SFT - INFO - COMP - PREFIX - TREE - 002

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:9c85ce9dbd865fa64f25eb57bac02d32b65155e26a6b88a7e34be8e2a083225a. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-003 product before observation access.

Boundary: Every positive finite COMP-003 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-003 uniquely retains source-bound-dictionary-expansion, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__source-bound-dictionary-expansion__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: 7b87415c6bf28c9a0245f6a7d0cb9444ff2fd168800653abc192503cba19f6fe; generality sha256: cca4a16975205d36e2afaac28546d183f60661620f6f38463910871977d365cb.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt sha256: d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256: e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256: 3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate sha256: b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal sha256: 8d849a78a5fc1a4affdb9be2aef190b3c583130535cdd8256d1a68c5a3d565ed. Source manifest sha256: d2c4a515e0c55625121b2f8a87fc1bab67938679e42df51a90525d456a718147. Engine receipt sha256: 9c85ce9dbd865fa64f25eb57bac02d32b65155e26a6b88a7e34be8e2a083225a at receipts/engine/model_admitted/SFT-INFO-COMP-DICTIONARY-003-9c85ce9dbd865fa6.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-003: dictionary

- exact observation: {"dictionary_entries":2,"exact":true,"reconstructed_symbols":9,"tokens":3}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:0ee9fd9da4a10149dad004f3fc9238f3df6e95b024e18b2ffe31014a9967d713; empirical validation sha256:aaf9ceb09fb9777e7602d946d5a930f8e4777ac236acaa4c1d9253e9e3bb1254;

external validation

sha256:e9c8fec04730cb4e38f8b57b5b38a6b86920d119741cd6b3528be6cf6566f73e; isolation

sha256:074e5f7616cd4b86ac320545a2413ff48d03c87bfd515e8331a3cd07000fc8b; custody

sha256:3ca9f43726156378c0961a71152f40fa1ff37fa74817e2a33cb4bf53fb3b05ca.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

86. Run and recurrence compression

Family: COMP Obligation: SFT - INFO - OBL - COMP - 004 Claim: SFT - INFO - COMP - RUN - RECURRENCE - 004

Question, law and forced result. Run and recurrence compression retains the first symbol plus exact positive run counts or same/change labels, each with a deterministic inverse over the declared alphabet.

COMP-004 uniquely retains run-count-and-change-record, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SIGNAL-COMPLETENESS-014
- SFT-INFO-COMP-DICTIONARY-003

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:b4a528eb3ab8ced04357e955ddb95a589902d586e753ed655926b40b02ca5a96. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-004 product before observation access.

Boundary: Every positive finite COMP-004 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-004 uniquely retains run-count-and-change-record, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__run-count-and-change-record__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or

-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:c7ae55c66f920c84b7871e2bcc5c89636ff0539d68382813df7e874395f1b00e; generality
sha256:9a4a3acae8300e87cdcd7c78351d0b1ff6b71408e25cca71fe3abfc73f1242a4.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt
sha256:d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256:e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256:3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate
sha256:b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal
sha256:75ffe2a1d9a6d1805c546113bbce98ad5eb0111944c2b879f18a41457ec5bbbbb. Source manifest
sha256:1157793f377ddbe38aa86e12ca2f1c0fa87d3a860416d934d947cb51d9f461fa. Engine receipt
sha256:b4a528eb3ab8ced04357e955ddb95a589902d586e753ed655926b40b02ca5a96 at receipts/engine/model_admitted/SFT-INFO-COMP-RUN-RECURRENCE-004-b4a528eb3ab8ced0.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-004: run_recurrence
- exact observation: {"both_invertible":true,"recurrence_rows":5,"run_rows":3,"source_symbols":6}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:50baacb874dc54f4f73d5f60df3ef55462ded66e0dad952e0e62e9943f303c87; empirical validation sha256:ea6527b33d7821633770aa0e59f966514e2e3e7868e4a87b19fee31f54678d07; external validation sha256:a7e1a01723d7554756326c8115b2ab8873c83958021b417acee2e522da98a69c; isolation sha256:939a0fb1e3a1d85571c0f34226718d4b5c4f63bd174822b2d8f3a7af697a92b4; custody sha256:e2293dba56100cb516ba3206aa868ad576d80e57fd8f41b22469461ba883c27a.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

87. Transform compression correspondence

Family: COMP Obligation: SFT - INFO - OBL - COMP - 005 Claim: SFT - INFO - COMP - TRANSFORM - 005

Question, law and forced result. Transform compression first applies a bijection on complete source support and then represents recurrence; any shortened record remains lossless only with the transform identity and inverse retained.

COMP-005 uniquely retains reversible-transform-representation, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SIGNAL - COMPLETENESS - 014
- SFT - INFO - COMP - RUN - RECURRENCE - 004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:ecb0fee425b7b45949d54d5cd7dd2b2573b22a0cf4c4d5fdf2061cc3532eb26d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-005 product before observation access.

Boundary: Every positive finite COMP-005 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-005 uniquely retains reversible-transform-representation, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__reversible-transform-representation__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:28ba00a1f6590798e82bae726129d2a277484215e8250da278afdc1522883da2; generality sha256:a69298087a8c9186da5c18f01e8987edff97094091c3975b247a3c9c3f5cd540.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt
sha256: d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256: e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256: 3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate

sha256: b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal

sha256: 6c42a65b15d93c73f11396307dc9372e170d4d52c69ff441d2a4d25df8970382. Source manifest

sha256: 943c7e5410b32efa4c79490c4c485bf4e2f9d5e3e54a8b0f7fefc7314ceb3122. Engine receipt

sha256: ec0fee425b7b45949d54d5cd7dd2b2573b22a0cf4c4d5fdf2061cc3532eb26d at

receipts/engine/model_admitted/SFT-INFO-COMP-TRANSFORM-005-ecb0fee425b7b459.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-005: transform
- exact observation: {"bijective":true,"reconstruction":true,"source_symbols":4,"transform_rows":4}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 437047f294fb657abd03c0bbcad61cb168073e409c60f802c38d140022e8542b; empirical

validation sha256: f269f39dfe5873807c347313a38f3776b294c4e0395cf078e2a9080e89b135c7;

external validation

sha256:1bffaafac2f86cbd272bbcfac727bfb2cf4168edb81bfda81be59a42e09e4ee; isolation
sha256:c291df417f1c80575d599d8ee62669ff1564bc249762ee572289f44a4841d5d1; custody
sha256:03bc1eb91f7730fa79b8cdc45956bd1d9cd4a93bebdabe945620deff29ef79a8.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

88. Source-model compression boundary

Family: COMP Obligation: SFT - INFO - OBL - COMP - 006 Claim: SFT - INFO - COMP - SOURCE - MODEL - BOUNDARY - 006

Question, law and forced result. A source model may define a code only when its grammar is frozen before outcome access and every model candidate, length and reconstruction result is preserved; the opened source cannot select the grammar.

COMP-006 uniquely retains frozen-model-comparison-ledger, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SIGNAL - COMPLETENESS - 014
- SFT - INFO - COMP - TRANSFORM - 005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:742ad4e7eb89095f56faf0252f1c8c852124a83c341b75c06b61c13197aac2c6. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-006 product before observation access.

Boundary: Every positive finite COMP-006 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-006 uniquely retains frozen-model-comparison-ledger, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__frozen-model-comparison-ledger__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:0944796a6858b8122b69819b8bd82c31de13962c27432572cae6e361033833b0; generality
sha256:26cd8aae2bd51cccc925c7d1afa7304a5e61d04d56f903e0d050b80bc498dce1.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt
sha256: d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256: e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256: 3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate

sha256: b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal

sha256: c2e8ea1a3c665310cb78be2f6215a73137635d4690d793688d8d44f2028b75a2. Source manifest

sha256: 21afbe7f999823c7c181ee0654c2ddfda6331a8f33f223ed8a6bffb7c0e4cde. Engine receipt

sha256: 742ad4e7eb89095f56faf0252f1c8c852124a83c341b75c06b61c13197aac2c6 at receipts/engine/model_admitted/SFT-INFO-COMP-SOURCE-MODEL-BOUNDARY-006-742ad4e7eb89095f.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-006: source_model_boundary
- exact observation: {"all_models_preserved":true,"code_lengths":[8,6],"models":2,"target_selected_model":false}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: ce62b361dcee9a0c3752259d7774d6ccc4be4c8d415a8b3eefe16a2ebaed69ce; empirical validation sha256: 8f8efe10a34f2e342bbf2db63a171bd20141a61e1a6d3aea01d879f7c1169f61;

external validation

sha256: c1e4375770f59bcf65bf92e4437183a3db85b9695261c1a42c974859ec1c616a; isolation

sha256: 916504f5019cf14262a4c879f4fbfc7bfa18fb17cc437d461690af4f31a1684b; custody

sha256: f4212c7af603aac5321338ccdf5176db07e59049443bccfcac6d08d279848e4.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

89. Redundancy as retained excess distinction

Family: COMP Obligation: SFT - INFO - OBL - COMP - 007 Claim: SFT - INFO - COMP - REDUNDANCY - 007

Question, law and forced result. Redundancy is the held excess of a reconstructing representation over the least reconstructing representation in one fixed generated grammar, never a negative scalar or imported entropy target.

COMP-007 uniquely retains held-excess-code-record, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SIGNAL - COMPLETENESS - 014
- SFT - INFO - COMP - SOURCE - MODEL - BOUNDARY - 006

The current reconciliation binds this obligation to the current model-admitted receipt
sha256: f850128e41642d307bf48ce21ea4391f943a081398cd427276ab4d6cc04606ce. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-007 product before observation access.

Boundary: Every positive finite COMP-007 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-007 uniquely retains held-excess-code-record, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__held-excess-code-record__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256: e804d9a6a4ffbbd6fd4832d1f9961a2f472b5d90dd5cc2ff0785b13bd6ca74ca; generality
sha256: f92f270ac7ab12e25c4c2bffd0181499e62da7d92a1aadb3e74fb01bcc01d4bc.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt
sha256: d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt

sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256:e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256:3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate

sha256:b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal

sha256:0d62994e0454af31bedd21777c4b9b0b997f44c3aab760e3fba75039d16cac8c. Source manifest

sha256:5e70477b0ca7b398054ee18af2279fea904afc2f426e5ed0acd587d54f28d0fe. Engine receipt

sha256:f850128e41642d307bf48ce21ea4391f943a081398cd427276ab4d6cc04606ce at

receipts/engine/model_admitted/SFT-INFO-COMP-REDUNDANCY-007-f850128e41642d30.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-007: redundancy
- exact observation: {"held_excess":2,"least_length":6,"long_length":8,"negative_scalar_used":false}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:45219e7cc972adcd5ff9f6774c2cc7d31951fc79818758a9582c833a6bfaa46; empirical

validation sha256:9e3a57a1d83901d7b7a1778c5176ec5f55aae2216745fdca46075dcf81880beb;

external validation

sha256:1024fdb68151b4ebd0b5464a55e664a2558122345568ddae3e4fa2036bc87938; isolation

sha256:77cae71985c1d97527bb12a30e3d80c6dc15f5950a7110b9b4945837bc866559; custody

sha256:c51011c185ba817a1a8474487666c179949a80a54b1a8dfb8c442f3b840f49bc.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source

- no failed route retires an obligation or changes protected authority

90. Minimum description within a fixed grammar

Family: COMP Obligation: SFT - INFO - OBL - COMP - 008 Claim: SFT - INFO - COMP - MINIMUM - DESCRIPTION - 008

Question, law and forced result. Minimum description is the unique least-cost source-reconstructing member of a completely enumerated fixed grammar; ties remain multiple minima and unrestricted grammars remain outside the claim.

COMP-008 uniquely retains exhaustive-least-reconstructing-description, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SIGNAL - COMPLETENESS - 014
- SFT - INFO - COMP - REDUNDANCY - 007

The current reconciliation binds this obligation to the current model-admitted receipt
sha256: 42735a957fd33924e42a8b258840f35eca3e5247b9a40a15d437211cea8dd463. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-008 product before observation access.

Boundary: Every positive finite COMP-008 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-008 uniquely retains exhaustive-least-reconstructing-description, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__exhaustive-least-reconstructing-description__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256: c7e1f5df2e3ab58b5639385837ee338dd2ccfd67bce5904ec1b369273f6bab84; generality
sha256: e05f2752c69df97b43b33fb13ef3fbc22d42547e681ecd5c656c153733f7646c.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt
sha256: d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural

absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
 sha256:e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256:3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate
 sha256:b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal
 sha256:5158d54d056d0c0252b41dea49a42709f1f872ab016085eee4ebee51bbb64c05. Source manifest
 sha256:738af4721332af845aa8dc5a447a2bd20712f26b5858e99c739dacabeab13aee. Engine receipt
 sha256:42735a957fd33924e42a8b258840f35eca3e5247b9a40a15d437211cea8dd463 at receipts/engine/model_admitted/SFT-INFO-COMP-MINIMUM-DESCRIPTION-008-42735a957fd33924.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-008: minimum_description
- exact observation: {"grammar_candidates":3,"least_units":6,"target_descriptions":2,"unique_least":"prefix"}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:2baaad8c0dc59027990f8b7b8d1965efed6c2f6233a34140dd201d72fa47451a; empirical validation sha256:6f67a2ca55538f594b3c486939945efdf69b4fe945f80030f76bb740658bc03e; external validation
 sha256:cf58dd8cd42325101af27cf6ae478ffcbfc37fefc75acc364d46826abf600c8e; isolation
 sha256:f3a1ca77fb3c04760bfe5003a93c24b9dd9229d5c89842cd94f597d20b279363; custody
 sha256:a0f24ffa5888bc63f4dd1cd2d5bfb9c9204ca70562bf0f7232002e638f91b4260.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

91. Lossy compression as declared coarsening

Family: COMP Obligation: SFT-INFO-OBL-COMP-009 Claim: SFT-INFO-COMP-LOSSY-COARSENING-009

Question, law and forced result. Lossy compression is a declared observation partition that maps multiple exact source forms to retained representatives while preserving every closed microform in the loss ledger.

COMP-009 uniquely retains source-microform-retaining-coarsening, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SIGNAL-COMPLETENESS-014
- SFT-INFO-COMP-MINIMUM-DESCRIPTION-008

The current reconciliation binds this obligation to the current model-admitted receipt sha256:a1ccda333ad3d3550b18ada3704c9ecc26d2104c1010dd9fd1a574d1819edbe4. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-009 product before observation access.

Boundary: Every positive finite COMP-009 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-009 uniquely retains source-microform-retaining-coarsening, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__source-microform-retaining-coarsening__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:a1c8e8ec38792a59bc2f62486b54672bd3a2cd3b61cb4c7424c8e02d6fca58cc; generality sha256:732fc79dcf99219bb35f594275d020efdc82a90bd40475c8f49041f18e1a33fa.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt sha256:d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256:3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate

sha256:b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal sha256:a0974dda74c5890c3a8632a0582f363adc1c4e82ab02d14baacacb30a2a5d7c7. Source manifest sha256:02d0e1ba19af3d8f698f62b121ad129859dd08ef227c21cb1952cbcadce008ba. Engine receipt sha256:a1ccda333ad3d3550b18ada3704c9ecc26d2104c1010dd9fd1a574d1819edbe4 at receipts/engine/model_admitted/SFT-INFO-COMP-LOSSY-COARSENING-009-a1ccda333ad3d355.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-009: lossy_coarsening
- exact observation: {"microforms":4,"microforms_retained":4,"representative_classes":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:97de7b9aeedc3fdbb33a71ea57fa3b0544b7657976d0a81401ab7cd654d4edc1; empirical validation sha256:ac6c3fb0e6acc7b10d7805124116f3993c549c0d8e469baf4a012bb68aac62eb; external validation

sha256:89fe80244e75d33ba1f2d0cf505ffecdcde180c9f7ae1c4936a664b3e1a7f9c1e; isolation sha256:4c37d93546fb5d1ce74114a34af1111fb8aed037143671b67b8747d6e59a94dc; custody sha256:50b4d75733c9c67e31bcde12e258b1fecc0867f7892083240a75dbed765a702f.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

92. Exact distortion relation

Family: COMP Obligation: SFT-INFO-OBL-COMP-010 Claim: SFT-INFO-COMP-DISTORTION-010

Question, law and forced result. Distortion is the exact part of common source positions whose reconstructed labels differ, with held source/image orientation and structural absence rather than numerical zero for exact agreement.

COMP-010 uniquely retains exact-position-mismatch-part, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SIGNAL-COMPLETENESS-014
- SFT-INFO-COMP-LOSSY-COARSENING-009

The current reconciliation binds this obligation to the current model-admitted receipt sha256:11f671de205911db7f08466a3ebefed3d2a35f2d77c374030e0d495a29ac30e7. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-010 product before observation access.

Boundary: Every positive finite COMP-010 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-010 uniquely retains exact-position-mismatch-part, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__exact-position-mismatch-part__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:94fff894d4f5f632ef1f7411819a00c87aa0f2ea606e6d3cdcce560705f08d48; generality sha256:60349460a678f00df9a918e330194ee8bc660cec7969c4aeef6983fe6415ff90.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt sha256:d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256:3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate

sha256:b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal sha256:3af69384f0eebdfdf8cdeed5d71cbf48391d21b4e2383e6c2a1c803e97c28ae4. Source manifest

sha256:796c8ec61aa7cc050074fdc428c1b0e4a949f5eeaa3ed4ec1e190e4b9868b9cc. Engine receipt
 sha256:11f671de205911db7f08466a3ebefed3d2a35f2d77c374030e0d495a29ac30e7 at
 receipts/engine/model_admitted/SFT-INFO-COMP-DISTORTION-010-11f671de205911db.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows
 preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-010: distortion
- exact observation: {"exact-reconstruction-distortion":"structural absence displayed
0","exact_distortion":"1/4","mismatches":1,"positions":4}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration
supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source
 identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:6cdf90e2bf334c87eb8e0edfb49cf2bc650373078de1cf25e3d98c7c907a2416; empirical
 validation sha256:1369153c3f99851c89bd61230da1595ac404166e2761851da7dbba54e6b4d2a7;
 external validation
 sha256:781e2e57c11dbdd8d04a4d30a1f4afbc296665ab77fd2642ebc12555d1f2ecd; isolation
 sha256:51b4a5e668b52a82562952f85417d2ede6b39e0b2151be9b9823117b28fd383e; custody
 sha256:efcedc5bef73bfbdea04deff381584c8af1285b73236c33008a2ad9dc316431b.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

93. Rate-distortion correspondence boundary

Family: COMP Obligation: SFT-INFO-OBL-COMP-011 Claim:

SFT-INFO-COMP-RATE-DISTORTION-BOUNDARY-011

Question, law and forced result. A rate-distortion frontier is the complete non-dominated set from a frozen finite
 code/partition grammar, retaining exact rate units and rational distortion parts without importing a continuum optimum.

*COMP-011 uniquely retains finite-nondominated-rate-distortion-ledger, complete compression custody, root
 forcing, post-registry observation and no extra rule.*

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares
 SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SIGNAL-COMPLETENESS-014

- SFT-INFO-COMP-DISTORTION-010

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:0f827a2d676d98d73c209c65c79885d8e50a7a6c099a75516866f15a1b4bd54d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-011 product before observation access.

Boundary: Every positive finite COMP-011 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-011 uniquely retains finite-nondominated-rate-distortion-ledger, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__finite-nondominated-rate-distortion-ledger__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:bdebad106d5fae6b4238bee918b741c53eb346ea2757dc32350ad2ab90c46012; generality
sha256:fb53e89fac66868f229cf1cb88ed80c715fa90c377e17f01eca61b45e088c81f.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt
sha256:d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256:e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256:3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate
sha256:b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal
sha256:d5e112db473cf71ed396b4fd0829610c127c06ba8de183f3f1ca1ad7c5af223a. Source manifest
sha256:9a65fead8d32e07be022221c606dfc5f6a748c8cfcd405cde72339473493a1b. Engine receipt
sha256:0f827a2d676d98d73c209c65c79885d8e50a7a6c099a75516866f15a1b4bd54d at receipts/engine/model_admitted/SFT-INFO-COMP-RATE-DISTORTION-BOUNDARY-011-0f827a2d676d98d7.json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-011: rate_distortion
- exact observation: {"continuum_optimum_claimed":false,"frontier_rows":[[4,"1/4"],[3,"1/2"],[2,"3/4"]],"rows":3}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:d139566452abb8af6340d7828b4b684108086f5325ea22b1858ed74da5e60c13; empirical validation sha256:a718f3119650492dfabf839d4fd0324f671c63341c54b359fa02cedd8728e8ec; external validation sha256:28acb6af878ed375c10d5d7ac840d2dfe0a91e0c11f702b62ce46b56ab4a249f; isolation sha256:8769e65aae9dc01886a5e22bc05b138c0a57d5987f8edcdf860af849f56ec67e; custody sha256:a605db5f8d22a81068657c8432c0726c39a573ca7321ca8b5532f3cab6a8eca3.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

94. Successive and layered refinement

Family: COMP Obligation: SFT-INFO-OBL-COMP-012 Claim: SFT-INFO-COMP-SUCCESSIVE-REFINEMENT-012

Question, law and forced result. Successive refinement transmits a coarse base record followed by retained distinction layers; concatenating every registered layer reconstructs the exact source and each prefix declares its closed distinctions.

COMP-012 uniquely retains base-plus-detail-reconstruction, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SIGNAL-COMPLETENESS-014
- SFT-INFO-COMP-RATE-DISTORTION-BOUNDARY-011

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:d2446fb05bf0591c284f9caf0cdb471bfdcef08f4e5f68d4c77f8841a92cc7a9. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-012 product before observation access.

Boundary: Every positive finite COMP-012 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-012 uniquely retains base-plus-detail-reconstruction, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__base-plus-detail-reconstruction__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: eb17db9ffa0ab315f4a3cbc1a3fce057b7f0fe40a12093ef84e827c8d5459f70; generality

sha256: b08e31c8890b0f9ecd780913ad87502a59ac573df6555b21726cdb17825ba615.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt
sha256: d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
sha256: e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256: 3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate

sha256: b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal

sha256: 42e059889962e1fd2f3fd7c10bf57a31929e56680c3ec801e9fbdd1c411c4599. Source manifest

sha256: 1f0718199f208d6d9c5f4de628ab5d742727b841d3261c2cfd7103e716369329. Engine receipt

sha256: d2446fb05bf0591c284f9caf0cdb471bfdcef08f4e5f68d4c77f8841a92cc7a9 at receipts/engine/model_admitted/SFT-INFO-COMP-SUCCESSIVE-REFINEMENT-012-d2446fb05bf0591c.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER

- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-012: successive_refinement
- exact observation: {"base_labels":4,"detail_labels":4,"exact":true,"reconstructed_forms":4}
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:6d623952de707606803a1ee25d79bbda7b089b69a8ca94f13bc0e3d88d3fbaf4; empirical validation sha256:4e198ac3ded7329e8915466693aca61e00cdc9ef24e1edbd9868d5bc6c5a3e86; external validation sha256:26e05dafbb788f1056584c5f774651de43ab47687ba7f56c2acea7ff3c648a9d; isolation sha256:524cc2759159345ad04f211646970d90f45f7a40c8f6b7eb250a5199fbf9be03; custody sha256:099c34acf28f77e6d2cd6eb6d107be41c0c6507a1e320be3d8af2e38e4d4f406.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

95. Compression under side information

Family: COMP Obligation: SFT-INFO-0BL-COMP-013 Claim: SFT-INFO-COMP-SIDE-INFORMATION-013

Question, law and forced result. Compression with side information transmits only the exact source/side relation not already retained by the side record; decoding binds both records and reconstructs the source without an independence prior.

COMP-013 uniquely retains side-bound-relation-code, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SIGNAL-COMPLETENESS-014
- SFT-INFO-COMP-SUCCESSIVE-REFINEMENT-012

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:82fece7fd52b43897da962940500e75ced12d34679f4e47e070cba852f766c94. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-013 product before observation access.

Boundary: Every positive finite COMP-013 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-013 uniquely retains side-bound-relation-code, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__side-bound-relation-code__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:8987c6e422c1ff965e0fe72f110e1eb7bcbe8bcebd7c70dfcd9bbab69a8f92b0; generality sha256:fce12baf64d2df725dd69ee7e9760bcfc4dc8f74e587d38a25d7c89bc11d1562.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt sha256:d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt sha256:e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation

sha256:3b5381ffe215bcd34953248086b6511c1c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate sha256:b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal sha256:41cda43e1ff34f84f29d67c553be6f0f557072b208942d4277fd9b8a7600b264. Source manifest sha256:74489b2874000eeceec576c5076eab6fb68b3a87cd0d9d39b54502f092259da46. Engine receipt sha256:82fece7fd52b43897da962940500e75ced12d34679f4e47e070cba852f766c94 at receipts/engine/model_admitted/SFT-INFO-COMP-SIDE-INFORMATION-013-82fece7fd52b4389.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-013: side_information
- exact observation: {"independence_prior_used":false,"reconstructed_positions":4,"relation_labels":4,"side_positions":4}

- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:cc4d223816fd7c73254e4c454af9d9bee4c72f7b737dfa418401105ca38f65ff; empirical validation sha256:300ec93ceeab514848d3ccba154c336dab2ddc18614e8e9b39dabae360863524; external validation

sha256:0a70aeb0d6d83962cda25a21d3fd9164e51785b974c73494df6f6f7180f5d93; isolation sha256:c94b03eef08673077d5ac8e02eb4c8910c3010628edef513487d26103d5bdab2; custody sha256:a84ebd2aa01073277ce096d72dd63765459c5627f26bbc54bd7a6204475598a1.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

96. Compression completeness and adverse-control certificate

Family: COMP Obligation: SFT - INFO - OBL - COMP - 014 Claim: SFT - INFO - COMP - COMPLETENESS - 014

Question, law and forced result. Compression-family completeness is the one-to-one reconciliation of all fourteen frozen obligations with reconstruction, distortion, boundary, adverse-control and ownership records.

COMP-014 uniquely retains fourteen-compression-obligation-ledger, complete compression custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SIGNAL - COMPLETENESS - 014
- SFT - INFO - COMP - SIDE - INFORMATION - 013

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:c8eb7350bc06e0e4c3b2ff9c1c3bbfcbc2dd84fd48c09ed457ccf8c6f9a3d644. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COMP-014 product before observation access.

Boundary: Every positive finite COMP-014 source, code, reconstruction, distortion, resource ledger and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COMP-014 uniquely retains fourteen-compression-obligation-ledger, complete compression custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source__fourteen-compression-obligation-ledger__retained-reconstruction-and-loss-ledger__exact-code-units-and-parts__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: e9aa7969f1a8654de7cc38206897d24767aa104a998f3dae45d9171caf56c8c1; generality
 sha256: ea5ea3c76ba257e31ace74bac2bc4e02cab235c5388764bd7707c58c5f13dfa9.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: Partial support changes the compression problem.; receipt
 sha256: d8c903637cda8764e6e10cffa9f0a60c0a02fceb6a07473f2b19bcc284663755.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported entropy model, rate-distortion formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no fitted dictionary, hidden model selection, erased microform or unregistered code grammar; no sampled candidate family or completed-infinite source; no failed route retires an obligation or changes protected authority; receipt
 sha256: e2c1a26141109748478c34b76296826e2268ecbcf00fd0a49fa71e163c5e190b.

Independent reconstruction and engine admission. Independent implementation
 sha256: 3b5381ffe215bcd34953248086b6511c11c0de41231b05fc45750593bf0d414d regenerated the declared census and sole survivor; independent certificate
 sha256: b3157922e3d8d5e1b332b3b470f8f8b5ea7de7215f0c8c0ea562e8e460046845. Derivation seal
 sha256: db6aa1a775ee66c532a0049e9ce9c07ae745aa84aece2b81186c7db688bb227f. Source manifest
 sha256: 71f6d97355526250a5a67c350b696ff4a7b54ba691f9081f5057670ca264158f. Engine receipt
 sha256: c8eb7350bc06e0e4c3b2ff9c1c3bbfcbcd2dd84fd48c09ed457ccf8c6f9a3d644 at `receipts/engine/model_admitted/SFT-INFO-COMP-COMPLETENESS-014-c8eb7350bc06e0e4.json`. Closure scope `depth_independent`. External status `empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COMPRESSION-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- COMP-014: `compression_completeness`
- exact observation: `{"duplicate_owners":0,"observation_rows":14,"omitted_owners":0,"registered_obligations":14}`
- complete family observation custody: 14 records; all rows preserved
- complete finite source, codeword, dictionary, recurrence, reconstruction, distortion, layer and side-information enumeration supplies the observation; no imported entropy optimum, fitted dictionary or erased predecessor enters

Falsification condition: Reject if the value-free registry, complete comp census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: ae2c9dadd71d27c5ab4fe633256be63b7ebfbf4acc473e58da96787241d78ea1; empirical validation sha256: aeb99b89dc3a31526599b1d4b70cb6a73a8fab706073734aca6f7ce6287325fd; external validation sha256: f603639bb3b7ff67173915210b1794427ab1ba295f77d1221a7a91f053b2f514; isolation sha256: 3b924001d433c4bba3ba20177434563a8ab0f1c3c2cd23404dfe58c3b03a3989; custody sha256: a2917c704541b47f22794e9d14457602d956b59c6fd3250b970d7b1318ae3ecb.

Prohibited inputs.

- no axiom, imported entropy model, rate-distortion formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no fitted dictionary, hidden model selection, erased microform or unregistered code grammar
- no sampled candidate family or completed-infinite source
- no failed route retires an obligation or changes protected authority

Complete-field family: CHAN

97. Channel as an exact input-output relation

Family: CHAN Obligation: SFT - INFO - OBL - CHAN - 001 Claim: SFT - INFO - CHAN - INPUT - OUTPUT - RELATION - 001

Question, law and forced result. A finite channel is a complete canonical input carrier, output carrier and source-bound relation with at least one retained output row for every input.

CHAN-001 uniquely retains complete-total-input-output-relation, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COMP - COMPLETENESS - 014

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 9b6d6289033145ac5a9fda6fd1bcd4fa3562793b11a757faa6e32b2e4104580c. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-001 product before observation access.

Boundary: Every positive finite CHAN-001 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-001 uniquely retains complete-total-input-output-relation, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__complete-total-input-output-relation__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: a935d74f9c5b9c0fe765b6989869d95a2c7970bc6dd2bc7ddb0f3493ef298ba; generality sha256: 36b44919d941346a457cd484f759f35a2547c82e2fd9199f8f4f33eb417ddd49.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit

downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt
sha256: 0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256: 25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256: 8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate

sha256: 7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal

sha256: 596131aa82c9e666e725bdfc85a3b1dd9f77b76f4ea369de5d66b5892e7db19a. Source manifest

sha256: 9c8d4fd68ac0f4f6023c0f16836e1b21161bf786a890825a2c32d7bf3bea5c6e. Engine receipt

sha256: 9b6d6289033145ac5a9fda6fd1bcd4fa3562793b11a757faa6e32b2e4104580c at receipts/engine/model_admitted/SFT-INFO-CHAN-INPUT-OUTPUT-RELATION-001-9b6d6289033145ac.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-001: channel_relation
- exact observation: {"all_inputs_covered":true,"inputs":3,"outputs":2,"relation_rows":4}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: e63a261bc34648211c13a4a0d50f00432a318285dc3fc1146688e8f20a04c4e2; empirical validation sha256: 3234b8b59d7d894b176996ae6dbdb1888495f46b13f3ea519f672aa28ff5c528; external validation

sha256:a6cd39b8c61555a59617b7bf6df69ebaa5a02f167ad2e3713097b9b380dee6fa; isolation
 sha256:2280e80137d36fda37bf1eaf8d9c328dc79045a27902b364ecbcf508002f1ad7; custody
 sha256:38e5b9eb9b19003e8697c751cee9bcd9cba4f0947f5d19e9e678990f7dd211ea.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

98. Deterministic channel transport

Family: CHAN Obligation: SFT-INFO-0BL-CHAN-002 Claim:
 SFT-INFO-CHAN-DETERMINISTIC-TRANSPORT-002

Question, law and forced result. A deterministic channel is the unique single-valued member of the complete relation grammar: every input has exactly one retained output and composition preserves provenance.

CHAN-002 uniquely retains total-single-valued-channel-map, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COMP-COMPLETENESS-014
- SFT-INFO-CHAN-INPUT-OUTPUT-RELATION-001

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:22e6d3a79fed38097a99abb5528d2ea2871c46408c454e1ee66cd500f5ec4fa3. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-002 product before observation access.

Boundary: Every positive finite CHAN-002 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-002 uniquely retains total-single-valued-channel-map, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__total-single-valued-channel-map__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:f362721040abaf6d90c1206164a6e8b03bb8b507157f6a5060a36366a808ec7c; generality
 sha256:34ecaebe79f98f2be6e998f9398e47fe6463eeb4dc0b5a340b148e945d062d55.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt
sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate
sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal
sha256:89a738ac71413d56652d7d4666a871f9adddd0e1eb8a2af9b4c4f2b0e5e1a024. Source manifest
sha256:de3d0cd434de42a40584d18261266fdae18f02c38fb1f2cf726304f7076717e3. Engine receipt
sha256:22e6d3a79fed38097a99abb5528d2ea2871c46408c454e1ee66cd500f5ec4fa3 at receipts/engine/model_admitted/SFT-INFO-CHAN-DETERMINISTIC-TRANSPORT-002-22e6d3a79fed3809. json. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-002: deterministic_transport
- exact observation: {"inputs":3,"outputs_per_input":1,"single_valued":true}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:6f0aac2e11ed281ae93bea5b8b720ad3b0a7a47fed6147b8170a40468e24e6cc; empirical validation sha256:661e1cf10c08dafa05a249c24be0a0269556606d7085944684fc53e05a03d7b1; external validation
sha256:8dd17fc68ba311ee173286af845dffda1f6e20f1808ad790aec2b316ad51a0d7; isolation
sha256:cfb7f3af26180f974439c84b3a1be308ce9a2bd666870d1d0f6028389f8bd116; custody
sha256:88ee0d38b3185c20a66ed1db1b6a7d50225c60c96e1366c624b5566ce457b99b.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

99. Observation-relative channel equivalence

Family: CHAN Obligation: SFT - INFO - OBL - CHAN - 003 Claim:
SFT - INFO - CHAN - OBSERVATION - EQUIVALENCE - 003

Question, law and forced result. Channel inputs are equivalent relative to an output observation exactly when their output images lie in one retained observation class; source microforms remain distinct.

CHAN-003 uniquely retains output-observation-equivalence-classes, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COMP - COMPLETENESS - 014
- SFT - INFO - CHAN - DETERMINISTIC - TRANSPORT - 002

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:d2c46be7dd3eae8df49ca261ec873c568ce21aed7e20072d166643fed55a75d5. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-003 product before observation access.

Boundary: Every positive finite CHAN-003 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-003 uniquely retains output-observation-equivalence-classes, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__output-observation-equivalence-classes__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:106ffab94cde0a7ac5e9927264bb421f14f0994dc07ba2bff9298b2d90ade0c3; generality
sha256:058906f244e52afc0257efd6c1e6b0c2e66aa354a285611164b5aab389732d65.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt

sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate
sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal
sha256:160e02672a0c7cc1e2094cef2d966812c9466d1a06732322818f34047fa7f27a. Source manifest
sha256:a0b581ff96c002d706482e89f5bd07f6063c49fe59b043e74acc997afcaaed4e. Engine receipt
sha256:d2c46be7dd3eae8df49ca261ec873c568ce21aed7e20072d166643fed55a75d5 at receipts/engine/model_admitted/SFT-INFO-CHAN-OBSERVATION-EQUIVALENCE-003-d2c46be7dd3eae8d.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-003: observation_equivalence
- exact observation: {"coarse_classes":2,"fine_classes":3,"source_microforms":3}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:b51463e669b7eb0b2c8499091a10fe137e172a821338c9e623f7f405ec782dea; empirical validation sha256:7d09b7642919aebbb43812397603eef47cfad7d88238c0abcb9076197c8e626; external validation
sha256:e15fe2c54bff2194551b3b5dd2daa8c7c365f8ff25a2a2accc00a764d956c6a2; isolation
sha256:c282624710ed78d6431a6da50d162a77b5038540964b7e6378232fe63e9e1b4b; custody
sha256:173c794debb8a988b417b29acc97506ef48aadb65d428b4c7eb9d3ca95ca8395.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result

- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

100. Single-user channel capacity

Family: CHAN Obligation: SFT - INFO - OBL - CHAN - 004 Claim: SFT - INFO - CHAN - SINGLE - USER - CAPACITY - 004

Question, law and forced result. Single-use capacity is the largest completely enumerated input codebook whose channel images remain pairwise distinguishable at the declared output observation, reported as exact code forms per use.

CHAN-004 uniquely retains maximum-distinguishable-single-use-codebook, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COMP - COMPLETENESS - 014
- SFT - INFO - CHAN - OBSERVATION - EQUIVALENCE - 003

The current reconciliation binds this obligation to the current model-admitted receipt sha256:53834d9a99266ce3af489d5ded0e1685a02188d4dbbffd07ada9a30e4fd2c7d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-004 product before observation access.

Boundary: Every positive finite CHAN-004 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-004 uniquely retains maximum-distinguishable-single-use-codebook, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__maximum-distinguishable-single-use-codebook__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:35e5e7f6ea02e22ec0d445e5537dfcb63b332cd419c6ce6df6aade3d097153b6; generality sha256:be29297e5c446985b25cd2b3010987e059e6edb3db3f1c29c69b6e987ac553c0.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate

sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal
sha256:c4e14bcc37d1d0fa754438f1c8cdb8ce6432e690960fb6935dede988a8a4b533. Source manifest
sha256:57a6a20192fd26b6c1cbb75118d25b5edfaa98265e52c16b2be4502584f15cb4. Engine receipt
sha256:53834d9a99266ce3af489d5ded0e1685a02188d4dbbfeffd07ada9a30e4fd2c7d at receipts/engine/model_admitted/SFT-INF0-CHAN-SINGLE-USER-CAPACITY-004-53834d9a99266ce3.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-004: single_user_capacity
- exact observation: {"channel_inputs":3,"distinguishable_code_forms":3,"unit":"forms per one registered use"}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:b1de8448ae67c8a9832875903024c892edb9235cf2afe19d655040ed7f665e76; empirical validation
sha256:2ed1ad1ba821d2ffe5a55d0bb9423ac4580b750db196ab6004ab8fc10a2ea283; external validation
sha256:866472111393927e43b913e76cbbb6acb0e8f39ceb17178cbac55b9326a9298e; isolation
sha256:e512788c5e3127df2ab91217707957d47957aa4291374645ecdd625ac0428eee; custody
sha256:773d2aa0c83760cc4cdc711e9d5a82e38e8c12b2fffec4c3546a074bdd9c635b.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit

- no failed route retires an obligation or changes protected authority

101. Resource-bounded channel capacity

Family: CHAN Obligation: SFT - INFO - OBL - CHAN - 005 Claim: SFT - INFO - CHAN - RESOURCE - CAPACITY - 005

Question, law and forced result. Resource-bounded capacity is the largest distinguishable codebook inside a frozen use, input, depth or retained-record budget; every excluded candidate retains its failed resource or distinction condition.

CHAN-005 uniquely retains registered-resource-codebook-maximum, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COMP - COMPLETENESS - 014
- SFT - INFO - CHAN - SINGLE - USER - CAPACITY - 004

The current reconciliation binds this obligation to the current model-admitted receipt sha256:2f8912021f96bde3a55df4b3ee714d73e80e3fe2a88441fa5d0cf123727ad293. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-005 product before observation access.

Boundary: Every positive finite CHAN-005 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-005 uniquely retains registered-resource-codebook-maximum, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__registered-resource-codebook-maximum__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:3d73ebce4d308b5e4b80cbd0335de3c570d41843cadf44d23000cb167371df04; generality sha256:9f43c13c7148b1e1610cb9f9096823ec085b4568c3bd14752a1ef3cb2b175b28.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbdade4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural

absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt sha256: 25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256: 8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate
 sha256: 7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal
 sha256: 167bb7a2d5f24b970d9bcff1f86effd1d64979619e78a2039c0ba205cb300645. Source manifest
 sha256: c828364138ed323b68fb285a63ee16c311590141e404c2f1ee8c380c2a5c9ce0. Engine receipt
 sha256: 2f8912021f96bde3a55df4b3ee714d73e80e3fe2a88441fa5d0cf123727ad293 at receipts/engine/model_admitted/SFT-INFO-CHAN-RESOURCE-CAPACITY-005-2f8912021f96bde3.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-005: resource_capacity
- exact observation: {"admissible_inputs":2,"distinguishable_code_forms":2,"resource":"registered two-input codebook"}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: b4fa35e702cd305a8b86bf5f7db8f35909538cd0e745f3d5e62757d0da214610; empirical validation sha256: 689f87e29c648de7651a2e76cf1f55c2854d5201832a0af7642f09efa1258e2e; external validation
 sha256: 2d65ff86129fc909b35f548ba83ff422c2cfb17f3a8ba6a74cd6ccc080fcb054; isolation
 sha256: a69c7a51b9d3c7b34efb3b0bafabb54f785f085410ca15c679be6c7128a0b36a; custody
 sha256: 06e3f2cee342c960e8ec075eacd9b22c7402e3d6413aa36efd3adfc10738233c.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

102. Noiseless channel composition

Family: CHAN Obligation: SFT-INFO-OBL-CHAN-006 Claim: SFT-INFO-CHAN-NOISELESS-COMPOSITION-006

Question, law and forced result. Noiseless channel composition pairs each upstream output with its downstream input and retains the unique composed source-terminal relation and both component traces.

CHAN-006 uniquely retains exact-relational-channel-composition, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COMP-COMPLETENESS-014
- SFT-INFO-CHAN-RESOURCE-CAPACITY-005

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:88b8cbb3c5dc90be0d264f9bc30f027729aeac0ec680897eade8ee56f68d9bb1. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-006 product before observation access.

Boundary: Every positive finite CHAN-006 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-006 uniquely retains exact-relational-channel-composition, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__exact-relational-channel-composition__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:d8c6303f06c5818fc23556b75ffa5dc5d36511549a6d93253b18c977bea5252b; generality
sha256:6fa79412f4dcb838d3876098af7f8149893e56d5a27119ae3de73920368ed6.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt
sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate

sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal sha256:d6cfa80ef4879e46aee38a3f68d419284921305407af43a17667cdf978a7a208. Source manifest sha256:2c6d42e9ff6c4f9aadf4162b775bc2a03a48b4f690f5837b794b17d7783d09dd. Engine receipt sha256:88b8cbb3c5dc90be0d264f9bc30f027729aeac0ec680897eade8ee56f68d9bb1 at receipts/engine/model_admitted/SFT-INFO-CHAN-NOISELESS-COMPOSITION-006-88b8cbb3c5dc90be.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-006: noiseless_composition
- exact observation: {"component_channels":2,"inputs":3,"terminal_images":3,"trace_retained":true}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:2c85bbc6a17272d79a4d7180f1dcea70c3c81fa9cf99dac3b88367aef9ba6ce5; empirical validation sha256:d472641b1007f85f476a134e04e441272ebc71ba4a4edee4ed94c15f0121a167; external validation

sha256:8243b977ab984c71260c29975d345fdde70022771e51697609ac1cc7f48e4cb5; isolation sha256:8aeeb3ece70f60ccaf5daf4e8e425c871b60c693d3f8bd088d4fe6bd2a70e1bb; custody sha256:eb2b2b10079583eaae9ca89f34cceca4e26bc1332072777df76601f6fab61e4b.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

103. Cascaded channel capacity boundary

Family: CHAN Obligation: SFT-INFO-OBL-CHAN-007 Claim: SFT-INFO-CHAN-CASCADE-BOUNDARY-007

Question, law and forced result. A deterministic cascade can retain no more distinguishable forms than occur in its terminal image; exact enumeration of the composed relation supplies the boundary without a logarithmic premise.

CHAN-007 uniquely retains terminal-image-cascade-boundary, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COMP-COMPLETENESS-014
- SFT-INFO-CHAN-NOISELESS-COMPOSITION-006

The current reconciliation binds this obligation to the current model-admitted receipt sha256:b1dee0c023acc5b27d8c1cee1f22d8e2b4f31bd9c5d5297d94e97516c7e96049. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-007 product before observation access.

Boundary: Every positive finite CHAN-007 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-007 uniquely retains terminal-image-cascade-boundary, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__terminal-image-cascade-boundary__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:38bf98966e24d1d2479c91ade4a401154602fb65ff76de44956a7ac2439cabfe; generality sha256:bf129912b41537ee609a06abc3f3f5ccb8189bac1137f696ff0a6c42a09d4ef0.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal sha256:7d68d22d1a01e6491600c3fc01282ae17ba01880d2c94760b3d27c913d890ad0. Source manifest

sha256:3d2686726e2579ef6f9e6f482df3f9919f157685acc5503a60181d47fa7e7448. Engine receipt sha256:b1dee0c023acc5b27d8c1cee1f22d8e2b4f31bd9c5d5297d94e97516c7e96049 at receipts/engine/model_admitted/SFT-INFO-CHAN-CASCADE-BOUNDARY-007-b1dee0c023acc5b2.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-007: cascade_boundary
- exact observation: {"boundary_met":true,"terminal_distinguishable_forms":2,"upstream_forms":3}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:5b5da80b0b90790c7738118016f3a956eaf96cadba18303e4bfd7ca6f2aa1c07; empirical validation sha256:0c8627d9c8c56aad6a36668e5ddd6de2d7de9458aa15bec457b759a94091d2b8; external validation sha256:260003cab8eac2a722721db78f19ec3908925908dce66717ba08f08bf833ee3f; isolation sha256:d5de1e03b846a8156f01e94367e8b8f201a45751958bffe2a03ea5f11e3c45c3; custody sha256:327e8ca2b143077346acec86fd5ea00e3416f3c04bb1db5b74a9cc984537c92b.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

104. Parallel channel composition

Family: CHAN Obligation: SFT-INFO-OBL-CHAN-008 Claim: SFT-INFO-CHAN-PARALLEL-COMPOSITION-008

Question, law and forced result. Parallel channels generate the complete ordered product of inputs and outputs, and their distinguishable code forms compose multiplicatively while component use units remain attached.

CHAN-008 uniquely retains complete-product-channel-support, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COMP-COMPLETENESS-014
- SFT-INFO-CHAN-CASCADE-BOUNDARY-007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:a668da33982d6faeea901f928f58ed885205c0e8b5cf761eeeca1a1610a0ad22. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-008 product before observation access.

Boundary: Every positive finite CHAN-008 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-008 uniquely retains complete-product-channel-support, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__complete-product-channel-support__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:9d00f0b682b6ff6b90a573c1ddb4d7bf7f66ec4e2adac970c117afd971b8d972; generality

sha256:6cf45bd6951b741b0ef72f74171821ca496a965072104e603855808dcaa18eb1.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt
sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate

sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal

sha256:e9ebb0362aa63ca709ff75fffc92a8c2eda85e1ca2156f13e2792acf74d3a18. Source manifest

sha256:86dc86dea3be6afc99183a0c27b20be92b7678d08ebe690220caff7b309daaef. Engine receipt

sha256:a668da33982d6faeea901f928f58ed885205c0e8b5cf761eeeca1a1610a0ad22 at receipts/engine/model_admitted/SFT-INFO-CHAN-PARALLEL-COMPOSITION-008-a668da33982d6fae.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-008: `parallel_composition`
- exact observation: `{"all_cross_pairs":true,"joint_forms":6,"left_forms":2,"right_forms":3}`
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:c550839244ed09171378e468b461d02f6c9ef95b393672fb33f61cbca18ff7ca; empirical validation sha256:82c298bb8fcd4722c66cca4d53064faefa4c4c01c803970edfd6eeb13e8fd428; external validation

sha256:ef1899f8a910855c7ee4ea5655947a8e66cbf099cd64488f1b86ef0f96baf022; isolation sha256:e7fbd42efe1ebb22c802ec5d64fd7d4ae4383c5cd094f93552ca07ba454a64ec; custody sha256:d83b7c4769396ea94d4958aea18563525f9582af1617efd01eb962f5e57c76.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

105. Feedback channel correspondence

Family: CHAN Obligation: SFT-INFO-OBL-CHAN-009 Claim: SFT-INFO-CHAN-FEEDBACK-009

Question, law and forced result. Feedback augments each later input record with retained prior outputs; its exact finite-use support is enumerated without assuming that feedback adds or removes capacity.

CHAN-009 uniquely retains history-retaining-feedback-support, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COMP-COMPLETENESS-014
- SFT-INFO-CHAN-PARALLEL-COMPOSITION-008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:a2680a6acd2ea50085b0136025dfbc6ddc68fce71c2daea4a5a18be49d39c40. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-009 product before observation access.

Boundary: Every positive finite CHAN-009 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-009 uniquely retains history-retaining-feedback-support, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__history-retaining-feedback-support__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256: dd89afbf7ecba653d4d6aed47af5b7b65e4b2ebca49038edda271debc99cd5e2; generality
sha256: 7dabca010e70c2105f7f9ad5c23640029c8202bb7a93b6565919f4088b8b2084.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt
sha256: 0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256: 25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256: 8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate
sha256: 7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal
sha256: 455cf6a2bae865ffb4c8be3dcfb0e898677ada29642b3c7cdbe76ecc34dd07b7. Source manifest
sha256: e5b4ba67f2ca10eb74ac9e56fd747892ad59fa32361a4ee9e013c3ef6b7741c6. Engine receipt
sha256: a2680a6acd2ea50085b0136025dfbc6cddc68fce71c2daea4a5a18be49d39c40 at
receipts/engine/model_admitted/SFT-INFO-CHAN-FEEDBACK-009-a2680a6acd2ea500.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER

- SFT - V1 - V2 - INFORMATION - OBSERVATION - CORPUS

Observed, adverse and boundary records:

- CHAN-009: feedback
- exact observation: {"distinct_terminal_words":4,"hidden_forms_added":false,"histories":4,"uses":2}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:87da372aedb1b1e7ee23134da4e32fde30dba207628470e0a925addf73adbeae; empirical validation sha256:6e19c227b26acee2079dfcb3f9d072027077b9b3f2ec44cd400e0bddafe0b156; external validation

sha256:66fdb6b055492b4c871d8ad8458552d61b189b7017eb6e6652cd41021c86ea4b; isolation sha256:f0f82f6d5d5f0e15bf478603e6c9aa99719aaf61b375ef0bf10e0e14b6b43416; custody sha256:566196d7fb469112761d951bdf38580536de8705f51a1c1a393a51bae8b51d75.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

106. Multiple-access channel support

Family: CHAN Obligation: SFT - INFO - OBL - CHAN - 010 Claim: SFT - INFO - CHAN - MULTIPLE - ACCESS - 010

Question, law and forced result. A multiple-access channel has a complete ordered product of sender inputs and one retained receiver output relation for every joint input, preserving each sender coordinate.

CHAN-010 uniquely retains joint-sender-single-output-relation, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COMP - COMPLETENESS - 014
- SFT - INFO - CHAN - FEEDBACK - 009

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:8e3b9e23e378669305eb5758f7c5c12037ca9e4db86bac395618464dc96ff007. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-010 product before observation access.

Boundary: Every positive finite CHAN-010 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-010 uniquely retains joint-sender-single-output-relation, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__joint-sender-single-output-relation__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: c6f4d23ea3b67034140f0d95c5e0a23af0816894d149ca2e5e604910bd1b3bc7; generality sha256: 059538cdb85998978f5345acc7d252611b6d30b7b080f7fe7569acf87ac33e8b.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt sha256: 0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt sha256: 25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256: 8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate sha256: 7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal sha256: b078ae1c31b70e796c68beb75b2bd0bb290d572afc9a0e93b4110005d3fc77e4. Source manifest sha256: 30a01f055e7dc54989be64fbd4af4e3e9685c91ee10e35a203fe33907d346b6b. Engine receipt sha256: 8e3b9e23e378669305eb5758f7c5c12037ca9e4db86bac395618464dc96ff007 at receipts/engine/model_admitted/SFT-INFO-CHAN-MULTIPLE-ACCESS-010-8e3b9e23e3786693.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-010: multiple_access
- exact observation: {"joint_inputs":4,"output_classes":2,"sender_forms":[2,2]}

- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 34b9a98747b2b8479d9cc026bd3670e8cb9b9607c6d5be21b51e4a3c98770264; empirical validation sha256: 7cdae5a87d7a7f1b8624da3499efeb4f61037ba2079a8300fd272af53046fc8d; external validation

sha256: 214c328f6086e42aa547f338fad2ee133243b7c7bfa36655af2bc8c6ca2ba2e2; isolation sha256: ce0b07556c42db9c71183bd875a5f418bb130a941d5b93a8320c160578101e5b; custody sha256: 35f722f67fb1d764c813eeaf7fd06fef60eeac09ade7b6f1ee09e0ef5d6160ee.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

107. Broadcast channel support

Family: CHAN Obligation: SFT - INFO - OBL - CHAN - 011 Claim: SFT - INFO - CHAN - BROADCAST - 011

Question, law and forced result. A broadcast channel maps each source input to a complete ordered receiver-output tuple, retaining every receiver coordinate and their shared source provenance.

CHAN-011 uniquely retains single-input-joint-receiver-relation, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COMP - COMPLETENESS - 014
- SFT - INFO - CHAN - MULTIPLE - ACCESS - 010

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 215e043f018808b473b08f9113b2900b1948a728324f1990fdaffb48b4aa6969. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-011 product before observation access.

Boundary: Every positive finite CHAN-011 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-011 uniquely retains single-input-joint-receiver-relation, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__single-input-joint-receiver-relation__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observatio

`n__finite-successor-or-explicit-boundary.`

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
`sha256:14eb5ac594756e7ceec0b54c0459ca6bbe57acfc8daf53a1bc8049206ec9d77`; generality
`sha256:97eeeda0c8abe5ed8aba60e47a847bfae532606b381ea4d560be76e67a569837`.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt
`sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373`.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
`sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
`sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
`sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738`.

Independent reconstruction and engine admission. Independent implementation

`sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58` regenerated the declared census and sole survivor; independent certificate
`sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302`. Derivation seal
`sha256:d9a39d0e99d6ab18b004919d836909826ae7a09631d189e3ec05ede3457e4abc`. Source manifest
`sha256:9e6e9699a389276bf08dba1e5fb3e02d81551ac25d5bd18d1a12e09158390567`. Engine receipt
`sha256:215e043f018808b473b08f9113b2900b1948a728324f1990fdaffb48b4aa6969` at
`receipts/engine/model_admitted/SFT-INFO-CHAN-BROADCAST-011-215e043f018808b4.json`.
 Closure scope `depth_independent`. External status
`empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-011: broadcast
- exact observation: `{"joint_output_rows":2,"receivers":2,"source_forms":2}`
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:8d169191baeb3e2d65e0e0938f8d29f4670c8798d69f17b29a7830ce66223ebb; empirical validation sha256:f84ee86c4a4d9824ebdf53ac1bdd89882bd3dc7f568641856bec92a09f348acf; external validation sha256:a41d6bba9c406363fcaab3a04e338cac8bc6f8a3a98302845b157e94a925cdba; isolation sha256:a2d095854a4d877b6395eea47f1853230d3d5665234fafb9488fabdc62572a6b; custody sha256:170f142eb239fd7511a342bc8a2cbe46e710b3d6aef00da92e69aee49841aa40.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

108. Relay channel support

Family: CHAN Obligation: SFT - INFO - OBL - CHAN - 012 Claim: SFT - INFO - CHAN - RELAY - 012

Question, law and forced result. A relay channel retains each source-relay and relay-terminal transition and enumerates every complete source-terminal route rather than collapsing routes into terminal reachability.

CHAN-012 uniquely retains source-relay-terminal-path-ledger, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COMP - COMPLETENESS - 014
- SFT - INFO - CHAN - BROADCAST - 011

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:df62ef27fa5b444b13adccfc7c2f9537b756a724591476dc885662cf562b0d6f. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-012 product before observation access.

Boundary: Every positive finite CHAN-012 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-012 uniquely retains source-relay-terminal-path-ledger, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__source-relay-terminal-path-ledger__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:e0deee2d1d975224cad7847d5ae81b08d8b7387efff41fb81a42d32abde3e065; generality sha256:6edb76f0e2f44cc17b2a6d7ebfc546702f34b86a7eebac9d87b7a6344563d161.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt
sha256: 0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256: 25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256: 8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate

sha256: 7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal
sha256: 9530b589c51a18b0320a88da7a7ee8775c9f3803fa48712f22404561f2f64581. Source manifest
sha256: 3f0825bfe7c5047d8bc6e1851d31123a94d95d251514955ae62a498c6b00313e. Engine receipt
sha256: df62ef27fa5b444b13adccfc7c2f9537b756a724591476dc885662cf562b0d6f at
receipts/engine/model_admitted/SFT-INFO-CHAN-RELAY-012-df62ef27fa5b444b.json. Closure
scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-012: relay
- exact observation: {"all_routes_retained":true,"relay_nodes":2,"source_terminal_paths":2}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: eadcef08efebff9250b5eb78e140b21aca1670fea38cad288e2bf1a3de1fbdddf; empirical
validation sha256: 464340acc3346e407716a209f41483d6de7fd8749b8be4ecb2a053fd013fcaee;

external validation

sha256:c00a3123d89330eedee6a6ec3af0f204895cb42b1cec8a09ba065f368c06b46f; isolation
sha256:c8bf306dfc1128dd1618375c2366003e67585309c0a24e9ad46a180f6dcd3dcf; custody
sha256:6a6702e673d5e8d45ebbd5b1d0f11be6c243274e8e23c05773f37a36b59ef31e.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

109. Interference-channel support

Family: CHAN Obligation: SFT - INFO - OBL - CHAN - 013 Claim: SFT - INFO - CHAN - INTERFERENCE - 013

Question, law and forced result. An interference channel is a complete joint-input/joint-output relation in which at least one receiver image depends on another sender coordinate; all cross-dependencies are retained.

CHAN-013 uniquely retains cross-dependent-joint-channel-relation, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COMP - COMPLETENESS - 014
- SFT - INFO - CHAN - RELAY - 012

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:431428161759b67f87b339a89148bfe1602ec4be735ea07b7308cf3096896224. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-013 product before observation access.

Boundary: Every positive finite CHAN-013 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-013 uniquely retains cross-dependent-joint-channel-relation, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__cross-dependent-joint-channel-relation__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:782705b060345196ff13e1c19f1334180852ff0e94448bc6aa1fd11d18d9de1c; generality
sha256:528bb5a6c5e4a8c453f319fe03a58abdddf58dc1954311254fdccaf2ca7b07b6.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt
sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate

sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal sha256:645c9e42c1552372437048c1932b1df7ad92189a483db2e400e552e64f3c6f36. Source manifest sha256:3c8122c434ad44eca4652866e9dc88425e2b2628bf707c7a0bd738b8612e4811. Engine receipt sha256:431428161759b67f87b339a89148bfe1602ec4be735ea07b7308cf3096896224 at receipts/engine/model_admitted/SFT-INFO-CHAN-INTERFERENCE-013-431428161759b67f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-013: interference
- exact observation: {"cross_dependency":true,"joint_inputs":4,"joint_outputs":4}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:b5d699e6bf68ea749943e582e19d5d91f877465c5a119a7ab4ec78c8aba51e3f; empirical validation sha256:cdcd419990cee19e61227160f9afe27337332141719024fb9a5559f51960f1c9; external validation

sha256:8d17599874275322e9f3314919ddf3ef8db25dbbe9c03f13e56da334c53d127c; isolation sha256:e31f0e722b0973b54cf5db0057c1acdc6e159ef62d3eaa6b26a575a0f6c4e436; custody sha256:55a5e7879dc1648bb322e25c71dcf10ca96c9f4b7fd9b22973c65572a9af988e.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

110. Bidirectional channel support

Family: CHAN Obligation: SFT - INFO - OBL - CHAN - 014 Claim: SFT - INFO - CHAN - BIDIRECTIONAL - 014

Question, law and forced result. A bidirectional channel retains both directed messages, their joint history and every ordered exchange state; neither direction is treated as unrecorded side information.

CHAN-014 uniquely retains two-way-history-bound-channel, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COMP - COMPLETENESS - 014
- SFT - INFO - CHAN - INTERFERENCE - 013

The current reconciliation binds this obligation to the current model-admitted receipt sha256: 4e22c4332825787a8cbfffb3676c73f37edc72076625863e07cfbbfc0137704bd. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-014 product before observation access.

Boundary: Every positive finite CHAN-014 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-014 uniquely retains two-way-history-bound-channel, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__two-way-history-bound-channel__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: ee413b18a32f0bc855d5f78b3e6912227a2f58025282e383e5202553e4100134; generality sha256: 0e74e9d4d8229e2a5db90fcaa642149f69af84c946031a88e8809d1ceb980211.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt sha256: 0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt

sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate

sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal

sha256:7e9a09038be0bb155508e4c507a863e75a8f0300fc519dd2b077e346fdb5b19b. Source manifest

sha256:2a77e57ebc8309ffa8d410dd490ff29281cc7f10cfc8776656b805e40e0470a6. Engine receipt

sha256:4e22c4332825787a8cbfffb3676c73f37edc72076625863e07cfbbfc0137704bd at receipts/e

ngine/model_admitted/SFT-INFO-CHAN-BIDIRECTIONAL-014-4e22c4332825787a.json. Closure

scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-014: bidirectional
- exact observation: {"both_histories_retained":true,"directional_alphabets":[2,2],"exchange_states":4}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:f86c13895e5c3a0faae049ff5bb60d013588b5f1cc710736170b985be4a148dc; empirical validation sha256:6f78d20dc39a0d4935ef04470a6e40aedd9b120d04a8ea46d4d77a92324d016; external validation

sha256:45f85b3ba8da74dc029066456b91589aa3c1660a29410a37409b2ef3a698b40b; isolation

sha256:4794dc6fd165ac25ceeba40c3acd0a784aa9913297dfbfedbb5234be4ca5d714; custody

sha256:0c0d80efdc40fa5292db1c03b917313b5c3de6f343d62b7b332c7db164e74df1.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit

- no failed route retires an obligation or changes protected authority

111. Network channel and cut boundary

Family: CHAN Obligation: SFT - INFO - OBL - CHAN - 015 Claim: SFT - INFO - CHAN - NETWORK - CUT - 015

Question, law and forced result. Network transport retains every source-terminal path; a cut is a held edge selection meeting every such path, and its exact crossing support bounds distinguishable transport.

CHAN-015 uniquely retains complete-path-and-cut-ledger, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COMP - COMPLETENESS - 014
- SFT - INFO - CHAN - BIDIRECTIONAL - 014

The current reconciliation binds this obligation to the current model-admitted receipt sha256:41d79da23d5b96253aabb7370098d045e18418d1e0f2242676731cdacf2fe74e. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-015 product before observation access.

Boundary: Every positive finite CHAN-015 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-015 uniquely retains complete-path-and-cut-ledger, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__complete-path-and-cut-ledger__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:7bdf013e7176edd02051d2b316c297c9fa09c3e14311a085721153994864527d; generality sha256:d34146ec5bdc85f235c0445d85efa2e4951e47011c2503329a4e3a4de24e2c0e.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural

absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt sha256: 25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256: 8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate

sha256: 7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal

sha256: 09533c907c899707c0edd7d914d7e0af3109456a6ed9f647252647c8b0d2ad0c. Source manifest

sha256: 85cb706cc160ee512548b0f97fc61a805c72094a972f74213eee05b419571d5b. Engine receipt

sha256: 41d79da23d5b96253aabb7370098d045e18418d1e0f2242676731cdacf2fe74e at

receipts/engine/model_admitted/SFT-INFO-CHAN-NETWORK-CUT-015-41d79da23d5b9625.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-015: network_cut
- exact observation: {"cut_closes_transport":true,"edge_disjoint_paths":2,"source_cut_edges":2}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: db858add4d983d972812f0da1a5f1042758149821843fcf2e2906a5c71225fc9; empirical validation sha256: ff3249949e5efcb16b5ced3babdf8dd8b00549ac25e8061440f73ecbbbdce9; external validation

sha256: e7128d0c55c3cb62d7dbd6fdb7caff1cbae6d6da4eb2795c3586d3dd44b424bb; isolation

sha256: 90a6c80af6339ffc6b60a0ec7ace1f01f599abeb18bae3a1a04d2f26c7d2697c; custody

sha256: 99cbe40f787e61961f10368b2242d3588a8f04bb03465c4c1bf9fda501d25a60.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

112. Finite-use capacity succession

Family: CHAN Obligation: SFT-INFO-OBL-CHAN-016 Claim: SFT-INFO-CHAN-FINITE-USE-SUCCESSION-016

Question, law and forced result. For a noiseless finite channel, adding one use forms the complete product of prior codewords with the single-use codebook, preserving exact use count and distinguishability.

CHAN-016 uniquely retains complete-multiuse-word-support, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COMP-COMPLETENESS-014
- SFT-INFO-CHAN-NETWORK-CUT-015

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:c7531830adf2893dd2b011aa3d8aaf18925a189bce01a40a665498597988a2df. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-016 product before observation access.

Boundary: Every positive finite CHAN-016 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-016 uniquely retains complete-multiuse-word-support, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__complete-multiuse-word-support__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:7a20f5e0ef146bb19a1e389de9ccda24265ac76815f8cd400c3a5cd50e3f9a43; generality
sha256:348d4283db8649b0202ff5822d5aa7e4562d80b9f3c3b982e364efb6e5f064aa.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt
sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate

sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal sha256:edcc4c638a4397d6f18a08d1fe17080b6639b9e39e7a61312d39b95032893e7f. Source manifest sha256:36e34be7ed9fe1960faf78e75b36f99b5f0419806ba8c2f60480b0650d3849a0. Engine receipt sha256:c7531830adf2893dd2b011aa3d8aaf18925a189bce01a40a665498597988a2df at receipts/engine/model_admitted/SFT-INFO-CHAN-FINITE-USE-SUCCESSION-016-c7531830adf2893d.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-016: finite_use
- exact observation: {"single_use_forms":2,"unit_retained":true,"uses":2,"word_forms":4}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:008fce4511249afe7d35601e09ee9601c74485054474ba8b693e3609871df471; empirical validation sha256:ceffd7cb3a59ab5ccadae1bf1179f831fec193288f792ba05271ef8ce6a4f139; external validation

sha256:c18b25a71f514f36b85c63bacd96176b7fe5ca945afb6a2b168f37f004a8426e; isolation sha256:ba208b72c02ff59fc1177295a02c7853d79faa03377feba5bc5e8cb00810c9c8; custody sha256:e98b402b3efd30c6672400c4c58908a971d8da14dd88e0d9fe27474fcdf7dbe8.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

113. Channel simulation correspondence

Family: CHAN Obligation: SFT-INFO-OBL-CHAN-017 Claim: SFT-INFO-CHAN-SIMULATION-017

Question, law and forced result. One channel simulates another at a frozen boundary exactly when a declared encoder, base channel and decoder compose to the target relation on every generated input.

CHAN-017 uniquely retains encoder-channel-decoder-equivalence, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COMP-COMPLETENESS-014
- SFT-INFO-CHAN-FINITE-USE-SUCCESSION-016

The current reconciliation binds this obligation to the current model-admitted receipt sha256:d7266c17d27cc6c694d8357154b0b4ed73a21be9e2c4ca9aaa351577e86abafb. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-017 product before observation access.

Boundary: Every positive finite CHAN-017 carrier, relation, observation, resource record, path family and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CHAN-017 uniquely retains encoder-channel-decoder-equivalence, complete channel custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__encoder-channel-decoder-equivalence__declared-output-observation-custody__exact-code-forms-per-registered-use__complete-declared-channel-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:30953b423021bae2eb20519c2fd8958e939774a6b4a1211443519a3db0740958; generality sha256:366743f2e25ca9361b3b0f509de983799f0a7fa38439bdf52aaba054481759ea.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Partial carriers change transport support.; receipt sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the declared census and sole survivor; independent certificate sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal sha256:3a418fe1fd7464a4659dc3962e9edcaf5811e48ebcf112a99fbfda8b87fdac52. Source manifest

sha256:34a96487f4f0e16bd63dda86071abc3ceabbb09a6a77385f8d76b19f03605a99. Engine receipt sha256:d7266c17d27cc6c694d8357154b0b4ed73a21be9e2c4ca9aaa351577e86abafb at receipts/engine/model_admitted/SFT-INFO-CHAN-SIMULATION-017-d7266c17d27cc6c6.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-017: channel_simulation
- exact observation: {"base_rows":2,"decoder_rows":2,"encoder_rows":2,"relations_equal":true,"target_inputs":2}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:ae6447f48e34c22dee8257228b1903b82c50bb21f31a327923ef69ca887e19ac; empirical validation sha256:5ece6fa80c41757c2664a1b10425d5dc19e09158998dc36e46ccf953ad658368; external validation sha256:df248afb96f8744b65bcec5ee52542678aeb26d76612a93b9f8d34e0804e8984; isolation sha256:c1f968db0c6ccafc48a3b46de6c3b19a1aa5037aeee998bc3c5303e15229c318; custody sha256:291bc0d6d0b3a5ae43a7c3716a551be823031cb4baad643c8d84881564e1dfbe.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

114. Channel-family completeness certificate

Family: CHAN Obligation: SFT-INFO-OBL-CHAN-018 Claim: SFT-INFO-CHAN-COMPLETENESS-018

Question, law and forced result. Channel-family completeness is the one-to-one reconciliation of all eighteen frozen obligations with exact relations, resource units, observations, controls and ownership boundaries.

CHAN-018 uniquely retains eighteen-channel-obligation-ledger, complete channel custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COMP-COMPLETENESS-014
- SFT-INFO-CHAN-SIMULATION-017

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:e3c2296a39a176c18bd68576bd1f623c3eb8e9757321974d2bc2f16a57ffaea1. A missing
 dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CHAN-018 product before observation access.

Boundary: Every positive finite CHAN-018 carrier, relation, observation, resource record, path family and registered
 successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly
 one survives and 255 are eliminated. The forced coordinates are:

- CHAN-018 uniquely retains eighteen-channel-obligation-ledger, complete channel
 custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-input-output-carriers__eighteen-channel-obligation-ledger__dec
 lared-output-observation-custody__exact-code-forms-per-registered-use__complete-dec
 lared-channel-product__root-bound-forward-forcing__post-registry-exact-observation_
 _finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:572433c17cb6d85dde5d5d8681c715b7dafa3e6b668a51d3e9fc1a396107b15a; generality
 sha256:1f240a70866746b36d591dceeb44d637ca53d39f67f124707137a78519ac0dc4.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained
 distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit
 downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted
 magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT
 numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed:
 Partial carriers change transport support.; receipt
 sha256:0c122000995f41a8ba53736f8fd0442055b9a9bec2fbddae4ce2483de9176373.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs
 from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated
 product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no
 axiom, imported stochastic channel, Shannon formula or target outcome selects the result; host 0 denotes structural
 absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar;
 no hidden input row, erased route, fitted channel distribution or unitless capacity; no sampled channel family or
 completed-infinite use limit; no failed route retires an obligation or changes protected authority; receipt
 sha256:25e460c6672e27741457bec93a1154c55ce069d08aef542dcd507abf84045738.

Independent reconstruction and engine admission. Independent implementation

sha256:8aee7ef2e7881a031673344b34dfe3522c20a0ea322c96ba8e4d66cc44033e58 regenerated the
 declared census and sole survivor; independent certificate
 sha256:7213d35647c6311095ee73ecd931de209b445ec4409f49277f247ed80296f302. Derivation seal
 sha256:6078e895c109d6e5e9fff2def1bbc7969e6b1c1a83c23255e8319345ed08f346. Source manifest
 sha256:8d236877ef15f796a2268fb1bcd259ce56dfc40dc8c41c298737999891544bf. Engine receipt
 sha256:e3c2296a39a176c18bd68576bd1f623c3eb8e9757321974d2bc2f16a57ffaea1 at receipts/e
 ngine/model_admitted/SFT-INFO-CHAN-COMPLETENESS-018-e3c2296a39a176c1.json. Closure scope
 depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-FINITE-CHANNEL-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CHAN-018: channel_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":18,"omitted_owners":0,"registered_obligations":18}
- complete family observation custody: 18 records; all rows preserved
- complete finite input, output, relation, observation, codebook, use-history, route, cut and simulation enumeration supplies the observation; no imported stochastic channel, logarithmic capacity formula or fitted distribution enters

Falsification condition: Reject if the value-free registry, complete channel census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 73987a18951845f3235f10c4781fd6f279d672b055ab28de36815e5b3de867d3; empirical validation sha256: 2149ba006287bdcba7bb7914c3c4df025b3bbe10734cd46cfdbdfaf298172918; external validation sha256: fd22929f5dc9581a242d3c951081f5240966f785233b97800674329c6c10a044; isolation sha256: 0ca10115de5be59f90c2831e698a393b315d182e7e017ccabde6899924ef330a; custody sha256: a4e60a1a84a3239e1d80a61c8ba0620123fa983b162421a1b9315e12025f390c.

Prohibited inputs.

- no axiom, imported stochastic channel, Shannon formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden input row, erased route, fitted channel distribution or unitless capacity
- no sampled channel family or completed-infinite use limit
- no failed route retires an obligation or changes protected authority

Complete-field family: NOISE

115. Noise as unrecorded distinction closure

Family: NOISE Obligation: SFT-INFO-OBL-NOISE-001 Claim: SFT-INFO-NOISE-DISTINCTION-CLOSURE-001

Question, law and forced result. Noise closes a distinction when multiple exact source/action paths share one received observation; every predecessor remains in the retained closed class.

NOISE-001 uniquely retains complete-predecessor-closure-ledger, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CHAN-COMPLETENESS-018

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 2c59fcb12bb0cd1f79cba8728fe3740b4e7e53d8890175ff1f2dc3dc966dd8fc. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-001 product before observation access.

Boundary: Every positive finite NOISE-001 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-001 uniquely retains complete-predecessor-closure-ledger, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__complete-predecessor-closure-ledger__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:08d11e6b01efbf04b352543100fda4fab44eff5c25d7b8c7ab31065bcd6b8755; generality sha256:c0750dad6fc891dd58b12e07145225e3c85c828994aa7cb6ce27678b1bbb9f39.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt sha256:c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256:dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate sha256:7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal sha256:b6d81d0dc55d905f9f0bbe442432edec6ad37235b1ad7958e10dd0b23a962963. Source manifest sha256:f284ba71c7a33443c6781e0a6ff2336b1ad796aa7c3f16ef8d13da4f3251eb7c. Engine receipt sha256:2c59fcb12bb0cd1f79cba8728fe3740b4e7e53d8890175ff1f2dc3dc966dd8fc at receipts/engine/model_admitted/SFT-INFO-NOISE-DISTINCTION-CLOSURE-001-2c59fcb12bb0cd1f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-001: distinction_closure
- exact observation: {"all_predecessors_retained":true,"closed_class_size":2,"received_images":1,"source_predecessors":2}
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:605a2b1306a700fc5ef76d2c4db7df5647a81578d73aaedc5047ee5abf950da2; empirical validation sha256:afadb660c5f851c30cd5c5cba4b4eb39deb931ee03dd70deae2169965ce41bcf; external validation

sha256:8779da73247ecae1409d69b245f8fdecdebfa60c84865ceed2ef3563a542ed12; isolation sha256:03462e13c30da548785be71aada0da205ce98fa7e8752cd416f8ba1aa27fc1ce; custody sha256:fe265bc1a8d35e914a5a2a99539bb43179c613aac300370cce0623387f881be.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

116. Error as source-output mismatch

Family: NOISE Obligation: SFT-INFO-OBL-NOISE-002 Claim:
SFT-INFO-NOISE-SOURCE-OUTPUT-MISMATCH-002

Question, law and forced result. Error is the complete positionwise mismatch trace between one retained source and received word, including both labels; identity has structural empty-One trace.

NOISE-002 uniquely retains complete-changed-position-trace, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CHAN-COMPLETENESS-018
- SFT-INFO-NOISE-DISTINCTION-CLOSURE-001

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:20fe9bbcc22854b2caf0877cb0da418006b97f0f4d3a940c7eddd588a206e3d7. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-002 product before observation access.

Boundary: Every positive finite NOISE-002 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-002 uniquely retains complete-changed-position-trace, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__complete-changed-position-trace__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:e6d783b20819b9d601773d2d161efabaf8eff7484336c0ed0c2404bed1f12743; generality

sha256:a1b5bd7aa2333d1de3664eab02edbc120209ad648dad9757ef66251328022778.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt
sha256:c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256:dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate

sha256:7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal

sha256:a98cd7eb41cbdc0dda9ced66a0cd8639e1a6918d88557d0f2eab021c382fba7d. Source manifest

sha256:652c094ae97948e6b6f19cc8263f1c00b22e07de4b6f28c410195f3a494382eb. Engine receipt

sha256:20fe9bbcc22854b2caf0877cb0da418006b97f0f4d3a940c7eddd588a206e3d7 at receipts/engine/model_admitted/SFT-INFO-NOISE-SOURCE-OUTPUT-MISMATCH-002-20fe9bbcc22854b2.json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-002: source_output_mismatch
- exact observation:
{"changed_positions":[1],"identity_trace":"empty-One","old_new_labels_retained":true,"word_positions":3}
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:f0f15c07aa428d505e755fe65d9cdc951f4ec46e760d70158602a2a541da28c0; empirical validation sha256:8821b085f6e027dbec8872a5c820b8c6de938aaf2f9abf3dda97154779ba3d4b; external validation sha256:836de21202c0b676feae98cd3770d3312ff231a94336a17b36517bb7801a9533; isolation sha256:55f477fc96e0d68db04264eff855a15bab8672ed2f806e1d5ac187ad101ed0e5; custody sha256:0d63f3432ff6fde908af6b86f0c807a60b42eafaca362c01b1fa814f7a78ff22.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

117. Deterministic noise-pattern support

Family: NOISE Obligation: SFT - INFO - OBL - NOISE - 003 Claim: SFT - INFO - NOISE - PATTERN - SUPPORT - 003

Question, law and forced result. Noise-pattern support is the complete generated family of exact held position/action masks inside a registered width and budget, with no probability attached.

NOISE-003 uniquely retains complete-generated-error-mask-support, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CHAN - COMPLETENESS - 018
- SFT - INFO - NOISE - SOURCE - OUTPUT - MISMATCH - 002

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:ca36620686e6ba42acc03080cda9f0e4e3468b429ed9b20ae7792558202d47f0. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-003 product before observation access.

Boundary: Every positive finite NOISE-003 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-003 uniquely retains complete-generated-error-mask-support, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__complete-generated-error-mask-support__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:ba71218ff96f6ed63faa169fbc6f97b677a42493182e5055b8eca18658f9a2c6; generality
 sha256:8687dbe5b7dfb49707012f6547def7944c11fe1769b1d6e2e5fdb02504085ab4.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt
 sha256:c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
 sha256:4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256:dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate
 sha256:7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal
 sha256:a7b50a97a543955d4295aee0a57b78c5874390b5df81a1be5c09b941c7817102. Source manifest
 sha256:44a2521c77bc4b8786be59efbfa00730c273cbcbdb20910bb88e6c85bb2d89d68. Engine receipt
 sha256:ca36620686e6ba42acc03080cda9f0e4e3468b429ed9b20ae7792558202d47f0 at receipts/engine/model_admitted/SFT-INFO-NOISE-PATTERN-SUPPORT-003-ca36620686e6ba42.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-003: pattern_support
- exact observation: {"budget":1,"distinct_images":3,"mask_count":3,"width":3}
- complete family observation custody: 12 records; all rows preserved

- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: e707626a191537dac8695724ae86ef607ceebd3e5efc72cced06907ae79e0a78; empirical validation sha256: 695b591012d995eb683b5b0e7d72687972bc2cc7093de6cc9bb7c965487b454d; external validation

sha256: bc89de8e9d3df50d4d3569a9a9835c1efd00c84277a5049d784a1aea1997108b; isolation sha256: b4926e4c23a3a51bbafa8455df9961b9cfea6eb2d539ac61690d78c2fcb7c90f; custody sha256: f832e8e397d5f94300f78a0abb2c0f3e386e6e728fb1a9f2823c5e906411a498.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

118. Noise transport through composition

Family: NOISE Obligation: SFT - INFO - OBL - NOISE - 004 Claim: SFT - INFO - NOISE - COMPOSITION - 004

Question, law and forced result. Composed noise transport applies retained actions in order and binds the terminal image to every intermediate action and the complete source-terminal mismatch trace.

NOISE-004 uniquely retains ordered-action-trace-composition, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CHAN - COMPLETENESS - 018
- SFT - INFO - NOISE - PATTERN - SUPPORT - 003

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: f411d0dd446f5e8aa8fec496d859398646ec02613011b5d75d7d57d8fe9cc9d1. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-004 product before observation access.

Boundary: Every positive finite NOISE-004 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-004 uniquely retains ordered-action-trace-composition, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__ordered-action-trace-composition__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: 987f66279379b3d1c5e0df755c44d13c98e5783eadcd76081c9c307ed19440d1; generality
 sha256: 1243241409b609071ad7a7b6092ffec1791f771bce50edad96e79c8ab867e702.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt
 sha256: c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
 sha256: 4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation
 sha256: dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate
 sha256: 7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal
 sha256: 9d6039203c63585644be41fb9b673fe0fde2ef4ecfe63fb82956b41df38eab88. Source manifest
 sha256: d87c5c60e123542d1097fba5828c1b8851fc5204f2c07b73be10bb94010f0907. Engine receipt
 sha256: f411d0dd446f5e8aa8fec496d859398646ec02613011b5d75d7d57d8fe9cc9d1 at receipts/engine/model_admitted/SFT-INFO-NOISE-COMPOSITION-004-f411d0dd446f5e8a.json. Closure scope
`depth_independent`. External status `empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-004: composition
- exact observation: `{"actions": [[1],[3]], "intermediate_actions_retained": true, "terminal_changed_positions": [1,3]}`
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:5fb70c89ceb538e7047e8ad53065fd6ef7757d4b5a26d39387dfe10253a22459; empirical validation sha256:409ebff53e988debc7347c366f38bba75f277522bb62d243a786de2004bb1c6d; external validation sha256:9feb4f4faa7d65cc23a2851843b04c72e172ad96ec7cb531c16969d1c6587175; isolation sha256:4a5dea7712da855658b493d9be6906d755644d4a6bacebfff5c73a6824574ce5b; custody sha256:26699861fec30bad086d4cefb111ec2e1626ba9e32a4e915d0ac60446c2ab7e2.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

119. Error detection condition

Family: NOISE Obligation: SFT - INFO - OBL - NOISE - 005 Claim: SFT - INFO - NOISE - DETECTION - 005

Question, law and forced result. An error is structurally detected exactly when the received image is outside the complete valid code support; no fitted distance or probability threshold is needed.

NOISE-005 uniquely retains invalid-code-image-detection, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CHAN - COMPLETENESS - 018
- SFT - INFO - NOISE - COMPOSITION - 004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:ff443b32efff524ab75589e0b024af5fd9561d6ea07801e84c12a8a08b03563d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-005 product before observation access.

Boundary: Every positive finite NOISE-005 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-005 uniquely retains invalid-code-image-detection, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__invalid-code-image-detection__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:df0b51ce1184eddd31fec08a7181127e4ed9b22b5f89380e95f6dd1856998322; generality sha256:7be43b91cc5e469cb1d7534cc07fcaeb3b65c38030d46bf01e42bebd6b9d68d9.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted

magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt
sha256: c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256: dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate

sha256: 7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal

sha256: 476dda1e79690a5db1f2e236e7844bafc73fb956de3ba20299dff8dee8d93cc9. Source manifest

sha256: 29f47db3826e71f6ecda1db308eede74ae45bf1a2d7ac514eb0632ec50441d87. Engine receipt

sha256: ff443b32efff524ab75589e0b024af5fd9561d6ea07801e84c12a8a08b03563d at

receipts/engine/model_admitted/SFT-INFO-NOISE-DETECTION-005-ff443b32efff524a. json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-005: detection
- exact observation: {"all_outside_code":true,"codewords":2,"one-change_images":6,"threshold_used":false}
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 031ccbdcc758c2dcb4fc25eb9b2f37cf2c2b658ff9a93a4d8c7accd58c97bd059; empirical

validation sha256: 0ac9b68a9f9eb7713f5cc4f0981c9895458e0e83c2eadb0b9e7f336ee7f388ce; external validation

sha256: af42cc778ec5febe82e1c51e5d40c10816a42373de841c505460667c51cfa59b; isolation

sha256:a996765e0903ac3c980d225826d49a95192fa4f94d7189a8527e8562f6e850c7; custody
 sha256:a8b97c2445cb1710c3fc318b1565135ef0406829d4a9f8318accaffc3c91daf6.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

120. Error localization condition

Family: NOISE Obligation: SFT - INFO - OBL - NOISE - 006 Claim: SFT - INFO - NOISE - LOCALIZATION - 006

Question, law and forced result. An error is localized exactly when the retained source/image comparison or syndrome relation identifies one changed-position set among the complete registered mask support.

NOISE-006 uniquely retains unique-changed-position-record, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CHAN - COMPLETENESS - 018
- SFT - INFO - NOISE - DETECTION - 005

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:25f5f7d192ad767fd4f5ee6d74b4392c2b73b0273c4bce8fdb616b3e33ae5a4. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-006 product before observation access.

Boundary: Every positive finite NOISE-006 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-006 uniquely retains unique-changed-position-record, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__unique-changed-position-record__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:4305eae9de8a2a923b8d980c2055fe9f122b8870821c4b416139912c7a819cab; generality
 sha256:b79520cfeb1a3ed177e0b377e46320226ee96ef3d112f6620bd6fdf085d5d1a4.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt
sha256:c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256:dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate
sha256:7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal
sha256:dcd087314eeae397aebdb057370e9e8c8ae2d9f3fecafd411636ad661433900f. Source manifest
sha256:7662f0040c8e9cae7720cbdaa4e062fb3f0c5d877fa38fff01725a0d779a274e. Engine receipt
sha256:25f5f7d192ad767fd4f5ee6d74b4392c2b73b0273c4bce8fdb616b3e33ae5a4 at receipts/engine/model_admitted/SFT-INFO-NOISE-LOCALIZATION-006-25f5f7d192ad767f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-006: localization
- exact observation: {"all_unique":true,"localized_positions":[1,2,3],"masks":3}
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:1f591bd681606aa3e9199a4e89cf34897c5fe09027ef05dcf6f3e932ac31dc80; empirical validation sha256:1bd4f4bdc072393ccd6d89bcad30b5e5ed52bec982cb9f7315e95a44adfa8e53; external validation
sha256:f6b7ea75a024bb0a3a61a0f0ef5adea3380c19184aac70232e59d867ee6c8f2a; isolation
sha256:a1f289b179a7f4e44d5aa3dfd4b0497cb5f8998814770a9d054f469ed8f06ed0; custody
sha256:5d89dabe196ca3bdb528e6efe452345d366d97e0cd5393cde0b8678659d75653.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result

- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

121. Error estimation condition

Family: NOISE Obligation: SFT - INFO - OBL - NOISE - 007 Claim: SFT - INFO - NOISE - ESTIMATION - 007

Question, law and forced result. Error estimation returns the complete source/action predecessor class consistent with the received image; a singleton is exact and multiple members remain unresolved.

NOISE-007 uniquely retains complete-predecessor-estimate-class, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CHAN - COMPLETENESS - 018
- SFT - INFO - NOISE - LOCALIZATION - 006

The current reconciliation binds this obligation to the current model-admitted receipt sha256:80035d32fec7e6432c5dab60ca700b90ce1c7cc157fa2e044e8a2fac1bd83a64. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-007 product before observation access.

Boundary: Every positive finite NOISE-007 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-007 uniquely retains complete-predecessor-estimate-class, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__complete-predecessor-estimate-class__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:0b55bb26612dbddfa046ba9e26c16e3bf4a8839bb01c6ce86447bc76ccdc13a5; generality sha256:9739cff643e0b03987ccd63a6d9671f3bd9ed3ad7c02d1a7834c19011723ffa9.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt sha256:c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256:dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate

sha256:7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal
sha256:50abbdffc6df50fb40c216d009fca19ba36e1718da3553d6ddf9681b00c4cc875. Source manifest
sha256:73c5e23e89b6b55609d6ea7ea8a0c8bac926e00943d4b22e2c552c1e57653b72. Engine receipt
sha256:80035d32fec7e6432c5dab60ca700b90ce1c7cc157fa2e044e8a2fac1bd83a64 at
receipts/engine/model_admitted/SFT-INFO-NOISE-ESTIMATION-007-80035d32fec7e643.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-007: estimation
- exact observation: {"likelihood_prior_used":false,"received_images_checked":6,"singleton_predecessors":6}
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:f69290cf395179d5eb3179e148f4d9f2aa7a7f9c1aa53d16901aeba3ab705bb9; empirical validation
sha256:69bb911c989e430dd868a0fe234a70c7cd6c8bc8637fab21b81a8f2cb3f98d2b; external validation

sha256:4a7480cd07fd42b6ea918432323dfe52e7c4f2886ff06bca88039dcefa655b45; isolation
sha256:2a0d279e06ec1c900b2e62f1c6f64f453c9734d2fffa9df529fd8e0c31d65cd2; custody
sha256:3daa9482760295d3924b3e6c9818af931a0c566e023d98c2fe9cad560d0d19e4.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner

- no failed route retires an obligation or changes protected authority

122. Erasure and substitution distinction

Family: NOISE Obligation: SFT - INFO - OBL - NOISE - 008 Claim:

SFT - INFO - NOISE - ERASURE - SUBSTITUTION - 008

Question, law and forced result. Erasure closes the received label while retaining its source position; substitution retains a different received label. Their action types and ledgers may not be collapsed.

NOISE-008 uniquely retains typed-erasure-substitution-actions, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CHAN - COMPLETENESS - 018
- SFT - INFO - NOISE - ESTIMATION - 007

The current reconciliation binds this obligation to the current model-admitted receipt sha256:b7bd41ec0ec0c0bef55b91fb73331a1f0d3bcced6873707b656343ca52da8130. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-008 product before observation access.

Boundary: Every positive finite NOISE-008 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-008 uniquely retains typed-erasure-substitution-actions, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__typed-erasure-substitution-actions__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:a3f7979e3c261e4537edcc00af151a99ca09a23c532528e15b87be64eb0486ba; generality sha256:97a37a4daa94d19a9e7ee8acce7f087102989c7bdc48d2ed5912c1aa1e4a8a5d.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt sha256:c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: 4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256: dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate
 sha256: 7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal
 sha256: 1a724f79d626cdc0b1afc937b620b98335e6eb232e53c0eba5422ba582deb979. Source manifest
 sha256: a3b4051c0e35b7b63c8642308c163b0fdee95689dd48dfdd74778814c8cb408b. Engine receipt
 sha256: b7bd41ec0ec0c0bef55b91fb73331a1f0d3bcced6873707b656343ca52da8130 at receipts/engine/model_admitted/SFT-INFO-NOISE-ERASURE-SUBSTITUTION-008-b7bd41ec0ec0c0be.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-008: erasure_substitution
- exact observation: {"action_types":["erasure","substitution"],"erasure_received":"absent","substitution_received":"opposed label","types_distinct":true}
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: a2be1964d48b8dba1594927bd614fc0ddfd2837d1010dfa6dc98c1e7a76b04bd; empirical validation sha256: fb9201844158908a288a3d1dd9fed32ea728ef2cc64242a8a960f02528878cbe; external validation
 sha256: d2620ca259cd6b63a410cc0769e8907a6470eca44b87518b22854670a1e1bbe1; isolation
 sha256: a7a8e79614e618a3b4eb06e48283d4380f6a32871048f68ee6fd884cde62cc21; custody
 sha256: 4d856b47d695c182b629cdeefd4492c6f07bae0b75f61faaed1876f769519082.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

123. Burst and correlated error structure

Family: NOISE Obligation: SFT - INFO - OBL - NOISE - 009 Claim: SFT - INFO - NOISE - BURST - CORRELATION - 009

Question, law and forced result. A burst is a nonempty contiguous changed-position interval; correlated error is a declared relation among mask positions or actions, including noncontiguous patterns.

NOISE-009 uniquely retains contiguous-and-related-mask-structure, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CHAN - COMPLETENESS - 018
- SFT - INFO - NOISE - ERASURE - SUBSTITUTION - 008

The current reconciliation binds this obligation to the current model-admitted receipt
sha256: bcdb898d47724bcf7de94e045b234bf837df6bc470adc054329c5995b76d24b6. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-009 product before observation access.

Boundary: Every positive finite NOISE-009 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-009 uniquely retains contiguous-and-related-mask-structure, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__contiguous-and-related-mask-structure__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256: 532238c0579886cee23448f84c48e197ac77b43573f306ebcf754f5e4f1e20b6; generality
sha256: 50ff827e8380a51009f6e1bde2175a479b5c2452bbc267daa31b7b1dc996c380.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt
sha256: c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism

imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
 sha256: 4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256: dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the
 declared census and sole survivor; independent certificate

sha256: 7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal
 sha256: 6ff7d9243f8b88fce7ab87980cf6d6f9e0b8026025e6c47c903c3543770dcbe1. Source manifest
 sha256: 096848c5e073b76b90fae6ac8358398d3b8937ba2fec859a8e6a0d6cf9b5feb9. Engine receipt
 sha256: bcdb898d47724bcf7de94e045b234bf837df6bc470adc054329c5995b76d24b6 at receipts/e
 ngine/model_admitted/SFT-INFO-NOISE-BURST-CORRELATION-009-bcdb898d47724bcf.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows
 preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-009: burst_correlation
- exact observation: {"contiguous_bursts":2,"separated_correlated_masks":1,"width_two_masks":3}
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the
 observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact
 observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing,
 changed, duplicated or opened out of order.

Measurement receipt

sha256: de782e351cab329cffa4b931d12bfc477a2efd26f8750b2989d927b4abd3e27; empirical
 validation sha256: 72ae059eacff616f93fd18fcbcd5465d5275e6dc3ae74ad46d86381b9bce4efff;
 external validation
 sha256: 611e665eacd5362d19cefd6f33bef3609650b8ed1f3795b203e4c9e08f7c91ea; isolation
 sha256: 568ca9a6ece8093568522b062915ceb82f0064b83b4bf3813f9e87c62a9fdeeb; custody
 sha256: 9fde8be9ca1b16348a06f1337e71ec775274de3521acae234921783bc81a309d.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

124. Adversarial error support

Family: NOISE Obligation: SFT-INFO-OBL-NOISE-010 Claim:
 SFT-INFO-NOISE-ADVERSARIAL-SUPPORT-010

Question, law and forced result. Adversarial noise support contains every action mask allowed by an exact registered resource budget and no distribution or favorable sampling assumption.

NOISE-010 uniquely retains complete-resource-bounded-adversary-masks, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CHAN-COMPLETENESS-018
- SFT-INFO-NOISE-BURST-CORRELATION-009

The current reconciliation binds this obligation to the current model-admitted receipt sha256:5a50c16fcc7ecd835dcf922434f8e4ccdfb25c89cfc384af932d0eea79792ad. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-010 product before observation access.

Boundary: Every positive finite NOISE-010 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-010 uniquely retains complete-resource-bounded-adversary-masks, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__complete-resource-bounded-adversary-masks__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:e5533ae5e74409c19cd5d4ec4285924678250ca3d200c531af575198207dfffb5; generality sha256:44f837adec17effa8d4b07dfa2014d53e31ea5918fd08ed25d160d90b3ef5efc.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt sha256:c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256:dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate

sha256:7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal sha256:d93e03ce89d75ef021acf41dc4c03f133f5782c34cede4c91710f90651651736. Source manifest sha256:35d79fce7186f8c8c7a3eea2f73b6877021af0aeb1305a3c02fc931430c4ed. Engine receipt sha256:5a50c16fcc7ecdf835dcf922434f8e4ccdfb25c89cfc384af932d0eea79792ad at receipts/engine/model_admitted/SFT-INFO-NOISE-ADVERSARIAL-SUPPORT-010-5a50c16fcc7ecdf8.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2- INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-010: adversarial_support
- exact observation: {"budget":1,"complete_masks":4,"stochastic_prior_used":false,"width":4}
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:51f9a26477dbe65a36116242b7219492e8836c48f69ca97122fb30d11e6098b6; empirical validation sha256:1185ac3ebb63a887481fa7fc357e1e4cea7d2f0e1af99788945b8a7a9e759b47; external validation

sha256:0a0fb88e78de68b605740dd0f084379e36b9e7d3fe06e6a937a52461590dac73; isolation sha256:8ac9f549d39e0379735f871baef3dc70a751d85e3c3040f1412ecdfb6ab49290; custody sha256:e3e6c017fdae80f2fe277b0c5cb8f7b1a77dc7d19a32b654a201d597a8d32652.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

125. Noise budget and exact error ledger

Family: NOISE Obligation: SFT-INFO-OBL-NOISE-011 Claim: SFT-INFO-NOISE-BUDGET-LEDGER-011

Question, law and forced result. A noise budget is an exact positive limit on retained changed-position/action rows; every within-budget mask is admitted and every exceeding mask is rejected with its trace.

NOISE-011 uniquely retains changed-position-budget-custody, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CHAN-COMPLETENESS-018
- SFT-INFO-NOISE-ADVERSARIAL-SUPPORT-010

The current reconciliation binds this obligation to the current model-admitted receipt sha256:c291041e45974beb105ec02a65ae1f50562ef81e75edd429ac13cd02e70a5417. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-011 product before observation access.

Boundary: Every positive finite NOISE-011 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-011 uniquely retains changed-position-budget-custody, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__changed-position-budget-custody__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:f18064e622af31528d30be8f9a3369385ba2d8c6fc8d13a77cfb1a10779b282e; generality sha256:e79e3590c7028e07818849b80d2e702191b993aec7974a7ef0327c48f867268c.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt sha256:c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256:dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate

sha256:7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal sha256:2940735eaa965c580cf2d3ff207965c653353b7f2986c2bcd2784949f355c9a9. Source manifest

sha256:8803c586f748662ca1e1f0330344abcae417bcc56603096c9425e01475e847ce. Engine receipt sha256:c291041e45974beb105ec02a65ae1f50562ef81e75edd429ac13cd02e70a5417 at receipts/engine/model_admitted/SFT-INFO-NOISE-BUDGET-LEDGER-011-c291041e45974beb.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-011: budget_ledger
- exact observation: {"over_budget_control":[1,3],"over_budget_rejected":true,"within_budget_masks":4}
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:b663e18bfeac61d2c4be80a79de47b87f1c7de7b069bc2d0d93ff201b46ace54; empirical validation sha256:1f70ce5936f1bf9a33bf46b420a29b9090f8eabd4c854131628809498b6a9085; external validation

sha256:1550f0150ab1409076e9c06af45f8c333df2b6114464da9b33c175254bce174b; isolation sha256:350881078c0833a335d139eb2398285d634a0ac98c9c9ccc0a0f3847144bde9c; custody sha256:7b85ac51cd8d8dfbfda05581efbbd9d45ea205a0f2ff2f08ef25a3a0243f6588.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

126. Noise-family completeness certificate

Family: NOISE Obligation: SFT-INFO-OBL-NOISE-012 Claim: SFT-INFO-NOISE-COMPLETENESS-012

Question, law and forced result. Noise-family completeness is the one-to-one reconciliation of all twelve frozen obligations with exact patterns, predecessor classes, budgets, adverse controls and ownership boundaries.

NOISE-012 uniquely retains twelve-noise-obligation-ledger, complete error custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CHAN-COMPLETENESS-018
- SFT-INFO-NOISE-BUDGET-LEDGER-011

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:be51f31b7e7846f52494eb93132aa6ca588b62e40e282f9750304563011314ee. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis NOISE-012 product before observation access.

Boundary: Every positive finite NOISE-012 source, image, action mask, predecessor class, budget and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- NOISE-012 uniquely retains twelve-noise-obligation-ledger, complete error custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__twelve-noise-obligation-ledger__complete-position-action-trace__complete-predecessor-class__complete-declared-mask-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:3aa3ad6ce8e84796f499571dd831e8f3a344ae3b9834affb1518cc871d8de84f; generality
 sha256:e3f9275aa2b929dbcbdb8a0002b7d470588344654a1e758c6610db4a9a273d383.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted sources hide predecessors.; receipt
 sha256:c73032aa35f7b22888998127a6948f1acc6ddc4d4d43f1f1770d41d7cf8cfedd.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption; no physical noise mechanism imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
 sha256:4c8287af2dc9326c557e1b91b9379373bb1acdafdb340629f9c8830ed087cd4b.

Independent reconstruction and engine admission. Independent implementation

sha256:dc092d32706e86cbcecb985f5f6c2c37cd076690865837ff340144d40b0b1447 regenerated the declared census and sole survivor; independent certificate
 sha256:7135de13f6dde44ce56216e0802a8c62842755609174b7655d01a2abbe3f36ca. Derivation seal
 sha256:25604d1b91eac9a31b844df25c498af2d5e8f0725514b51348d1760dcf712114. Source manifest
 sha256:ce3b6f86b3f2169235be1e70e58c67f89cdfef7aefec1f54d057f8cd8845408c. Engine receipt
 sha256:be51f31b7e7846f52494eb93132aa6ca588b62e40e282f9750304563011314ee at receipts/engine/model_admitted/SFT-INFO-NOISE-COMPLETENESS-012-be51f31b7e7846f5.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-DETERMINISTIC-NOISE-OBSERVER
- SFT-V1-V2-INFORMATION-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- NOISE-012: noise_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":12,"omitted_owners":0,"registered_obligations":12}
- complete family observation custody: 12 records; all rows preserved
- complete deterministic mask, trace, predecessor, detection, localization, estimation and budget enumeration supplies the observation; no stochastic distribution, fitted error rate, likelihood prior or hidden threshold enters

Falsification condition: Reject if the value-free registry, complete deterministic error-mask and predecessor census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:ed89eddb10c02822e4cb1a5fe1aa2132be14f1b2c9780732e4681ed0015560f; empirical validation sha256:7e123f3e9a922dff3367bf286ab3e4d3570c87af837108c82e57d95f96b9f641; external validation

sha256:934bbbe5258b6b4a690a5eb483ee8ebb11740eb5dfe5855ab7f444cd7ca48780; isolation sha256:908354b4824de6af7605214dbbc46a24746bb931c5d89b9ddaa77f62d2ed1a46; custody sha256:44879dd77871ef0d0a6518705e9ebd1972f0c7e0fc33597f205d02b730289f31.

Prohibited inputs.

- no axiom, stochastic noise distribution, fitted error rate or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, sampled mask family, likelihood choice or nearest-neighbor assumption
- no physical noise mechanism imported into the information-law owner
- no failed route retires an obligation or changes protected authority

Complete-field family: CODE

127. Code as injective representation relation

Family: CODE Obligation: SFT-INFO-OBL-CODE-001 Claim:

SFT-INFO-CODE-INJECTIVE-REPRESENTATION-001

Question, law and forced result. A code is a complete source-to-codeword relation that preserves every source distinction by assigning distinct generated words and retaining the inverse decoding record.

CODE-001 uniquely retains complete-injective-code-relation, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-NOISE-COMPLETENESS-012

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:8ee1db307c63266882a05b5847914fffd99607833496076f4e64cffe8938c3291. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-001 product before observation access.

Boundary: Every positive finite CODE-001 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-001 uniquely retains complete-injective-code-relation, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__complete-injective-code-relation__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: e69a65d4e8370cf16808add26cd73d601395fc2a749107daf9869dfab437f94e; generality sha256: 868434ecea8f6952781cf918fad31440b6eb6ce20ea68d4de1e5b9c590bd6a2.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt sha256: c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256: 34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate sha256: 43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal sha256: cad47266c2a612e6fc7a67b191a0897c8cdde6d1fafc054bffa09f233e528c8d. Source manifest sha256: 477e1db4dc1873cc49e4104ce62e1c0b0715237941313f9201a764ae3319fde1. Engine receipt sha256: 8ee1db307c63266882a05b5847914ffdf99607833496076f4e64cffe8938c3291 at receipts/engine/model_admitted/SFT-INFO-CODE-INJECTIVE-REPRESENTATION-001-8ee1db307c632668.js on. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-001: injective_representation
- exact observation: {"codeword_width":2,"distinct_codewords":2,"source_forms":2}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:918d9ad01a7ba7faba9275097932089098c9436007384e9070162cc3c73f9d9c; empirical validation sha256:c10180798bc850846c9ec18e90a3412dacee65400ab204862a338338d27c8bc4; external validation

sha256:88795e3f51d42551082b807f498909a64bb356b2b3be0e2041d266128ec471c5; isolation sha256:9749656bd2997a345957498ae3b394990c90a69d8aab2e6f0027774300db42eb; custody sha256:fe4b76b8353f3e617a154e39ff4b5b44d3fc683e7cf8ee7756f58f35c8509862.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

128. Minimum separation and correctability

Family: CODE Obligation: SFT-INFO-OBL-CODE-002 Claim: SFT-INFO-CODE-SEPARATION-CORRECTION-002

Question, law and forced result. Minimum code separation is the least exact changed-position count among distinct valid words; radius-bounded correction is unique exactly when every received support cell has at most one valid predecessor.

CODE-002 uniquely retains minimum-separation-correction-ledger, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-NOISE-COMPLETENESS-012
- SFT-INFO-CODE-INJECTIVE-REPRESENTATION-001

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:bad76227c373f70cbf44a67af4d068dc22bfa62392550b69f95d2ac1025e8904. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-002 product before observation access.

Boundary: Every positive finite CODE-002 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-002 uniquely retains minimum-separation-correction-ledger, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__minimum-separation-correction-ledger__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:3bd617aad78d73c9d12558fb8f8554190ad844fad1dccc7bcecd7170319e1447; generality sha256:068781585f36359dea894e8364d059313ecf93b0524277897b0545f328078bf3.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal sha256:c6df8c12e5709521036ecbf9edb46428a3727cbe68de3571a0e008348876137d. Source manifest sha256:dc1de29effb4f9264fa4f36e4a6b79d48c86d12ec8c57e0dcc017bdae11c07ff. Engine receipt sha256:bad76227c373f70cbf44a67af4d068dc22bfa62392550b69f95d2ac1025e8904 at receipts/engine/model_admitted/SFT-INFO-CODE-SEPARATION-CORRECTION-002-bad76227c373f70c.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-002: separation_correction
- exact observation: {"minimum_separation":3,"radius":1,"single_change_images_checked":6,"unique_predecessors":6}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:73ad38d9733231fe5eee01fec5ace36d5ef11e69aa6e0dc80557266f290b3757; empirical validation sha256:c3aa1791b942f32506f0a4e5e096a31cb0258748d75dcbe7b57a138789f05775; external validation sha256:574d9f2c20233b671e1ad6675d7dab45af2cdd8f3c7682dab1d53dc85eb6230f; isolation sha256:80bb0c7966d0bd3435aaaaa9652e7608bc52c1f8d2ea1b3efbea310454c34441; custody sha256:23750efa69698eabbc0ef6978e844f987064963455a519bfb1a0504a7d64abb5.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

129. Block-code structure

Family: CODE Obligation: SFT - INFO - OBL - CODE - 003 Claim: SFT - INFO - CODE - BLOCK - STRUCTURE - 003

Question, law and forced result. A block code is a finite injective code relation whose complete valid-word support has one registered positive width and explicit word boundaries.

CODE-003 uniquely retains equal-width-block-code-support, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - SEPARATION - CORRECTION - 002

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:905e921ab972c4348f0aea9764c9945d1159b7b8c1d4942624809b15144dd2c7. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-003 product before observation access.

Boundary: Every positive finite CODE-003 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-003 uniquely retains equal-width-block-code-support, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__equal-width-block-code-support__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:0f6b1c3bceac1a791d669d98c95b5c8f648ef98933e696aebec25d2b3504719e; generality
sha256:fe9262718e573ab6286670af87a1a1ac611de03bb2c1b8ece1de01af6c6407c9.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate
sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal
sha256:dc616194cb3f67a0d10ecb36f293125a62ab428be2f623c6ef0617c9a62a4172. Source manifest
sha256:84efc284b3382f70314f1814a290c093cd41920ed5191d1159df00fa0a221391. Engine receipt
sha256:905e921ab972c4348f0aea9764c9945d1159b7b8c1d4942624809b15144dd2c7 at receipts/engine/model_admitted/SFT-INFO-CODE-BLOCK-STRUCTURE-003-905e921ab972c434.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-003: block_structure
- exact observation: {"block_count":2,"common_width":3,"word_boundaries_retained":true}
- complete family observation custody: 18 records; all rows preserved

- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:586caf0612794421ec1dc757a9641ee8d75640ef9803133cb3456329b9608b13; empirical validation sha256:8f2e711ce616a5f893dc9644463b30565edddd435eb0402d67d34f2ef27d673b; external validation

sha256:a063bb9c61592fa4ebc736f65be9f742548867df19c73de23f707c888216e3a0; isolation sha256:3b440dec39d9356b287bd93fac7bf2942ac3faabfc1b6f465b265e432f6b33f6; custody sha256:17f8ef60e2959f7bec29bbfa8b6796d8444089567762607d9fe136baecd2b6c2.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

130. Linear-code correspondence

Family: CODE Obligation: SFT - INFO - OBL - CODE - 004 Claim: SFT - INFO - CODE - LINEAR - CORRESPONDENCE - 004

Question, law and forced result. Linear-code correspondence holds when a codeword support contains the identity word and is closed under the exact coordinatewise period-two Fold-label action, without importing a field axiom.

CODE-004 uniquely retains fold-label-composition-closure, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - BLOCK - STRUCTURE - 003

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:b283fd299b33287ea5c2a445ae2f8f1c17966bc1a215641f97c9b2629fcb8e62. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-004 product before observation access.

Boundary: Every positive finite CODE-004 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-004 uniquely retains fold-label-composition-closure, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__fold-label-composition-closure__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: e4281a0e6f5fe64b8f375426fd1ef8aaf073abaaaf3e426276a1645f873188778; generality
 sha256: 43f78bc8f4ca340670b7f4d5a57f79bc98252f80b0290e2c7de931cdc24604fe.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
 sha256: c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
 sha256: d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256: 34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate

sha256: 43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal

sha256: d4324fbb6366933949363b5ba1ed4c11d02cb141a82c1927e0f95ec54ee03cbf. Source manifest

sha256: 76fea0d8a01dcd8aeb0476320ac765c939d407b0a9bda8fc13348a6318469f2a. Engine receipt

sha256: b283fd299b33287ea5c2a445ae2f8f1c17966bc1a215641f97c9b2629fcb8e62 at `receipts/engine/model_admitted/SFT-INFO-CODE-LINEAR-CORRESPONDENCE-004-b283fd299b33287e.json`.

Closure scope `depth_independent`. External status

`empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-004: linear_correspondence
- exact observation: {"closure_pairs_checked":16,"identity_word_retained":true,"support_words":4}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:465eb81bd63e1501dc4080516872b9dd6837c37053466b4d276d668c2288cd6f; empirical validation sha256:7fedee006d0845c6b21b0f3e727b40579870b0d6dafffa33ec42b054ac3882e63; external validation sha256:4caf3618fd06d74b64c35eaf7ca374dbdcccc7d0d153a88935f0cf9681fad5fe; isolation sha256:297cf4d2b0cb5f9fe9f3acf29f226a8607dfa25bfc90045ee4f389a28d373bd9; custody sha256:82383648866ee7af7065c338cc0fcfca892a2260ea6bcc7144c44908c59fc14f.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

131. Parity and syndrome records

Family: CODE Obligation: SFT - INFO - OBL - CODE - 005 Claim: SFT - INFO - CODE - PARITY - SYNDROME - 005

Question, law and forced result. A parity check is an exact relation among held coordinates; a syndrome is the complete ordered check-result word and localizes only distinctions separated by the complete check map.

CODE-005 uniquely retains complete-check-syndrome-record, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - LINEAR - CORRESPONDENCE - 004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:bb6fce2cb9f68313d7ace83002d0da5f299aac40b0e8b8f2b0a21d806375af59. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-005 product before observation access.

Boundary: Every positive finite CODE-005 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-005 uniquely retains complete-check-syndrome-record, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__complete-check-syndrome-record__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:d6eb46720afa1bc79dd8ec5cb69b70e3d60d70ae713767dee6bd51946c724bc4; generality sha256:643a0ab690c3b3e8e20ddc0a5de581d2aea454ca1a7b88462da89aeffe7ee267.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
sha256: c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256: d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256: 34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate

sha256: 43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal
sha256: ae57927bd22680e423314255af1d601a12d5885ba8e291bd509845ddd34cc494. Source manifest
sha256: 91407bf08ffe1572b065d4735f038dcf4bf3878371a237733b27c8b1eb124cb6. Engine receipt
sha256: bb6fce2cb9f68313d7ace83002d0da5f299aac40b0e8b8f2b0a21d806375af59 at receipts/engine/model_admitted/SFT-INFO-CODE-PARITY-SYNDROME-005-bb6fce2cb9f68313.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-005: parity_syndrome
- exact observation: {"check_count":2,"localized_position":1,"syndrome":["different","same"]}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: aa7568cc6e48cf45052d9aa0841539e24de93002cb0ba430b70f2171c89c69b1; empirical validation sha256: e61854292a6d42e46d7dd99fdc0e258e28802fd796299a12c083dbf5c1cc80ec;

external validation

sha256:65f327aa5871632b0459112b82d6c9ff24658f48702a4d81b417df245a3cba22; isolation
sha256:d7718ef5d36c7e01b46e86f65b97a2cfc317d9ef65285f77bc41a35734b9ae8a; custody
sha256:5163d91d92e13a8ac1184239386aa3bbf572fc31c5353c69c294c077f679aa50.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

132. Repetition and majority structure

Family: CODE Obligation: SFT - INFO - OBL - CODE - 006 Claim: SFT - INFO - CODE - REPETITION - MAJORITY - 006

Question, law and forced result. The width-three repetition code is forced as the least odd repeated-label support that retains one source label after every one-position substitution; exact majority is its unique complete decoder.

CODE-006 uniquely retains odd-repetition-majority-recovery, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - PARITY - SYNDROME - 005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:ac46498d78be4d3779b05e688659877eb01456d859eecd51858cf493a9d47678. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-006 product before observation access.

Boundary: Every positive finite CODE-006 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-006 uniquely retains odd-repetition-majority-recovery, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__odd-repetition-majority-recovery__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:3199dd0f505cc0407e4a284e437aaf5ba3afd9c2968e3942985fb72d750df235; generality
sha256:3c0786d60846f498d52736dfbc1c1631c4f8fcdcd576ce587a3522a7d670998e.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
sha256: c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256: d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256: 34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate

sha256: 43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal

sha256: e33b9a7235021d285fa19c0f292a749d4840a382098cde6a7aa6ff6d77dc5275. Source manifest

sha256: d31f8077b58de4371aa827aa201f3e86c73cd0f7496f29954c29bf3fe9865f81. Engine receipt

sha256: ac46498d78be4d3779b05e688659877eb01456d859eecd51858cf493a9d47678 at receipts/engine/model_admitted/SFT-INFO-CODE-REPETITION-MAJORITY-006-ac46498d78be4d37.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-006: repetition_majority
- exact observation: {"codewords":2,"single_substitution_cases":6,"successful_recoveries":6,"width":3}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 2c5b3240c7e11d59bdec59027dfef15ebfa02a07ba5e9e4ec182253586adbe5d; empirical validation sha256: 48edcae65b7dbb0bde6f719b89ae72ef7932455e371b9344f6ce72915cb32e59; external validation

sha256: a8a3b41e7500eedc44ecc9b37555e4d4cdb53f6b0102993ea577def1668eee60; isolation

sha256: bc79de4641c5d52cfe23b414db718f384137fd9940a28c446a71c0b4cc88d041; custody

sha256: f35e076c61061203590175467226e32cfe05b9cf71d556d2fd08907e960c938e.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

133. Erasure-correcting structure

Family: CODE Obligation: SFT - INFO - OBL - CODE - 007 Claim: SFT - INFO - CODE - ERASURE - CORRECTION - 007

Question, law and forced result. A typed erasure is corrected exactly when every admissible erased-position record leaves a unique valid codeword compatible with all retained labels.

CODE-007 uniquely retains typed-erasure-reconstruction, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - REPETITION - MAJORITY - 006

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:85e62758213d1c9a18f068bdfd412c13a237be46f00a426f65d0f0a75452d398. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-007 product before observation access.

Boundary: Every positive finite CODE-007 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-007 uniquely retains typed-erasure-reconstruction, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__typed-erasure-reconstruction__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:8ccd02a4fbee0d5e65fffa93aeb1a6f41084e49f8b032a60c6143de16cb556e9; generality
sha256:6d00af80c70ae0fb8af0f7af2f265cccf4d01fe592f7a4a687c59c6e6c860654.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate

sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal

sha256:ad97f3771646af7ac17f8e6900635d77311abd6ea5bf50c3129eb86bf443fe21. Source manifest

sha256:8260d1aeeda733e6ccceded57681cd5b8fd0ee3b0a8fbae4401bf68f7ad086e7e. Engine receipt

sha256:85e62758213d1c9a18f068bdfd412c13a237be46f00a426f65d0f0a75452d398 at receipts/engine/model_admitted/SFT-INFO-CODE-ERASURE-CORRECTION-007-85e62758213d1c9a.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-007: erasure_correction
- exact observation: {"erasure_cases":6,"retained_labels_per_case":2,"unique_reconstructions":6}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e6a6cf8757b1149938e0b8aae7b50f46c1e03b0fd97effbe297b078df157c634; empirical validation sha256:d5f819c2307d9404e3a6f3bfc0c392e1f8e8760a116db4fc5c701d5f4e9d0215; external validation

sha256:50695efcead7659f0351849b47723737eb7883141fbae802c4be1ff608c9c458; isolation

sha256:d6c72addf00355c9eb51d91167d8746a759dadfacfdd88d6692ab6ef3f1d33e6; custody

sha256:77ba7e022c9a8b8a04d4d1704e81ec3138d6ebd95c1b6fab34720256329f200f.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar

- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

134. Substitution-error correction

Family: CODE Obligation: SFT - INFO - OBL - CODE - 008 Claim:

SFT - INFO - CODE - SUBSTITUTION - CORRECTION - 008

Question, law and forced result. A substitution pattern is corrected exactly when complete predecessor enumeration within the registered change budget returns one valid codeword.

CODE-008 uniquely retains unique-substitution-predecessor, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - ERASURE - CORRECTION - 007

The current reconciliation binds this obligation to the current model-admitted receipt sha256:506db5408d690b248e30fa4d958785d0aa07699594aa62dfc0e815bd136858e6. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-008 product before observation access.

Boundary: Every positive finite CODE-008 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-008 uniquely retains unique-substitution-predecessor, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__unique-substitution-predecessor__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:2d7d67c196273b42134c95aab71d0893167540bc511a2534d7643b1407a0ae5f; generality sha256:0096b1ab3a31105b074167e297b84f73455c022b544866ddae530df767dedc55.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate
sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal
sha256:b7b7191f6b4f973466ac5fad8aeee19de284cc873bd3841875998f9f6edbf270. Source manifest
sha256:d5df1c9d0b9024feb528393e5c84362a9a1e6f9b07bba32f47e7512b6ca4a007. Engine receipt
sha256:506db5408d690b248e30fa4d958785d0aa07699594aa62dfc0e815bd136858e6 at receipts/engine/model_admitted/SFT-INFO-CODE-SUBSTITUTION-CORRECTION-008-506db5408d690b24.js
on. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-008: substitution_correction
- exact observation: {"radius":1,"substitution_cases":6,"unique_predecessors":6}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:bfdcf0c525cf0ecbf4c48ba686bf83a5ea8d061a0acedb64ddb47cbe088b1fca; empirical validation
sha256:77db0ab6e54b8ddb11e69a167c2196d93b90e4860d548f3d0c4942b07ddcde0c; external validation
sha256:680f64cf84a6f91f6c8067d5fba4d73566f8bc1bc1877cf8083cb132c3361438; isolation
sha256:6b445f5326cd21d7e02b65a9323e6f8a0f2d85b189e04eaac0471d3720f1f7e6; custody
sha256:ef4a55629af50e0a9d40b9055f7b6a82b0070f7fe2da418c8156dc18a6169744.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner

- no failed route retires an obligation or changes protected authority

135. Burst-error correction

Family: CODE Obligation: SFT - INFO - OBL - CODE - 009 Claim: SFT - INFO - CODE - BURST - CORRECTION - 009

Question, law and forced result. A burst code corrects the declared positive contiguous change interval exactly when every generated burst image has one valid source predecessor under the registered decoder.

CODE-009 uniquely retains contiguous-burst-recovery, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - SUBSTITUTION - CORRECTION - 008

The current reconciliation binds this obligation to the current model-admitted receipt sha256:101335c069192c7d670895d0862b72a8d4f2e77ed4f5158644fc57b51c56ffe2. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-009 product before observation access.

Boundary: Every positive finite CODE-009 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-009 uniquely retains contiguous-burst-recovery, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__contiguous-burst-recovery__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:4b75b6e9942017673afc52ee423f5c98efa4c73d871f8242177a8968cdf6f861; generality sha256:c77a6abcb8bd8a4a0c9d7b92cbe8df1e26c56a33ed33a4ef6b09141a5bcf8296.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence,

index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
 sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate
 sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal
 sha256:89e4fe003b47d23f97ec5d9084feb19b529dbf94bc7a7b4228a82ab178668e1f. Source manifest
 sha256:aedef83ca0c45ca2774f5c480582299059152d19b3b0f7076c9f24b834bca96f. Engine receipt
 sha256:101335c069192c7d670895d0862b72a8d4f2e77ed4f5158644fc57b51c56ffe2 at receipts/engine/model_admitted/SFT-INFO-CODE-BURST-CORRECTION-009-101335c069192c7d.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-009: burst_correction
- exact observation: {"burst_cases":8,"burst_width":2,"successful_recoveries":8,"word_width":5}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:b199636f18f01062a7b5fad4e3ad720787127e9fdb1a4095ff2f9199d0663aa1; empirical validation sha256:3252453c7b32e06f782a84775508e62190776dfd8c7cc7d0bceef37a295beeac; external validation
 sha256:77d495007f153069f3c70683801e6e81570ffd3e5113bea7f1fa7ef8adba801f; isolation
 sha256:00328bc8b5b281e51993787aab46be1af32d76ac9e1cef4e4b2ac5385b7e6380; custody
 sha256:ef69264dbea79969ca07b3dcf849146b4c3be7d474daf13224e402144278a324.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

136. Convolutional-code correspondence

Family: CODE Obligation: SFT-INFO-OBL-CODE-010 Claim:

SFT-INFO-CODE-CONVOLUTIONAL-CORRESPONDENCE-010

Question, law and forced result. Convolutional correspondence is a finite Fold process whose output row is forced by the current input and retained predecessor state, with every state transition and output coordinate preserved.

CODE-010 uniquely retains state-retaining-convolution-code, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-NOISE-COMPLETENESS-012
- SFT-INFO-CODE-BURST-CORRECTION-009

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:72487799dc7d67f12cdbc3f6789fed882aaa259d11071a12a656c74e7ec277c2. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-010 product before observation access.

Boundary: Every positive finite CODE-010 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-010 uniquely retains state-retaining-convolution-code, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__state-retaining-convolution-code__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:ab374a78a938ca7b3752555138806ee342a03f0de89c68180d4febac7979e215; generality
sha256:7b9f18a4e5a8531d7a53be2a005b6b1e4b86048862ec6db28c3dfaa8e4e3da97.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate

sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal sha256:08940e38880131abed96dce36e1a60afb6842810ce99067066ea1b9408b1bf0c. Source manifest sha256:9d5462f134fc90c30164f14a4875e4358e789d71bae42f11d0cd8f80e6d47de3. Engine receipt sha256:72487799dc7d67f12cdbe3f6789fed882aaa259d11071a12a656c74e7ec277c2 at receipts/engine/model_admitted/SFT-INFO-CODE-CONVOLUTIONAL-CORRESPONDENCE-010-72487799dc7d67f1.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-010: convolutional_correspondence
- exact observation: {"input_rows":3,"output_labels_per_row":2,"predecessor_states_retained":3}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e459c29695fc28cbbd4d2ea8d8f5255a27934472a6edaab577f3e4ea2a64573c; empirical validation sha256:43fbfffd34ed56ed3332a4c97a9c94683dcac7dfe4af614abf98e309f5e2101f0; external validation

sha256:82bd721e2e8c526ddaba82abeb7dc9d3730b8064b9fb178f7febc1e9577591cf; isolation sha256:4ce8aae151e942b194af97892cb7133e629028699d969fc6f6ede9c675725964; custody sha256:6e3bbf39bc017fb8748e26635c6f0fe2e4b204480d37866e98a240a91589d597.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

137. Tree and sequential decoding

Family: CODE Obligation: SFT-INFO-OBL-CODE-011 Claim:
SFT-INFO-CODE-TREE-SEQUENTIAL-DECODING-011

Question, law and forced result. Tree decoding enumerates every generated code prefix; sequential decoding may select a terminal word only when all competing prefix paths and their exact mismatch records have been retained.

CODE-011 uniquely retains complete-prefix-tree-decoding, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-NOISE-COMPLETENESS-012
- SFT-INFO-CODE-CONVOLUTIONAL-CORRESPONDENCE-010

The current reconciliation binds this obligation to the current model-admitted receipt sha256:d815342fed0191dc4195e8cd0134a851bb3acb86cc5820b5dee5b32c4af33551. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-011 product before observation access.

Boundary: Every positive finite CODE-011 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-011 uniquely retains complete-prefix-tree-decoding, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__complete-prefix-tree-decoding__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:0ff0264496a5f9315a8e774b06ad41a7378075de34ab705b60f98a3bafcd77bf; generality sha256:4d3f61faa192d83f6a7a9e0d762ba612fd212956d4e9a4dcdb8ebf18be392e38.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal

sha256:562d058609f86a522fd2dab9522e892aef04a424e521835a59e45300e192056c. Source manifest sha256:4fbdbf8e10fbbd68b88e72184df08a57de62c47647cd9caf1ea3e366186a4bec. Engine receipt sha256:d815342fed0191dc4195e8cd0134a851bb3acb86cc5820b5dee5b32c4af33551 at receipts/engine/model_admitted/SFT-INFO-CODE-TREE-SEQUENTIAL-DECODING-011-d815342fed0191dc.js on. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-011: tree_sequential_decoding
- exact observation: {"complete_terminal_paths":4,"matching_terminal_paths":1,"prefix_depth":2}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:645497347ee8f63cce962e192fba82b9e20acc2c5beb9c690c5d65901b3869b4; empirical validation sha256:aded59761b9e3ca2f8a87c29614c3bc8c9d949b5d3138b6ffaa9e0048ae698d0; external validation sha256:9ff6d03af31f3953bbfae1bbabdc2add7b82c8434be22e3cd77c1b623bfc00cb; isolation sha256:258f97859380c1a27530e7d735af5154fea56f40f65f57713d3b4f2f12429b76; custody sha256:4610571ad17cb221af200eb7b9e95903ca238a2f44936d2fe7ed2b050b13f94a.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

138. Product and concatenated codes

Family: CODE Obligation: SFT-INFO-OBL-CODE-012 Claim: SFT-INFO-CODE-PRODUCT-CONCATENATED-012

Question, law and forced result. Product coding generates the complete ordered row-column support and concatenation composes injective encoders while retaining each component boundary, width and decoder trace.

CODE-012 uniquely retains product-concatenated-code-composition, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-NOISE-COMPLETENESS-012
- SFT-INFO-CODE-TREE-SEQUENTIAL-DECODING-011

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:290655a4d8a33e94bb941d8829a425a03ebbd12b83673edce73edd086b9b6edf. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-012 product before observation access.

Boundary: Every positive finite CODE-012 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-012 uniquely retains product-concatenated-code-composition, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__product-concatenated-code-composition__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:a4cf28cca13ac6a9671cb3f14aceec2ab6f30cf2df69bac7085c3622efde4af9; generality
 sha256:361ac7f9ceae06ca9bc1803a1e016d1cfdc5c3bce7b739d775bf7feb1060a97b.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
 sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
 sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate
 sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal
 sha256:d3441dbc974f85b2b8832e5c8b27102010174e2de95c60d09910b813647a5c2c. Source manifest
 sha256:c7368170ccbf08135f6cc151e91a8f960fc342a5c4d40f8a575945f9f5aa5b52. Engine receipt
 sha256:290655a4d8a33e94bb941d8829a425a03ebbd12b83673edce73edd086b9b6edf at receipts/engine/model_admitted/SFT-INFO-CODE-PRODUCT-CONCATENATED-012-290655a4d8a33e94.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-012: product_concatenated
- exact observation: {"concatenated_width":6,"product_columns":3,"product_rows":2,"source_forms":2}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:b4b29206798862f737e6c3eaf51b3ec1c83b8ee5719e2c87fea45d2da507a87a; empirical validation sha256:077c50c3fc6557d7e9e8926b00ceafa07c7022b3444ef1c093a60f164b4370af; external validation sha256:725e278806620339e9564d23d4a9191a34ddaa72dcc0d854ab994f0abf02dc4; isolation sha256:a8e8152a46c3cbb67f80f5ff606e4083e9e2daedf1c3adac5fe1b4e750bf3a2c; custody sha256:e4107307526faccab41e8ee481b2b783d77925107fcee5a13072a0e485e60594.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

139. Sparse-check code correspondence

Family: CODE Obligation: SFT-INFO-OBL-CODE-013 Claim: SFT-INFO-CODE-SPARSE-CHECK-013

Question, law and forced result. Sparse-check correspondence is a bipartite held-coordinate/check relation with explicit finite incidence support; accepted words satisfy every retained local check.

CODE-013 uniquely retains sparse-held-coordinate-check-graph, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-NOISE-COMPLETENESS-012
- SFT-INFO-CODE-PRODUCT-CONCATENATED-012

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:486e1fce3d13ba85cdfb02a817d17ff810de95ed0ec2bf7652109105dba55a3. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-013 product before observation access.

Boundary: Every positive finite CODE-013 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-013 uniquely retains sparse-held-coordinate-check-graph, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__sparse-held-coordinate-check-graph__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:b34d51f65387d5b7e392e15f870aa70b653e51c9324d2d652492c9c54b58f85b; generality
sha256:6878dc0f0dd9cced4e4f1b7776a5289e1cce5f4820abf5abeba8d333044ac2a3.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate

sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal

sha256:8c3456e51e68ef37eb1b8e5422d70771427be5505eddac55831e32371e54a8f1. Source manifest

sha256:87970cfbe1f127c132abb117368671548ce923235b1c8acfb3c2b8ad32a80a3b. Engine receipt

sha256:486e1fce3d13ba85cdfb02a817d17ff810de95ed0ec2bf7652109105dba55a3 at receipts/engine/model_admitted/SFT-INFO-CODE-SPARSE-CHECK-013-486e1fce3d13ba85.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER

- SFT - V1 - V2 - CODING - OBSERVATION - CORPUS

Observed, adverse and boundary records:

- CODE-013: sparse_check
- exact observation: {"accepted_words":2,"check_degree":2,"local_checks":3,"words_enumerated":8}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:4ec0c3e25f5e2f408773f8cc355b4dbcaea36b1172857b0e4e5655a50a2a8193; empirical validation sha256:f40787cbe3b029af6c5e874c105c87e5b77e6be69e56bd6c56e03f4339873c57; external validation

sha256:a3e1aa97351d13a829c5343906a0eba2b2fdeefb8314d28bc88a728686297d29; isolation sha256:45adb6cdfa8f6d6f2bca5c6829b3094f93aafb9cdc1e391f02d27ca3264f62b; custody sha256:5b6d4a346d08186396ee37d7416bc3895f64410e038bd50278494709bc26f00b.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

140. Network coding structure

Family: CODE Obligation: SFT - INFO - OBL - CODE - 014 Claim: SFT - INFO - CODE - NETWORK - 014

Question, law and forced result. Network coding assigns exact source-dependent codewords to edges and is valid only when every receiver reconstructs its demanded source distinctions from its complete incoming and side-information records.

CODE-014 uniquely retains network-edge-code-custody, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - SPARSE - CHECK - 013

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:2e044cbe04a0e7547d1ca01b3ac08ba58a9fe6b7aa39aafa94cb578b9fb7e646. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-014 product before observation access.

Boundary: Every positive finite CODE-014 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-014 uniquely retains network-edge-code-custody, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__network-edge-code-custody__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: f371fd7a19ea0e0babf0064e5918b0395a04a2daa6f8ef48cefee54b14d95f89; generality sha256: 11e1ebbf7fd759c2d60a6eb40b5410619ae91c336aa3fe1b76fc5756fdd83134.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt sha256: c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256: 34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate

sha256: 43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal

sha256: a2fcbab89ab196332fdbeba8f2a7ee6f550c411f651031f5785ca72db904a152f. Source manifest

sha256: d133a0d51b0be4422d2538955b2cbf1ab4a85bac5fdd514c22001a30a55796f0. Engine receipt

sha256: 2e044cbe04a0e7547d1ca01b3ac08ba58a9fe6b7aa39aafa94cb578b9fb7e646 at

receipts/engine/model_admitted/SFT-INFO-CODE-NETWORK-014-2e044cbe04a0e754.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-014: network_code

- exact observation: {"joint_inputs":4,"network_labels":2,"receiver_reconstructions":4}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:04e71bbcc546aea1f5c78b3c15cbd8cab30efa271db9dde6d7f6b3d253ce631; empirical validation sha256:a34ab156e68a7d8225e763760ca6e3cee4c04a45f73eb93be0023e29bf4cf958;

external validation

sha256:1218f41f8877e34aa86055689559548f37e46a3e7ab43091349d65aa5b8b029b; isolation

sha256:fea7363453d4e336f6e33dd1462c8ffd6c048a911ebc638964e49128ad37ad34; custody

sha256:2d82be8a5f6b220f9060fddd36208fd472d7d5100977abbf4b69ce2c4a3d2d40.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

141. List-decoding boundary

Family: CODE Obligation: SFT - INFO - OBL - CODE - 015 Claim: SFT - INFO - CODE - LIST - DECODING - 015

Question, law and forced result. List decoding returns every valid codeword within an exact registered changed-position radius; uniqueness is not asserted when the complete predecessor list has multiple members.

CODE-015 uniquely retains complete-radius-list-boundary, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-NOISE-COMPLETENESS-012
- SFT-INFO-CODE-NETWORK-014

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:1ab737f33f40f9d470ce5f700f24a0c6a39fcfd531a153b42c6cfe6aa569323d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-015 product before observation access.

Boundary: Every positive finite CODE-015 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-015 uniquely retains complete-radius-list-boundary, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__complete-radius-list-boundary__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-

explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:4eff8025be391741a6fa1261006c4efdf18480c918ac6ed4e21cc5a2b9d0772f; generality sha256:5560b4618fd0c4d23958b2fef7e19fba79fdd19f8a693faa2219f95a9151dd4e.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal sha256:3fd96af853526a7ab9b36091dd1083f60e8862367b0cc25e3aa9cd1c92e0195f. Source manifest sha256:d0fe0c163e5b5a75b0dd2cbe1612ecef0fc86fe9ecceba6d408d006ba63a2a57. Engine receipt sha256:1ab737f33f40f9d470ce5f700f24a0c6a39fcfd531a153b42c6cfe6aa569323d at receipts/engine/model_admitted/SFT-INFO-CODE-LIST-DECODING-015-1ab737f33f40f9d4.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-015: list_decoding_boundary
- exact observation: {"radius_one_list":1,"radius_two_list":2,"received_width":3}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:8ec71096f98e5facd07733334948b4469d1b257ddb30ff6293cac21a85447a90; empirical validation sha256:8ba0acb49d51a680ce2ed268d54b71d4770651391cbf215f9c9513025cc78c08; external validation sha256:50eae06e72caabe5af3c369897e4a92fad93239f2aff50cff960dc0f15d73e29; isolation sha256:b808d29876724ce163c88a421c5c2df225c9224b565b4f228bab22cc74a80ef9; custody sha256:a49f4cfd9f2ee122e2494472f42ab1698caa23ab3080bdf02018ff6758e6442d.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

142. Adversarial coding boundary

Family: CODE Obligation: SFT - INFO - OBL - CODE - 016 Claim: SFT - INFO - CODE - ADVERSARIAL - BOUNDARY - 016

Question, law and forced result. An adversarial correction claim must exhaust every mask within its positive resource budget and preserve the first explicit exceeding-budget control; no stochastic or favorable-mask premise is admissible.

CODE-016 uniquely retains complete-budget-adversary-boundary, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - LIST - DECODING - 015

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:48ca65c2688eaf13fc9142b6a5a0b8539006ef92a83e6f9a3154168dfa09f1fb. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-016 product before observation access.

Boundary: Every positive finite CODE-016 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-016 uniquely retains complete-budget-adversary-boundary, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__complete-budget-adversary-boundary__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:391833b3c380800e173ff5815be6f7776cc1906f1e5a9fca5a8c4ae2972d7528; generality sha256:0f8f8d0dcf00ba806c6138c496492c1b5ef1cc84ca4fe2fb8db64124d4a82535.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
sha256: c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256: d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256: 34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate

sha256: 43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal

sha256: 29fa87d883f04aa519c8d1325fd9709e34580ff86f14dc8c9c01568cd2fd806e. Source manifest

sha256: 64dd0455cbc30dbd4a4f03a93a866afa203fff00083e1606afdbbc9181960c13f. Engine receipt

sha256: 48ca65c2688eaf13fc9142b6a5a0b8539006ef92a83e6f9a3154168dfa09f1fb at receipts/engine/model_admitted/SFT-INFO-CODE-ADVERSARIAL-BOUNDARY-016-48ca65c2688eaf13.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-016: adversarial_boundary
- exact observation: {"corrected_one_position_masks":3,"first_exceeding_mask_width":2,"word_width":3}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 6d742a5f69f3dccdebb18951bb7413011f14476950f121e8e4ad150dbc958a2d; empirical

validation sha256:5e620e6be2530d0e974f6ed06044bf5936966b52c816c6526701a9e9424d01d2;
 external validation
 sha256:6c0b2fc92be1f16b1c5d5c2b098f9e5b4fa707fc65ed9ccc1f10252e7d425b9b; isolation
 sha256:42d4cd5c687e31dda19a88e836f8415d41a0c3338b0df8ace55de0c71dab1b23; custody
 sha256:589d431b91c5baee1fc026dcf56acb13024a1bf4e1e12af17fc056bd64a64d91.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

143. Code composition and rate custody

Family: CODE Obligation: SFT - INFO - OBL - CODE - 017 Claim: SFT - INFO - CODE - COMPOSITION - RATE - 017

Question, law and forced result. Code composition retains every encoder width and source-form count; rate is an exact ordered part of source distinctions per transmitted positions, never a fitted floating scalar.

CODE-017 uniquely retains exact-source-width-rate-parts, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - ADVERSARIAL - BOUNDARY - 016

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:de4ee63380b53b014cb1a58b74555f20c6ff56fd3fd12400dae8c6a569acbbe1. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-017 product before observation access.

Boundary: Every positive finite CODE-017 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-017 uniquely retains exact-source-width-rate-parts, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__exact-source-width-rate-parts__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:eedc9ecf0fb13042277a6a96d7b98ecdf790a4567d2a10c9bbc706cc87daf371; generality
 sha256:3e254392550f9ae18164fafb78eab86000cbd756bfff331801ce5d105044d1b2.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
sha256: c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256: d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256: 34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate

sha256: 43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal
sha256: 2aa81e6a368cc7bb616819eb776eb8030e95f1cd33b5ee23feb4ed7c214bfa4f. Source manifest
sha256: f1acc0fbd6674d034500fb2db639ee9c38624dcb83ecef786581f20b0578f18. Engine receipt
sha256: de4ee63380b53b014cb1a58b74555f20c6ff56fd3fd12400dae8c6a569acbce1 at receipts/engine/model_admitted/SFT-INFO-CODE-COMPOSITION-RATE-017-de4ee63380b53b01.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INDEPENDENT- EXACT- CODING- OBSERVER
- SFT-V1-V2- CODING- OBSERVATION- CORPUS

Observed, adverse and boundary records:

- CODE-017: composition_rate_custody
- exact observation: {"concatenated_rate_part": [1,6], "inner_rate_part": [1,3], "source_forms": 2}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: cd668dc35bd88d7aa5f64f94637b8392e497d020959cfb388fd99b346cf80413; empirical validation sha256: 1026543b6ecef9a941f43bfc86220f6fb728ba62a823d04c8ea4aa5b4503c0e7; external validation

sha256: 1be43186d2e9fda6f68db3aabfdd1047f4a4bbb494cc114802a3e59030844b139; isolation sha256: 16ce05b36c21b367a3579826d91ebe05e7f88ef4900e32e5c0069eaa0adcc300; custody sha256: def77e92beb702098174f7c6d89943478079c14bee3536d9a9de164cfa07ffbe.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner
- no failed route retires an obligation or changes protected authority

144. Coding-family completeness certificate

Family: CODE Obligation: SFT - INFO - OBL - CODE - 018 Claim: SFT - INFO - CODE - COMPLETENESS - 018

Question, law and forced result. Coding-family completeness is the one-to-one reconciliation of all eighteen frozen obligations with exact code supports, error masks, decoding records, controls and ownership boundaries.

CODE-018 uniquely retains eighteen-coding-obligation-ledger, complete coding custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - NOISE - COMPLETENESS - 012
- SFT - INFO - CODE - COMPOSITION - RATE - 017

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:3229c52eba142879eb841c1149af7581c4ffa8b1bed3702a9f4bfe4eef53cd5c. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CODE-018 product before observation access.

Boundary: Every positive finite CODE-018 source, codeword, error mask, decoder record, resource part and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CODE-018 uniquely retains eighteen-coding-obligation-ledger, complete coding custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-code-support__eighteen-coding-obligation-ledger__complete-typed-error-support__complete-predecessor-decoder__complete-declared-code-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:2423d41214327f81eef81a515309cbe308e1682d974355ab597728727a62a232; generality
sha256:c973c44d765edc208a7d4d9cd249479b56a71afefac1f65d7647c169657586e0.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: A partial codebook cannot establish separation or decoding.; receipt
sha256:c164a751a3062176f5e01b6e75545df019df62ac4bca06bd4ee2486bdda09f0c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt

sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported coding formula, stochastic error law or target outcome selects the result; host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold; no physical error mechanism or quantum gate law is imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:d98a24a09f0c0cca4ee752081712eeb0a8a3f2f56e78b51dd07e6140ac5eb13c.

Independent reconstruction and engine admission. Independent implementation

sha256:34fcd74e8d70c68c560146c8ff8912358127df5556c4aff6829a0292b8ebbbd1 regenerated the declared census and sole survivor; independent certificate

sha256:43e710eab36dd6e513e12d1b025e1926be4454ad19dedae67482c4bea9f51381. Derivation seal
sha256:5f141521a8879295eaa17e28414d76b7daeccffce87fce9afe0c2f8aa3ffd44. Source manifest
sha256:ecf630ddc99143c3387eadfbf6410b8e3d8433d4f5a5406a5d64e8e6d0ed7428. Engine receipt
sha256:3229c52eba142879eb841c1149af7581c4ffa8b1bed3702a9f4bfe4eef53cd5c at receipts/engine/model_admitted/SFT-INFO-CODE-COMPLETENESS-018-3229c52eba142879.json. Closure scope
depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CODING-OBSERVER
- SFT-V1-V2-CODING-OBSERVATION-CORPUS

Observed, adverse and boundary records:

- CODE-018: coding_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":18,"omitted_owners":0,"registered_obligations":18}
- complete family observation custody: 18 records; all rows preserved
- complete finite source, codeword, error-mask, check, predecessor, decoder, path and rate-part enumeration supplies the observation; no imported stochastic error law, likelihood decoder, fitted threshold or hidden codeword enters

Falsification condition: Reject if the value-free registry, complete code census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:18f64c5410db110e624287ea2eaa44b5aa4e083969170bc9fb7ed120053735bf; empirical validation
sha256:06d2d46884d14c90214be5d27cc94e321b01572658a9261eed2e1667526c62ac; external validation

sha256:30b781cc0fe4fd1a51bdd26699dce6c62beeddc148f9a2b0c16a3d5931719057; isolation
sha256:a4ad1bea22f4989783c8c0830a63c8b36aaa8820e8ba6975d6f05cdf2052f08d; custody
sha256:0d6debc8d46258cb6e8141ed8ceaaf48ea50271d640ea98755a5daabe8df2a0.

Prohibited inputs.

- no axiom, imported coding formula, stochastic error law or target outcome selects the result
- host 0 denotes structural absence, index origin or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no sampled mask family, hidden codeword, likelihood decoder or fitted correction threshold
- no physical error mechanism or quantum gate law is imported into the information-law owner

- no failed route retires an obligation or changes protected authority

Complete-field family: REL

145. Joint information support

Family: REL Obligation: SFT - INFO - OBL - REL - 001 Claim: SFT - INFO - REL - JOINT - SUPPORT - 001

Question, law and forced result. Joint information support is the complete canonical set of co-occurring Fold records with every coordinate projection and joint provenance retained.

REL-001 uniquely retains complete-ordered-joint-support, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CODE - COMPLETENESS - 018

The current reconciliation binds this obligation to the current model-admitted receipt sha256:00ba20826fa7886f7011c7e6e0984c59f5813a4a84e2ebfe55ad990bcc6c4519. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-001 product before observation access.

Boundary: Every positive finite REL-001 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-001 uniquely retains complete-ordered-joint-support, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__complete-ordered-joint-support__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:943e4595ae88affde06633ab2e6c24eabc620507c47648f6355d5b8c96647bf; generality sha256:deef3748cf24d96c284667f7f06e9107d962ec3c464e0b8eeae326afef60af53.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt sha256:3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256:1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256:84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate

sha256:b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal

sha256:b37c907f84503344defd7ee7af64111fdded05ef443d879c28b9c75627051ce1. Source manifest

sha256:dd991d87e7bbd1516827d962beac0686d6b19e1c98288e73b18d0679b793e16d. Engine receipt

sha256:00ba20826fa7886f7011c7e6e0984c59f5813a4a84e2ebfe55ad990bcc6c4519 at receipts/engine/model_admitted/SFT-INFO-REL-JOINT-SUPPORT-001-00ba20826fa7886f.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-001: joint_support
- exact observation: {"joint_rows":4,"projection_forms_each":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e6248a4ccd39c1bb0e4746b5cb80d43bca30d051f8464dda99432ed4fe569698; empirical validation sha256:bcbf8fbb336daea4b17c24124c8b71dd76db895b8fb65d515794fd769ac50c493; external validation

sha256:6f46c9a5f0372befe04d29d2561abcfeb2dba10d181d907ad27ffb381b51c1d1; isolation

sha256:7502136ed85aee56757447dc4bdd3907c89ce839e9fb42a9f3853c626def048c; custody

sha256:34bea0c0af9feb6e4c0409326adc70c42c5ae954b51e4493e5930999ae11b1ae.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

146. Conditional information by exact restriction

Family: REL Obligation: SFT - INFO - OBL - REL - 002 Claim: SFT - INFO - REL - CONDITIONAL - RESTRICTION - 002

Question, law and forced result. Conditional information is the exact support remaining after restricting a complete joint record to one explicitly held condition; no conditional probability is presumed.

REL-002 uniquely retains held-condition-support-restriction, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CODE - COMPLETENESS - 018
- SFT - INFO - REL - JOINT - SUPPORT - 001

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:9823edaf1f8c1c55b687027fdfa5fcca80b3feeb6d156dea59641b5d2015518. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-002 product before observation access.

Boundary: Every positive finite REL-002 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-002 uniquely retains held-condition-support-restriction, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__held-condition-support-restriction__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:76ea22570e2c9c7ace34ffbd3454ad721fc7ffde3c654189125badfd6036ff52; generality

sha256:26771e3f9f13949e4e0dd9fea80f0cafee37b1f866072bb083c81eae46f24439.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt
sha256:3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes

protected authority; receipt

sha256:1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256:84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate

sha256:b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal

sha256:38c3294e5a0937f7af355e712334053e14b67e863a1c9d682682e64d7cd86825. Source manifest

sha256:de0091d7649efb1f2c519bc6768b4b57fb650a851fded4220b3cf0a682b057d1. Engine receipt

sha256:9823edaf1f8c1c55b687027fdfa5fccaa80b3feeb6d156dea59641b5d2015518 at receipts/engine/model_admitted/SFT-INFO-REL-CONDITIONAL-RESTRICTION-002-9823edaf1f8c1c55.json

. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-002: conditional_restriction
- exact observation: {"compatible_rows":2,"held_conditions":1,"joint_rows":4}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:7019cc863a4f042fa7b6576ed53b3a1b072cfb0e07428621aa472aeb38dc3821; empirical validation sha256:a449c2105957f6acead6b87ff2c4a3aa38072fecc1c7625bb2744883c1f0cc16; external validation

sha256:dd7c7562f51c95e58f70d74e1402c9e257a514fe5890363ec24e7c3e27dd98ad; isolation

sha256:c00ac0e60ed25c8f412db6cca9721515f4c7d28b6af0eb68b80f922d2bb31a9d; custody

sha256:9b8d5c9353812eaac0852ae324bae8979d8169d6a058fdee406838e644680d85.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

147. Mutual information by shared distinctions

Family: REL Obligation: SFT-INFO-OBL-REL-003 Claim: SFT-INFO-REL-MUTUAL-SHARED-003

Question, law and forced result. Mutual information is the exact family of distinctions in either coordinate that determine retained classes in the other at the declared observation.

REL-003 uniquely retains shared-determination-class-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CODE-COMPLETENESS-018
- SFT-INFO-REL-CONDITIONAL-RESTRICTION-002

The current reconciliation binds this obligation to the current model-admitted receipt sha256:f88c636c56eb29db885aef3de87e6e335ddca74a4ce8f5b4ef597c3137ce10ee. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-003 product before observation access.

Boundary: Every positive finite REL-003 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-003 uniquely retains shared-determination-class-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__shared-determination-class-ledger__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:e7070a77b13d89011b07af88d7ada164bf016695fb7ce3ccd5b81a402e89d8a6; generality sha256:2c826516dc964303a575dcaee309277990f8e8a0c2f80bfe62ae78a3cba95ceb.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt sha256:3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt sha256:1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256:84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the

declared census and sole survivor; independent certificate

sha256:b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal sha256:cdfb2c3640ff1ace9d6eec548a9b7fbe35e34543927c6470b5460e04f57bbce8. Source manifest sha256:13f6098f6ec0ec49be101364901ce56c067519aeacb19fc7795904db8cab0a48. Engine receipt sha256:f88c636c56eb29db885aef3de87e6e335ddca74a4ce8f5b4ef597c3137ce10ee at receipts/engine/model_admitted/SFT-INFO-REL-MUTUAL-SHARED-003-f88c636c56eb29db.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-003: mutual_shared
- exact observation: {"independent_fibre_width":2,"perfect_fibre_width":1,"perfect_shared_classes":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:d323b80b4db6ce1e457895e12909fb1571485d7b2137b1a3a0a5f31387cf020b; empirical validation sha256:cc2eea3a5457379bef3da0a8509b1f808c7bca4019afdb236f4a32e8b4598f9a; external validation

sha256:aa3a8c3906acfffb47d5c47e241f8a4a362f1914775486ca6c106716057d52fa9; isolation sha256:059c96995581d4df586ebe3b1241aa554166283177c6b2b0236a2d1942f52c48; custody sha256:97cfd58c3d02007d7bfcd512c5fd0e0386f4dea2197f930f557f468dc801a7a.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

148. Chain rule for retained information

Family: REL Obligation: SFT-INFO-OBL-REL-004 Claim: SFT-INFO-REL-CHAIN-RULE-004

Question, law and forced result. The information chain rule is the one-to-one decomposition of complete joint support into first-coordinate forms and each form's conditional continuation support.

REL-004 uniquely retains joint-as-marginal-conditional-composition, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CODE-COMPLETENESS-018
- SFT-INFO-REL-MUTUAL-SHARED-003

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:e81a57da54a45508607392bd274da76eb730b4f2d0dc4e9ae4df33ada9aec2c4. A missing
dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-004 product before observation access.

Boundary: Every positive finite REL-004 joint row, restriction, class, path, representation record and registered successor
boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly
one survives and 255 are eliminated. The forced coordinates are:

- REL-004 uniquely retains joint-as-marginal-conditional-composition, complete
relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__joint-as-marginal-conditional-composi
tion__explicit-held-condition-record__complete-typed-distinction-ledger__complete-d
eclared-relational-product__root-bound-forward-forcing__post-registry-exact-observa
tion__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:3e9b4efa7fbf822f20d7a13525f23161979f5a067d4b130c7b7be83350c84f76; generality
sha256:24a9a2e6a4a9b32d1d678e9568bb983f2e46a3391ce9d8ec22c53bbd3b1b4bc0.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained
distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit
downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted
magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT
numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed:
Missing rows change every relational conclusion.; receipt
sha256:3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs
from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated
product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no
axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural
absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar;
no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning,
cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes
protected authority; receipt
sha256:1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256:84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the
declared census and sole survivor; independent certificate
sha256:b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal
sha256:c04e6c3fa8eea2edf9be9eea9cdf630dd1efa3c790bb0e02c049f4807f65b6279. Source manifest
sha256:d0cf6d4e67a66828714d7588cda82bf741d82329655a65b16069d8201aaa79b. Engine receipt
sha256:e81a57da54a45508607392bd274da76eb730b4f2d0dc4e9ae4df33ada9aec2c4 at

receipts/engine/model_admitted/SFT-INFO-REL-CHAIN-RULE-004-e81a57da54a45508.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-004: chain_rule
- exact observation: {"conditional_continuations_each":2,"first_forms":2,"joint_forms":4}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:821db4a7b97fd1da7f5d76634b3eb44575a4cfe61dcc0e510bc6ecb5244435eb; empirical validation sha256:742b14fb03cf234df5e3342ab9b1527731a87a6c190a443e07381fc1355ae127; external validation sha256:8365abde9ecd0cf43f7ae75e5ee4fc7faa6fae84e9db585af20eaf9062e61276; isolation sha256:503be8573a93f634d590602c7a01cb32490cc3110be33d836509da5d7d444c84; custody sha256:564a219101d1c22c0ee52c4e1c64f061b8f4638e4bfcaf19fff23e535a916304.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

149. Data-processing monotonicity

Family: REL Obligation: SFT-INFO-OBL-REL-005 Claim: SFT-INFO-REL-DATA-PROCESSING-005

Question, law and forced result. A deterministic information transform cannot create source distinctions: each terminal observation class is a union of intermediate classes and complete image support cannot increase.

REL-005 uniquely retains image-distinction-monotonicity, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CODE-COMPLETENESS-018
- SFT-INFO-REL-CHAIN-RULE-004

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:9307af947c9efb9363a3b5c28c8d40f4e699ccd949f4c08f5b3e883e80418e75. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-005 product before observation access.

Boundary: Every positive finite REL-005 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-005 uniquely retains image-distinction-monotonicity, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__image-distinction-monotonicity__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:ed582aa4077eda7bdd6bb2361d7aa501ba703295f3416bbad41de2af417423e2; generality
 sha256:10a285494f00042bd56466520dac9f18af0b716428bfb5ede2a06212e3ebae1.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt
 sha256:3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
 sha256:1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256:84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate
 sha256:b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal
 sha256:722ba0390330c881e18ebe5ae4241bfebf31a22fc1a49886ba2bbab35f55635d4. Source manifest
 sha256:b2f2439b6f8cb24b5934cc1e0f9a35ecf4c270550e42001019af5d72c98ac761. Engine receipt
 sha256:9307af947c9efb9363a3b5c28c8d40f4e699ccd949f4c08f5b3e883e80418e75 at receipts/engine/model_admitted/SFT-INFO-REL-DATA-PROCESSING-005-9307af947c9efb93.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-005: data_processing
- exact observation: {"intermediate_images":3,"source_forms":3,"terminal_images":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:31ed72775b276514b66653ad183dae5d1ccbfec220351c87569fd74b67a5296a; empirical validation sha256:d547f5c58cfa3d037537d4c3c7605fe350fb5cfe1f3e577a86f0ff9751063fd2; external validation

sha256:342e505d6aabfa2cd9ebd90d30c4e6d9f3a7e900548cd75c127606bf9615caba; isolation sha256:ece1a77c935f486000a6f4b39bc855cb6cf52a14a70ad9c765df59e0550c759c; custody sha256:43c6b6b9c3af8ab03df36cc02d166143d0292c84e0dd786180343e3f36375aa6.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

150. Conditional independence support

Family: REL Obligation: SFT-INFO-OBL-REL-006 Claim: SFT-INFO-REL-CONDITIONAL-INDEPENDENCE-006

Question, law and forced result. Two coordinates are conditionally independent exactly when, inside every held condition class, their complete joint support equals the ordered product of their restricted projections.

REL-006 uniquely retains held-condition-product-factorization, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CODE-COMPLETENESS-018
- SFT-INFO-REL-DATA-PROCESSING-005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:dda9fdc3e893dd8e503cc7630fde3eadb73fda5d7e125d56ff9f85cc22083efb. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-006 product before observation access.

Boundary: Every positive finite REL-006 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-006 uniquely retains held-condition-product-factorization, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__held-condition-product-factorization__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:9cb679e173936753232652d3b62f9940c6b8ca89cd2268f59d07991768f8e1a1; generality

sha256:8cd5ae5d1651bb63c513eed7100abe2371b9940372ae3de25ace43abb8cb4240.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt
sha256:3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256:1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256:84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate

sha256:b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal

sha256:e2fd74ec4a80ac9d805f840f382adeb59c98563ee8754e99ec5bd49a4449725f. Source manifest

sha256:b48c2c4214130651a3faaa37dfdc3857b683a468907fc924ff0e8f980ef3e9d2. Engine receipt

sha256:dda9fdc3e893dd8e503cc7630fde3eadb73fda5d7e125d56ff9f85cc22083efb at receipts/e

ngine/model_admitted/SFT-INFO-REL-CONDITIONAL-INDEPENDENCE-006-dda9fdc3e893dd8e.json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-006: conditional_independence
- exact observation: {"condition_classes":2,"joint_pairs_each":4,"product_factorization":true}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:6e3d8abf83e54dd3e32b80746641e56045f26df8692bd7885b436a3485979a40; empirical validation sha256:324ca47db2d0ea9bea07607a87c68c530d914e4d0ff50d913d4cc9d4377f660c; external validation

sha256:750bc53e9c5c76027a85a4c2d25eff8e3dd60d13f75eb28baa39781e6dd9c0ef; isolation sha256:71256d1683273f6f8f50590f927d37d4354395dc4777bffe3229740263baf977; custody sha256:b79fe315b91f60d9bb1f80d8e99f38d6b468118d73f7707e9e2e37fa3a00d353.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

151. Common information correspondence

Family: REL Obligation: SFT-INFO-OBL-REL-007 Claim: SFT-INFO-REL-COMMON-INFORMATION-007

Question, law and forced result. Common information is a retained class label recoverable exactly from either observation alone across the complete joint support.

REL-007 uniquely retains recoverable-common-class, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CODE-COMPLETENESS-018
- SFT-INFO-REL-CONDITIONAL-INDEPENDENCE-006

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:dae36a9b6029365ebe6ec148303a56e91d1a971082cfb9f46aed353ced0a4a2c. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-007 product before observation access.

Boundary: Every positive finite REL-007 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-007 uniquely retains recoverable-common-class, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__recoverable-common-class__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: ca0aee12c83af115ce431c53b75a9c8905dcb367a3895373ec74e1d92c066d2e; generality

sha256: 7a8487ff85135fe5b305465cc22e41f6d0b230769f7eb1a5a5e6f3485047c50c.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt
sha256: 3f5d9f9eeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256: 84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate

sha256: b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal

sha256: 6088fa1594c751b99067588c28183e3943c5c09f876e0468c6942c45c89a760d. Source manifest

sha256: d8957f1af8ca661287373fe1375c415ff74df8fd18366ddc146f36d9224f6fc. Engine receipt

sha256: dae36a9b6029365ebe6ec148303a56e91d1a971082cfb9f46aed353ced0a4a2c at receipts/engine/model_admitted/SFT-INFO-REL-COMMON-INFORMATION-007-dae36a9b6029365e.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-007: common_information
- exact observation: {"common_classes":2,"recoverable_from_either":true}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e3920b43dd4eddc6eca9ef5de588d99386311ace6099589f8283681462db5412; empirical validation sha256:15fe31202c8a6a8bdc6e9ce244eb2912dd0f7fa8fbc28ad65cae7a15d09fb8c9; external validation sha256:a9313dc02ca746ee2c8a9113695b4ba72425238fa29c7a8d67d068fdff4797fc; isolation sha256:19f5221f5d172c70253a5a772772bb3c791d213b6c07e33838c9aade7716b8ff; custody sha256:a8c04450489494a1e81477577ce2d9748642900efbc85f207c4c4287b0055ef3.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

152. Directed information on ordered records

Family: REL Obligation: SFT - INFO - OBL - REL - 008 Claim: SFT - INFO - REL - DIRECTED - INFORMATION - 008

Question, law and forced result. Directed information is the exact source distinctions carried into successive output records under a declared causal order, with prior histories retained and future inputs excluded.

REL-008 uniquely retains ordered-causal-distinction-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CODE - COMPLETENESS - 018
- SFT - INFO - REL - COMMON - INFORMATION - 007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:933936e04c713c7e0b37eb9b883ed4d9bb5ab8fd523288db234566e9f74da58f. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-008 product before observation access.

Boundary: Every positive finite REL-008 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-008 uniquely retains ordered-causal-distinction-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__ordered-causal-distinction-ledger__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:e9a5925237a197290be16a28759d28512b22df100ba78c79a31de0bc93058461; generality
sha256:c64a18ef3ac9a11b62d1517690f1446dda5fd4785ad3f0af1814c017e1fb5418.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt
sha256:3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256:1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256:84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate
sha256:b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal
sha256:572376709aec6d05a3b3e9c2fcd60eb527ac39449dd9ba3045d48f69388ec7af. Source manifest
sha256:db308556919563c0a94eb9fc5e264a8652e3efd4e3503bd68a49a893fee16966. Engine receipt
sha256:933936e04c713c7e0b37eb9b883ed4d9bb5ab8fd523288db234566e9f74da58f at receipts/engine/model_admitted/SFT-INFO-REL-DIRECTED-INFORMATION-008-933936e04c713c7e.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-008: directed_information
- exact observation: {"future_inputs_imported":false,"ordered_steps":2,"source_histories":4}

- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 4f30a1736ff7a2dcff37342eefbb839411ec1ae91fc070aefaa50731bff6e3e6; empirical validation sha256: bebc77370d05186627e074aa36949ce9d9f3537cd9fa990915b4ae662db51246; external validation

sha256: 7d48b030c4e9da576d68783466ea879e36d283b4aa81af0f4104daa4631b1d15; isolation sha256: bca8bf493cb7f29808c8ef38353805a380bf464caeb97cf30169261bd5e772df; custody sha256: 129c60386ca6137f70d43fac51f3977ed06723cd0686fd4564bd80c289813f87.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

153. Interaction information correspondence

Family: REL Obligation: SFT - INFO - OBL - REL - 009 Claim: SFT - INFO - REL - INTERACTION - 009

Question, law and forced result. Interaction-information correspondence records distinctions whose retention or closure changes only when three complete coordinate supports are compared jointly rather than pairwise.

REL-009 uniquely retains three-way-shared-closure-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CODE - COMPLETENESS - 018
- SFT - INFO - REL - DIRECTED - INFORMATION - 008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: fd5b38b110dce54b1ccdf9496204f79aa354525dd69d49c55cbc15c8d0966172. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-009 product before observation access.

Boundary: Every positive finite REL-009 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-009 uniquely retains three-way-shared-closure-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__three-way-shared-closure-ledger__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__fini

te-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256: 24fa9435d77a15e7c2a42c7b06da07bd7b998957f1bf7d5331e430551c600d2b; generality
sha256: 19d90bece4bc55ec797dc363d01ef918fa7173959796406e495cd883088a60d1.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt
sha256: 3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256: 84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate
sha256: b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal
sha256: a56fef7bd95b9e0eab7af83b675981e2e52ad35625d98b0d69ed66f389adfeac. Source manifest
sha256: f34eaea6c14887d45f1e8e3941ae5859a028222897323b63f3456feef1659386. Engine receipt
sha256: fd5b38b110dce54b1ccdf9496204f79aa354525dd69d49c55cbc15c8d0966172 at
receipts/engine/model_admitted/SFT-INFO-REL-INTERACTION-009-fd5b38b110dce54b.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-009: interaction_information
- exact observation: {"complete_triples":8,"equal_three_way_classes":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e18c89e661af97a3ae180c09adbce50e31bc58096adbfa775bbfb721be0deaa8; empirical validation sha256:af8a4cdab2c93c8b545f1b596dbba6cca815f6a3b071231514a478b287c9aa95; external validation sha256:af75d422f858b93cd0ed4473b7293e56f648e4acc59689ea9a55e40da7adfd2b; isolation sha256:a3af251bd507e6eb3ea05dcfa7ebc3969996a0a5850aae8f54794a0d41d85e05; custody sha256:85c5c6288d0b7778fc7bc8063246e5c10b4508bda2368f34a08ed2716ef2b20c.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

154. Multi-information and total dependence

Family: REL Obligation: SFT - INFO - OBL - REL - 010 Claim: SFT - INFO - REL - MULTI - TOTAL - DEPENDENCE - 010

Question, law and forced result. Total dependence is the exact distinction between complete joint support and the ordered product generated by its marginal supports, retained as support rows rather than a signed scalar.

REL-010 uniquely retains joint-versus-marginal-product-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CODE - COMPLETENESS - 018
- SFT - INFO - REL - INTERACTION - 009

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:3613adf822c9822cb6d05e3411ed89b1de4cbffd0595524c5b1d1cc2de28a167. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-010 product before observation access.

Boundary: Every positive finite REL-010 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-010 uniquely retains joint-versus-marginal-product-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__joint-versus-marginal-product-ledger__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:dd532bbfbef90192fc08254f523c1cf15cb091e7d446a366c86500058915b541c; generality sha256:4644d3d49b41b39ddb08e4bb00cc6fffb029ba306e13ffd97aa4b54822e343cc.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt
sha256: 3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256: 84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate

sha256: b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal

sha256: f8073a47a3c71701e1e3172a93379752dad0f8ab48ecf99c5944ddd60e438c81. Source manifest

sha256: 3e9fb0082352fe6d2091902a7ef86c7c5129840a125c8f0713a1cf6a35ab8547. Engine receipt

sha256: 3613adf822c9822cb6d05e3411ed89b1de4cbffd0595524c5b1d1cc2de28a167 at receipts/engine/model_admitted/SFT-INFO-REL-MULTI-TOTAL-DEPENDENCE-010-3613adf822c9822c.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-010: multi_total_dependence
- exact observation: {"joint_forms":8,"marginal_forms_each":2,"marginals":3}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:7f0e2b27d8a7364c09d10db0a66cc104e61728569fa903c79b08ae3515402ae4; empirical validation sha256:0d6728e09b13df6f6abc27780ffa3a2d24532b659429091a45317160c2bba157; external validation sha256:992bd2fd60eef76994ab2ae7855c4c8990de594c2e990d0c56de055dcc3d5231; isolation sha256:be11f63d7553a252ead42fe413a97efa95a84e1afdb7c01d565ffd65fdbba2245; custody sha256:352f207a3395d034e66752c2baafe439bb798c7696f958931dfec7292137ffea.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

155. Shared, unique and synergistic decomposition boundary

Family: REL Obligation: SFT - INFO - OBL - REL - 011 Claim: SFT - INFO - REL - PARTIAL - DECOMPOSITION - 011

Question, law and forced result. Shared, source-unique and jointly available distinctions require separate typed records; a scalar decomposition is admitted only after its allocation rule is itself structurally forced.

REL-011 uniquely retains typed-shared-unique-synergy-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CODE - COMPLETENESS - 018
- SFT - INFO - REL - MULTI - TOTAL - DEPENDENCE - 010

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:f66b4ac88b7fa7a98574bc566451edf5034078fe587f4ac0663848461f0d198e. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-011 product before observation access.

Boundary: Every positive finite REL-011 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-011 uniquely retains typed-shared-unique-synergy-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__typed-shared-unique-synergy-ledger__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:adabb3b5a9a63e68e4597b667937ced78fa7f728b6efbd70e09eea0eebd85adc; generality sha256:51fa93ba15a7f271ac250920fa316491724000d554d1da1f56117cb9725b3513.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted

magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt
sha256: 3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256: 84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate

sha256: b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal

sha256: cf908b95810a96466f4a1933ae7066990d1e2447fb166dd9b838bebc4476eba9. Source manifest

sha256: 1cc32270b4ef1dad41c1cff5c0beac6afe0904791a321116ff7429c6d63ace3b. Engine receipt

sha256: f66b4ac88b7fa7a98574bc566451edf5034078fe587f4ac0663848461f0d198e at receipts/engine/model_admitted/SFT-INFO-REL-PARTIAL-DECOMPOSITION-011-f66b4ac88b7fa7a9.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-011: partial_decomposition_boundary
- exact observation: {"shared_types":1,"synergy_types":1,"unique_types":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: b54c57a2349e0a973cffe64fc5b3617eb928f895546bcb83f664a9322f353947; empirical

validation sha256: fb45312d224427155b4f1d59ea4c72f2caff0fad64387f19b2252cb20da99a23;

external validation

sha256:1b50be18cad31d6834f6b4574f7d158d5ec65e0d45f635295a24d8b583a5ced8; isolation
 sha256:09c81e91934cd57b06e31fbfce0e1b70670520e1ee41a5c7c657779130f615e6; custody
 sha256:785e3550412d8c29d29a211518f1327b4c3b0360c58c48038ae4251858778d2d.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

156. Information flow on causal paths

Family: REL Obligation: SFT - INFO - OBL - REL - 012 Claim: SFT - INFO - REL - CAUSAL - FLOW - 012

Question, law and forced result. Information flow retains every directed source-terminal path, transition and observation class; terminal association alone cannot establish a causal path.

REL-012 uniquely retains complete-directed-path-custody, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CODE - COMPLETENESS - 018
- SFT - INFO - REL - PARTIAL - DECOMPOSITION - 011

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:bdc251e3f98224ab9c7a08c616e51fda4f449bb626984661ed201517c6df8f66. A missing
 dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-012 product before observation access.

Boundary: Every positive finite REL-012 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-012 uniquely retains complete-directed-path-custody, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__complete-directed-path-custody__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:e777f142c3f766128969157ee64243c0ab959f2a7a5b2ac8a529a24bbc0be004; generality
 sha256:344164990e46ecd172d5761fa451ac5a679ba6c6faf8d3922d5b91e17709f780.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt
sha256: 3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256: 84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate

sha256: b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal

sha256: 8fd8bb9aac41dd9131586298cb59c36d00b7bf4acc197f973ed990d7d2ef8b78. Source manifest

sha256: 62029af197fb52e4daf2f9b999a88e2b96f839227dd85757c5bbb37d37b4f20b. Engine receipt

sha256: bdc251e3f98224ab9c7a08c616e51fda4f449bb626984661ed201517c6df8f66 at

receipts/engine/model_admitted/SFT-INFO-REL-CAUSAL-FLOW-012-bdc251e3f98224ab.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-012: causal_flow
- exact observation: {"paths_after_middle_removal":1,"source_terminal_paths":2,"terminal_association_only":false}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 4f2f5a0200740cb2dd036757d8ac830d3dbba27800a315f21b3c4062dab1595a; empirical validation sha256: aa5ceb21a8a1bdb190ec2aeadeb944ce2f10c53f241ed7ada5bc42190355393f;

external validation

sha256: de2f41da6f03d710dc0644388c9ef625a00b0de32825616a4793816a7e43d5d0; isolation

sha256: e8a6857a511abb35bde248c9fd3775df601971dce7beb18de81584ed9ec45bda; custody

sha256: b7c07241ba3d51503a1501247c0984736ff84d8e8d0c90f74e1e42fc6f8b410a.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

157. Relative information under representation change

Family: REL Obligation: SFT - INFO - OBL - REL - 013 Claim: SFT - INFO - REL - REPRESENTATION - CHANGE - 013

Question, law and forced result. Bijective representation change preserves every relative distinction and inverse record; a non-injective change closes exactly the merged classes.

REL-013 uniquely retains bijection-invariant-relative-support, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CODE - COMPLETENESS - 018
- SFT - INFO - REL - CAUSAL - FLOW - 012

The current reconciliation binds this obligation to the current model-admitted receipt sha256:e0e7ae12c6ccaadfb7281fdde62bd2cf442dd4928905cb1a09ccdce3bf107f25. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-013 product before observation access.

Boundary: Every positive finite REL-013 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-013 uniquely retains bijection-invariant-relative-support, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__bijection-invariant-relative-support__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:cc22ba8aa0e637bdc239053a1b0d6914728c2b30d8a515e28dd94bbc82983c52; generality sha256:a6a3369390466832aa8b1947b39dbfd55590113e75b3330869efc33a92661580.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt sha256:3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256:1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256:84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate

sha256:b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal

sha256:8bf13fbbf1bda42e3b56c56df9a3af024d2cb7fd5c695361ad7b857269bd9309. Source manifest

sha256:0ea5dfdbb87956a1e6d88ff2ddb0b86274cb06cd25ce5353868ccca6a5fe2c33. Engine receipt

sha256:e0e7ae12c6ccaadf7281fdde62bd2cf442dd4928905cb1a09ccdce3bf107f25 at receipts/engine/model_admitted/SFT-INFO-REL-REPRESENTATION-CHANGE-013-e0e7ae12c6ccaadf.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-013: representation_change
- exact observation: {"inverse_reconstructions":2,"preserved_classes":2,"representation_bijective":true}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:212f039a09bf65a4f5037a3dd9845e04f64fec981475a44af9db609e345d7b9a; empirical

validation sha256:0e5c15d6528f588425f204e6332334a1ebe44dbb323f9f31cc82421eef17e0ce; external validation

sha256:a57a4c395323c08894fbf9ae40983e5cfde49a98f4bcc3f2fbe22e2988d83857; isolation

sha256:de2eb7eb07671b25afbdbbe32d86c60ea8de1d10de17e17071558e384987052be; custody

sha256:ec95ae5fe297b0f38b6f729edca53ef9215b4ba30b250db69cb61bb4eb0da427.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object

- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold
- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

158. Relational-information completeness certificate

Family: REL Obligation: SFT - INFO - OBL - REL - 014 Claim: SFT - INFO - REL - COMPLETENESS - 014

Question, law and forced result. Relational-information completeness is the one-to-one reconciliation of all fourteen frozen obligations with exact joint, conditional, shared, causal and representation records.

REL-014 uniquely retains fourteen-relational-obligation-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CODE - COMPLETENESS - 018
- SFT - INFO - REL - REPRESENTATION - CHANGE - 013

The current reconciliation binds this obligation to the current model-admitted receipt sha256:224b10cd37c046fa40ce6981d82d55bbabedfe79b44dffca0213159c94898ef6. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis REL-014 product before observation access.

Boundary: Every positive finite REL-014 joint row, restriction, class, path, representation record and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- REL-014 uniquely retains fourteen-relational-obligation-ledger, complete relational custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-joint-support__fourteen-relational-obligation-ledger__explicit-held-condition-record__complete-typed-distinction-ledger__complete-declared-relational-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:2c2bcd51d4bc6b5a7c8a9e19ac72c463f4a67850a64bd3e8b44567e46642c45d; generality sha256:ee48b5d333d7e3a78591bc91c2bad1c5c792c82f41f096b97b8af72e80f6b273.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing rows change every relational conclusion.; receipt sha256:3f5d9feeadc74dc7ae61aa463e75737fcc14230a0affa5b32f883826ab2be615.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability distribution, logarithmic formula or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold; no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256:1749cd95e21f5c374892a1679553e40a163b80340c55ced59e104b1107f10009.

Independent reconstruction and engine admission. Independent implementation

sha256:84dc4959da3d69a978e3fe2339fe630f71ee779af14f5cc20fbdec9bc13b7d1c regenerated the declared census and sole survivor; independent certificate

sha256:b0abdc6db75d27ede0580836038e0a9371bde83e1c6d7874bf79cd15d3e4ef0b. Derivation seal

sha256:c4339dfa99362d747e43701a6314619917ee72311aeb5dbd87859698ff1d1399. Source manifest

sha256:ae8a0b2af6959697af4652739dbc773281d95e3c7eb8e5150db59a8704914ed9. Engine receipt

sha256:224b10cd37c046fa40ce6981d82d55bbabedfe79b44dffca0213159c94898ef6 at

receipts/engine/model_admitted/SFT-INFO-REL-COMPLETENESS-014-224b10cd37c046fa.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RELATIONAL-OBSERVER
- SFT-V1-V2-RELATIONAL-INFORMATION-CORPUS

Observed, adverse and boundary records:

- REL-014: relational_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":14,"omitted_owners":0,"registered_obligations":14}
- complete family observation custody: 14 records; all rows preserved
- complete finite joint, conditional, shared, directed, causal-path and representation enumeration supplies the observation; no imported distribution, signed-information premise, fitted allocation or hidden joint row enters

Falsification condition: Reject if the value-free registry, complete relational census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:2f0144411a3a76aeb416206f62bed93a7f674225dcf66a444600f64b6a64256c; empirical validation sha256:cc45a2af5c009147f62b5269868f3c29e7520796be299246b98603c4e6b4cbd3; external validation

sha256:5dfeef9600622dda591a20fd71dd78b0784af4d9ccd989df73012d2d527ec368; isolation

sha256:38550d5b22ea4f7bde0ef2e0759cfe54bf8b17e8def4eff131c50d9312b48c65; custody

sha256:557f4237cfae3799c0c2ccf070029f0fae5b3727a3eec830003adab4ec1d798c.

Prohibited inputs.

- no axiom, imported probability distribution, logarithmic formula or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden joint row, signed-information premise, likelihood allocation or fitted causal threshold

- no semantic meaning, cryptographic secrecy or physical mechanism imported into this owner
- no failed route retires an obligation or changes protected authority

Complete-field family: COARSE

159. Sufficient record for a declared observation

Family: COARSE Obligation: SFT - INFO - OBL - COARSE - 001 Claim:
SFT - INFO - COARSE - SUFFICIENT - RECORD - 001

Question, law and forced result. A summary record is sufficient for a declared observation exactly when that observation is constant on every complete summary fibre.

COARSE-001 uniquely retains observation-sufficient-fibre-record, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - REL - COMPLETENESS - 014

The current reconciliation binds this obligation to the current model-admitted receipt sha256:6c95f59d6ede0708d99a596971812733d6fe2dc7b4ba20851d4541467ea493aa. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-001 product before observation access.

Boundary: Every positive finite COARSE-001 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-001 uniquely retains observation-sufficient-fibre-record, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__observation-sufficient-fibre-record__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:9156adef226235482344ba79a364b218d691f517da62fccc9c3355e44ef423cc; generality sha256:95cd0882a478324f8da821da36c9858019b6523fcfbfd300fdb67e0647332157.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt sha256:640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt

sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256:92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate

sha256:f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal

sha256:ab2ed277a87a0de214877a9b1e4b3fae5ed6f437d618b6c3084db10e1a194f51. Source manifest

sha256:12d66879d87361f5a2993a1c87d1161cf69d71ec8f3f4a10ee1b626564965f15. Engine receipt

sha256:6c95f59d6ede0708d99a596971812733d6fe2dc7b4ba20851d4541467ea493aa at receipts/engine/model_admitted/SFT-INFO-COARSE-SUFFICIENT-RECORD-001-6c95f59d6ede0708.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-001: sufficient_record
- exact observation: {"source_forms":4,"summary_cells":2,"target_classes_retained":2}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:3a689b6df8771f3c7179011552def53218bab652dafbd0e0da8ab4a6e6936e7; empirical validation sha256:cb4ee007a3e26e77963610d46527b0f46402963d5b0a6cc04e73e6a3e10a1029; external validation

sha256:87e1536c5f8ff1bc3009ed79080c1412609bf6d513221380bfc376f0d1385109; isolation

sha256:98b5400eff64fa129311dbf4cc8e03a534db6c53ffa05fee23088cbe9daf6ec1; custody

sha256:5cfa055406bddff3f7a74e9d66f29425fcc368926d1e6f6dfd8efbe9ef891b6b.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

160. Minimal sufficient record

Family: COARSE Obligation: SFT - INFO - OBL - COARSE - 002 Claim:

SFT - INFO - COARSE - MINIMAL - SUFFICIENT - 002

Question, law and forced result. A minimal sufficient record is a sufficient partition for which every strictly coarser generated partition closes at least one declared observation distinction.

COARSE-002 uniquely retains least-sufficient-partition, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - REL - COMPLETENESS - 014
- SFT - INFO - COARSE - SUFFICIENT - RECORD - 001

The current reconciliation binds this obligation to the current model-admitted receipt sha256:db19424a38e4035ad03da8a402288566459711ce4db9c31d7c6522ba5b8c599f. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-002 product before observation access.

Boundary: Every positive finite COARSE-002 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-002 uniquely retains least-sufficient-partition, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__least-sufficient-partition__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:10860d5fb20776c0b0b815558d0d1d7aff4412630ce21079ac298f16ba975d6a; generality sha256:9687fe842a16115a861e6ec470cc1547c74dedde4e5686a40a4c9f6670873ac4.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt sha256:640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden

source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: 80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256: 92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate

sha256: f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal

sha256: 5799f4072b6664a37216b00e5c9cd0edfc79e256fa05e7113ba55ccbff530ebf. Source manifest

sha256: acbc286d6b6daec39554ff92ee3b63aca8d7bbe3c822a517ac6e0dc4288b242b. Engine receipt

sha256: db19424a38e4035ad03da8a402288566459711ce4db9c31d7c6522ba5b8c599f at receipts/engine/model_admitted/SFT-INFO-COARSE-MINIMAL-SUFFICIENT-002-db19424a38e4035a.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-002: minimal_sufficient
- exact observation: {"least_sufficient_cells":2,"one_cell_control_sufficient":false}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: e3873b1a41ebc8c95d0125f7815a19fa6c30f21b8cd04fc03effd0a89f767be7; empirical validation sha256: 46e76869acb911e33bc3bf0c2cdc73206134c76409778da19a91064b6c14351b; external validation

sha256: 10d9fdc5054990ba7dd9396c1ae72de1113352f64122fda64e5dd6c2c8a4af21; isolation

sha256: 8700e9c73f7921358a542d342d69c75ef9da3df2837f4d7ef2e054b3356f6bef; custody

sha256: 109d7cb39dd61894556315fb7608eaaaa1e9c4087956b2a2e4c02401aa92a34c.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

161. Information bottleneck as lawful coarsening

Family: COARSE Obligation: SFT-INFO-OBL-COARSE-003 Claim: SFT-INFO-COARSE-BOTTLENECK-003

Question, law and forced result. A lawful information bottleneck is the least generated coarsening that retains every registered target distinction and explicitly records all other closed distinctions.

COARSE-003 uniquely retains least-target-preserving-coarsening, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-REL-COMPLETENESS-014
- SFT-INFO-COARSE-MINIMAL-SUFFICIENT-002

The current reconciliation binds this obligation to the current model-admitted receipt sha256:a8e223eaed37a7481f3b4d7e86c64f73874b5342ef0d762e22c97bcf35a1597e. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-003 product before observation access.

Boundary: Every positive finite COARSE-003 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-003 uniquely retains least-target-preserving-coarsening, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__least-target-preserving-coarsening__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:f13e5b53f01d8785fbbec71dea8a5f029dbe08eeddca35d58bd0ff48eb21cd1f; generality sha256:8572b259d3d713023f79718fc0c4561031fdbafdfd840401d80fa575e95382ef.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt sha256:640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256:92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate

sha256:f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal sha256:2e0ac83f0d892970f33ebefcd47c271f4f13071be20487c38dc59fd4ee0f2bc5. Source manifest sha256:da56f2de4f3751e1f99453f348b3d3f6fa4a90494b0b190fa01ced7bb8d4cfe9. Engine receipt sha256:a8e223eaed37a7481f3b4d7e86c64f73874b5342ef0d762e22c97bcf35a1597e at receipts/engine/model_admitted/SFT-INFO-COARSE-BOTTLENECK-003-a8e223eaed37a748.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-003: bottleneck
- exact observation: {"merged_within_target_pairs":2,"retained_target_classes":2}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:7bcf4f25351c58886bae520a256ed55db2ed3a792013574fce7b027f986a59b2; empirical validation sha256:7d99c4575558e0fd709848a331a98275827def1e98754f8b52afa50f59ed3d0a; external validation

sha256:02631a14ac3fd0f5a975478b311cf39c8f1ebb154bf07cc09bf3b4e63ada4501; isolation sha256:33ac51a77354d0aea67637363f97e044427b7544813253de64dd50db37d9a152; custody sha256:2bd0d0dd21648295cae4f56e7cb8c6bf567674b61b1dc1e5240e7c8e8a7a1e79.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

162. Coarse-graining partition structure

Family: COARSE Obligation: SFT-INFO-OBL-COARSE-004 Claim: SFT-INFO-COARSE-PARTITION-004

Question, law and forced result. Every coarse map forces a disjoint exhaustive partition of the source support by equal images, with no source omitted or duplicated.

COARSE-004 uniquely retains complete-disjoint-exhaustive-partition, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-REL-COMPLETENESS-014
- SFT-INFO-COARSE-BOTTLENECK-003

The current reconciliation binds this obligation to the current model-admitted receipt sha256:53298df77ca5b49459e6f736cce7e341a10de8459dacefe92fd41119c71b16b8. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-004 product before observation access.

Boundary: Every positive finite COARSE-004 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-004 uniquely retains complete-disjoint-exhaustive-partition, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__complete-disjoint-exhaustive-partition__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:cf585e45f20ee1cd54c07b5a4e3553bef17b16f7c2d49b1ed3b38002533852e3; generality sha256:b167ce8316e8c89700a62cf6d6303c7eed255f022d99b14baa835661fafdba2.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt sha256:640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation sha256:92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate sha256:f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal sha256:79e5ed583ef1d5acc2aa3c051982c7a4e1d07aa48cd4ab254ee5d773256867d8. Source manifest

sha256:3731c5010f4c068024d2c0a0d9f7d817f7e936f2557b3d19f591b960bde1cb83. Engine receipt sha256:53298df77ca5b49459e6f736cce7e341a10de8459dacefe92fd41119c71b16b8 at receipts/engine/model_admitted/SFT-INFO-COARSE-PARTITION-004-53298df77ca5b494.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-004: partition
- exact observation: {"disjoint_cells":2,"exhaustive_sources":4}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:db7c95310da5f9097049e39ebc88ba474aa7b9ad386b4447f8c6e0db88d1eaeaf; empirical validation sha256:941473853f7c3b869d88c4b0be0320a198d30abee63c9c3afff9abf3c03a5ed1; external validation sha256:647722cf1cd275ada0ee709e35e9210f3b577135da9fd2ea470b3a65ea678a6b; isolation sha256:73e2b39fd80d1a0e9667291f90e593684fd98eea8622e8260f619b14ec1521f1; custody sha256:03993bd6b9fab2358c129d5fba5c96317090280ffd858fe43d9cb9204b7bb390.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

163. Refinement and abstraction order

Family: COARSE Obligation: SFT-INFO-OBL-COARSE-005 Claim: SFT-INFO-COARSE-REFINEMENT-ORDER-005

Question, law and forced result. One information organization refines another exactly when each fine cell lies wholly inside one coarse cell; composition preserves this abstraction order.

COARSE-005 uniquely retains partition-refinement-order, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-REL-COMPLETENESS-014

- SFT-INFO-COARSE-PARTITION-004

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:854bcb61f6e1991169d6a533f0d115b7ddf4cb3124f870501a0fb431c2c61877. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-005 product before observation access.

Boundary: Every positive finite COARSE-005 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-005 uniquely retains partition-refinement-order, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__partition-refinement-order__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:1ca5facad38d079a99b05b6aecfca2636179c43a6986995862d79e819be60c4b; generality
sha256:09278d802df8b45d4e14a5b231bae6727f3667fbe8bdd12ea8776f42383376eb.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt
sha256:640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256:92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate
sha256:f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal
sha256:762a2b321add2e7bd85389dc2b33b15bbe964689d5e1793538aa113b6a8f22ac. Source manifest
sha256:e41f77194fd065fcbcb4342094d4f29e0c299cf286aee879e8e33e65247053ead. Engine receipt
sha256:854bcb61f6e1991169d6a533f0d115b7ddf4cb3124f870501a0fb431c2c61877 at receipts/engine/model_admitted/SFT-INFO-COARSE-REFINEMENT-ORDER-005-854bcb61f6e19911.json.
Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-005: refinement_order
- exact observation: {"fine_cells":4,"intermediate_cells":2,"terminal_cells":1}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:be201901329f1f4ceb2a3ead0825f94e452b63fd1cd5c3ce915462c8878078ff; empirical validation sha256:6ddabf2dcfe8d6ccb15b0436939252571864fac5cdfbe7c954da9f85fd6239db; external validation

sha256:e4ab6b12eabcfa4b8ad78614e2b3e5b48151949519b61855730cbeea607ba8a8; isolation sha256:3596f7dcbb82b3c707d8379c033c4035e458973b8edba10fc2156ce38c269b7; custody sha256:86c3e458e5f9bd9c22ed9269de4663fc9ca2049e129fb5388b1ef4e41847df94.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

164. Loss ledger for aggregation

Family: COARSE Obligation: SFT-INFO-OBL-COARSE-006 Claim: SFT-INFO-COARSE-LOSS-LEDGER-006

Question, law and forced result. The loss ledger is the complete set of previously distinguishable source pairs mapped into one aggregate, retaining their source identities rather than reporting only a scalar loss.

COARSE-006 uniquely retains complete-closed-pair-ledger, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-REL-COMPLETENESS-014
- SFT-INFO-COARSE-REFINEMENT-ORDER-005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:7271f4e4aac9fba5c1d1fb9e9f4f83468773b8de1b2ffd627b9028fbd09d77bb. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-006 product before observation access.

Boundary: Every positive finite COARSE-006 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-006 uniquely retains complete-closed-pair-ledger, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__complete-closed-pair-ledger__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: 637ea3dcab7e30be60db4d93cd6a65266d9862d14b10b871f11a6779bc5d8abf; generality sha256: 11e479980de9e6ef58decdf44658afdee5b8f54ded7c98cf0e81843cc59e49cd.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt sha256: 640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: 80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256: 92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate sha256: f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal sha256: 53fdd3c68731c61d0d7428b6ebc32f203cb4b3668e83d5bc70319c8f2d574eef. Source manifest sha256: 5997b95cc8a5b1d77544c8c848d5090636bc0e134895f4c848819e10fb4bb89e. Engine receipt sha256: 7271f4e4aac9fba5c1d1fb9e9f4f83468773b8de1b2ffd627b9028fbd09d77bb at receipts/engine/model_admitted/SFT-INFO-COARSE-LOSS-LEDGER-006-7271f4e4aac9fba5.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER

- SFT - V1 - V2 - COARSE - INFORMATION - CORPUS

Observed, adverse and boundary records:

- COARSE-006: loss_ledger
- exact observation: {"closed_source_pairs":2,"source_forms":4}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:1c3bda7fe94176a2ee04775180b8976f0f264228b312546ec5a98baaef2322f7; empirical validation sha256:685ad30734e5b4b696302c23d6d3468157c9a66918a8b89191dbe1a25337b50e; external validation

sha256:41157309b90e896d07e973820e13c5b1d8a014e9c7d210ce32a2d9d6c12e3f7a; isolation sha256:0cb43a2e7120e5826cef6f41dff2e1bef7315b91a8b046ecf1dc2cf710f55879; custody sha256:444c1b6c361480b622674c6b71dba84f339a64dc85568125711328b7729483e9.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

165. Feature selection by retained distinction

Family: COARSE Obligation: SFT - INFO - OBL - COARSE - 007 Claim:

SFT - INFO - COARSE - FEATURE - SELECTION - 007

Question, law and forced result. A feature is admissible exactly when its fibres retain every registered target distinction; minimal selection eliminates every feature coordinate not required for that sufficiency.

COARSE-007 uniquely retains target-distinction-feature-selection, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - REL - COMPLETENESS - 014
- SFT - INFO - COARSE - LOSS - LEDGER - 006

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:6f6034b8862bc5167fb3441ba76b71cb16aaac7795d4446fc05209400c4141d2. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-007 product before observation access.

Boundary: Every positive finite COARSE-007 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-007 uniquely retains target-distinction-feature-selection, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__target-distinction-feature-selection__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: 2d5b0d1d34604a36a9b498777719586568a0bb281630b679c98faf571df94c16; generality sha256: b2afccdab51e8c7667f3c017b50fb4db5bd10091349b5d71b913100cb2ef9acb.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt sha256: 640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: 80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256: 92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate sha256: f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal sha256: fd707bc5a27221336208611aee101b019371a8b61eb674dbe3d8dda9dd327695. Source manifest sha256: 6705cc4458e46f55ee57ec81a858460df1682227e431b38725e16fe89dbddb09. Engine receipt sha256: 6f6034b8862bc5167fb3441ba76b71cb16aaac7795d4446fc05209400c4141d2 at receipts/engine/model_admitted/SFT-INFO-COARSE-FEATURE-SELECTION-007-6f6034b8862bc516.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-007: feature_selection

- exact observation: {"constant_feature_sufficient":false,"selected_feature_classes":2}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:3dda7de3c613bc71608f282ee203a5889017248ac4dae6994abbcfcb8d759679; empirical validation sha256:c0bdfdf990bdb867ab1b41841d5a895c7b2491366c9ee8ee9ea8b5620e990ef31; external validation

sha256:cd77588906c9d75a36881275d18ff33edb660614eb724028d44334740856ba76; isolation sha256:afb5289df9ecb3aa7f3a12cbadcfc39d47f2b726ff770d6db503e45bc690efdb; custody sha256:24432147637405f6cbd408d8c6cbb8b36629b5cf460f6bd387894d0875f85fbb.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

166. State aggregation and exact lumping

Family: COARSE Obligation: SFT - INFO - OBL - COARSE - 008 Claim: SFT - INFO - COARSE - STATE - LUMPING - 008

Question, law and forced result. State aggregation is exactly lumpable when all states in each aggregate have identical successor-aggregate support, forcing a well-defined quotient transition.

COARSE-008 uniquely retains transition-consistent-state-lumping, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - REL - COMPLETENESS - 014
- SFT - INFO - COARSE - FEATURE - SELECTION - 007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:c4419bef376017a10c6f4c09e1f6cacc6e2675645cb04a897984bcf4d96e6f83. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-008 product before observation access.

Boundary: Every positive finite COARSE-008 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-008 uniquely retains transition-consistent-state-lumping, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__transition-consistent-state-lumping__registered-observation-distinction__complete-closed-distinction-ledger__complete-d

eclared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:e167e82c984266b1c493560db7008397b476e527714e1fe3b01f7c221504543c; generality sha256:7cb75e41b07fbd96b9b2550c5f0eeb1ef4b6b73ab3cc2aaf681c5543b42a77c5.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt sha256:640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256:92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate sha256:f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal sha256:7a21dfe0cb5d43cc4c194354b0512ad2266c16cd7b76f6c15cebca5a0a244df6. Source manifest sha256:a678490c69f75d2a76cb33ab3623b2b2f983d0d927b10a06706db383737388ae. Engine receipt sha256:c4419bef376017a10c6f4c09e1f6cacc6e2675645cb04a897984bcf4d96e6f83 at receipts/engine/model_admitted/SFT-INFO-COARSE-STATE-LUMPING-008-c4419bef376017a1.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-008: state_lumping
- exact observation: {"aggregate_states":2,"microstates":4,"quotient_well_defined":true}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 4674227e40f61e75b0254483d233c6ba4c926bc297697e9893745863a08c805b; empirical validation sha256: 9a00c3be49a9443887ba4ac441f9d8cb19c9db7a77a026fd1f17c8c565bafb5f; external validation sha256: a1aa175ac6e13a6abf04cea9672cf36d479ed1af02faa13f46fd414fe78dc623; isolation sha256: af1c6b2292a66a473643869b14299d1325adc424b2a1d15c1fa7bb3c73dccc90; custody sha256: 1b5b008f9e9b51270b853389807701f3c4ee397a1ceb0e0250eda7d067498140.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

167. Summary record reconstruction boundary

Family: COARSE Obligation: SFT - INFO - OBL - COARSE - 009 Claim: SFT - INFO - COARSE - RECONSTRUCTION - BOUNDARY - 009

Question, law and forced result. A summary reconstructs a source exactly only for singleton fibres; otherwise its complete preimage is the lawful unresolved reconstruction class.

COARSE-009 uniquely retains complete-summary-preimage-boundary, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - REL - COMPLETENESS - 014
- SFT - INFO - COARSE - STATE - LUMPING - 008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 8b53ea67fcfb48054fa3dc7e7cb03681b0cc30535c8f679c69a9fc4c99580f71. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-009 product before observation access.

Boundary: Every positive finite COARSE-009 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-009 uniquely retains complete-summary-preimage-boundary, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__complete-summary-preimage-boundary__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: 499f685193dcef8c5b700f8f11fae696b340725e0e266332e5404124c4893c6d; generality

sha256:8090ebd1c20e14275b13d558ba56b87f0e8ef10f845f6db89d6de4a49c43143d.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt
sha256:640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256:92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate

sha256:f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal

sha256:54e06b45b5d396e55b618d58c251b1e8e77846725c73301edb8a1e680c4c6c70. Source manifest

sha256:fc4f6ec564116ac10f844d61a7a5b6ae3b3a3c63534819452961b2bc6e5a3a21. Engine receipt

sha256:8b53ea67fcfb48054fa3dc7e7cb03681b0cc30535c8f679c69a9fc4c99580f71 at receipts/engine/model_admitted/SFT-INFO-COARSE-RECONSTRUCTION-BOUNDARY-009-8b53ea67fcfb4805.json. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-009: reconstruction_boundary
- exact observation: {"fibre_size_each":2,"unique_reconstruction":false}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:c7baf8f2d8b68bc8b49947f5f4723d8e70beed433490c16a3270252e23c5d952; empirical validation sha256:349f4a4bb2710f69e6fd89c8513033d45bfadc4bca258f3ca9f9aff219903508; external validation sha256:0d1d1c41b0beb1e7b93754fa8942410e589359981c7fa2144de2d39dcb10ce96; isolation sha256:0d0abde2fd9ec0d9135c3e7433436e059d922d8b32c985ddecfb57c401b4798; custody sha256:5812cacba70055e6d7b7e6235e87276b3b128e667721d8fb3f91c076f74389a2.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

168. Multi-stage coarse-graining composition

Family: COARSE Obligation: SFT - INFO - OBL - COARSE - 010 Claim: SFT - INFO - COARSE - COMPOSITION - 010

Question, law and forced result. Multi-stage coarse-graining composes image maps exactly while retaining every intermediate partition and cumulative loss ledger.

COARSE-010 uniquely retains exact-coarsening-composition, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - REL - COMPLETENESS - 014
- SFT - INFO - COARSE - RECONSTRUCTION - BOUNDARY - 009

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:04e613932405f3bd3840ee2ad0012433d18876dd48ceef853726af654c860c7. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-010 product before observation access.

Boundary: Every positive finite COARSE-010 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-010 uniquely retains exact-coarsening-composition, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__exact-coarsening-composition__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:f9371afb8d419657a19079e3a95937269f2c83cf1394284341e5096297f89cea; generality sha256:1e6434deff3b2a67515d3c29d5fa3ba6a128af87d85754cc77b3c843ac2137a3.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted

magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt
sha256: 640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256: 92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate

sha256: f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal
sha256: 204afa364b351c2fed20cbc72a32265c64c0209b4bf0ff4397bf3dfd88ea5e19. Source manifest
sha256: accbf644762f11631f5ed7233db0c5b3c455d8033ce30aaa479ac954503e56. Engine receipt
sha256: 04e613932405f3bd3840ee2ad0012433d18876dd48ceeeef853726af654c860c7 at receipts/engine/model_admitted/SFT-INFO-COARSE-COMPOSITION-010-04e613932405f3bd.json. Closure
scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-010: composition
- exact observation: {"composition_stages":2,"source_forms":4,"terminal_cells":1}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 5622a9e91108bc49209d99d9379e9bea67dc17ce93c7e38da6f0631adf714dbe4; empirical
validation sha256: fd79c5228fc3d88efd6d62c537c47c45acdfe4bfe2625ab5cf760da52e8725a9;
external validation

sha256: 9c08c6333f622019e43ff39c5310c8f3aceba4214ad45faa6c62f64a190c87c6; isolation
sha256: 0a82a948127b165bfec181849a143362e3b9eb74c88cde2133c613d4f55cafcc1; custody

sha256:86497dd56c9c8509767617efbc2a8216a3f324b5f1eae5edef8a7bfea35c032.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

169. Coarse-graining reversibility condition

Family: COARSE Obligation: SFT - INFO - OBL - COARSE - 011 Claim: SFT - INFO - COARSE - REVERSIBILITY - 011

Question, law and forced result. A coarse-graining is reversible from its output alone exactly when every image fibre is singleton; non-singleton fibres require retained source labels.

COARSE-011 uniquely retains singleton-fibre-reversibility, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - REL - COMPLETENESS - 014
- SFT - INFO - COARSE - COMPOSITION - 010

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:d98b026ead033fe633b15866110646cba156f06f9434e7dc464f2ea64429a71e. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-011 product before observation access.

Boundary: Every positive finite COARSE-011 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-011 uniquely retains singleton-fibre-reversibility, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__singleton-fibre-reversibility__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:43bdfbbcca1635af80f7a7243a71f18ff29dd93656d0ea01fa5c07b0306d18f5; generality sha256:441b33261c62ed0afbea1f517f9c2049078876464831224d514da4c0b2a16ec7.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt

sha256:640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256:92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate

sha256:f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal

sha256:6fc41b0f88d539021c023dcec37e3469710f06c1be8e58f8d7188380ba3a88e3. Source manifest

sha256:15d394e66a902e741353cc759a320ef83b4a400b74cfd2a8a712a89331f4cd5d. Engine receipt

sha256:d98b026ead033fe633b15866110646cba156f06f9434e7dc464f2ea64429a71e at receipts/engine/model_admitted/SFT-INFO-COARSE-REVERSIBILITY-011-d98b026ead033fe6.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-011: reversibility
- exact observation: {"identity_fibre_size":1,"pair_fibre_size":2,"retained_label_required":true}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:2b3627ff7d23be3c6d2511f20f71c099f113ec58c29e499d995dfc4134bb62e1; empirical validation sha256:60900bc12ec9e914cc85acebab40dff104cd91839a26eec65ce5456466e188f; external validation

sha256:d3939b91a51a7a6e8abb5fc928d1776f497af95247f2fb677d806d55fb81a49a; isolation

sha256:bec67684f0478cc7b5363b92ca448d8c6593df7df029ae74f8e3504591daaa57; custody

sha256:badb6219b02eeb7fc21620bb1d2b0d582181c587996b2f6d5de586038ee687a7.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object

- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

170. Sufficiency and bottleneck completeness certificate

Family: COARSE Obligation: SFT - INFO - OBL - COARSE - 012 Claim: SFT - INFO - COARSE - COMPLETENESS - 012

Question, law and forced result. Family completeness is the one-to-one reconciliation of all twelve frozen sufficiency, bottleneck, partition, loss and reconstruction obligations.

COARSE-012 uniquely retains twelve-coarse-obligation-ledger, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - REL - COMPLETENESS - 014
- SFT - INFO - COARSE - REVERSIBILITY - 011

The current reconciliation binds this obligation to the current model-admitted receipt sha256: 2a994334a9f3743447e8ee7d38a84b77d1217bc1b599dff7722f758d63e0f963. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis COARSE-012 product before observation access.

Boundary: Every positive finite COARSE-012 source, partition, target class, loss row, reconstruction fibre and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- COARSE-012 uniquely retains twelve-coarse-obligation-ledger, complete coarsening custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-canonical-source-support__twelve-coarse-obligation-ledger__registered-observation-distinction__complete-closed-distinction-ledger__complete-declared-partition-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: ff28e4f024086e83f2eaaccb7a4d0cdaaa139ba3d8e60a5a7c96ab7beb218b61; generality sha256: 41c7636b285cfaed4cd99ccb8b8db7271970aa6c4034bb08b468cef14c4f39a8.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing sources change every fibre.; receipt sha256: 640a62a94b8a9ac52ed88d1e1d84312b69c4a2fee26caf41ab771fd81c11190a.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported statistic, stochastic distribution or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair; no semantic, learning or physical criterion imported into this information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:80d9d9eba728543287f2956f52df2c0806de3939587df8ac292aded74950aa0a.

Independent reconstruction and engine admission. Independent implementation

sha256:92a02490d29e0b39e6e9646aea143021c76a693e26b974bc9174bd5c1e9b98aa regenerated the declared census and sole survivor; independent certificate

sha256:f6dfe6fa6fc6a12d9939f76faeb9714bc965eebb8a04a6569beee4d1263941f5. Derivation seal

sha256:a987958fa1cf1d8dd9d8d11fd045b85735d28ce2d594b993f1a10cd38faf4caf. Source manifest

sha256:d7f71b41e7f8b3eb31dd8920dcac8173b3f7bfbce5f849071cfcccd09057ff. Engine receipt

sha256:2a994334a9f3743447e8ee7d38a84b77d1217bc1b599dff7722f758d63e0f963 at receipts/engine/model_admitted/SFT-INF0-COARSE-COMPLETENESS-012-2a994334a9f37434.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-COARSENING-OBSERVER
- SFT-V1-V2-COARSE-INFORMATION-CORPUS

Observed, adverse and boundary records:

- COARSE-012: coarse_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":12,"omitted_owners":0,"registered_obligations":12}
- complete family observation custody: 12 records; all rows preserved
- complete source, partition, target-class, loss-pair, quotient-transition and reconstruction-fibre enumeration supplies the observation; no imported statistic, fitted bottleneck weight, chosen feature score or hidden source enters

Falsification condition: Reject if the value-free registry, complete source-partition and reconstruction census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:825804c82674e7bfd35903bcb76689fc9864b4965c4199213108f26cac643889; empirical validation sha256:8290ff7b1311784c49c45461f6108a3fc3a4ecbb10d363c1643e63e0fe89a259; external validation

sha256:c26ab9d82e95aab251b76bd90bdbbc5809c139b55e0d6d269c6ad70e1524ee59; isolation

sha256:aabd45a524c8ae9e49834cd76bf0e36ac1888aca791096ad8eca0c2da2140e93; custody

sha256:ce34d74897277c76b3ef035beca586e298128172d354237459aaeb4cfd55eb29.

Prohibited inputs.

- no axiom, imported statistic, stochastic distribution or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden source, chosen feature score, fitted bottleneck weight or unrecorded closed pair
- no semantic, learning or physical criterion imported into this information-law owner
- no failed route retires an obligation or changes protected authority

Complete-field family: RETR

171. Index as a source-bound location relation

Family: RETR Obligation: SFT - INFO - OBL - RETR - 001 Claim: SFT - INFO - RETR - INDEX - LOCATION - 001

Question, law and forced result. An index is the complete canonical relation from each generated key to every source-bound location at which that key occurs, with source and version custody retained.

RETR-001 uniquely retains complete-source-location-index, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COARSE - COMPLETENESS - 012

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:3094bedd21d986cf9b7f63ba112c136a6f81d95ea86f3bb2b180f7f79af39b46. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-001 product before observation access.

Boundary: Every positive finite RETR-001 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-001 uniquely retains complete-source-location-index, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__complete-source-location-index__preregistered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:ea75baf8c6768edf0848128baf826c790d00b551131ac89dfa1d110919b1dfef; generality
sha256:8232ef0b952313b96b6157ac66b589829ec8bf13a27c2ee7fa9b4e2554931748.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt
sha256:5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating

proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
 sha256:0cccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256:cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate
 sha256:f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal
 sha256:b78c813353844657db9a42f1ca2ebfbd52d002f7d8401e6eebe76400da048fa1. Source manifest
 sha256:c14a701423f4ae161332055c2ccd95ef968c8992f67cccccab33701c35c19a705. Engine receipt
 sha256:3094bedd21d986cf9b7f63ba112c136a6f81d95ea86f3bb2b180f7f79af39b46 at receipts/engine/model_admitted/SFT-INFO-RETR-INDEX-LOCATION-001-3094bedd21d986cf.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-001: index_location
- exact observation: {"index_keys":3,"location_rows":5,"source_documents":3}
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:b06fa1e67561fe7fe1d9d2dcabddba323a07ff6d57561851364757d0de4f3c0a; empirical validation sha256:c125f0fc3194f7cd9b7a4b077f9734bda78c633faf59c0208120526a58b7d109; external validation
 sha256:80708d6db078c74d49882dfdc0fe3a948bd5a7dbcb293ce8de59f68e83305937; isolation
 sha256:c7e965e5af0a1f2b7646f9dc7e062f7336231355fc6deccde2f58a15084ef520; custody
 sha256:7583bbec50577055b361868aad09752e2543fc1cf0e371453ef85e657d633171.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

172. Exact retrieval and membership

Family: RETR Obligation: SFT-INFO-OBL-RETR-002 Claim: SFT-INFO-RETR-EXACT-MEMBERSHIP-002

Question, law and forced result. Exact retrieval returns all and only locations related to the declared query key; a key outside support returns structural empty-One rather than a numerical value.

RETR-002 uniquely retains complete-key-membership-retrieval, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COARSE-COMPLETENESS-012
- SFT-INFO-RETR-INDEX-LOCATION-001

The current reconciliation binds this obligation to the current model-admitted receipt sha256:9e2ca2e612e6a0462fc5fe76c9e1900cb806254bdbe760e3c34b27bfe387ddbd. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-002 product before observation access.

Boundary: Every positive finite RETR-002 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-002 uniquely retains complete-key-membership-retrieval, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__complete-key-membership-retrieval__preregistered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:dd949413378d464ec9e24568cae52f02ce77ec9de936ee549ae8dc649df4fb36; generality sha256:5b0d5774add1a0d677c53ed07dc85e3969208ecc432d8dc112ce8ddc2800e591.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt sha256:5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256:cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate

sha256:f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal sha256:be3735d6e2aa9a175dbe3cb892795a44844ffcf446e519eb8ec5ba2a6d9157db. Source manifest sha256:07215f7cc21e839c5efce329658a6d748abdf4b2b40c71ea09e7d65bc586d65e. Engine receipt sha256:9e2ca2e612e6a0462fc5fe76c9e1900cb806254bdbe760e3c34b27bfe387ddbd at receipts/engine/model_admitted/SFT-INFO-RETR-EXACT-MEMBERSHIP-002-9e2ca2e612e6a046.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-002: exact_membership
- exact observation: {"absent_query":"empty-One","beta_locations":2,"false_members":0}
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:4b1541b82a76a14ec28d5436ce9abbb8d96279964a3f067546c0c4103234226c; empirical validation sha256:26e6fa95c34cada5aee8d2c5bc4c0c9529fb31d5a158514c4eea399313558c4c; external validation

sha256:116f1f595fbd9d99d00e06c015a98b48725fbb9a1f56053236a1aaf73e3dfb63; isolation sha256:cf9023a1a22c0b6d54702d846a8a7d0fef303dfa260a62f69e5abe6a8f1b7d92; custody sha256:a3c9ed9eff8c9143d372c2917877a214e66374c019b7bcdad0e94f1f903ecfc1.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

173. Inverted and forward index correspondence

Family: RETR Obligation: SFT-INFO-OBL-RETR-003 Claim: SFT-INFO-RETR-FORWARD-INVERTED-003

Question, law and forced result. Forward and inverted indexes correspond exactly when exhaustive inversion and reconstruction preserve every document-key incidence and provenance record.

RETR-003 uniquely retains invertible-forward-inverted-ledger, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COARSE-COMPLETENESS-012
- SFT-INFO-RETR-EXACT-MEMBERSHIP-002

The current reconciliation binds this obligation to the current model-admitted receipt sha256: ae60d6a6c7cc6e341bb8f9d9ce9405d94303cf5d57cf4dbf6503ec3b5259d4ce. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-003 product before observation access.

Boundary: Every positive finite RETR-003 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-003 uniquely retains invertible-forward-inverted-ledger, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__invertible-forward-inverted-ledger__pre-registered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: 87fc813a22bd5c79f879ba7e0409770ef8919ee471de99048d693bf40d54815c; generality sha256: 67ba472e2ccc53c3af95f5888a6e4daae77f418e1a564b774f99a5684b7f1a5.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt sha256: 5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: 0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256: cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate

sha256: f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal

sha256:bc004ada6492c3cf796e0108dba72716d5276aa08bf57c980833d0db5e1cf4c0. Source manifest sha256:f06c5399d17345ba9ea5626ea8e5fcf282eee2218b9adb320d87d8df50930497. Engine receipt sha256:ae60d6a6c7cc6e341bb8f9d9ce9405d94303cf5d57cf4dbf6503ec3b5259d4ce at receipts/engine/model_admitted/SFT-INFO-RETR-FORWARD-INVERTED-003-ae60d6a6c7cc6e34.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-003: forward_inverted
- exact observation: {"documents_reconstructed":3,"incidences_preserved":5}
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:a3c1461b5e11a7bff05606be11b2f03d9467541b7763ab9ec3d4f4fe59522734; empirical validation sha256:1bb35eedfdc1e28f3c5eb0f13787a713d4a0af3b889446367b35585704b1a760; external validation sha256:f929eb09b5ca404f7a4ecdd266d5131de2481fa203d935b9c10c7a1d653640a1; isolation sha256:c23dea137b8957859c515ea276f73ffe4711fa4a657b36e117306d05c22c9023; custody sha256:8d5bb8eee77da1f707057e2b3695e3989bfa4994e44da1da5cbe317a93db897b.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

174. Query as a declared observation

Family: RETR Obligation: SFT-INFO-OBL-RETR-004 Claim: SFT-INFO-RETR-QUERY-OBSERVATION-004

Question, law and forced result. A query is a declared observation over indexed support; its operators, key scope and combination rule must be frozen before result access.

RETR-004 uniquely retains registered-query-observation-map, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COARSE-COMPLETENESS-012
- SFT-INFO-RETR-FORWARD-INVERTED-003

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:1b61a4d68015955f8847b4cf7cd8dbbc819bf41dcd3bfe74499791fa86b56001. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-004 product before observation access.

Boundary: Every positive finite RETR-004 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-004 uniquely retains registered-query-observation-map, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__registered-query-observation-map__preregistered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:5fcdd011266f258ff145a8994725c8631614272b999e628c27013685ae26cdc6; generality
 sha256:1d261a24164aeb3441e0566ada131279d72156c70600f74a3b3e83f4a18b45d4.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt
 sha256:5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
 sha256:0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256:cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate

sha256:f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal
 sha256:7ebb4066b847aeaca6547a60e5d20860123ed48f5434e31aa0cd5db676fd80b7. Source manifest
 sha256:5c120b6cbe80d89ceaf8a1af6b6f2350f5b8ce465415491eb396b1fe17b86dc2. Engine receipt
 sha256:1b61a4d68015955f8847b4cf7cd8dbbc819bf41dcd3bfe74499791fa86b56001 at receipts/engine/model_admitted/SFT-INFO-RETR-QUERY-OBSERVATION-004-1b61a4d68015955f.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-004: query_observation
- exact observation: `{"conjunction_keys":2,"matching_documents":1,"query_frozen":true}`
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e74c477f41ace4d6ac363608f06405934b31ac33cbc47e9522c6f4443980e0ac; empirical validation sha256:9ad53e45eb48b31a677548ffeea767b1a81300bd9629d8bc0ce76a4b29f47587; external validation

sha256:1a7cfff700246c040db9f74ab35b1ccdec3c6708d60dde95fcd5dcaa7601cebf8; isolation sha256:3dff538768f8461d8aee4ac38b56c3933bb921b816f6799242d7fe27f14f1a6a; custody sha256:2343cd8f61c8b8bacdff26d9968de11d6eb38e42f29e2171d923cae794bd77db.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

175. Ranking as an exact finite order

Family: RETR Obligation: SFT-INFO-OBL-RETR-005 Claim: SFT-INFO-RETR-RANKING-ORDER-005

Question, law and forced result. Ranking is an exact finite preorder or partial order over retrieved records, retaining comparison witnesses and never importing an unregistered floating score.

RETR-005 uniquely retains exact-finite-ranking-order, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COARSE-COMPLETENESS-012
- SFT-INFO-RETR-QUERY-OBSERVATION-004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:69dccac7fcf3a3c65390d4e04026a4fbf21ed31f3ef1f54b33280b0263b001e2. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-005 product before observation access.

Boundary: Every positive finite RETR-005 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-005 uniquely retains exact-finite-ranking-order, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__exact-finite-ranking-order__preregistered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:b205c3bb6b232b381a6e44c29cf1b338469c7cf1819d0cc0dc46c6831087e81e; generality
sha256:45506529ab3f5bb57134887a82ec1925672e080ab34166e35a7965d41817a08b.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt
sha256:5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256:cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate
sha256:f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal
sha256:ef5c220f2e40678afd0abca28306bc556d52deec527abfecf7af193c7a7cbb3a. Source manifest
sha256:eaf83d43e760922cff0968b89b194f312a15b04ef7cd0053a3d658b8c4223de4. Engine receipt
sha256:69dccac7fcf3a3c65390d4e04026a4fbf21ed31f3ef1f54b33280b0263b001e2 at receipts/engine/model_admitted/SFT-INFO-RETR-RANKING-ORDER-005-69dccac7fcf3a3c6.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER

- SFT - V1 - V2 - RETRIEVAL - CORPUS

Observed, adverse and boundary records:

- RETR-005: ranking_order
- exact observation: {"ordered_records":3,"positive_rank_levels":3,"ties":0}
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:dc8d2a47cd476b164d3286297e22dc8445d521b39c618b268469347fae622fd1; empirical validation sha256:6e63b9e73d0aab6433c7954624f618773d8dbb6961dfc42c7e6758cd4897770e; external validation

sha256:cb2d2c67adc2a5624a8b090d60e2b137252a08e7beecd3d23ee1103c1c03dff8; isolation sha256:92866174571ac85e3a2034a54875db66be774efef0096e50e120fd946c1ff107; custody sha256:eee8ecf588e1892217e2fba33795acc688bc293f5c172bb13388118acabe1ecd.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

176. Tie and incomparability custody

Family: RETR Obligation: SFT - INFO - OBL - RETR - 006 Claim: SFT - INFO - RETR - TIE - INCOMPARABILITY - 006

Question, law and forced result. Equal ranking evidence forces an explicit tie class; absent comparison evidence forces incomparability. Neither may be broken silently.

RETR-006 uniquely retains tie-and-incomparability-classes, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COARSE - COMPLETENESS - 012
- SFT - INFO - RETR - RANKING - ORDER - 005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:ee4180014b8786d4ecad3cd98ab4e5cd392a1b4a599803c11ac32f3c70b0900b. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-006 product before observation access.

Boundary: Every positive finite RETR-006 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-006 uniquely retains tie-and-incomparability-classes, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__tie-and-incomparability-classes__preregistered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-success-or-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: fd59af0648ab118dca2a349eea199c1804c1793871a9d39f59b0034f0801dfb9; generality sha256: f6443944774076f5d6ac5d6437122f512ce2dc13fdaf03c0771b5538546691fe.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt sha256: 5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: 0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256: cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate sha256: f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal sha256: 59bccbe109c4beaf976685ce3d71a07b3bcc0f1954a0806606d9dea90dd44f90. Source manifest sha256: 7efa296448189fdb69900b08b4e2f7aa5d79e9c2ee1bac8163fd29df0c4c186c. Engine receipt sha256: ee4180014b8786d4ecad3cd98ab4e5cd392a1b4a599803c11ac32f3c70b0900b at receipts/engine/model_admitted/SFT-INFO-RETR-TIE-INCOMPARABILITY-006-ee4180014b8786d4.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-006: tie_incomparability

- exact observation: {"incomparable_control_retained":true,"tie_members":2}
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:b0ca6e78e6faabf5b717a79b3cc8b4e2b62a98a562ab24f2dd7876719661d311; empirical validation sha256:27de40636b156471ed6b5649d8f1e6c04a4a444b480db91fdaefe0a4f69d9ef3; external validation

sha256:bba050bfd46b471d89b0e178723ab5489776e14ad8723ed31d29b32ef54b73b4; isolation sha256:74ae2aa128b00aea9bd1d586f978646a23a7aa9a0791412438a8c5c0e04f495a; custody sha256:554b3f0d7a84f0a327268587568256cd161cacda0f8c1e8f711c5c22eb3d1684.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

177. Relevance as a registered relation

Family: RETR Obligation: SFT - INFO - OBL - RETR - 007 Claim: SFT - INFO - RETR - RELEVANCE - RELATION - 007

Question, law and forced result. Relevance is an independently registered query-record relation and cannot be inferred from retrieval order, popularity or evaluator preference.

RETR-007 uniquely retains query-record-relevance-registry, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COARSE - COMPLETENESS - 012
- SFT - INFO - RETR - TIE - INCOMPARABILITY - 006

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:c550607fac9287a53ec4ced76deff4b98df4b93944267917ad7b4c35a1a65a47. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-007 product before observation access.

Boundary: Every positive finite RETR-007 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-007 uniquely retains query-record-relevance-registry, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__query-record-relevance-registry__preregistered-query-observation__exact-order-class-custody__complete-declared-retrieval-p

roduct__root-bound-forward-forcing__post-registry-exact-observation__finite-success
or-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:3067cc08eeeb5a17ea40565d40d8c79a74ba459db8467595f97b68c8977d9448; generality
sha256:5ddcf185a75a15dec55b3a76fd86d95e58b0636e7473f8ba28c1add104eb71fd.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt
sha256:5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256:cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate
sha256:f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal
sha256:2a5213ffa64b7e0b3a26a18e33af1fd1f5fc7f255f803ed6c4096099d2429c23. Source manifest
sha256:70d4649fd3f412035cccb03cb7e112da09537565acd717b5d8fd4c41f99cd9c4. Engine receipt
sha256:c550607fac9287a53ec4ced76deff4b98df4b93944267917ad7b4c35a1a65a47 at receipts/engine/model_admitted/SFT-INFO-RETR-RELEVANCE-RELATION-007-c550607fac9287a5.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-007: relevance_relation
- exact observation: {"queries":1,"rank_used":false,"registered_relevant_records":2}
- complete family observation custody: 12 records; all rows preserved

- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:8ebd1b582b4ce736532c30e131f3d3bf285dba5546e80920ac1ea24e0ff2ac45; empirical validation sha256:37ec9921b3b26fd8dbdadc50e9aa4b5001ddcbda19a93b7c5233fec6ff73f8d0; external validation

sha256:3cb95b0872f076a5f5256013651304367a0707c41b1c400d2b5c05e1cb890c3c; isolation sha256:f41f82f15378aac639a4f078d38f2b4978fa60b1864e7363bd310377e29e822a; custody sha256:366df9c1bde527a3c9fc43dcf5c7f1259f63851b5622cf4ba5ed409f3e868b6d.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

178. Precision and recall exact-part correspondence

Family: RETR Obligation: SFT - INFO - OBL - RETR - 008 Claim: SFT - INFO - RETR - PRECISION - RECALL - 008

Question, law and forced result. Precision and recall are exact ordered parts: retained relevant retrieved forms over retrieved forms, and retained relevant retrieved forms over registered relevant forms.

RETR-008 uniquely retains retrieved-relevant-exact-parts, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COARSE-COMPLETENESS-012
- SFT-INFO-RETR-RELEVANCE-RELATION-007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:a57b6ba88d5f0fb00942324999058aa7562753f755d1c7ef33c58c5acde62bd5. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-008 product before observation access.

Boundary: Every positive finite RETR-008 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-008 uniquely retains retrieved-relevant-exact-parts, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__retrieved-relevant-exact-parts__preregistered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: f27742cb7c3bc3ff2f882bb3ca71b55c7a459f627c848dcbb2c14abd5ce57018; generality
 sha256: 441acba0b0be0eeee8cfc6ba9c91ba2ad0c819ace00c6e7c5d2a21e871e61c68.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt
 sha256: 5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
 sha256: 0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256: cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate

sha256: f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal
 sha256: 6af76b859b469b777a1c155bbf713a3a1f95b2d0f4b915c68e98aa96a8b7e5ba. Source manifest
 sha256: a535b0fef9412dc33bd5041d741991c35df5d5a9b559bdda3c4b3a5fd38439b0. Engine receipt
 sha256: a57b6ba88d5f0fb00942324999058aa7562753f755d1c7ef33c58c5acde62bd5 at receipts/engine/model_admitted/SFT-INFO-RETR-PRECISION-RECALL-008-a57b6ba88d5f0fb0.json. Closure
 scope `depth_independent`. External status `empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-008: precision_recall
- exact observation: {"intersection":1,"precision_part":[1,2],"recall_part":[1,2]}
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed,

duplicated or opened out of order.

Measurement receipt

sha256: f8c9922cedc99cb568b44bd1f8594d20d92580a646216b886ea1c97f1c5126c3; empirical validation sha256: 4dd3f5599d046412fc55b85760fc18626824e56bf2bba7ed7027cd27891be037; external validation

sha256: bbea7ddd362aed031e831c944787f3cb6ca729effa3a46226cded44205e12aa4; isolation sha256: 0c6c6caee2d20e40b45d83ade5bb789fb620809bf3de7b4d35d293d60ace86b3; custody sha256: 3e375dd2fe3466adb918b3f293cda943a6d82bcfb7cc64d1d24e234f08ed588.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

179. Knowledge organization and taxonomy

Family: RETR Obligation: SFT - INFO - OBL - RETR - 009 Claim: SFT - INFO - RETR - TAXONOMY - 009

Question, law and forced result. A taxonomy is a complete typed parent-child relation with roots, internal classes, leaves and multiple-parent records explicitly retained.

RETR-009 uniquely retains complete-parent-child-taxonomy, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COARSE - COMPLETENESS - 012
- SFT - INFO - RETR - PRECISION - RECALL - 008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 8150318cc24387a8e2d1bebb8908d558604af3ba4c08dbaa128e50c4e44282ad. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-009 product before observation access.

Boundary: Every positive finite RETR-009 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-009 uniquely retains complete-parent-child-taxonomy, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__complete-parent-child-taxonomy__preregistered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: 79f13763bb8275d58697bff5206a826c96994ffbc2343721f6eb420cd1e9c5b8; generality sha256: 45ffbfd8e264abfcac0e644d25f872feb98cd5181250c0292fe94393fe375b9c.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt
sha256: 5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256: cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate

sha256: f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal

sha256: aecbe0cd5a9f8f2feeed957338d016958cfad591d24f29ae9e334dd42de56e7b. Source manifest

sha256: b9d4798c109621b9ce671828f3a29ab239487e2069c906af8162528b18e2380d. Engine receipt

sha256: 8150318cc24387a8e2d1bebb8908d558604af3ba4c08dbaa128e50c4e44282ad at

receipts/engine/model_admitted/SFT-INFO-RETR-TAXONOMY-009-8150318cc24387a8.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-009: taxonomy
- exact observation: {"edges":4,"information_path_depth":2,"roots":1}
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: a3dd3e35b04287ab907512af5709666ed2c0063eb502e7f18af10c26496ccc93; empirical validation sha256: 273fcc35c52a32720d6990caab74e997521294da17ed5cefce8bf78d1bc0b5e; external validation sha256: 37aab54bb6c8738de1e66ecfac45b46448218b5153d5fbde695458e00d901280; isolation sha256: 31277984e773f41bf07f65b5437109890c15201e90f11efd8a1d8b817949546e; custody sha256: 5b399f0aa8f83a76ba17b3489dc6eb93dd822629cc2b0cb430a140469793f220.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

180. Cross-reference and citation graph

Family: RETR Obligation: SFT - INFO - OBL - RETR - 010 Claim: SFT - INFO - RETR - CROSS - REFERENCE - GRAPH - 010

Question, law and forced result. Cross-reference organization is a complete directed source-target graph retaining every edge, path, cycle, unreachable pair and version identity.

RETR-010 uniquely retains complete-directed-reference-graph, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COARSE - COMPLETENESS - 012
- SFT - INFO - RETR - TAXONOMY - 009

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: d2245c1b9469cc2f88abb75687e89ae76f94736b2a9f254372692fd4adba2394. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-010 product before observation access.

Boundary: Every positive finite RETR-010 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-010 uniquely retains complete-directed-reference-graph, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__complete-directed-reference-graph__preregistered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: 15da27ecb13efc54bcbd28a0b4c01e02241f37d5f61650787228b409a4422c50; generality sha256: 1dad3bb0dcd35c6c5877a7e584e6662eaa391114d9f001c6afe786883084cbd5.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted

magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt
sha256:5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256:cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate

sha256:f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal

sha256:4dff3c8d35233d1fb2c4cceb1d0dfb760a0acd2a557c876cd0689a6f827e7b99. Source manifest

sha256:39ffee562fb1c0c0331ad12a7bd343c54720c7841f891d1990c08925accbd6e2. Engine receipt

sha256:d2245c1b9469cc2f88abb75687e89ae76f94736b2a9f254372692fd4adba2394 at receipts/engine/model_admitted/SFT-INFO-RETR-CROSS-REFERENCE-GRAPH-010-d2245c1b9469cc2f.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-010: reference_graph
- exact observation: {"citation_edges":4,"cycle_control_retained":true,"source_target_paths":2}
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:87cb2535d858b65a74eb827fbdde780652f9dd68bd2a987e58e71d64a52ef51a; empirical

validation sha256:fd7c2bdb9967aa348a0ac7b237b1d2ea5ad7228c1afb896bb6953361e01aacf6;

external validation

sha256:8ab6ab444ee3bffdaf76d27ed6111154a6167b8c745308c5b2f4cd40729fa481; isolation
 sha256:07cad0df566b77e402a8f4af7a21e51e81e7b80db7b35dc2b22eeeee68970f91; custody
 sha256:d39b4fdbb5924db6097a944de9fc58a87dbe5a19ad6af7af747493d1a0d41bb9.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

181. Update and stale-index boundaries

Family: RETR Obligation: SFT-INFO-OBL-RETR-011 Claim: SFT-INFO-RETR-UPDATE-STALE-BOUNDARY-011

Question, law and forced result. An index is current only for the exact source version from which it was generated; any source-incidence change makes the prior affected index stale and forces halt or reconstruction.

RETR-011 uniquely retains source-version-index-consistency, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-COARSE-COMPLETENESS-012
- SFT-INFO-RETR-CROSS-REFERENCE-GRAPH-010

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:d6fd5ac22dd4ce19e5594b4f56e332bb946f3a7e1beb405baffea696a369b11d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-011 product before observation access.

Boundary: Every positive finite RETR-011 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-011 uniquely retains source-version-index-consistency, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__source-version-index-consistency__preregistered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:93f9ba59c0354490eb7763b159fafbfc31e88c8be0041222e3f9e8f279467f5; generality
 sha256:a91abd9f39714f85fd54a34997115b9376923ba354e2798b2aab511a301f987a.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt
sha256: 5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256: cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate

sha256: f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal

sha256: 21c72491e7413cf8777d8fb06d7bd7e3746f820e12949e36ebd622b71b73a05b. Source manifest

sha256: eae4bd5d1941852a3aa552f99ffe034334e0ecf8866dc966c8831bc722ef2ed4. Engine receipt

sha256: d6fd5ac22dd4ce19e5594b4f56e332bb946f3a7e1beb405baffea696a369b11d at receipts/engine/model_admitted/SFT-INFO-RETR-UPDATE-STALE-BOUNDARY-011-d6fd5ac22dd4ce19.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-011: stale_boundary
- exact observation: {"new_alpha_locations":3,"prior_alpha_locations":2,"stale_difference":1}
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: aa74add7de8db9f285b2c7e23bf1303843dfa0d2b7917f412f213cd92368eed6; empirical

validation sha256: 8c49df7c817719a1b68f2e901181ffbb2922b6c9fafc1073c635660096539492; external validation

sha256: c36263e72b1c45c3340a904f9ca85d5c45d55d039918ff3886f0ca431db15a3c; isolation

sha256: d1e47508a9581b82bd569d8151331cf8dccdd591093d2d284f0d26d5fa1c5c66; custody

sha256: ee69edb3e35027acd1da4ee15d5714bad9d4b1083250b09329c4af1f242e1bcc.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

182. Retrieval-family completeness certificate

Family: RETR Obligation: SFT - INFO - OBL - RETR - 012 Claim: SFT - INFO - RETR - COMPLETENESS - 012

Question, law and forced result. Retrieval-family completeness is the one-to-one reconciliation of all twelve frozen indexing, query, ranking, relevance, organization and update obligations.

RETR-012 uniquely retains twelve-retrieval-obligation-ledger, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - COARSE - COMPLETENESS - 012
- SFT - INFO - RETR - UPDATE - STALE - BOUNDARY - 011

The current reconciliation binds this obligation to the current model-admitted receipt sha256:31e9b8db1a4f93fc1b36ea5adc18f7a85c5153ad3f6e81ebeca87da8a4c8955c. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis RETR-012 product before observation access.

Boundary: Every positive finite RETR-012 record, key, location, query, order class, reference path and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- RETR-012 uniquely retains twelve-retrieval-obligation-ledger, complete retrieval custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-source-record-support__twelve-retrieval-obligation-ledger__pre-registered-query-observation__exact-order-class-custody__complete-declared-retrieval-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:b4214caf8b576d9eb6079873a21cea343031316deac4f6210879cf08578c69a9; generality sha256:79e41a504623575be1f5e0f717aa1b5952d57dbc6247e9f3f0b06d113be59f5b.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing records change retrieval results.; receipt sha256:5018e03b5b1241a73bc438b80405c50ab75701267f25867053679a9af85cc287.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden document, omitted location, silent tie break, fitted score or stale source version; no semantic meaning or learning objective imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt
sha256:0ccc6a85d7772730a4ddb619eec0eea95d60df1641596dab486cd025f784608b.

Independent reconstruction and engine admission. Independent implementation

sha256:cff62a3b7175f7bc46eb0e2f9b1439e6c88ebcde14ada5fcac993d88d488d6ba regenerated the declared census and sole survivor; independent certificate
sha256:f9ec0865fb6ee9f3f225c5bc85d028f68082e1dd1c54a90e2d787b4c4895a72d. Derivation seal
sha256:493d2d664c871766e8d90114fa3e029b4208c24632825c0442a5378c1bcb4731. Source manifest
sha256:2f64b4a5733b56289f4de79136010a04eba673280ebfbb117439ea090c0ea9dd. Engine receipt
sha256:31e9b8db1a4f93fc1b36ea5adc18f7a85c5153ad3f6e81ebeca87da8a4c8955c at receipts/engine/model_admitted/SFT-INFO-RETR-COMPLETENESS-012-31e9b8db1a4f93fc.json. Closure scope
depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-RETRIEVAL-OBSERVER
- SFT-V1-V2-RETRIEVAL-CORPUS

Observed, adverse and boundary records:

- RETR-012: retrieval_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":12,"omitted_owners":0,"registered_obligations":12}
- complete family observation custody: 12 records; all rows preserved
- complete source, key, location, query, result, order-class, relevance and reference-edge enumeration supplies the observation; no imported relevance judgment, fitted rank score, silent tie break or stale source version enters

Falsification condition: Reject if the value-free registry, complete source-index-query-result census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:5e2574a8b7d1b1af7dc830cec830d8e9325acddca8d3a48f7c9ff11e6e43c41b; empirical validation sha256:dd5e46269058dfb1da095ed58913ecf6a4c445c63353fcd9d866204a973aed0f; external validation
sha256:736e082fee98afed0e5b8fdc42714b767471b7012ed4c9251a8c9944534b1ecf; isolation
sha256:e2bea1ccf13aba36829ea42b378918c9cedd04569445cf3bf99c873de682230c; custody
sha256:cb3fba713ec8527236a5bce9128bfd206fcacaf3358f5948b85e56f145e2316b.

Prohibited inputs.

- no axiom, imported relevance judgment, stochastic ranking model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar

- no hidden document, omitted location, silent tie break, fitted score or stale source version
- no semantic meaning or learning objective imported into the information-law owner
- no failed route retires an obligation or changes protected authority

Complete-field family: INFER

183. Inference as support restriction

Family: INFER Obligation: SFT - INFO - OBL - INFER - 001 Claim:

SFT - INFO - INFER - SUPPORT - RESTRICTION - 001

Question, law and forced result. Inference restricts a complete registered hypothesis support to every and only hypothesis compatible with the retained evidence record; no posterior distribution is presumed.

INFER-001 uniquely retains evidence-compatible-support-restriction, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RETR - COMPLETENESS - 012

The current reconciliation binds this obligation to the current model-admitted receipt sha256:5a05c6e4b4280b01bed0a1dc1abc4fd89a4b184670827653acc73bf724e3a546. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-001 product before observation access.

Boundary: Every positive finite INFER-001 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-001 uniquely retains evidence-compatible-support-restriction, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__evidence-compatible-support-restriction__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:34d48b2d1eea12db78a4cb69d5e20177d0aa471d4c5712024e1cd61a41e0a7de; generality sha256:fa40d576858bc08eb5295b009a528b0aee4f06f3e1ae25d5eda86e027c5bc639.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt sha256:ad8b63321fd1752e677525e3865b7139ffdd8913fff7f7e92dcdbf5bc72e81f7.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256:22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256:c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate

sha256:8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal sha256:fc3bd7fc3953c604f2c95e7d4ab35f58c497f5fd14ce38a51f2e09ebbc7aea. Source manifest sha256:1a6d7e0ead62dbc90d29e2b22d3152b25889556e2f5cf473b8ee24a1039f4d5e. Engine receipt sha256:5a05c6e4b4280b01bed0a1dc1abc4fd89a4b184670827653acc73bf724e3a546 at receipts/engine/model_admitted/SFT-INFO-INFER-SUPPORT-RESTRICTION-001-5a05c6e4b4280b01.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENCE-OBSERVER
- SFT-V1-V2-INFERENCE-CORPUS

Observed, adverse and boundary records:

- INFER-001: support_restriction
- exact observation: {"initial_hypotheses":3,"retained_hypotheses":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e87c01a24ba73eb44eb944b61ddb3447d2b9a7782936b3101baaba4b6ef61; empirical validation sha256:f13c80d2e3ae731be7543aa80b02723d538de74b0a6e14f216c2b82a5adffa9b; external validation

sha256:edc156cdd77bcd7844ac1e3d7424bec22b4f0f0ad3fbc0407c56dfffb93f01ed3; isolation sha256:55238085d43789fd5f6875f1c3952aca8ac5044962d8a72bb27c036b4afef554; custody sha256:4343c2b2929df44ff1c628496e259d3bff6b274aad64d9d435b680742d279be0.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class

- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

184. Detection as observation-class decision

Family: INFER Obligation: SFT - INFO - OBL - INFER - 002 Claim: SFT - INFO - INFER - DETECTION - 002

Question, law and forced result. Detection assigns an observed record to a preregistered observation class by exact membership and retains unresolved classes rather than importing a fitted score.

INFER-002 uniquely retains declared-observation-class-detection, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RETR - COMPLETENESS - 012
- SFT - INFO - INFER - SUPPORT - RESTRICTION - 001

The current reconciliation binds this obligation to the current model-admitted receipt sha256:99b0c51a3c228b310adea6ba2f0acfb43cb5416bab58876bc797ab2219ed9ea1. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-002 product before observation access.

Boundary: Every positive finite INFER-002 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-002 uniquely retains declared-observation-class-detection, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__declared-observation-class-detection__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:2ae315356ec7f92025d6bea015d7491916f9919852dcfdd85f3ebdac86d9a1c5; generality sha256:409de789dd3e27a2c718922de135d750adb0e3f8fdd445c9d51efe8051f63615.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt sha256:ad8b63321fd1752e677525e3865b7139ffdd8913ffff7f7e92dcdbf5bc72e81f7.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256: c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate

sha256: 8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal sha256: 355fda7287bfc0b3e5e868d569e1d6c2cc3d4a58d0b65975222c6a85ae59ee91. Source manifest sha256: e0b3f767ed0a95062d9816434acfe247ab176aaa48002fd9f55b5a910040a38f. Engine receipt sha256: 99b0c51a3c228b310adea6ba2f0acfb43cb5416bab58876bc797ab2219ed9ea1 at receipts/engine/model_admitted/SFT-INFO-INFER-DETECTION-002-99b0c51a3c228b31.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENCE-OBSERVER
- SFT-V1-V2-INFERENCE-CORPUS

Observed, adverse and boundary records:

- INFER-002: detection
- exact observation: {"absence_forms":1,"observations":3,"signal_forms":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 4a97cc72246d77688ca8dbccbe4748853755ed20f7e35f5292dec682150d08f; empirical validation sha256: b6018be60ca3190adc2434aee2a7280d0edbb118b96a962099ca3a73935de248; external validation

sha256: a709560968269fcd53ff5e50e74a3c674c1a83806735d5173c189404e7c1805d; isolation sha256: 86f13a177a7a42978a2d4595f47e3a3b833103f251dbe38fd8bf2b30c5a56ae2; custody sha256: 90f7be561f7cb899ca119851c640d51772a67a812e317ebf0a3c94c0d692515b.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

185. Estimation as retained representative selection

Family: INFER Obligation: SFT - INFO - OBL - INFER - 003 Claim: SFT - INFO - INFER - ESTIMATION - 003

Question, law and forced result. Estimation selects a representative only when the declared loss-free relation forces one member; otherwise the complete compatible class remains unresolved.

INFER-003 uniquely retains unique-or-unresolved-representative, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RETR - COMPLETENESS - 012
- SFT - INFO - INFER - DETECTION - 002

The current reconciliation binds this obligation to the current model-admitted receipt sha256:cca1bb73b97dbc67d792394b0b0579f60d7c71bf84ff0a6dd32a0011aac46ac7. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-003 product before observation access.

Boundary: Every positive finite INFER-003 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-003 uniquely retains unique-or-unresolved-representative, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__unique-or-unresolved-representative__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:38bc0d7b38f5a8daf66852ae4848299676c1de9b968dc73e753061b4c12a4831; generality sha256:4f3e3b0a32509075b788d144825623355f68459019cd46dffc39afbba00c14e7.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt sha256:ad8b63321fd1752e677525e3865b7139ffdd8913fff7f7e92dcdbf5bc72e81f7.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective,

semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt
 sha256: 22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256: c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate
 sha256: 8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal
 sha256: 7a8e407ca363f08f194588430443860985b10d4c178b6d009ef6a7e032e1fa72. Source manifest
 sha256: b3ae4e2dbc8ec7c2929ec0927b0280da88d4f9f6b34922a65f62f780971524d7. Engine receipt
 sha256: cca1bb73b97dbc67d792394b0b0579f60d7c71bf84ff0a6dd32a0011aac46ac7 at
 receipts/engine/model_admitted/SFT-INFO-INFER-ESTIMATION-003-cca1bb73b97dbc67.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERERENCE-OBSERVER
- SFT-V1-V2-INFERERENCE-CORPUS

Observed, adverse and boundary records:

- INFER-003: estimation
- exact observation: {"singleton_selected":true,"two_member_class_unresolved":true}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: ad0126f973cbc48496155cef160038b932ebda296cec5ec2df09c69d71929671; empirical validation sha256: 1e01939636c167029093b3b555c33ba2bf9ee574a2c47cbdf31d78ef8a85fb5f; external validation
 sha256: 2867e9cd8b74014c7f1208989b56a97dcc7d4803716f1bce8d7260c30aae635d; isolation
 sha256: 02d1a32f705cb28a123df0e95e7703ef44adef7d967d2356493ca07259cbd3c6; custody
 sha256: 1ee1f06c687d09da8c7bbf528c807140cddea9c7acd95de9f8dee28b0c810692.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

186. Filtering as sequential record update

Family: INFER Obligation: SFT-INFO-OBL-INFER-004 Claim: SFT-INFO-INFER-FILTERING-004

Question, law and forced result. Filtering is the ordered restriction of current support by each newly observed record, retaining every intermediate support and evidence identity.

INFER-004 uniquely retains ordered-evidence-support-update, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RETR-COMPLETENESS-012
- SFT-INFO-INFER-ESTIMATION-003

The current reconciliation binds this obligation to the current model-admitted receipt sha256:8d8f08164eb9cecfaf066daf07808217a325464a55a8a6ecd6d666c3917940b28. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-004 product before observation access.

Boundary: Every positive finite INFER-004 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-004 uniquely retains ordered-evidence-support-update, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__ordered-evidence-support-update__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:61f276453f0a192e1cd70ae01795b35b4cd05bc0f2041f53b8bcb0d1e9908349; generality sha256:bab4dd7427e44284050b2d8aa8985ad7592aac616f848d23c44a87d12eca5380.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt sha256:ad8b63321fd1752e677525e3865b7139ffdd8913fff7f7e92dcdbf5bc72e81f7.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt sha256:22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256:c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate

sha256:8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal sha256:f2a14d6e471245b6eb242185504bc6d2c5e60f62f4582988fd0f184755c7e49b. Source manifest sha256:2bf3a7c166c0dccc586d96aa3e436529245e276cc87a1cf669173b884c9a663e. Engine receipt sha256:8d8f08164eb9cecf066daf07808217a325464a55a8a6ecd6d666c3917940b28 at receipts/engine/model_admitted/SFT-INFO-INFER-FILTERING-004-8d8f08164eb9cecf.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENC-OBSEVER
- SFT-V1-V2-INFERENC-CORPUS

Observed, adverse and boundary records:

- INFER-004: filtering
- exact observation: {"evidence_steps":2,"intermediate_support":2,"terminal_support":1}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:203f68cb31e71fd40554cdf8ba3866257f254519a9ebfec1be5e673cb68b3131; empirical validation sha256:34443acfcdfc84d0d115be36eddc94136ba3b13e827fd8d1b41d5a686bb8b3be; external validation

sha256:d217d74d14aa8159b9d9bb3072a87189d5001f23cc6f22f8507e9a1799dfaf25; isolation sha256:3ad650da415cb288bb7eea086be7145d96dc89168afa36717728be6135bbc2ee; custody sha256:09d12c9825c22c7b9a7bd1567ab98009a75272e5f9cb8e048286227703508767.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

187. Smoothing as whole-record reconstruction

Family: INFER Obligation: SFT-INFO-OBL-INFER-005 Claim: SFT-INFO-INFER-SMOOTHING-005

Question, law and forced result. Smoothing reconstructs earlier support using the complete retained record, without erasing the earlier forward-only class or importing future evidence into its historical trace.

INFER-005 uniquely retains whole-record-support-reconstruction, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RETR-COMPLETENESS-012
- SFT-INFO-INFER-FILTERING-004

The current reconciliation binds this obligation to the current model-admitted receipt sha256:51b87b53d8caff87ca4d4743ed0b0af0c9da636c6f778fd36b021491d08f9ae3. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-005 product before observation access.

Boundary: Every positive finite INFER-005 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-005 uniquely retains whole-record-support-reconstruction, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__whole-record-support-reconstruction__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:83d8fd2a511c4789f60450a7a44ba868cb1269c81f468d2fdaa43f396d573c25; generality sha256:13c11bbb1b64a0323528db170b4967a054a56c6a585e60160ae6a709e1ccd477.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt sha256:ad8b63321fd1752e677525e3865b7139ffdd8913fff7f7e92dcdbf5bc72e81f7.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt sha256:22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256:c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate sha256:8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal

sha256:2b17ede82a699420edeb3a8d306c7f4a3331b77b480ee0a96e2aea9198531400. Source manifest sha256:8bda310ed9802db93a95ccc7cf88f49d5c5fb9c9a722faa4c6ab7ab45aa5ba57. Engine receipt sha256:51b87b53d8caff87ca4d4743ed0b0af0c9da636c6f778fd36b021491d08f9ae3 at receipts/engine/model_admitted/SFT-INFO-INFER-SMOOTHING-005-51b87b53d8caff87.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENC-OBSEVER
- SFT-V1-V2-INFERENC-CORPUS

Observed, adverse and boundary records:

- INFER-005: smoothing
- exact observation: {"earlier_class_retained":2,"terminal_reconstruction":1}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:758da05034f3365cf0317ced4428073c482bee965545dba703a9fbb519c2f9c1; empirical validation sha256:2d51712923bd59d98b4a2d0064ecd3ef1425c762a17f8372112272a7e0f58b00; external validation

sha256:5f0034b6f232df34c3587ae8ed8e9f719be9532cde3382e9a8bb73c1de561ceb; isolation sha256:fc7a2a3a782f8ef5700462478cc69aeacf45d4cfd6fc7169bd783b63de0fd02; custody sha256:ae838a26f8d37ee9b7675bc7b6da1f64c7ca546c68b00da1e0f2cb702aa5bfc7.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

188. Prediction as registered transition support

Family: INFER Obligation: SFT-INFO-OBL-INFER-006 Claim: SFT-INFO-INFER-PREDICTION-006

Question, law and forced result. Prediction is the complete successor support generated by a frozen state-transition relation; it is a standing support claim until a later observation is opened.

INFER-006 uniquely retains registered-successor-support, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RETR-COMPLETENESS-012

- SFT-INFO-INFER-SMOOTHING-005

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:de5f48db576b5a0f6e31af57fab4fe81b4fbe5022603a58862a466d4b5dba69e. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-006 product before observation access.

Boundary: Every positive finite INFER-006 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-006 uniquely retains registered-successor-support, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__registered-successor-support__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:f439cf1d8c0e69e7ff0a9e737dad51adf486e15820e5fbc9d7e8c9a9b3c98ee; generality
sha256:83d37a86720fa62af126a4ffbe4df9ec30be2f8546b1e167320106742eb0c215.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt
sha256:ad8b63321fd1752e677525e3865b7139ffdd8913ffff7f7e92dcdbf5bc72e81f7.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256:22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256:c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate

sha256:8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal

sha256:1fafac9c52187ceb3eb79767ac59583331116644e4d85c91eeb0e7f14473637b. Source manifest

sha256:58a83429d70b4fd8c785bbfd1b7c691f43599613171e1d5e822f13d43ba6ed2d. Engine receipt

sha256:de5f48db576b5a0f6e31af57fab4fe81b4fbe5022603a58862a466d4b5dba69e at
receipts/engine/model_admitted/SFT-INFO-INFER-PREDICTION-006-de5f48db576b5a0f.json.

Closure scope `depth_independent`. External status `empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENCE-OBSERVER
- SFT-V1-V2-INFERENCE-CORPUS

Observed, adverse and boundary records:

- INFER-006: prediction
- exact observation: `{"registered_states":2,"standing_support":true,"successor_rows":3}`
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 451a0d1d4a1de906b8307ad404c459d2a2c98da44873536aee6f77de606d4291; empirical validation sha256: 3324653348aaa37445f8cfffac928981efa6a3c1815a33a99b6b309aff752203d; external validation

sha256: 305e3742ad86e19dc4995b72dc986172f650712e7553bfac9a27734a39135649; isolation sha256: cdeb06b954ff00cdf6316216090aa2a3baa748b58fb56c96d45bf65c96a90040; custody sha256: 81cd248b9a9ba4849cef16f961ba0dc1b44ba1847e2570497935268e5752cdc4.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

189. Evidence accumulation

Family: INFER Obligation: SFT-INFO-OBL-INFER-007 Claim: SFT-INFO-INFER-EVIDENCE-ACCUMULATION-007

Question, law and forced result. Evidence accumulation is exact successive intersection of compatible supports; each record may retain or close distinctions but cannot silently add hypotheses.

INFER-007 uniquely retains monotone-compatible-support-intersection, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RETR-COMPLETENESS-012
- SFT-INFO-INFER-PREDICTION-006

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:3dfafa28b5e96b811e725a521080c4ec557601297fad97f08d5650808c296373. A missing
 dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-007 product before observation access.

Boundary: Every positive finite INFER-007 hypothesis, evidence item, observation class, decision row, error row and
 registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly
 one survives and 255 are eliminated. The forced coordinates are:

- INFER-007 uniquely retains monotone-compatible-support-intersection, complete
 inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__monotone-compatible-support-int
 ersection__source-bound-evidence-record__structural-support-decision__complete-decl
 ared-inference-product__root-bound-forward-forcing__post-registry-exact-observation
 __finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:1259abc5ba71481c330c590430ccc8977a2d293b68e45639b280c8634e11bc06; generality
 sha256:3f9f63a69f830974094b4fb739020a8a738b44e400807d6393621439db05f250.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained
 distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit
 downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted
 magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT
 numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed:
 Omitted hypotheses falsify inference closure.; receipt
 sha256:ad8b63321fd1752e677525e3865b7139ffdd8913ffff7f7e92dcdbf5bc72e81f7.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs
 from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated
 product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no
 axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence
 or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no
 hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective,
 semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes
 protected authority; receipt
 sha256:22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256:c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the
 declared census and sole survivor; independent certificate
 sha256:8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal
 sha256:507b302adb0d00bf3d859447ba53c8dd64c16ce191de679c175644ae84fcdc58. Source manifest
 sha256:40496420a3d625f814d569a436b71d77e67cd26331f79d7383dfcc0fe064e30e. Engine receipt
 sha256:3dfafa28b5e96b811e725a521080c4ec557601297fad97f08d5650808c296373 at receipts/e
 ngine/model_admitted/SFT-INFO-INFER-EVIDENCE-ACCUMULATION-007-3dfafa28b5e96b81.json
 . Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENCE-OBSERVER
- SFT-V1-V2-INFERENCE-CORPUS

Observed, adverse and boundary records:

- INFER-007: evidence_accumulation
- exact observation: {"first_support":2,"initial_support":3,"second_support":1}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 275e4983bc5c7538e2c24188d806ed3b256585d618fce000259c67c1e5d3703b; empirical validation sha256: 5242c91b2ca55fc1d1991f42c7be5757a7ef4e75a0766cfb87d3d43c0ba06d99; external validation sha256: 634499205b0ff2a54fdac2447e853efe938fcddd182982f39faaec08931a638b; isolation sha256: 40841709698b8e556162b2b71a010fa80da8a8cacf902fbd7e7dad8aee702598; custody sha256: b728fd16cc38d9122f8429c13ff0063b92279ff3a9e8ca9d5dc3619a79f508e4.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

190. Likelihood correspondence on exact support

Family: INFER Obligation: SFT-INFO-OBL-INFER-008 Claim:

SFT-INFO-INFER-LIKELIHOOD-CORRESPONDENCE-008

Question, law and forced result. Likelihood correspondence begins with exact evidence-compatible multiplicities; normalized stochastic likelihoods are typed translations only when their measure law is separately forced.

INFER-008 uniquely retains compatibility-multiplicity-correspondence, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RETR-COMPLETENESS-012
- SFT-INFO-INFER-EVIDENCE-ACCUMULATION-007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 4439c042519652aa49a8c876702cf8815d7140074c716c2acea73d38259e40d4. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-008 product before observation access.

Boundary: Every positive finite INFER-008 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-008 uniquely retains compatibility-multiplicity-correspondence, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__compatibility-multiplicity-correspondence__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:13a9d81f89ac8926a915ea12fd224ac229e86ec6f7005acc84402b3a84c2de82; generality sha256:e141acc10e9976f57586da43c72b037171b255969af6b49ad9751e30a4359ca7.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt sha256:ad8b63321fd1752e677525e3865b7139ffdd8913ffff7f7e92dcdbf5bc72e81f7.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt sha256:22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256:c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate sha256:8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal sha256:c7fb72a7678a568f673f62efe007232d53d040ff9ad373a20c80b58843a28f8a. Source manifest sha256:1bcaef8a9265f2b792b60bab7edea32c4676504c47a8bea33ada2a3ff83e66ef. Engine receipt sha256:4439c042519652aa49a8c876702cf8815d7140074c716c2acea73d38259e40d4 at receipts/engine/model_admitted/SFT-INFO-INFER-LIKELIHOOD-CORRESPONDENCE-008-4439c042519652aa.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENCE-OBSERVER
- SFT-V1-V2-INFERENCE-CORPUS

Observed, adverse and boundary records:

- INFER-008: likelihood_correspondence
- exact observation: {"compatibility_multiplicities":[2,1,1],"normalization_imported":false}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:13b1b7026f8e1978563d98f6cfafaa86839d31fae17b46be92e41d7d993c423f; empirical validation sha256:f4d5b9c8d2e0c6245bff6ac08726acf2f6e3537ac9e6a9a30b61ff0a57c14093; external validation

sha256:05f899e598ccbc2e01b669a4c41705c480b089595b928ffda6bd6cf42086c42c; isolation sha256:6a8b46c11bfd4a3f8f1d28451e71863bd26c84e2d49cf0c36bb60b70adec58bf; custody sha256:9f1474c5743b3d6c22ecdf6bb826e74a1e8f587d2dc1be4dec9179db9c1affa3.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

191. Decision threshold without fitted scalar

Family: INFER Obligation: SFT-INFO-OBL-INFER-009 Claim:

SFT-INFO-INFER-STRUCTURAL-THRESHOLD-009

Question, law and forced result. The parameter-free decision boundary is structural: decide only at a singleton compatible class, otherwise retain every alternative.

INFER-009 uniquely retains singleton-structural-decision-boundary, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-RETR-COMPLETENESS-012
- SFT-INFO-INFER-LIKELIHOOD-CORRESPONDENCE-008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:72f4c8072030e763982c4166aeb472fb02467bc3d41e178b8595b67b1119a8e5. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-009 product before observation access.

Boundary: Every positive finite INFER-009 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-009 uniquely retains singleton-structural-decision-boundary, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__singleton-structural-decision-boundary__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:dfc79c14e6e3f59eb074cde6613455e0732a35c80e562cc485bcd45abfa8e55f; generality
sha256:e5b2348abaf62cd3ecac8063353934e2f54469e9bd583c061a9f5a6a37fb115f.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt
sha256:ad8b63321fd1752e677525e3865b7139ffdd8913ffff7f7e92dcd5bc72e81f7.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256:22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256:c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate
sha256:8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal
sha256:48cee54489d2f5df4c0265863b500b240b06231b5e2a7ef66801331ca75685d8. Source manifest
sha256:02d66619730c86159484b8931ba525718e4abddb0095071721f120fc811c41e8. Engine receipt
sha256:72f4c8072030e763982c4166aeb472fb02467bc3d41e178b8595b67b1119a8e5 at receipts/engine/model_admitted/SFT-INFO-INFER-STRUCTURAL-THRESHOLD-009-72f4c8072030e763.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT -V3 - INDEPENDENT - EXACT - INFERENCE - OBSERVER
- SFT -V1 -V2 - INFERENCE - CORPUS

Observed, adverse and boundary records:

- INFER-009: structural_threshold
- exact observation: {"fitted_scalar_used":false,"singleton_decisions":1,"unresolved_class_size":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:19265441cd49cbf8bf09513b0bb86485eec8ced0a588ea2bde48d4cb2887785; empirical validation sha256:4d7c819ac0354a1776eca05c3118cd02009533941dccad9ea6e6f140a62e65c3; external validation

sha256:4bd5e9abea7b840ed741cf8cf148b3cbe974cc7085048d9b67a6ce7cebf88c8c; isolation sha256:1a324bc0d99be9127cfd43d0446ca15a90d045b895a9e8dd4e7d42cf7271e1ac; custody sha256:a158efd4798fb4d1b37eba8aa367e54709dadab6f6a574d6369ae94c76575320.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

192. False-positive and false-negative custody

Family: INFER Obligation: SFT - INFO - OBL - INFER - 010 Claim: SFT - INFO - INFER - ERROR - CUSTODY - 010

Question, law and forced result. Validation retains true-positive, false-positive, false-negative and true-negative rows separately for every registered decision and observation pair.

INFER-010 uniquely retains complete-decision-outcome-ledger, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RETR - COMPLETENESS - 012
- SFT - INFO - INFER - STRUCTURAL - THRESHOLD - 009

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:29a3735b9dd799d72a1f1513f6de32df493747f163a0a572c97d0f8dbfffc0716. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-010 product before observation access.

Boundary: Every positive finite INFER-010 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- **INFER-010** uniquely retains complete-decision-outcome-ledger, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__complete-decision-outcome-ledger__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:e23f06d48faeba3a5ea94351b7dfe876534d76b310f4f4d359bdaa6dc7f4a02a; generality

sha256:3ffd9df669c6abf45d8c2aaf658cb814174511a6f365ee17bc145386256d26b2.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt
sha256:ad8b63321fd1752e677525e3865b7139ffdd8913fff7f7e92dcdbf5bc72e81f7.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256:22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256:c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate

sha256:8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal

sha256:5f92866f7d4997e4609f297db8bc099c1854190d805f1233861fbc81de37e282. Source manifest

sha256:ef9db9bb577c07b45ba86257d0ebd91b86050db7e0b35a1fc95b243baf4578b5. Engine receipt

sha256:29a3735b9dd799d72a1f1513f6de32df493747f163a0a572c97d0f8dbfffc0716 at receipts/engine/model_admitted/SFT-INFO-INFER-ERROR-CUSTODY-010-29a3735b9dd799d7.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENC-OBSEVER
- SFT-V1-V2-INFERENC-CORPUS

Observed, adverse and boundary records:

- INFER-010: error_custody
- exact observation: {"false_negative":1,"false_positive":1,"true_negative":1,"true_positive":1}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:2abc018b09744998bcde6c2a193252c80dc537e23fbb94be19edd9c343ee962d; empirical validation sha256:938233e26dd1c92d3d005cf9642517f4e2ed12f43706268df8d3791e7682d418; external validation sha256:99642cb878c208301df78d57a9f8c10c09815793c609fd6142a504a8efaea7f2; isolation sha256:7380e9ebd0ced00dac921c391957ce1d0a8aced75b921e8f26b0cf4f5d95df8f; custody sha256:e72fc3ed96131872e0ac3ec04230d566ff3371a02e63026feec291f2f4596a91.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

193. Identifiability and observational equivalence

Family: INFER Obligation: SFT - INFO - OBL - INFER - 011 Claim: SFT - INFO - INFER - IDENTIFIABILITY - 011

Question, law and forced result. Hypotheses are identifiable exactly when the registered observation family separates them; indistinguishable hypotheses remain one explicit equivalence class.

INFER-011 uniquely retains complete-observation-equivalence-classes, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RETR - COMPLETENESS - 012
- SFT - INFO - INFER - ERROR - CUSTODY - 010

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:85783c389a0eb5a4026e0333b60c7ba90abf6912f7fc20a4fe70c8b74d84467c. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-011 product before observation access.

Boundary: Every positive finite INFER-011 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-011 uniquely retains complete-observation-equivalence-classes, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__complete-observation-equivalenc
e-classes__source-bound-evidence-record__structural-support-decision__complete-decl
ared-inference-product__root-bound-forward-forcing__post-registry-exact-observation
__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:24466c957198dbef465583795fa07ffddc21487a04017b2b4e39bb5796379dad; generality
sha256:7a231a34245b4300f95a7602dbc1c24f03bab6d12f2c1535218ba89588158974.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt
sha256:ad8b63321fd1752e677525e3865b7139ffdd8913fff7f7e92dcd5bc72e81f7.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256:22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256:c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate
sha256:8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal
sha256:453c7f6946dbbe2fc6d0b19a50b59f72dbc8474fe1bf57392e55baf977076263. Source manifest
sha256:1e4ad07066b1f220bae8da96c4d1dd6d4aac9719a809cdc322dc13c63c8f61fd. Engine receipt
sha256:85783c389a0eb5a4026e0333b60c7ba90abf6912f7fc20a4fe70c8b74d84467c at receipts/e
ngine/model_admitted/SFT-INFO-INFER-IDENTIFIABILITY-011-85783c389a0eb5a4.json. Closure
scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENC-OBSEVER
- SFT-V1-V2-INFERENC-CORPUS

Observed, adverse and boundary records:

- INFER-011: identifiability
- exact observation: {"equivalent_hypotheses_before":2,"identifiable_after":1}
- complete family observation custody: 14 records; all rows preserved

- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: fd60738c8af3dfacbd971b06d784cb6f4f9e0ecd0c6e6b3c3f623e7e5e6a9ecb; empirical validation sha256: 1bf66fd17cf6fa466e2aa420a8fb95e200fbbfac9d9a58aeb9578dc456ffd37e; external validation

sha256: 054b43b733905c0a325db2016e633b501a95bde41c61140dedab4badaea236d9; isolation sha256: 5848328775db2757e6306470ae6bc92d4376d407da0a50d9ee19a8b6619de45f; custody sha256: f45d75c142bf4835da340a71fed97619be09697a8bea501f4e41a83abe63b752.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

194. Sequential detection and stopping boundary

Family: INFER Obligation: SFT - INFO - OBL - INFER - 012 Claim:

SFT - INFO - INFER - SEQUENTIAL - STOPPING - 012

Question, law and forced result. Sequential detection stops at the first retained evidence prefix that forces the registered decision and records every earlier nonterminal prefix.

INFER-012 uniquely retains first-forced-decision-stopping-record, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RETR - COMPLETENESS - 012
- SFT - INFO - INFER - IDENTIFIABILITY - 011

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 2d7d620ce1c627a1bf992164b1cb7b651248b88576dc99fb5b28494ce3232c58. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-012 product before observation access.

Boundary: Every positive finite INFER-012 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-012 uniquely retains first-forced-decision-stopping-record, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__first-forced-decision-stopping-record__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: 67b4d05c3ce82721e676387539c305f3a76d5d24c895186e837285fbb01412bd; generality
 sha256: 5852c1997782c835e79850a3c72c9d72884d31597cb284b481b91636963f743f.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt
 sha256: ad8b63321fd1752e677525e3865b7139ffdd8913ffff7f7e92dcdbf5bc72e81f7.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt
 sha256: 22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256: c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate
 sha256: 8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal
 sha256: 143b98d20cf36c5a6f0f3c118117ac02c77a2864dbf77eae33277e78e9402d6. Source manifest
 sha256: 8c441ba6477390fc60d07e939a6694c859168f4d386151d02b5bd4fb2791ad2e. Engine receipt
 sha256: 2d7d620ce1c627a1bf992164b1cb7b651248b88576dc99fb5b28494ce3232c58 at receipts/engine/model_admitted/SFT-INFO-INFER-SEQUENTIAL-STOPPING-012-2d7d620ce1c627a1.json.
 Closure scope `depth_independent`. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENC-OBSEVER
- SFT-V1-V2-INFERENC-CORPUS

Observed, adverse and boundary records:

- INFER-012: sequential_stopping
- exact observation: {"continue_prefixes":1,"first_forced_stop_step":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:2da4b4c26b14b9d5db1bed83955b9df122bb20dc4389b6cdefe8150eaff8d8ae; empirical validation sha256:249868c9a64b3c60fc3453a8829702aee383f5df37cf64c430670a4e6d1d9383; external validation sha256:b3f349dafeb84a3145f4404621c401bbe064d380caa6bbe7868cc9e057181bd6; isolation sha256:83d96a80dc9a1b116b45de1efd760bb8225d071cb6d13b40c39b44af65422b32; custody sha256:744cca26880530b05677346026771070c177f0b7bde89917fc509211074a5424.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

195. Multi-sensor information fusion

Family: INFER Obligation: SFT - INFO - OBL - INFER - 013 Claim: SFT - INFO - INFER - MULTISENSOR - FUSION - 013

Question, law and forced result. Multi-sensor fusion intersects source-bound compatible supports while retaining each sensor record, its observation class and any conflict or empty intersection.

INFER-013 uniquely retains source-bound-support-fusion, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RETR - COMPLETENESS - 012
- SFT - INFO - INFER - SEQUENTIAL - STOPPING - 012

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:f174110d0bd0aa794981eee40c481bd4bfff6606f74c7730019b4aad2548eb141. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-013 product before observation access.

Boundary: Every positive finite INFER-013 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-013 uniquely retains source-bound-support-fusion, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__source-bound-support-fusion__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:701dbd853ce5c8ab84d8b5c80b6384095e2299bf2db52154221cd7d8ad56426d; generality sha256:759a4be7d6bc047dab3823b33ca745c573a9c11162751a7231f072eb2e9f6fad.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt
sha256: ad8b63321fd1752e677525e3865b7139ffdd8913fff7f7e92dcdbf5bc72e81f7.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256: c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate

sha256: 8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal

sha256: 3b35830e58eb8cb36e6b0f9e50f09eac83981530a03108d33fc80c6234bf55b6. Source manifest

sha256: d8f1dfdca278d7f24a0fb98df785a7201c230617699a85bcf9245301f54dfa64. Engine receipt

sha256: f174110d0bd0aa794981eee40c481bd4bfff6606f74c7730019b4aad2548eb141 at receipts/engine/model_admitted/SFT-INFO-INFER-MULTISENSOR-FUSION-013-f174110d0bd0aa79. json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENC-OBSEVER
- SFT-V1-V2-INFERENC-CORPUS

Observed, adverse and boundary records:

- INFER-013: multisensor_fusion
- exact observation: {"fused_support":1,"sensor_supports":2,"sensors":2}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: def32ee669536a81d7266bbecc295e279dfb4842fe375cfff1dab29cdd6a371; empirical validation sha256: 8eae5609970380ce60c5031f3eae5bd0caf4acaf1895e0a7487a3a20bad06f1f; external validation sha256: d10297837f9bf74412f8a1652a353811480fdce07a28609bc9b7129f55fd8223; isolation sha256: d6a5faf026079867b49e2338db1c6694cdee89be0e8fdb7e883bb0065634b539; custody sha256: 091604ee1aee4406c9b8fc2c51bb8c1a0f231173d047a8560bd26e2c0a349c48.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

196. Inference-family completeness certificate

Family: INFER Obligation: SFT - INFO - OBL - INFER - 014 Claim: SFT - INFO - INFER - COMPLETENESS - 014

Question, law and forced result. Inference-family completeness is the one-to-one reconciliation of all fourteen frozen inference, detection, estimation, filtering and validation obligations.

INFER-014 uniquely retains fourteen-inference-obligation-ledger, complete inference custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - RETR - COMPLETENESS - 012
- SFT - INFO - INFER - MULTISENSOR - FUSION - 013

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: f2972f07a838cfda163177fd0520b6273828f220ab0d32e26a6713090235aedd. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis INFER-014 product before observation access.

Boundary: Every positive finite INFER-014 hypothesis, evidence item, observation class, decision row, error row and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- INFER-014 uniquely retains fourteen-inference-obligation-ledger, complete inference custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-registered-hypothesis-support__fourteen-inference-obligation-ledger__source-bound-evidence-record__structural-support-decision__complete-declared-inference-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: 090a8b480dc5da6bfd0e624fe2aa573baf0c4db8da27fe3eb333c003ca4e7e41; generality sha256: a63c5c57a9372af88483e02bc4ffef91a78258198f101a4310e26adddb140a2.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted

magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted hypotheses falsify inference closure.; receipt
sha256: ad8b63321fd1752e677525e3865b7139ffdd8913ffff7f7e92dcd5f5bc72e81f7.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported prior distribution, fitted likelihood or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class; no learning objective, semantic meaning or physical sensor law imported into this owner; no failed route retires an obligation or changes protected authority; receipt
sha256: 22d370210860de504b38f8c2b933cbe1bf96fb314a33669c1ef6ebca82b38a6f.

Independent reconstruction and engine admission. Independent implementation

sha256: c4b39a95532ec1bcb804507e418ccd1a7f245fba37fcf4d7729baa335762ba34 regenerated the declared census and sole survivor; independent certificate

sha256: 8b2293d2a57e34c270886b96c5b9d6a1704328f12c50527f489126a5e82d1bbd. Derivation seal
sha256: dc0974ae17fbf233923a731479b813c0e46c994882bdaa26ff5c1e7e3f4205a2. Source manifest
sha256: e36eb423e390c45dec97eb03aff901a6d0e06f07c72b768254d037b005496cf. Engine receipt
sha256: f2972f07a838cfda163177fd0520b6273828f220ab0d32e26a6713090235aedd at receipts/engine/model_admitted/SFT-INFO-INFER-COMPLETENESS-014-f2972f07a838cfda.json. Closure
scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFERENCE-OBSERVER
- SFT-V1-V2-INFERENCE-CORPUS

Observed, adverse and boundary records:

- INFER-014: inference_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":14,"omitted_owners":0,"registered_obligations":14}
- complete family observation custody: 14 records; all rows preserved
- complete finite hypothesis, evidence, support, decision, error-class, stopping and sensor-record enumeration supplies the observation; no imported prior, fitted likelihood, chosen threshold or hidden hypothesis enters

Falsification condition: Reject if the value-free registry, complete inference census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 008a7cc7dc79617ba15d5cb9710f873db8c791ab07c91aa8b1f172540371bdb9; empirical
validation sha256: 70eba4d8791a19c3bc5d98d74e28077ff04420dc6a6b94468d486a35ea710c01;
external validation
sha256: b8738965aacbd10475ed731cebb68d27764f586d1230675a51da63369b0f6428; isolation

sha256:9953f9754548250f30997abc93c294db7b2b946d70a794c684a0e6fa9ff76da47; custody
sha256:c666d9b312688b84bdf950b0f4dcecaf270089323f71d6285c247a9979417f85.

Prohibited inputs.

- no axiom, imported prior distribution, fitted likelihood or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden hypothesis, favorable evidence sample, fitted decision threshold or erased error class
- no learning objective, semantic meaning or physical sensor law imported into this owner
- no failed route retires an obligation or changes protected authority

Complete-field family: PRIV

197. Information leakage as retained adversary distinction

Family: PRIV Obligation: SFT - INFO - OBL - PRIV - 001 Claim: SFT - INFO - PRIV - LEAKAGE - 001

Question, law and forced result. Information leakage is the complete set of protected source distinctions retained by a declared adversary observation, with every still-closed predecessor class preserved.

PRIV-001 uniquely retains adversary-observation-distinction-ledger, complete privacy custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - INFER - COMPLETENESS - 014

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:d5e1ed5a1b5fe01be6df5bb205feae224c7d37f497e52a7376b7e8a6614ccf65. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis PRIV-001 product before observation access.

Boundary: Every positive finite PRIV-001 protected source, adversary view, side record, disclosure pair, query composition and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- PRIV-001 uniquely retains adversary-observation-distinction-ledger, complete privacy custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-protected-source-support__adversary-observation-distinction-ledger__registered-side-information-custody__complete-leaked-distinction-ledger__complete-declared-observation-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:9e5217b37e63092594a4a8d0844bb06a1968d9d6a798175b58c91ec112564123; generality
sha256:f852bef9ff7585b9e1e58da180fc635bd14458773fd087ac147d7d34b330e5aa.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted protected forms falsify leakage.; receipt
sha256: f16da4f07a844ddd92ba94baa1c875d63c2b0221ec1a8fbefbf70a8085430534.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden identity, undeclared side information, fitted privacy weight or favorable adversary; no cryptographic hardness assumption imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: 8d3f8fb4bf448ce1a674d5e85b8c1ff15ff395e63548f99ddcf2e20d882a4243.

Independent reconstruction and engine admission. Independent implementation

sha256: fef54f4cf3b5c9676a199862e80ce4eb98271fdc36ccdf7e27e7855db36e5483 regenerated the declared census and sole survivor; independent certificate

sha256: afb3df3089bc2cef832a83457d894d2a5571bfa1d51f0fe37141b83f57d08369. Derivation seal

sha256: 5589748e5528a8ee113b69b2db7702a87922736f789e6a9b3be9a2c883f69560. Source manifest

sha256: 4250a27a0b33f0e02cc8752a21a0b73e63651ec9868490bf88a3d4999dc58cca. Engine receipt

sha256: d5e1ed5a1b5fe01be6df5bb205feae224c7d37f497e52a7376b7e8a6614ccf65 at

receipts/engine/model_admitted/SFT-INFO-PRIV-LEAKAGE-001-d5e1ed5a1b5fe01b.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-PRIVACY-OBSERVER
- SFT-V1-V2-PRIVACY-CORPUS

Observed, adverse and boundary records:

- PRIV-001: leakage
- exact observation: {"constant_view_leaked_distinctions":0,"identity_view_leaked_distinctions":1,"protected_forms":2}
- complete family observation custody: 10 records; all rows preserved
- complete protected-source, adversary-view, side-information, predecessor-class, query and leaked-distinction enumeration supplies the observation; no imported privacy metric, secret prior, fitted privacy weight or hidden side channel enters

Falsification condition: Reject if the value-free registry, complete protected-source/adversary-view census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: b1748db3decc7008b1707d5862411dc169869a6b6bdd10c37e75a1d3c59c6e04; empirical validation sha256: 053ebfc0569480c42ac8dae91886abc9ae399c88644eb8b0f9fe3bc45f0b030f; external validation

sha256: 578ffcb5d3a3a76c3d9f1d5b80c483ab6486aeb0880a903c69809ae7a1b02548; isolation

sha256: ddccad5749ddfd08e7f4d8cadd550fdd6c0a5f0c6a9f0e23faaff4b34dd9276ec; custody

sha256: 4f48ff596a239bbb27181a52a9d8c9f27d880585e2a70ed6c6d7d9533d11751d.

Prohibited inputs.

- no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden identity, undeclared side information, fitted privacy weight or favorable adversary
- no cryptographic hardness assumption imported into the information-law owner
- no failed route retires an obligation or changes protected authority

198. Perfect secrecy correspondence

Family: PRIV Obligation: SFT - INFO - OBL - PRIV - 002 Claim: SFT - INFO - PRIV - PERFECT - SECRECY - 002

Question, law and forced result. Perfect-secrecy correspondence holds when every public observation has the same complete secret predecessor support; no probability distribution is required for the support law.

PRIV-002 uniquely retains secret-independent-observation-support, complete privacy custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - INFER - COMPLETENESS - 014
- SFT - INFO - PRIV - LEAKAGE - 001

The current reconciliation binds this obligation to the current model-admitted receipt sha256:394da7d7a96dc6428ac5071d9a010365f35e8da0a70e3e5b25d7aa2ddf97039d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis PRIV-002 product before observation access.

Boundary: Every positive finite PRIV-002 protected source, adversary view, side record, disclosure pair, query composition and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- PRIV-002 uniquely retains secret-independent-observation-support, complete privacy custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-protected-source-support__secret-independent-observation-support__registered-side-information-custody__complete-leaked-distinction-ledger__complete-declared-observation-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:bedc22abc07633bbfad1d3742a501f3c49255408fd48670f45ca154f4aac424c; generality sha256:1f06675ed09e4e9b5017ab4007a6681d21f256dd3ec497e30612b3224db215e4.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted protected forms falsify leakage.; receipt sha256:f16da4f07a844ddd92ba94baa1c875d63c2b0221ec1a8fbebff70a8085430534.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt

sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden identity, undeclared side information, fitted privacy weight or favorable adversary; no cryptographic hardness assumption imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:8d3f8fb4bf448ce1a674d5e85b8c1ff15ff395e63548f99ddcf2e20d882a4243.

Independent reconstruction and engine admission. Independent implementation

sha256:fef54f4cf3b5c9676a199862e80ce4eb98271fdc36ccdf7e27e7855db36e5483 regenerated the declared census and sole survivor; independent certificate

sha256:afb3df3089bc2cef832a83457d894d2a5571bfa1d51f0fe37141b83f57d08369. Derivation seal

sha256:26d1a76c65a1ce5247871294641c3c7cb8459f0ccffbea1a8598b764f32e911d. Source manifest

sha256:9cd115ce52d0889a7c5d067da2ab2377d294a847bffb6972e58c1a491e69faa1. Engine receipt

sha256:394da7d7a96dc6428ac5071d9a010365f35e8da0a70e3e5b25d7aa2dddf97039d at receipts/engine/model_admitted/SFT-INFO-PRIV-PERFECT-SECREC-002-394da7d7a96dc642.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-PRIVACY-OBSERVER
- SFT-V1-V2-PRIVACY-CORPUS

Observed, adverse and boundary records:

- PRIV-002: perfect_secrecy_support
- exact observation: {"ciphertexts":2,"possible_secrets_each":2,"secret_prior_used":false}
- complete family observation custody: 10 records; all rows preserved
- complete protected-source, adversary-view, side-information, predecessor-class, query and leaked-distinction enumeration supplies the observation; no imported privacy metric, secret prior, fitted privacy weight or hidden side channel enters

Falsification condition: Reject if the value-free registry, complete protected-source/adversary-view census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:9d1af1b9b0a90f8d02e2ff81806fc854bd8e595640b0b59b85b3bee97e85454b; empirical validation sha256:d9ad60e6e3f9d6be3d5f76b544c6da3af0cb62106b9f8f5ff7a0b38c622005d0; external validation

sha256:9451d0f10179cc8dd31d5d4f34aece54f4851d61c7097d31f611c66d0f765bfff; isolation sha256:317919e57bb41fb67cac4b8457c9f5b2d94f1381e8c0771ca62ac4402b00c984; custody sha256:42f50657b7dbbf6d7094ccf5ef17df09794c6dac4dd50bbfe23e0bbb11693104b.

Prohibited inputs.

- no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden identity, undeclared side information, fitted privacy weight or favorable adversary
- no cryptographic hardness assumption imported into the information-law owner

- no failed route retires an obligation or changes protected authority

199. Side-information and disclosure boundary

Family: PRIV Obligation: SFT - INFO - OBL - PRIV - 003 Claim: SFT - INFO - PRIV - SIDE - INFORMATION - 003

Question, law and forced result. Side information refines adversary predecessor classes by exact intersection; every newly separated protected distinction is entered in the disclosure ledger.

PRIV-003 uniquely retains side-information-refined-predecessor-class, complete privacy custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - INFER - COMPLETENESS - 014
- SFT - INFO - PRIV - PERFECT - SECRECY - 002

The current reconciliation binds this obligation to the current model-admitted receipt sha256: 4463e1dc2d066c1746222cdddefdcbeaa42bb0d228ff7e6e1156a345edede63. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis PRIV-003 product before observation access.

Boundary: Every positive finite PRIV-003 protected source, adversary view, side record, disclosure pair, query composition and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- PRIV-003 uniquely retains side-information-refined-predecessor-class, complete privacy custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-protected-source-support__side-information-refined-predecessor-class__registered-side-information-custody__complete-leaked-distinction-ledger__complete-declared-observation-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: f1c5083f79bb625fea5f4d0235b0cdfba1e2ffd439558699060a390fc874b901; generality sha256: 219e72b64bd9efa8897804b5171965327332f76836afb19413dbd2c9f95b8593.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted protected forms falsify leakage.; receipt sha256: f16da4f07a844ddd92ba94baa1c875d63c2b0221ec1a8fbebff70a8085430534.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result; host 0 denotes structural

absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden identity, undeclared side information, fitted privacy weight or favorable adversary; no cryptographic hardness assumption imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:8d3f8fb4bf448ce1a674d5e85b8c1ff15ff395e63548f99ddcf2e20d882a4243.

Independent reconstruction and engine admission. Independent implementation

sha256:fef54f4cf3b5c9676a199862e80ce4eb98271fdc36ccdf7e27e7855db36e5483 regenerated the declared census and sole survivor; independent certificate
 sha256:afb3df3089bc2cef832a83457d894d2a5571bfa1d51f0fe37141b83f57d08369. Derivation seal
 sha256:a56ce34d41c03b8bdb89579413aed7688cabfd3ef80681ca1e8d8e036d71d637. Source manifest
 sha256:1c19cada80c31b2dbaf5a15550020b0d7b3edf6ad6ecf64b4577f99349e00e1e. Engine receipt
 sha256:4463e1dc2d066c1746222cdddefdccbeaa42bb0d228ff7e6e1156a345ededede63 at receipts/engine/model_admitted/SFT-INFO-PRIV-SIDE-INFORMATION-003-4463e1dc2d066c17.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-PRIVACY-OBSERVER
- SFT-V1-V2-PRIVACY-CORPUS

Observed, adverse and boundary records:

- PRIV-003: side_information
- exact observation: {"cipher_only_predecessors":2,"cipher_plus_key_predecessors":1}
- complete family observation custody: 10 records; all rows preserved
- complete protected-source, adversary-view, side-information, predecessor-class, query and leaked-distinction enumeration supplies the observation; no imported privacy metric, secret prior, fitted privacy weight or hidden side channel enters

Falsification condition: Reject if the value-free registry, complete protected-source/adversary-view census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:949a521b7d21bbdde039543702cd6db3f7e50a03c274b765d319f97c6f4b6b88; empirical validation sha256:3627de669a1332fcc11be5e482c00b59d8f1dbe1c3fafd5cebcbbdf14dd391089; external validation
 sha256:b955653bf19fbfa34976e3e32dfc3a4b65c5e91af2c131bfaa3b3fe322f7cdc8; isolation
 sha256:034d7ca2af3b8fd9336d9bf62b3a1806a2919418349a192ac984ba308ecf1240; custody
 sha256:3ce95595fd1258a8d569b6a4fe36341cb379a3faee6a29ccfe7b2f3f2c5f9a1d.

Prohibited inputs.

- no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden identity, undeclared side information, fitted privacy weight or favorable adversary
- no cryptographic hardness assumption imported into the information-law owner
- no failed route retires an obligation or changes protected authority

200. Anonymity and indistinguishability classes

Family: PRIV Obligation: SFT-INFO-OBL-PRIV-004 Claim: SFT-INFO-PRIV-ANONYMITY-004

Question, law and forced result. An anonymity class is the complete set of identities with one adversary-visible observation; class membership is observation-relative and source identities remain retained.

PRIV-004 uniquely retains complete-adversary-equivalence-classes, complete privacy custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-INFER-COMPLETENESS-014
- SFT-INFO-PRIV-SIDE-INFORMATION-003

The current reconciliation binds this obligation to the current model-admitted receipt sha256:1a437ee1b6cc8e81ecfb50465503ac76ea71b02fecbe6a84aab13782b442eb82. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis PRIV-004 product before observation access.

Boundary: Every positive finite PRIV-004 protected source, adversary view, side record, disclosure pair, query composition and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- PRIV-004 uniquely retains complete-adversary-equivalence-classes, complete privacy custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-protected-source-support__complete-adversary-equivalence-classes__registered-side-information-custody__complete-leaked-distinction-ledger__complete-declared-observation-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:0d9b6c55eaa363f3f58931ddc714b00efa2ba406fdfa4a66e047266617af9781; generality sha256:71fadd65b80302df28370cdcce04f5f467c5e7e470bc4b3e634c9e9678079d66.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted protected forms falsify leakage.; receipt sha256:f16da4f07a844ddd92ba94baa1c875d63c2b0221ec1a8fbebff70a8085430534.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden identity, undeclared side information, fitted privacy weight or favorable adversary; no cryptographic hardness assumption imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:8d3f8fb4bf448ce1a674d5e85b8c1ff15ff395e63548f99ddcf2e20d882a4243.

Independent reconstruction and engine admission. Independent implementation

sha256: fef54f4cf3b5c9676a199862e80ce4eb98271fdc36ccdf7e27e7855db36e5483 regenerated the declared census and sole survivor; independent certificate

sha256: afb3df3089bc2cef832a83457d894d2a5571bfa1d51f0fe37141b83f57d08369. Derivation seal sha256: a32be3dde1c642dbc9654a8410f9e26d7f2ca75285a4b11c24df98240542d2ea. Source manifest sha256: a882a3ba2b21aaeff5a8365cf31e9bdef31056f44041020cf503ed15cc353907. Engine receipt sha256: 1a437ee1b6cc8e81ecfb50465503ac76ea71b02fecbe6a84aab13782b442eb82 at receipts/engine/model_admitted/SFT-INFO-PRIV-ANONYMITY-004-1a437ee1b6cc8e81.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-PRIVACY-OBSERVER
- SFT-V1-V2-PRIVACY-CORPUS

Observed, adverse and boundary records:

- PRIV-004: anonymity
- exact observation: {"constant_anonymity_class":3,"role_classes":2}
- complete family observation custody: 10 records; all rows preserved
- complete protected-source, adversary-view, side-information, predecessor-class, query and leaked-distinction enumeration supplies the observation; no imported privacy metric, secret prior, fitted privacy weight or hidden side channel enters

Falsification condition: Reject if the value-free registry, complete protected-source/adversary-view census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: d9ab90773618f055b5d6afa8959b3451b164fe7480934e4033e7af7592942245; empirical validation sha256: 9a4ae03b31dc10452f10396ce62f669f134cabd247811dbe5a58e23184bab531; external validation

sha256: 9d8fc2ca857ab142922107a09ac3b987b54a0b972465790a2d3ce3a33487d3ae; isolation sha256: 6fdd676422720e35fac25bbc06b7aa5a9c543d06810dcd97af361152b71474d; custody sha256: 932b313f1d27fca36dd67ea0f1d14cccb683989fb9e3c17988d8d800978111a9.

Prohibited inputs.

- no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden identity, undeclared side information, fitted privacy weight or favorable adversary
- no cryptographic hardness assumption imported into the information-law owner
- no failed route retires an obligation or changes protected authority

201. Access observation and least disclosure

Family: PRIV Obligation: SFT-INFO-OBL-PRIV-005 Claim: SFT-INFO-PRIV-LEAST-DISCLOSURE-005

Question, law and forced result. Least disclosure exposes exactly the fields required by a frozen access observation and retains every withheld field and authorization record.

PRIV-005 uniquely retains role-bound-least-disclosure-map, complete privacy custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-INFER-COMPLETENESS-014
- SFT-INFO-PRIV-ANONYMITY-004

The current reconciliation binds this obligation to the current model-admitted receipt sha256:3b43fddbe433f91254544278a9b64ed60b222c09a9dd51c94ae3024b15d36e97. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis PRIV-005 product before observation access.

Boundary: Every positive finite PRIV-005 protected source, adversary view, side record, disclosure pair, query composition and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- PRIV-005 uniquely retains role-bound-least-disclosure-map, complete privacy custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-protected-source-support__role-bound-least-disclosure-map__registered-side-information-custody__complete-leaked-distinction-ledger__complete-declared-observation-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:6fde00b8ca8cb5c6de1a07940b512c990ab59b1d57d6f8c16cb28b89fb048265; generality sha256:2c9317b4d69760bf02d7e168f5e65ef311fe4f27b8257eaf4319a3cae9fb8bf7.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted protected forms falsify leakage.; receipt sha256:f16da4f07a844ddd92ba94baa1c875d63c2b0221ec1a8fbebfff70a8085430534.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden identity, undeclared side information, fitted privacy weight or favorable adversary; no cryptographic hardness assumption imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:8d3f8fb4bf448ce1a674d5e85b8c1ff15ff395e63548f99ddcf2e20d882a4243.

Independent reconstruction and engine admission. Independent implementation

sha256:fef54f4cf3b5c9676a199862e80ce4eb98271fdc36ccdf7e27e7855db36e5483 regenerated the declared census and sole survivor; independent certificate

sha256:afb3df3089bc2cef832a83457d894d2a5571bfa1d51f0fe37141b83f57d08369. Derivation seal sha256:0becb7cf110aace9b3045ec2e97c2806850b854f5a1a2ae9cbab49233619553d. Source manifest

sha256:f84fe9302118ed79f18e98ae64678f3abf4210612f56e32faa1d8799b3824c9e. Engine receipt sha256:3b43fddbe433f91254544278a9b64ed60b222c09a9dd51c94ae3024b15d36e97 at receipts/engine/model_admitted/SFT-INFO-PRIV-LEAST-DISCLOSURE-005-3b43fddbe433f912.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-PRIVACY-OBSERVER
- SFT-V1-V2-PRIVACY-CORPUS

Observed, adverse and boundary records:

- PRIV-005: least_disclosure
- exact observation: {"editor_fields":2,"reader_fields":1,"withheld_records_retained":true}
- complete family observation custody: 10 records; all rows preserved
- complete protected-source, adversary-view, side-information, predecessor-class, query and leaked-distinction enumeration supplies the observation; no imported privacy metric, secret prior, fitted privacy weight or hidden side channel enters

Falsification condition: Reject if the value-free registry, complete protected-source/adversary-view census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:ec777e2ea6510e800a467b70587ec850ee76c959d500cb99b47953367ff2a616; empirical validation sha256:e687244257231993abc923fa5a8a2d9d55ee536f75ed1a7cb52f388d285afbe6; external validation sha256:d86eeb983e0348bfa88fc160d6e98e23bf806b968e8a7922915eac98c548e93b; isolation sha256:888ccc3d072217eeaa8db5415045e69d8a41ff29b1bc58a023600dd80a4faa00; custody sha256:843626bbdc559c94403016da0e472f59344bb680bc2196285e7536c66cbb1547.

Prohibited inputs.

- no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden identity, undeclared side information, fitted privacy weight or favorable adversary
- no cryptographic hardness assumption imported into the information-law owner
- no failed route retires an obligation or changes protected authority

202. Composition of privacy losses

Family: PRIV Obligation: SFT-INFO-0BL-PRIV-006 Claim: SFT-INFO-PRIV-COMPOSITION-006

Question, law and forced result. Composed disclosure is the common refinement of all released observations; its privacy loss is the union of exact source distinctions separated by any release.

PRIV-006 uniquely retains composed-observation-refinement-ledger, complete privacy custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-INFER-COMPLETENESS-014
- SFT-INFO-PRIV-LEAST-DISCLOSURE-005

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:b1a70772a665f79302bee0d8b1824f7e0480a91e76fec9af54fbb674fdddee5. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis PRIV-006 product before observation access.

Boundary: Every positive finite PRIV-006 protected source, adversary view, side record, disclosure pair, query composition and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- PRIV-006 uniquely retains composed-observation-refinement-ledger, complete privacy custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-protected-source-support__composed-observation-refinement-ledger__registered-side-information-custody__complete-leaked-distinction-ledger__complete-declared-observation-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:cfacf9b0dbfe7e5250d879e785645d29ce051c3634f98b7f592f48be82a106dd; generality
 sha256:ae94e46e2f66991362784cea95134b42e6de6db30034a25f178197ca4f195516.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted protected forms falsify leakage.; receipt
 sha256:f16da4f07a844ddd92ba94baa1c875d63c2b0221ec1a8fbebff70a8085430534.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden identity, undeclared side information, fitted privacy weight or favorable adversary; no cryptographic hardness assumption imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:8d3f8fb4bf448ce1a674d5e85b8c1ff15ff395e63548f99ddcf2e20d882a4243.

Independent reconstruction and engine admission. Independent implementation

sha256:fef54f4cf3b5c9676a199862e80ce4eb98271fdc36ccdf7e27e7855db36e5483 regenerated the declared census and sole survivor; independent certificate
 sha256:afb3df3089bc2cef832a83457d894d2a5571bfa1d51f0fe37141b83f57d08369. Derivation seal
 sha256:a2a4d5cd95ce8a3b47dee7eb51063d13f04e4fa37db998aca540fcce7142fe83. Source manifest
 sha256:d9cb55bfbf9e2488a1eae0c996afde40313cd460ae5901a5bf91219c0b79e85a. Engine receipt
 sha256:b1a70772a665f79302bee0d8b1824f7e0480a91e76fec9af54fbb674fdddee5 at
 receipts/engine/model_admitted/SFT-INFO-PRIV-COMPOSITION-006-b1a70772a665f793.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-PRIVACY-OBSERVER
- SFT-V1-V2-PRIVACY-CORPUS

Observed, adverse and boundary records:

- PRIV-006: composition
- exact observation: `{"composed_distinctions":6,"protected_forms":4,"release_views":2}`
- complete family observation custody: 10 records; all rows preserved
- complete protected-source, adversary-view, side-information, predecessor-class, query and leaked-distinction enumeration supplies the observation; no imported privacy metric, secret prior, fitted privacy weight or hidden side channel enters

Falsification condition: Reject if the value-free registry, complete protected-source/adversary-view census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:93e10ddf3f9d1c91eb38a0a02ac95abcb24f6ab66e002f85f1a99a9e6fa1d776; empirical validation sha256:d06d7c81f6177536d5273d5f73cdef5ae39268fb2d5c42a711d33f1fac14e4bc; external validation

sha256:e8bc6cca8a665e51a1a2654a6e33ec0064ab84bc21a1df45057139635553f0; isolation sha256:32fd8b342b03bc391cdc8172f4306a059af1898a1812a009b82626e7752839cc; custody sha256:0be962dff99bb60384a43dbdc2e3524f6244bfca55dc3a8c4d2566c969e8679e.

Prohibited inputs.

- no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden identity, undeclared side information, fitted privacy weight or favorable adversary
- no cryptographic hardness assumption imported into the information-law owner
- no failed route retires an obligation or changes protected authority

203. Query disclosure and reconstruction risk

Family: PRIV Obligation: SFT-INFO-OBL-PRIV-007 Claim: SFT-INFO-PRIV-QUERY-RECONSTRUCTION-007

Question, law and forced result. Query reconstruction risk is the complete source predecessor class under all released query answers; singleton composition reconstructs while non-singleton classes remain private.

PRIV-007 uniquely retains complete-query-composition-preimages, complete privacy custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-INFER-COMPLETENESS-014
- SFT-INFO-PRIV-COMPOSITION-006

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:37d77397646cd879646a1d1c7e8bb548734757510e6a3c082430656463076e98. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis PRIV-007 product before observation access.

Boundary: Every positive finite PRIV-007 protected source, adversary view, side record, disclosure pair, query composition and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- PRIV-007 uniquely retains complete-query-composition-preimages, complete privacy custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-protected-source-support__complete-query-composition-preimages__registered-side-information-custody__complete-leaked-distinction-ledger__complete-declared-observation-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:e81f8723474faa4ff4588cfd72f233d961605a097f50012f46122ea46335dfb; generality

sha256:c0a6dcb25173107830aaa4553c3982fb755002a53d5ed24994a82ea82bbe223a.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted protected forms falsify leakage.; receipt
sha256:f16da4f07a844ddd92ba94baa1c875d63c2b0221ec1a8fbefb70a8085430534.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden identity, undeclared side information, fitted privacy weight or favorable adversary; no cryptographic hardness assumption imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:8d3f8fb4bf448ce1a674d5e85b8c1ff15ff395e63548f99ddcf2e20d882a4243.

Independent reconstruction and engine admission. Independent implementation

sha256:fef54f4cf3b5c9676a199862e80ce4eb98271fdc36ccdf7e27e7855db36e5483 regenerated the declared census and sole survivor; independent certificate

sha256:afb3df3089bc2cef832a83457d894d2a5571bfa1d51f0fe37141b83f57d08369. Derivation seal

sha256:081528137d7dab099cfc467e7135d0b78608c3940fc39b6ad81ffd26257b1166. Source manifest

sha256:fe37ace66cc60dcf5168643677f2adb0b4e85ce843b4965dac4ffa179c15635f. Engine receipt

sha256:37d77397646cd879646a1d1c7e8bb548734757510e6a3c082430656463076e98 at receipts/engine/model_admitted/SFT-INFO-PRIV-QUERY-RECONSTRUCTION-007-37d77397646cd879.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-PRIVACY-OBSERVER

- SFT - V1 - V2 - PRIVACY - CORPUS

Observed, adverse and boundary records:

- PRIV-007: query_reconstruction
- exact observation: {"joint_records":4,"single_query_predecessors":2,"two_query_predecessors":1}
- complete family observation custody: 10 records; all rows preserved
- complete protected-source, adversary-view, side-information, predecessor-class, query and leaked-distinction enumeration supplies the observation; no imported privacy metric, secret prior, fitted privacy weight or hidden side channel enters

Falsification condition: Reject if the value-free registry, complete protected-source/adversary-view census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:bec227569e2de07f3fb725b38cbda53cf67a1bc7acea54b02c959773289e582e; empirical validation sha256:c256692fd043b78e14c00d6f462e2006edd6f52dc79bc7cf6a4102074e1faa93; external validation

sha256:ab9112b7149339bb31a6610cbf3cac27dbebe6e5ebbb33f4ce8c0680387c0427; isolation sha256:79a5af77944e02fb83d66e1b419abbe32fb22f7e17720d3e28399b96859a3e22; custody sha256:466fc013e51ce8a25317fa12bb59d14e6b23fb0b7f87e6111c0d5000edaf3249.

Prohibited inputs.

- no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden identity, undeclared side information, fitted privacy weight or favorable adversary
- no cryptographic hardness assumption imported into the information-law owner
- no failed route retires an obligation or changes protected authority

204. Privacy-utility exact trade boundary

Family: PRIV Obligation: SFT - INFO - OBL - PRIV - 008 Claim: SFT - INFO - PRIV - UTILITY - BOUNDARY - 008

Question, law and forced result. The exact privacy-utility boundary is the least generated disclosure partition sufficient for every registered utility distinction, with all additional leaked distinctions rejected.

PRIV-008 uniquely retains least-utility-sufficient-disclosure, complete privacy custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - INFER - COMPLETENESS - 014
- SFT - INFO - PRIV - QUERY - RECONSTRUCTION - 007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:56e3cbc2c8b0503cbcc009d80fbe36b95875742aed1a5fe90e6fb9c35f66ffad. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis PRIV-008 product before observation access.

Boundary: Every positive finite PRIV-008 protected source, adversary view, side record, disclosure pair, query composition and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- PRIV-008 uniquely retains least-utility-sufficient-disclosure, complete privacy custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-protected-source-support__least-utility-sufficient-disclosure__registered-side-information-custody__complete-leaked-distinction-ledger__complete-declared-observation-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: 9f90f561e25c903780461a7d25d935c6c536f5c3ec165cb152555d389fc539e1; generality sha256: 36e4b2c8c97abd899c753e6a59557d98badb152ae8d9684f60b9a0d3e9281180.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted protected forms falsify leakage.; receipt sha256: f16da4f07a844ddd92ba94baa1c875d63c2b0221ec1a8fbebff70a8085430534.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden identity, undeclared side information, fitted privacy weight or favorable adversary; no cryptographic hardness assumption imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: 8d3f8fb4bf448ce1a674d5e85b8c1ff15ff395e63548f99ddcf2e20d882a4243.

Independent reconstruction and engine admission. Independent implementation

sha256: fef54f4cf3b5c9676a199862e80ce4eb98271fdc36ccdf7e27e7855db36e5483 regenerated the declared census and sole survivor; independent certificate sha256: afb3df3089bc2cef832a83457d894d2a5571bfa1d51f0fe37141b83f57d08369. Derivation seal sha256: ff1b38085e483af306d943c5567ecb8288de46d895a850829266fd7d81b7a886. Source manifest sha256: 05df0d9efad742e136e56593fc93cc9f5acf70bf29d93bebd984ad9ecd08c989. Engine receipt sha256: 56e3cbc2c8b0503cbcc009d80fbe36b95875742aed1a5fe90e6fb9c35f66ffad at receipts/engine/model_admitted/SFT-INFO-PRIV-UTILITY-BOUNDARY-008-56e3cbc2c8b0503c.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-PRIVACY-OBSERVER
- SFT-V1-V2-PRIVACY-CORPUS

Observed, adverse and boundary records:

- PRIV-008: utility_boundary
- exact observation: {"identity_cells":4,"least_sufficient_cells":2,"one_cell_sufficient":false}

- complete family observation custody: 10 records; all rows preserved
- complete protected-source, adversary-view, side-information, predecessor-class, query and leaked-distinction enumeration supplies the observation; no imported privacy metric, secret prior, fitted privacy weight or hidden side channel enters

Falsification condition: Reject if the value-free registry, complete protected-source/adversary-view census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: acf3fca3465d6b9ad0287785654c014e879b7515066e61be9f4be2ea95ddb01a; empirical validation sha256: 1d8f76940b4292eb13056001fe028a916248aae5a7235a7828c3c98625e35cc5; external validation

sha256: fbbffee77ae23d33d4acfa5f27fc695b127b6c3c364a8a40aa3ba1501406e26a; isolation sha256: 69c1fa491eb8f88f7c541d24626b48b30d20d4a7693443bb476a1ee2fd0f41f6; custody sha256: 32f13d9b55515297fd9be3601175e09c2444e9bd7d6114519b13ac293f508a3a.

Prohibited inputs.

- no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden identity, undeclared side information, fitted privacy weight or favorable adversary
- no cryptographic hardness assumption imported into the information-law owner
- no failed route retires an obligation or changes protected authority

205. Cryptographic ownership handoff

Family: PRIV Obligation: SFT - INFO - OBL - PRIV - 009 Claim: SFT - INFO - PRIV - CRYPTO - HANDOFF - 009

Question, law and forced result. Information Science owns observation classes and leakage ledgers; Cryptography owns adversary resources, one-wayness, computational secrecy and protocol constructions.

PRIV-009 uniquely retains privacy-law-cryptography-interface, complete privacy custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - INFER - COMPLETENESS - 014
- SFT - INFO - PRIV - UTILITY - BOUNDARY - 008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 8d2c7635bdd22d24732063dccb833a9c146aa331fef2ba202dd6eb42a7b88f8d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis PRIV-009 product before observation access.

Boundary: Every positive finite PRIV-009 protected source, adversary view, side record, disclosure pair, query composition and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- PRIV-009 uniquely retains privacy-law-cryptography-interface, complete privacy custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-protected-source-support__privacy-law-cryptography-interface__registered-side-information-custody__complete-leaked-distinction-ledger__complete-declared-observation-product__root-bound-forward-forcing__post-registry-exact-observ

ation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:3de0b5991caab1d27733a00f68ad9edad0bbc7f9504e1e2c3410b77277b76098; generality
 sha256:79924441d24c00030df6b08b10737151663ddff628ad3560b86698a14e37f1b9.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted protected forms falsify leakage.; receipt
 sha256:f16da4f07a844ddd92ba94baa1c875d63c2b0221ec1a8fbefbf70a8085430534.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden identity, undeclared side information, fitted privacy weight or favorable adversary; no cryptographic hardness assumption imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256:8d3f8fb4bf448ce1a674d5e85b8c1ff15ff395e63548f99ddcf2e20d882a4243.

Independent reconstruction and engine admission. Independent implementation

sha256:fef54f4cf3b5c9676a199862e80ce4eb98271fdc36ccdf7e27e7855db36e5483 regenerated the declared census and sole survivor; independent certificate
 sha256:afb3df3089bc2cef832a83457d894d2a5571bfa1d51f0fe37141b83f57d08369. Derivation seal
 sha256:6d0a4fb98b2084ef74a4857db75ff2126ffd251cd2d7cf13c17725a5d9bd5ce2. Source manifest
 sha256:5b4ceb4c13dce4f7ed966752457706b6b256fa96b22bd8ff05d15ad4c7474d20. Engine receipt
 sha256:8d2c7635bdd22d24732063dcc8b33a9c146aa331fef2ba202dd6eb42a7b88f8d at receipts/engine/model_admitted/SFT-INFO-PRIV-CRYPTO-HANDOFF-009-8d2c7635bdd22d24.json. Closure
 scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-PRIVACY-OBSERVER
- SFT-V1-V2-PRIVACY-CORPUS

Observed, adverse and boundary records:

- PRIV-009: cryptographic_handoff
- exact observation: {"downstream_owner":"cryptophy","exported_information_records":3,"ownership_overlap":0}
- complete family observation custody: 10 records; all rows preserved
- complete protected-source, adversary-view, side-information, predecessor-class, query and leaked-distinction enumeration supplies the observation; no imported privacy metric, secret prior, fitted privacy weight or hidden side channel enters

Falsification condition: Reject if the value-free registry, complete protected-source/adversary-view census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:557fd5e0c9a16e6a31820f3ccfd7433111df0ac836b8cade4c0e086242492bb2; empirical validation sha256:e43ff31d8af91c4650d264479e7c159f41a771f0f6924fa7486cb49b03a4b8d9; external validation sha256:9d36c6015c6c62e8b3f88552299887c4550b0a977e9349bf574bbd721147faff; isolation sha256:b26db5fee25bd867259c6a02f78c37bc4e2241ed620fe42c112359232f940863; custody sha256:8d8fc8fd88aedd08649877c70037331d6f0a9bddfa350b298a4a4c9771d30471.

Prohibited inputs.

- no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden identity, undeclared side information, fitted privacy weight or favorable adversary
- no cryptographic hardness assumption imported into the information-law owner
- no failed route retires an obligation or changes protected authority

206. Privacy-interface completeness certificate

Family: PRIV Obligation: SFT - INFO - OBL - PRIV - 010 Claim: SFT - INFO - PRIV - COMPLETENESS - 010

Question, law and forced result. Privacy-interface completeness is the one-to-one reconciliation of all ten frozen leakage, secrecy-support, anonymity, disclosure and handoff obligations.

PRIV-010 uniquely retains ten-privacy-obligation-ledger, complete privacy custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - INFER - COMPLETENESS - 014
- SFT - INFO - PRIV - CRYPTO - HANDOFF - 009

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:6a7f4ec296e1eb4ac71c80ecde7709c7ed090ee9ef8326130ed6ea5b5b4c4721. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis PRIV-010 product before observation access.

Boundary: Every positive finite PRIV-010 protected source, adversary view, side record, disclosure pair, query composition and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- PRIV-010 uniquely retains ten-privacy-obligation-ledger, complete privacy custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-protected-source-support__ten-privacy-obligation-ledger__registered-side-information-custody__complete-leaked-distinction-ledger__complete-declared-observation-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:97a2b18d2d351e11c4d31b490e1a1be2917823b9f10fa6eca2002c16b5f172a1; generality sha256:1c9715e423595055d2d9b778173d87e78d3ca6aa944e4f4805c4ecdf0dde34a.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted protected forms falsify leakage.; receipt
sha256: f16da4f07a844ddd92ba94baa1c875d63c2b0221ec1a8fbefbf70a8085430534.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden identity, undeclared side information, fitted privacy weight or favorable adversary; no cryptographic hardness assumption imported into the information-law owner; no failed route retires an obligation or changes protected authority; receipt sha256: 8d3f8fb4bf448ce1a674d5e85b8c1ff15ff395e63548f99ddcf2e20d882a4243.

Independent reconstruction and engine admission. Independent implementation

sha256: fef54f4cf3b5c9676a199862e80ce4eb98271fdc36ccdf7e27e7855db36e5483 regenerated the declared census and sole survivor; independent certificate

sha256: afb3df3089bc2cef832a83457d894d2a5571bfa1d51f0fe37141b83f57d08369. Derivation seal
sha256: a2769c4f7681a5f9e6a0abb8e4246764d8c7d828fba5bea888fc9fad5610d5c4. Source manifest
sha256: dfdd208fb07f2e7e0c888776bbbed9c9fa68eba9c346c648cc672614d4bc941. Engine receipt
sha256: 6a7f4ec296e1eb4ac71c80ecde7709c7ed090ee9ef8326130ed6ea5b5b4c4721 at receipts/engine/model_admitted/SFT-INFO-PRIV-COMPLETENESS-010-6a7f4ec296e1eb4a.json. Closure scope
depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-PRIVACY-OBSERVER
- SFT-V1-V2-PRIVACY-CORPUS

Observed, adverse and boundary records:

- PRIV-010: privacy_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":10,"omitted_owners":0,"registered_obligations":10}
- complete family observation custody: 10 records; all rows preserved
- complete protected-source, adversary-view, side-information, predecessor-class, query and leaked-distinction enumeration supplies the observation; no imported privacy metric, secret prior, fitted privacy weight or hidden side channel enters

Falsification condition: Reject if the value-free registry, complete protected-source/adversary-view census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 952992eef20e679b111a98907e33a7c3c6e59bf492fae9761b85da19fca258fa; empirical
validation sha256: e67c325ca7d1a5cc329185231a1d816f32143f06da16b73a529e1d0e987f3710;

external validation

sha256:28b7eec7cbdefef9e0e449d6f7a9c26843b22ae4fcb0e9c38f3518c717937874; isolation
sha256:cae0f53cb3fc6d8147dea842c64cb0a5f480c99d9570eeb565ef2bd1405f88dc; custody
sha256:de255082238f908ae6cacc46f79cb287236ce2f8e4ab2643a493b5944ecb91a0.

Prohibited inputs.

- no axiom, imported privacy metric, stochastic secret prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden identity, undeclared side information, fitted privacy weight or favorable adversary
- no cryptographic hardness assumption imported into the information-law owner
- no failed route retires an obligation or changes protected authority

Complete-field family: THERM

207. Information record as a physical custody obligation

Family: THERM Obligation: SFT - INFO - OBL - THERM - 001 Claim: SFT - INFO - THERM - PHYSICAL - CUSTODY - 001

Question, law and forced result. A retained information distinction requires a physically distinguishable custody record, while carrier material, energy and dynamics remain explicit downstream variables.

THERM-001 uniquely retains distinguishable-carrier-custody, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - PRIV - COMPLETENESS - 010

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:2d0295b2cf23ff58d8f98e9b1c160d517a1b36b691b686a7d0c88cc816ac76a2. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis THERM-001 product before observation access.

Boundary: Every positive finite THERM-001 record form, merge, predecessor class, held label, reset cycle and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- THERM-001 uniquely retains distinguishable-carrier-custody, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-logical-record-support__distinguishable-carrier-custody__held-predecessor-custody__typed-physical-translation-boundary__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:c5810952a1df2faf7a94aaa65f0c44673c2eb20605835b392f0d71df1e2bef3e; generality
sha256:cae0f53cb3fc6d8147dea842c64cb0a5f480c99d9570eeb565ef2bd1405f88dc.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Missing record forms hide physical custody.; receipt
sha256: c68e5b08f48dd681dc64f7db47ee32a82e31a70765db70257c8b81a3d99d5088.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption; no physical work, heat or temperature law claimed without the Physics-owned translation; no failed route retires an obligation or changes protected authority; receipt
sha256: 7413bcd4f9bb55ffb7d53b1328ef1b1da8c95aea43c3126152ef6678f5662991.

Independent reconstruction and engine admission. Independent implementation

sha256: 3882b70af437c25da5cbe28586c1abd8aad8fb27a41af518c3a85c84eefc47bb regenerated the declared census and sole survivor; independent certificate

sha256: 31befc877d9a95ff132f54671282b4829a8a9d6f652d27d23ec778a72fa170b6. Derivation seal

sha256: 6f4a1b7a76f9853e29d3ef6b7d2e0b26523dd64138c99fa7f1b4ee7452256ed2. Source manifest

sha256: 0796cd60f2665f6d8f22bd9dee1cd83afad7d34cf60fedb56babce73b4cf6f17. Engine receipt

sha256: 2d0295b2cf23ff58d8f98e9b1c160d517a1b36b691b686a7d0c88cc816ac76a2 at receipts/engine/model_admitted/SFT-INFO-THERM-PHYSICAL-CUSTODY-001-2d0295b2cf23ff58. json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFO-THERM-OBSERVER
- SFT-V1-V2-INFORMATION-THERMODYNAMICS-CORPUS

Observed, adverse and boundary records:

- THERM-001: physical_custody
- exact observation: {"carrier_implementations_open":true,"distinguishable_carrier_labels":2,"logical_forms":2}
- complete family observation custody: 10 records; all rows preserved
- complete logical-record, merge, predecessor, held-label, measurement, reset and cycle enumeration supplies the observation; no imported physical constant, fitted energy value, erased provenance or information-energy identity enters

Falsification condition: Reject if the value-free registry, complete record/transition/predecessor custody census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 84464b71de1c2d8a2d1c6a6824eb9fda9b808dce26c0fdf3f201289e7a461301; empirical validation sha256: 1aee0e2a806a2dbd0f0eb32d8032609630003ce67e351cae364cd2697b0ce80f; external validation

sha256: 1312d42c2c6db71eb6ee8bc9f3ef8d0e4fa032abd0eb755c91b6f8a1c407fc73; isolation

sha256: 9578254fdeb57f9a534a48af862faaf16cdaa610c9a529201c3d9c7cc680c130; custody

sha256:bdb0a1d8386c4ac087fa95ffb0be4c5f1c5c9e409382e0451444ab64abdf8a6f.

Prohibited inputs.

- no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption
- no physical work, heat or temperature law claimed without the Physics-owned translation
- no failed route retires an obligation or changes protected authority

208. Irreversible merge and missing predecessor label

Family: THERM Obligation: SFT - INFO - OBL - THERM - 002 Claim: SFT - INFO - THERM - IRREVERSIBLE - MERGE - 002

Question, law and forced result. An irreversible merge is a many-source one-image map; reversal is underdetermined by exactly its complete predecessor class unless a source label is retained.

THERM-002 uniquely retains merged-predecessor-class-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-PRIV-COMPLETENESS-010
- SFT-INFO-THERM-PHYSICAL-CUSTODY-001

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:a99a0ae47e94bee5844a5e063d75dbc92dc9428b94be5ee661b29e6318455463. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis THERM-002 product before observation access.

Boundary: Every positive finite THERM-002 record form, merge, predecessor class, held label, reset cycle and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- THERM-002 uniquely retains merged-predecessor-class-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-logical-record-support__merged-predecessor-class-ledger__held-predecessor-custody__typed-physical-translation-boundary__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:d50704813e53fe8a1ee5d55288d9f8ff2d030f814745e4e9213d71a1575ad8f0; generality
sha256:d0cd95f7a958f192494ca1a1d8d896e9d65b946141aa6e33a317cf49d42a6ff2.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing record forms hide physical custody.; receipt

sha256:c68e5b08f48dd681dc64f7db47ee32a82e31a70765db70257c8b81a3d99d5088.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption; no physical work, heat or temperature law claimed without the Physics-owned translation; no failed route retires an obligation or changes protected authority; receipt
sha256:7413bcd4f9bb55ffb7d53b1328ef1b1da8c95aea43c3126152ef6678f5662991.

Independent reconstruction and engine admission. Independent implementation

sha256:3882b70af437c25da5cbe28586c1abd8aad8fb27a41af518c3a85c84eefc47bb regenerated the declared census and sole survivor; independent certificate

sha256:31befc877d9a95ff132f54671282b4829a8a9d6f652d27d23ec778a72fa170b6. Derivation seal

sha256:2ee46ba3e1a3c151c6a90a91da5e4d46b4011cfe23f3c754f50930a400fd58ac. Source manifest

sha256:a069020cffbfc4cbb0d4661e93115d5facaf1d651b1f04f6ca30c8aee201f0eb. Engine receipt

sha256:a99a0ae47e94bee5844a5e063d75dbc92dc9428b94be5ee661b29e6318455463 at receipts/engine/model_admitted/SFT-INFO-THERM-IRREVERSIBLE-MERGE-002-a99a0ae47e94bee5.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFO-THERM-OBSERVER
- SFT-V1-V2-INFORMATION-THERMODYNAMICS-CORPUS

Observed, adverse and boundary records:

- THERM-002: irreversible_merge
- exact observation: {"images":1,"predecessor_class":2,"source_forms":2}
- complete family observation custody: 10 records; all rows preserved
- complete logical-record, merge, predecessor, held-label, measurement, reset and cycle enumeration supplies the observation; no imported physical constant, fitted energy value, erased provenance or information-energy identity enters

Falsification condition: Reject if the value-free registry, complete record/transition/predecessor custody census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:5f1164d3b40acff7804d6eddf097daf34e8d1d199ab2ec0365102aec43b62e9; empirical validation sha256:af8d1dfacc995275d847a7f708f38c9a1f7d62d9d44c01ba924803231df1e62f; external validation

sha256:306afaf88bfefa2e6b07d206089801dd6612748dc0934582dcba681f8801a8f7; isolation

sha256:529a47954e434d0c10e47481dfc445adc57c4dab7a9f8d7db835315a3c4bb7d9; custody

sha256:7d8dc30a535a7266568e667da5b6854ab90ad582bbbd8fbb8c1dc058a24e33d1.

Prohibited inputs.

- no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result

- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption
- no physical work, heat or temperature law claimed without the Physics-owned translation
- no failed route retires an obligation or changes protected authority

209. Reversible record retention

Family: THERM Obligation: SFT - INFO - OBL - THERM - 003 Claim:
SFT - INFO - THERM - REVERSIBLE - RETENTION - 003

Question, law and forced result. Retaining the exact predecessor label alongside a merged image makes the transition injective on the extended record and forces exact reversal.

THERM-003 uniquely retains held-predecessor-reversal-record, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - PRIV - COMPLETENESS - 010
- SFT - INFO - THERM - IRREVERSIBLE - MERGE - 002

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:a28cf0219b9c74c8c98b35ff376ebd20c0da17c5993bea2d662a24d50d13ae33. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis THERM-003 product before observation access.

Boundary: Every positive finite THERM-003 record form, merge, predecessor class, held label, reset cycle and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- THERM-003 uniquely retains held-predecessor-reversal-record, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-logical-record-support__held-predecessor-reversal-record__held-predecessor-custody__typed-physical-translation-boundary__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:9364687011690d8c6a24072ccaebca3418fe13e44de59a4c556e52a7f652bc8e; generality
sha256:71162fc8681216e3f2bd7d7e1998ec3546ef58c0eeb203689569391f359abc0d.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing record forms hide physical custody.; receipt
sha256:c68e5b08f48dd681dc64f7db47ee32a82e31a70765db70257c8b81a3d99d5088.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt

sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt

sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption; no physical work, heat or temperature law claimed without the Physics-owned translation; no failed route retires an obligation or changes protected authority; receipt

sha256:7413bcd4f9bb55ffb7d53b1328ef1b1da8c95aea43c3126152ef6678f5662991.

Independent reconstruction and engine admission. Independent implementation

sha256:3882b70af437c25da5cbe28586c1abd8aad8fb27a41af518c3a85c84eefc47bb regenerated the declared census and sole survivor; independent certificate

sha256:31befc877d9a95ff132f54671282b4829a8a9d6f652d27d23ec778a72fa170b6. Derivation seal

sha256:867b7bf6200e9cfb56b9092d4d95ae14581a2736b273915c1e9501a808944aa6. Source manifest

sha256:30ebdc3d8eb9a7e9c12595c3b4d6530cabeb5a242c9f8b40ed90d2749dfda7e0. Engine receipt

sha256:a28cf0219b9c74c8c98b35ff376ebd20c0da17c5993bea2d662a24d50d13ae33 at receipts/engine/model_admitted/SFT-INFO-THERM-REVERSIBLE-RETENTION-003-a28cf0219b9c74c8.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFO-THERM-OBSERVER
- SFT-V1-V2-INFORMATION-THERMODYNAMICS-CORPUS

Observed, adverse and boundary records:

- THERM-003: reversible_retention
- exact observation: {"held_labels_per_merge":1,"reconstructed_sources":2}
- complete family observation custody: 10 records; all rows preserved
- complete logical-record, merge, predecessor, held-label, measurement, reset and cycle enumeration supplies the observation; no imported physical constant, fitted energy value, erased provenance or information-energy identity enters

Falsification condition: Reject if the value-free registry, complete record/transition/predecessor custody census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:9763dc4eff7b6caf2b7590ed928521a6f0e733c7a9b87f79f841cd6a6e7b72e0; empirical validation sha256:55013c33e925398d513c10ebd45c354e365cb0c3adb7271ec89fb52389f76868; external validation

sha256:265d7cf8494f257524719be6a0cff8bf9c1e1c2d55e6bcb181891abe090d2845; isolation

sha256:9b7897ca77ae81623f99a4d2b22a85a9d9e3a4f3e0b6610a2cacb124dce847b5; custody

sha256:a5ed05aaf9193d37d6b1fcce8f759ee1ac5b7adb87f32829253320a784ba69a.

Prohibited inputs.

- no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar

- no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption
- no physical work, heat or temperature law claimed without the Physics-owned translation
- no failed route retires an obligation or changes protected authority

210. Erasure correspondence boundary

Family: THERM Obligation: SFT - INFO - OBL - THERM - 004 Claim: SFT - INFO - THERM - ERASURE - BOUNDARY - 004

Question, law and forced result. Logical erasure is the closure of distinct record forms into one designated empty state; its physical work or heat correspondence is not selected by the information map alone.

THERM-004 uniquely retains logical-erasure-distinction-closure, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - PRIV - COMPLETENESS - 010
- SFT - INFO - THERM - REVERSIBLE - RETENTION - 003

The current reconciliation binds this obligation to the current model-admitted receipt sha256: cea4d5f9af27a9f39ae64961bf86cc0b432ef1ef4abff03d4cb08ad51be26976. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis THERM-004 product before observation access.

Boundary: Every positive finite THERM-004 record form, merge, predecessor class, held label, reset cycle and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- THERM-004 uniquely retains logical-erasure-distinction-closure, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-logical-record-support__logical-erasure-distinction-closure__held-predecessor-custody__typed-physical-translation-boundary__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: 8d123e18d4b4bce74cb268e2cf718793150cacd6fb43401759b8c8b209f1a8d0; generality sha256: dac244f27b968a75b14e70ea49faacb7fdf6262ca2ccaa853b29de8cd852d4f7.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing record forms hide physical custody.; receipt sha256: c68e5b08f48dd681dc64f7db47ee32a82e31a70765db70257c8b81a3d99d5088.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt

sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption; no physical work, heat or temperature law claimed without the Physics-owned translation; no failed route retires an obligation or changes protected authority; receipt

sha256:7413bcd4f9bb55ffb7d53b1328ef1b1da8c95aea43c3126152ef6678f5662991.

Independent reconstruction and engine admission. Independent implementation

sha256:3882b70af437c25da5cbe28586c1abd8aad8fb27a41af518c3a85c84eefc47bb regenerated the declared census and sole survivor; independent certificate

sha256:31befc877d9a95ff132f54671282b4829a8a9d6f652d27d23ec778a72fa170b6. Derivation seal

sha256:bf02df5c036aa8fab0536135c9ffd1e1d7a15d251ba89d6a59eef34bff261aea. Source manifest

sha256:c16156880eb4b157c8690ac2d43ffad5fb0326bab6c2808ded523cf8f9535ddd. Engine receipt

sha256:cea4d5f9af27a9f39ae64961bf86cc0b432ef1ef4abff03d4cb08ad51be26976 at receipts/engine/model_admitted/SFT-INFO-THERM-ERASURE-BOUNDARY-004-cea4d5f9af27a9f3.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFO-THERM-OBSERVER
- SFT-V1-V2-INFORMATION-THERMODYNAMICS-CORPUS

Observed, adverse and boundary records:

- THERM-004: erasure_boundary
- exact observation: {"closed_distinctions":1,"physical_energy_claimed":false,"terminal_forms":1}
- complete family observation custody: 10 records; all rows preserved
- complete logical-record, merge, predecessor, held-label, measurement, reset and cycle enumeration supplies the observation; no imported physical constant, fitted energy value, erased provenance or information-energy identity enters

Falsification condition: Reject if the value-free registry, complete record/transition/predecessor custody census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:ad5095b2eeeca9420dbca54df2db1505aa8549eb8400a96435e442c03e8918fe; empirical validation sha256:06376961221e17186af92eba186bf1e4ee4a0b3e2a70fee8ab75e2dd9cd69c23; external validation

sha256:d98c60d882b8d81c154d37b5b715eb7d8f89e2c8ca5fe97077f923de47a622e3; isolation

sha256:e2def53287ff4f83c7caaca5ff2a091dc25300e63ebb532f0ef006474f623b97; custody

sha256:86ec118592e266e918b7896362b0df882e23c2fc66ed54456b9a3278f161fab5.

Prohibited inputs.

- no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption
- no physical work, heat or temperature law claimed without the Physics-owned translation
- no failed route retires an obligation or changes protected authority

211. Measurement record cost

Family: THERM Obligation: SFT - INFO - OBL - THERM - 005 Claim: SFT - INFO - THERM - MEASUREMENT - RECORD - 005

Question, law and forced result. A measurement that remains usable after interaction requires a retained record distinguishing every declared observation class; this is an information-record cost, not an energy value.

THERM-005 uniquely retains observed-distinction-record-custody, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - PRIV - COMPLETENESS - 010
- SFT - INFO - THERM - ERASURE - BOUNDARY - 004

The current reconciliation binds this obligation to the current model-admitted receipt
sha256: dcf2b5c1949690047ee8c1ba6acd05b4f1f3fc9f13499925486ab57d98533051. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis THERM-005 product before observation access.

Boundary: Every positive finite THERM-005 record form, merge, predecessor class, held label, reset cycle and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- THERM-005 uniquely retains observed-distinction-record-custody, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-logical-record-support__observed-distinction-record-custody__held-predecessor-custody__typed-physical-translation-boundary__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256: 98e26a83f41cf781e74adb7cc7e314d61f062b704f587933d48be7cd4fbd4f69; generality
sha256: 2a20d1fe6922de68d0ac9ba925d724ffba7c1a47b5b911ac50d4498cd32a4855.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing record forms hide physical custody.; receipt
sha256: c68e5b08f48dd681dc64f7db47ee32a82e31a70765db70257c8b81a3d99d5088.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption; no physical

work, heat or temperature law claimed without the Physics-owned translation; no failed route retires an obligation or changes protected authority; receipt
 sha256: 7413bcd4f9bb55ffb7d53b1328ef1b1da8c95aea43c3126152ef6678f5662991.

Independent reconstruction and engine admission. Independent implementation

sha256: 3882b70af437c25da5cbe28586c1abd8aad8fb27a41af518c3a85c84eefc47bb regenerated the declared census and sole survivor; independent certificate
 sha256: 31befc877d9a95ff132f54671282b4829a8a9d6f652d27d23ec778a72fa170b6. Derivation seal
 sha256: b1ab96303a83e5408d40d21f598725e6f1ea56f4edadfce0a8c91cf7c4472914. Source manifest
 sha256: 7ace763e7adfd8cca47e8189f2c6fb3087109e38589c3c05a36d091c7c2be5b6. Engine receipt
 sha256: dcf2b5c1949690047ee8c1ba6acd05b4f1f3fc9f13499925486ab57d98533051 at receipts/engine/model_admitted/SFT-INFO-THERM-MEASUREMENT-RECORD-005-dcf2b5c194969004.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFO-THERM-OBSERVER
- SFT-V1-V2-INFORMATION-THERMODYNAMICS-CORPUS

Observed, adverse and boundary records:

- THERM-005: measurement_record
- exact observation: {"measurement_classes":2,"outcome_records":2,"records_per_observation":1}
- complete family observation custody: 10 records; all rows preserved
- complete logical-record, merge, predecessor, held-label, measurement, reset and cycle enumeration supplies the observation; no imported physical constant, fitted energy value, erased provenance or information-energy identity enters

Falsification condition: Reject if the value-free registry, complete record/transition/predecessor custody census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: dce3a8701f853f3678ef4b249f117f1aea6383f6212747bb9304ae6b57341013; empirical validation sha256: 81ee5168b9c53e1ad2506b50e4936f58f8b6a539676c8772eb419c49807bc359; external validation
 sha256: 73308e9dc71bed8396984777f45767ef130fdaa24d60edf9c9a5f953f9042dba; isolation
 sha256: fcc7c17ce2e25ef921a3ee51494a2bc5ecbabeec60bba345a21e13805166bb00; custody
 sha256: bdb4b8dbb9405ccd9c7be9f444a55c5b5fcfebbbcf21109b41031569af505125.

Prohibited inputs.

- no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption
- no physical work, heat or temperature law claimed without the Physics-owned translation
- no failed route retires an obligation or changes protected authority

212. Memory reset and provenance custody

Family: THERM Obligation: SFT-INFO-OBL-THERM-006 Claim: SFT-INFO-THERM-MEMORY-RESET-006

Question, law and forced result. Memory reset preserves reversibility only when each predecessor distinction is transferred to another retained record before the reset image is written.

THERM-006 uniquely retains reset-predecessor-transfer-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-PRIV-COMPLETENESS-010
- SFT-INFO-THERM-MEASUREMENT-RECORD-005

The current reconciliation binds this obligation to the current model-admitted receipt sha256:0d4926053fca5a826d3950cd0c2356836bdd803489a7ebc4e195594e851f82b0. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis THERM-006 product before observation access.

Boundary: Every positive finite THERM-006 record form, merge, predecessor class, held label, reset cycle and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- THERM-006 uniquely retains reset-predecessor-transfer-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-logical-record-support__reset-predecessor-transfer-ledger__held-predecessor-custody__typed-physical-translation-boundary__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:86aa3e62c5fec3efbf77e959e14987340efe0bfc1656c4ae22d68f8cb881d299; generality sha256:238e38f1896b7c2b2f757c8b9ba85b61b12ea8c5404a49b31c99a5f0b86c0797.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing record forms hide physical custody.; receipt sha256:c68e5b08f48dd681dc64f7db47ee32a82e31a70765db70257c8b81a3d99d5088.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption; no physical work, heat or temperature law claimed without the Physics-owned translation; no failed route retires an obligation or changes protected authority; receipt sha256:7413bcd4f9bb55ffb7d53b1328ef1b1da8c95aea43c3126152ef6678f5662991.

Independent reconstruction and engine admission. Independent implementation

sha256:3882b70af437c25da5cbe28586c1abd8aad8fb27a41af518c3a85c84eefc47bb regenerated the declared census and sole survivor; independent certificate

sha256:31befc877d9a95ff132f54671282b4829a8a9d6f652d27d23ec778a72fa170b6. Derivation seal sha256:18bfffac4307d83c74613684c98ed3625c4bf1814a9e3e814f3390110b5e32ee7. Source manifest sha256:5d7bfa1ca74947157878e859259b554df1a1a5baea55091c4b60c196ad686cde. Engine receipt sha256:0d4926053fca5a826d3950cd0c2356836bdd803489a7ebc4e195594e851f82b0 at receipts/engine/model_admitted/SFT-INFO-THERM-MEMORY-RESET-006-0d4926053fca5a82.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFO-THERM-OBSERVER
- SFT-V1-V2-INFORMATION-THERMODYNAMICS-CORPUS

Observed, adverse and boundary records:

- THERM-006: memory_reset
- exact observation: {"predecessors_archived":2,"reset_images":1}
- complete family observation custody: 10 records; all rows preserved
- complete logical-record, merge, predecessor, held-label, measurement, reset and cycle enumeration supplies the observation; no imported physical constant, fitted energy value, erased provenance or information-energy identity enters

Falsification condition: Reject if the value-free registry, complete record/transition/predecessor custody census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:f83862e93c4e6bd91b54d332091f306045506e5bb2a604078e1f8e76a42efb2a; empirical validation sha256:13fba3aab8850735f88abdfec325a1b0136b6284e63ed726972d7033a4a69f; external validation

sha256:e2bb5b455b403cbc7aee7c2ecfafdf2720936c7bbc0d30572443cb009e8d49e1; isolation sha256:1203a7c99b80d59d35fc321f236248a412648c9663499c22fb85dcc62ecf08cf; custody sha256:fcf189e76e5640ccb5ac88969b26735890fd5dacc522ff055c266e3fee325d2.

Prohibited inputs.

- no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption
- no physical work, heat or temperature law claimed without the Physics-owned translation
- no failed route retires an obligation or changes protected authority

213. Maxwell-demon information ledger

Family: THERM Obligation: SFT-INFO-OBL-THERM-007 Claim: SFT-INFO-THERM-MAXWELL-DEMON-007

Question, law and forced result. A demon cycle must retain observation, conditional action, memory state and reset predecessor; omitting any row hides rather than removes the information cost.

THERM-007 uniquely retains observation-action-reset-cycle-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-PRIV-COMPLETENESS-010
- SFT-INFO-THERM-MEMORY-RESET-006

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:ac479ce2e5d5d8d4ff3e3edf60147a75e60bafd9edb7e79dcaba2d303e880733. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis THERM-007 product before observation access.

Boundary: Every positive finite THERM-007 record form, merge, predecessor class, held label, reset cycle and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- THERM-007 uniquely retains observation-action-reset-cycle-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-logical-record-support__observation-action-reset-cycle-ledger__held-predecessor-custody__typed-physical-translation-boundary__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:da85712135c6151cdaf06194e79075785b1044a589798964398916b85cbdb05d; generality
sha256:7a7836d842ee13e6e23b20d99602b3af704f4634113eabc8ee309e5c1202aa18.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing record forms hide physical custody.; receipt
sha256:c68e5b08f48dd681dc64f7db47ee32a82e31a70765db70257c8b81a3d99d5088.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption; no physical work, heat or temperature law claimed without the Physics-owned translation; no failed route retires an obligation or changes protected authority; receipt
sha256:7413bcd4f9bb55ffb7d53b1328ef1b1da8c95aea43c3126152ef6678f5662991.

Independent reconstruction and engine admission. Independent implementation

sha256:3882b70af437c25da5cbe28586c1abd8aad8fb27a41af518c3a85c84eefc47bb regenerated the declared census and sole survivor; independent certificate
sha256:31befc877d9a95ff132f54671282b4829a8a9d6f652d27d23ec778a72fa170b6. Derivation seal

sha256:d9d260947a70e346d3567767015fba6fd3e83fc78651247468897305a8f087c2. Source manifest sha256:750998686724718c178cff4180a6e8cad65fd4a3138cbc37fded59c49461cc11. Engine receipt sha256:ac479ce2e5d5d8d4ff3e3edf60147a75e60bafd9edb7e79dcaba2d303e880733 at receipts/engine/model_admitted/SFT-INFO-THERM-MAXWELL-DEMON-007-ac479ce2e5d5d8d4.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFO-THERM-OBSERVER
- SFT-V1-V2-INFORMATION-THERMODYNAMICS-CORPUS

Observed, adverse and boundary records:

- THERM-007: maxwell_demon_ledger
- exact observation: {"conditional_actions":2,"cycle_rows":2,"reset_records":2}
- complete family observation custody: 10 records; all rows preserved
- complete logical-record, merge, predecessor, held-label, measurement, reset and cycle enumeration supplies the observation; no imported physical constant, fitted energy value, erased provenance or information-energy identity enters

Falsification condition: Reject if the value-free registry, complete record/transition/predecessor custody census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:5738c6b2495cbc4dc166998ee60932f214088445be51c1faccd2cdd8b82c960f; empirical validation sha256:4b2cb0be27e22133fa461a1ea9a46156f7102d506483f9c387de521d91151a46; external validation sha256:a027b59babcd47980d47588ff70453b1136117cef404f89915ee14d60ddab77e; isolation sha256:50bb3c6682b0b1d9212f94811a57839ab01e38c9dc5bcf11ff94968b3f568c89; custody sha256:9368e234849903a341c9a13dc87749002503cfd4dcb0a57372ecccc2f41e5788f.

Prohibited inputs.

- no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption
- no physical work, heat or temperature law claimed without the Physics-owned translation
- no failed route retires an obligation or changes protected authority

214. Landauer and Bennett correspondence boundary

Family: THERM Obligation: SFT-INFO-OBL-THERM-008 Claim: SFT-INFO-THERM-LANDAUER-BENNETT-008

Question, law and forced result. The Fold law forces the missing-label cost of irreversible logic and the retained-label route of reversible logic; dimensional Landauer factors belong to Physics and empirical thermodynamics.

THERM-008 uniquely retains irreversible-reversible-translation-boundary, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-PRIV-COMPLETENESS-010
- SFT-INFO-THERM-MAXWELL-DEMON-007

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:138ca042b0cfac3f6186893aaae10d777cfd53161a559ce274641d8da0b4349c. A missing
 dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis THERM-008 product before observation access.

Boundary: Every positive finite THERM-008 record form, merge, predecessor class, held label, reset cycle and registered
 successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly
 one survives and 255 are eliminated. The forced coordinates are:

- THERM-008 uniquely retains irreversible-reversible-translation-boundary, complete
 record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-logical-record-support__irreversible-reversible-translation-bo
 undary__held-predecessor-custody__typed-physical-translation-boundary__complete-dec
 lared-record-product__root-bound-forward-forcing__post-registry-exact-observation__
 finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:975f03f990e2197b685a137ab6cad3a5a63e48054ffef3d20a5f9f667093222f; generality
 sha256:523cf8030c4429ade1b934251fdd6cbd8cd3688e7376c4d305829d02b4f2054b.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained
 distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit
 downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted
 magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT
 numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed:
 Missing record forms hide physical custody.; receipt
 sha256:c68e5b08f48dd681dc64f7db47ee32a82e31a70765db70257c8b81a3d99d5088.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs
 from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated
 product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no
 axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result; host 0 denotes
 structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating
 proof scalar; no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption; no physical
 work, heat or temperature law claimed without the Physics-owned translation; no failed route retires an obligation or
 changes protected authority; receipt
 sha256:7413bcd4f9bb55ffb7d53b1328ef1b1da8c95aea43c3126152ef6678f5662991.

Independent reconstruction and engine admission. Independent implementation

sha256:3882b70af437c25da5cbe28586c1abd8aad8fb27a41af518c3a85c84eefc47bb regenerated the
 declared census and sole survivor; independent certificate
 sha256:31befc877d9a95ff132f54671282b4829a8a9d6f652d27d23ec778a72fa170b6. Derivation seal
 sha256:87114c6fbbc26ab13aa2b96559b715fd70a1ab86ed2df3f08ec9ebc737afa657. Source manifest
 sha256:7fb0801e5867873269fae249166d7d8ea0f3fa85cb694a9473faff08ee790364. Engine receipt
 sha256:138ca042b0cfac3f6186893aaae10d777cfd53161a559ce274641d8da0b4349c at receipts/e
 ngine/model_admitted/SFT-INFO-THERM-LANDAUER-BENNETT-008-138ca042b0cfac3f.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFO-THERM-OBSERVER
- SFT-V1-V2-INFORMATION-THERMODYNAMICS-CORPUS

Observed, adverse and boundary records:

- THERM-008: landauer_bennett_boundary
- exact observation: {"dimensional_factor_imported":false,"irreversible_predecessors":2,"reversible_reconstructions":2}
- complete family observation custody: 10 records; all rows preserved
- complete logical-record, merge, predecessor, held-label, measurement, reset and cycle enumeration supplies the observation; no imported physical constant, fitted energy value, erased provenance or information-energy identity enters

Falsification condition: Reject if the value-free registry, complete record/transition/predecessor custody census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: aaf675c3f0a029e80af0fa4a5c55e85501db59e11c40c2c64c1f8b1f36dffe38; empirical validation sha256: eacc2ffb8737123476de21e592f317de69fd7c705f9153844dbde7abaca493; external validation

sha256: 572b717ec9558ee15b7c8cea6a90c6c57637030bb89298351194e413cf3711e5; isolation sha256: 2e05da6adb163b27dabeffc17b2076f19e97edde339db3c039fe9c175a2e253d; custody sha256: e51cbac07d2c9d83dc1a0e22946da656a1717c39a289e0a17d5c7811de76b446.

Prohibited inputs.

- no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption
- no physical work, heat or temperature law claimed without the Physics-owned translation
- no failed route retires an obligation or changes protected authority

215. Information-energy non-equivalence boundary

Family: THERM Obligation: SFT-INFO-OBL-THERM-009 Claim: SFT-INFO-THERM-NON-EQUIVALENCE-009

Question, law and forced result. Information structure does not equal energy: the same distinction organization may be realized by different carriers and energy ledgers, requiring a separately validated physical translation.

THERM-009 uniquely retains carrier-independent-information-structure, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-PRIV-COMPLETENESS-010
- SFT-INFO-THERM-LANDAUER-BENNETT-008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: a08b20c3def670d95b04b75bc75d3e53e4dd4a6a6dce9ac61238252da4aa838e. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis THERM-009 product before observation access.

Boundary: Every positive finite THERM-009 record form, merge, predecessor class, held label, reset cycle and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- THERM-009 uniquely retains carrier-independent-information-structure, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-logical-record-support__carrier-independent-information-structure__held-predecessor-custody__typed-physical-translation-boundary__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256: 250e0a7ed191e2286d95d80c71f996337bf48445040c62573ad2b51120b68e22; generality

sha256: 6a30fcb5e1c1daf4346a625a1b01135a20ffb11a1fb107bec2475b83a693fd15.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing record forms hide physical custody.; receipt
sha256: c68e5b08f48dd681dc64f7db47ee32a82e31a70765db70257c8b81a3d99d5088.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption; no physical work, heat or temperature law claimed without the Physics-owned translation; no failed route retires an obligation or changes protected authority; receipt
sha256: 7413bcd4f9bb55ffb7d53b1328ef1b1da8c95aea43c3126152ef6678f5662991.

Independent reconstruction and engine admission. Independent implementation

sha256: 3882b70af437c25da5cbe28586c1abd8aad8fb27a41af518c3a85c84eefc47bb regenerated the declared census and sole survivor; independent certificate

sha256: 31befc877d9a95ff132f54671282b4829a8a9d6f652d27d23ec778a72fa170b6. Derivation seal

sha256: f7d9971b694503c7cd25ae0a45948efd35ace570ec4ef13acbb6055cce296a6e. Source manifest

sha256: 6f3ab6dbc3716428a0dce1ffaf85458e39cc9123101f07250ae7d3a6d6461018. Engine receipt

sha256: a08b20c3def670d95b04b75bc75d3e53e4dd4a6a6dce9ac61238252da4aa838e at receipts/engine/model_admitted/SFT-INFO-THERM-NON-EQUIVALENCE-009-a08b20c3def670d9.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFO-THERM-OBSERVER

- SFT - V1 - V2 - INFORMATION - THERMODYNAMICS - CORPUS

Observed, adverse and boundary records:

- THERM-009: non_equivalence
- exact observation: {"carrier_families":2,"information_equals_energy":false,"information_organizations":1}
- complete family observation custody: 10 records; all rows preserved
- complete logical-record, merge, predecessor, held-label, measurement, reset and cycle enumeration supplies the observation; no imported physical constant, fitted energy value, erased provenance or information-energy identity enters

Falsification condition: Reject if the value-free registry, complete record/transition/predecessor custody census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:e4cfc063de026ea5afc9fa96a7c13feb7330ea80efb9803fac604a65695c4a22; empirical validation sha256:8f5268820a96dfd5e1f452d687302cb08ac17bcff9e2cbef4f53ab312dc1e56f; external validation

sha256:a53a5bc6ff9eb00c11dedb59b6ff714861d46eca3c7047c84d05b2ec7880450e; isolation sha256:9ec25e3790f77ffd7ca5193365a6ccfb2908859201679f875b440d1fc1bb41a9; custody sha256:ebb599ca2061fa24cbdfecbda9c98c36e1fa24a13a507975e3abdb5b745798e9.

Prohibited inputs.

- no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption
- no physical work, heat or temperature law claimed without the Physics-owned translation
- no failed route retires an obligation or changes protected authority

216. Information-thermodynamics completeness certificate

Family: THERM Obligation: SFT - INFO - OBL - THERM - 010 Claim: SFT - INFO - THERM - COMPLETENESS - 010

Question, law and forced result. Family completeness is the one-to-one reconciliation of all ten frozen physical-custody, merge, erasure, measurement, reset and correspondence obligations.

THERM-010 uniquely retains ten-information-thermodynamics-obligation-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - PRIV - COMPLETENESS - 010
- SFT - INFO - THERM - NON - EQUIVALENCE - 009

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:b662bedffa07ff8ade8f3596731a298ca45959b594568080eca0f994b4f4ecf7. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis THERM-010 product before observation access.

Boundary: Every positive finite THERM-010 record form, merge, predecessor class, held label, reset cycle and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- THERM-010 uniquely retains ten-information-thermodynamics-obligation-ledger, complete record custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-logical-record-support__ten-information-thermodynamics-obligation-ledger__held-predecessor-custody__typed-physical-translation-boundary__complete-declared-record-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:3c4d01be847665031d99539649b1357a3060dad190b15f8489f483a70e1e9e91; generality sha256:c90b798c4f904992cd88a457e8a5a7c2446741b0da9c88f386ee16af4f44d5d5.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Missing record forms hide physical custody.; receipt sha256:c68e5b08f48dd681dc64f7db47ee32a82e31a70765db70257c8b81a3d99d5088.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption; no physical work, heat or temperature law claimed without the Physics-owned translation; no failed route retires an obligation or changes protected authority; receipt sha256:7413bcd4f9bb55ffb7d53b1328ef1b1da8c95aea43c3126152ef6678f5662991.

Independent reconstruction and engine admission. Independent implementation

sha256:3882b70af437c25da5cbe28586c1abd8aad8fb27a41af518c3a85c84eefc47bb regenerated the declared census and sole survivor; independent certificate sha256:31befc877d9a95ff132f54671282b4829a8a9d6f652d27d23ec778a72fa170b6. Derivation seal sha256:112821b5b5e95abd73feafc26a454ce412592e644b23e1ab28541ea2f77a1eb4. Source manifest sha256:f2e9cddb142cecc180f7640c27a9b76ac1bfb338d90f75bc98cb40ae8efb681d. Engine receipt sha256:b662bedffa07ff8ade8f3596731a298ca45959b594568080eca0f994b4f4ecf7 at receipts/engine/model_admitted/SFT-INFO-THERM-COMPLETENESS-010-b662bedffa07ff8a.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-INFO-THERM-OBSERVER
- SFT-V1-V2-INFORMATION-THERMODYNAMICS-CORPUS

Observed, adverse and boundary records:

- THERM-010: information_thermodynamics_completeness

- exact observation: {"duplicate_owners":0,"observation_rows":10,"omitted_owners":0,"registered_obligations":10}
- complete family observation custody: 10 records; all rows preserved
- complete logical-record, merge, predecessor, held-label, measurement, reset and cycle enumeration supplies the observation; no imported physical constant, fitted energy value, erased provenance or information-energy identity enters

Falsification condition: Reject if the value-free registry, complete record/transition/predecessor custody census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:facee41be74699ea0ef35b3a1b0db0739bbf4eb92d3f254ddab2667f41e1f87b; empirical validation sha256:9f4e7c16ae0952e363fe1295f496869df753322a6aaebb46b6580d1db28f3380; external validation

sha256:8a4ac5fa77d543a5a89838472ddb150abfc27028b273ec515446b10f2b77d285; isolation sha256:03cbbd4258b0d405933946b80871e01372abd4cbd3e32bee063634c2ebb43152; custody sha256:050242a3e56dcc443f1cae14834f803ad9c097675bf5cb0d1eed30d92bb5f544.

Prohibited inputs.

- no axiom, imported thermodynamic formula, dimensional constant or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden predecessor, erased provenance, fitted energy value or carrier-specific assumption
- no physical work, heat or temperature law claimed without the Physics-owned translation
- no failed route retires an obligation or changes protected authority

Complete-field family: CORR

217. Classical symbol-information correspondence

Family: CORR Obligation: SFT - INFO - OBL - CORR - 001 Claim: SFT - INFO - CORR - CLASSICAL - SYMBOL - 001

Question, law and forced result. A classical symbol alphabet corresponds to generated Fold labels through a declared bijection that preserves every distinction and inverse decoding record.

CORR-001 uniquely retains bijective-classical-symbol-correspondence, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - THERM - COMPLETENESS - 010

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:118ba8a08ee951156e3c7e9e436b7ce27523ac223143cdf1e804d15e43f0a5ec. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-001 product before observation access.

Boundary: Every positive finite CORR-001 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-001 uniquely retains bijective-classical-symbol-correspondence, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__bijective-classical-symbol-correspondence__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:6df790f842ea7fbd8dca85d83adf9b64cbced11b5f71d39fcc24b64298dd07dc; generality sha256:66cab6eb83a321d8d881f2d991d20572cadb2095c336381dbf32d9bc564d3a21.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt sha256:6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt sha256:295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256:e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate sha256:8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal sha256:3cced8dcc1e6ccbdba8b95c3a17f8ca30aea000dcff9c08e34fc7b8d1706e820. Source manifest sha256:520586a490b2d5b39bc7e6c1574c3b7f159dc7aa8a6e9fbd126325c8403e2469c. Engine receipt sha256:118ba8a08ee951156e3c7e9e436b7ce27523ac223143cdf1e804d15e43f0a5ec at receipts/engine/model_admitted/SFT-INFO-CORR-CLASSICAL-SYMBOL-001-118ba8a08ee95115.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-001: classical_symbol
- exact observation: {"classical_symbols":2,"fold_labels":2,"inverse_preserved":true}
- complete family observation custody: 16 records; all rows preserved

- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:5222534ca80b9c58595a234fe4f5dbcb1e9a94aa73064029a73c537ea119cf93; empirical validation sha256:d2087ac778fb03ea23a1188f05dce82a0271a118f5ef0c9c9d6555ab86ba05e6; external validation

sha256:e178dedfed730a434517776c0a5520c61519245a02f7c03c5df25b64ccef8087; isolation sha256:3dc6a125ac9762b5dfb99972cbcf395f46121cb2e7f82eb70dce794c9d882da2; custody sha256:568b363eb390ff3f3abacb51b3ae3952239c3653a553c9f9bb77f91f7f6eff00.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

218. Deterministic-support probability correspondence

Family: CORR Obligation: SFT - INFO - OBL - CORR - 002 Claim:

SFT - INFO - CORR - DETERMINISTIC - PROBABILITY - 002

Question, law and forced result. A deterministic outcome corresponds to a singleton exact support and the ordered part one of one; probability notation is a translation, not a premise.

CORR-002 uniquely retains singleton-support-certainty-correspondence, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - THERM - COMPLETENESS - 010
- SFT - INFO - CORR - CLASSICAL - SYMBOL - 001

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:89616d25a9baf24176aa4e95c638f9c4a4359b6d70440a3f179dcf1184f31dd8. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-002 product before observation access.

Boundary: Every positive finite CORR-002 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-002 uniquely retains singleton-support-certainty-correspondence, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__singleton-support-certainty-correspondence__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: 2b3cf53d417f14942148cfc764ef69e23cc1ea8a415f96ffc008703f6ab4dc46; generality
 sha256: 343a1c8c60e17fb37f31cd33da8b7a999fc14fc6fb626c01d8b3c607623471d6.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
 sha256: 6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
 sha256: 295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256: e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate
 sha256: 8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal
 sha256: f91e84574cf954d02509d7f1f87e1a3266450156219ca6497d2a264c685283d6. Source manifest
 sha256: 089a4e523f37cd360d7f73b0ffad06245cc43275d74d61579d9cee5ce88d3427. Engine receipt
 sha256: 89616d25a9baf24176aa4e95c638f9c4a4359b6d70440a3f179dcf1184f31dd8 at receipts/engine/model_admitted/SFT-INFO-CORR-DETERMINISTIC-PROBABILITY-002-89616d25a9baf241.json. Closure scope `depth_independent`. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-002: deterministic_probability
- exact observation: {"certainty_part": [1,1], "support_forms": 1}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:6e5c85f4d947063b321137657cce887d07787a481981b8f82c10cac45807f050; empirical validation sha256:ac0ff71d5ccb30705261854b9d3bc8074fa2e1c33eba8baa96e150a84c8356a8; external validation sha256:c475796b8cbd9339dcdd7b61d5a0ae2c40f6bbcd547831d687ad6be02b2d75af; isolation sha256:db9b8259885736c3701b541f7dd76828b25ca06330c4e73325f93e9bac25a128; custody sha256:de25b2041f69e74cc1e3703a947857ca7f5c7828436e27f586d4f65b85a84470.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

219. Probabilistic mixture as hidden-support description

Family: CORR Obligation: SFT - INFO - OBL - CORR - 003 Claim: SFT - INFO - CORR - PROBABILISTIC - MIXTURE - 003

Question, law and forced result. A mixture description retains a complete source-labelled alternative support while its declared public observation closes the source label; weights require a separately registered measure.

CORR-003 uniquely retains hidden-source-support-mixture, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - THERM - COMPLETENESS - 010
- SFT - INFO - CORR - DETERMINISTIC - PROBABILITY - 002

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:2522d7ab8bdd8abb71758e3167a246660634f72b16c1b88a4deaf1644fb638c3. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-003 product before observation access.

Boundary: Every positive finite CORR-003 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-003 uniquely retains hidden-source-support-mixture, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__hidden-source-support-mixture__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:a5aefe1677e10e5e6d7762f56abac31b2664b494fa96ee0de1734f2d5346a095; generality sha256:e0beb200bcd8b13429f3a6f77468eab4c4d8d483d4780ccca4d264282d126159.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
sha256: 6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
sha256: 295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256: e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate

sha256: 8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal

sha256: 63a4858113ccfaa28718edeec9907646413186f4241552b03f54ba75ce19f287. Source manifest

sha256: dfaa335173b86df04712a817c2f29df4f337fa1c330914d772dc92146c77ae13. Engine receipt

sha256: 2522d7ab8bdd8abb71758e3167a246660634f72b16c1b88a4deaf1644fb638c3 at receipts/engine/model_admitted/SFT-INFO-CORR-PROBABILISTIC-MIXTURE-003-2522d7ab8bdd8abb.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-003: probabilistic_mixture
- exact observation: {"hidden_sources":2,"public_classes":1,"weights_imported":false}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:cc1cdd60ee9448be13e3e7ae6749dabb496ddc18de051eae5007faa9d3ddcb74; empirical validation sha256:1a021a116ba77eaa6f8caafd271aae8e1cc550d55e0305589815cd6bca58e877; external validation sha256:e14499ced09d4b5be5ffa53f289cb89dcdf898da7ba9c5bef3b8843b35c85d2d; isolation sha256:322e22cd9716c75991c4279dbb20bed596a8995e9fb1dc370f301a396c75a760; custody sha256:3278f3d31c8d005e181cb89ea0cd927f870291c3877df3308ab6e7f6aca71a30.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

220. Quantum support correspondence

Family: CORR Obligation: SFT - INFO - OBL - CORR - 004 Claim: SFT - INFO - CORR - QUANTUM - SUPPORT - 004

Question, law and forced result. Quantum computational-basis support corresponds to the complete generated Fold-word support; amplitudes, normalization and dynamics are not inferred from support alone.

CORR-004 uniquely retains complete-basis-labelled-word-support, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - THERM - COMPLETENESS - 010
- SFT - INFO - CORR - PROBABILISTIC - MIXTURE - 003

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:bc3f2bf9ab87ee90bdadfc03a01864f7e07a8559a6956f48d4be341ca27e1bec. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-004 product before observation access.

Boundary: Every positive finite CORR-004 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-004 uniquely retains complete-basis-labelled-word-support, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__complete-basis-labelled-word-support__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:75007fe3f2fe878570f4a8c9f4eac0f57329786b3833b8fd7a0bf9a153a95a32; generality sha256:7a0f0b0b3be2bc8c1f43bf74c3c9124b26010fdeb38a0b25d1014c1a480e1862.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted

magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
sha256: 6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
sha256: 295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256: e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate

sha256: 8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal
sha256: 701ca47787fc689aaba4120a3ed2f9631852756284746d000f7ee55760b07aad. Source manifest
sha256: 830af2613f0b8d84e6cb0a5693004bb6923c48374c68e8c806491e6b9a80a53e. Engine receipt
sha256: bc3f2bf9ab87ee90bdadfc03a01864f7e07a8559a6956f48d4be341ca27e1bec at receipts/engine/model_admitted/SFT-INFO-CORR-QUANTUM-SUPPORT-004-bc3f2bf9ab87ee90.json. Closure
scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-004: quantum_support
- exact observation: {"basis_support_words":4,"word_width":2}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 5910b43acc6069aa4479f8296ab48dce2fb3c7f96a9742905d84061cfbb2cf3d; empirical
validation sha256: d486a9857fe8f6ccd214b4b0815eb2df05c236e960a023b474e4c4a1174d35fa;
external validation
sha256: d4bdb209e1841da4da2346f5e1aea20cac6cbf88ef837f405440952de22e4f3c; isolation

sha256:1cee37af95521e27d4c1c0f1527ed7cca1ba9e05eb8362ec8a9637bd32ba516a; custody
sha256:20bcbe1f03b41817da439c65b3aae03426809c50591cd3ba624def0f2c48c0a4.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

221. Quantum information unit correspondence

Family: CORR Obligation: SFT - INFO - OBL - CORR - 005 Claim: SFT - INFO - CORR - QUANTUM - UNIT - 005

Question, law and forced result. One quantum information unit has a two-form basis support corresponding to one forced Fold distinction, while phase-bearing state structure remains downstream.

CORR-005 uniquely retains one-distinction-two-label-support, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - THERM - COMPLETENESS - 010
- SFT - INFO - CORR - QUANTUM - SUPPORT - 004

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:554e1458658eb5e15d2d128beff721c6539819a0666e5ba44d6b6e16bb20fd09. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-005 product before observation access.

Boundary: Every positive finite CORR-005 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-005 uniquely retains one-distinction-two-label-support, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__one-distinction-two-label-support__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:21bd2cc510b488e1cc5a9a48fd4491917ceb3d1c80f099c03ff7d9c2d3112c14; generality
sha256:ab877ac1b7533e0725b886548aa7e1ea24426aa82ec065835cb8cdaf75766f32.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
sha256: 6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
sha256: 295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256: e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate

sha256: 8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal
sha256: 7fb5a729f13cbc8e4a9c8f8b175db15cef4ee4771b70e231c49482054756cedc. Source manifest
sha256: 68e2845a2c1a0ad031ce0050047802156b5e11553621962239e9074465ebaa8d. Engine receipt
sha256: 554e1458658eb5e15d2d128beff721c6539819a0666e5ba44d6b6e16bb20fd09 at receipts/engine/model_admitted/SFT-INFO-CORR-QUANTUM-UNIT-005-554e1458658eb5e1.json. Closure scope
depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-005: quantum_unit
- exact observation: {"distinctions":1,"support_labels":2}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: f63386d677bfe66537482bdc44e853a9842aa6f902e1363f3f956e2dea674f7d; empirical validation
sha256: cffff1352f36ad3dd7453415c477d070bde148a9a8f5a68cf5c3209a3ce3ee3b0; external validation

sha256: b71d2812fcb07ef3ecf211f98f4e94f25ecd292cdc06f262c7b4e5528bcded17a; isolation
sha256: fee144c50e75f3ba4d1e5acc4aacee2a4cae1fab67747587b53d41da92ef8815; custody
sha256: 5ad052fad05a7c159b93d2d664348e3a28e9493056413e92d308bfee78618904.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

222. Composite support and product-state correspondence

Family: CORR Obligation: SFT - INFO - OBL - CORR - 006 Claim: SFT - INFO - CORR - PRODUCT - STATE - 006

Question, law and forced result. Independent component supports compose as their complete ordered product; this is the information-support boundary of product-state correspondence.

CORR-006 uniquely retains complete-product-support-correspondence, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - THERM - COMPLETENESS - 010
- SFT - INFO - CORR - QUANTUM - UNIT - 005

The current reconciliation binds this obligation to the current model-admitted receipt
sha256: f0ff219046370d360d05ac520611db612ba5173cefec635a8e1d19000d352ef6. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-006 product before observation access.

Boundary: Every positive finite CORR-006 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-006 uniquely retains complete-product-support-correspondence, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__complete-product-support-correspondence__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256: 3f3083caf89fcb0e439e43eaaafd8754e8da7ae51e80359e8158809e61aa67fca; generality
sha256: 4923e5d61b0338738ed45554c8059c225283880ed2675ac2d8d1134211fabbd0.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
sha256: 6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt

sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
sha256:295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256:e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate

sha256:8b3621ada15f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal

sha256:0c23cbede23bae1292e24986d75d55b3ea1bd62029f22ef74f73269a715d8a6a. Source manifest

sha256:85057b6e0010c55ebd5ccbb465010d0d23107fbbdf3fc1d1f90ed9f2fd77e603e. Engine receipt

sha256:f0ff219046370d360d05ac520611db612ba5173cefec635a8e1d19000d352ef6 at receipts/e

ngine/model_admitted/SFT-INFO-CORR-PRODUCT-STATE-006-f0ff219046370d36.json. Closure

scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-006: product_support
- exact observation: {"component_support_each":2,"product_support":4}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:58a9d29394f7645c0541d90e62563056026e7fb76416d33fe0c3f95c468d9450; empirical validation sha256:b23d0f0ab3326cc9e004b14280ddaa10a4928a21e49682ba0fc8fa71933d4a17; external validation

sha256:6c6a2dd57587419dff33fc902449d4e6c4ed81e13533b736ebfaa7ca9f596cf5; isolation

sha256:3b057440b67d491c1910c670d2126ee61c9fd2db2470ee85b7879e8f3c8b8c8b; custody

sha256:d64fdcb1ac500b4598eb415bdea95379fe852e3d00307b2ec4fe5112a33451dd.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization

- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

223. Joint support and entanglement boundary

Family: CORR Obligation: SFT - INFO - OBL - CORR - 007 Claim: SFT - INFO - CORR - ENTANGLEMENT - BOUNDARY - 007

Question, law and forced result. A joint support smaller than the product of its complete marginals is nonproduct support; entanglement requires additional phase/amplitude and operational certificates.

CORR-007 uniquely retains nonproduct-joint-support-boundary, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - THERM - COMPLETENESS - 010
- SFT - INFO - CORR - PRODUCT - STATE - 006

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:5fa979b85852fe123af398b4dcb4d2d49051bfa7e42a8bf6aa9110932fdadeec. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-007 product before observation access.

Boundary: Every positive finite CORR-007 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-007 uniquely retains nonproduct-joint-support-boundary, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__nonproduct-joint-support-boundary__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:005bd1e8a0cde600cf4102051213c7bf7f5df1d66d35aa3b4e9c844687321fbd; generality
sha256:a49d798e4ad034df6ad5765b0dea8cef0b1e27abc075e08e74dc3368d9c44b8a.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
sha256:6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
sha256:295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission.

Independent implementation
sha256:e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate

sha256:8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal

sha256:9649d472bc5257c2bfbfe8b22dc0d8d6d4823ced3fe0b3f1f84cd9af2ca4e26d7. Source manifest

sha256:8f94eb048239afe19482562965168f403d85f713c4bceb75a8e21629018a069b. Engine receipt

sha256:5fa979b85852fe123af398b4dcb4d2d49051bfa7e42a8bf6aa9110932fdadeec at receipts/engine/model_admitted/SFT-INFO-CORR-ENTANGLEMENT-BOUNDARY-007-5fa979b85852fe12.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-007: entanglement_boundary
- exact observation: {"joint_support":2,"marginal_product_support":4,"operational_entanglement_claimed":false}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:3c4ea1778cb9c23492ed3ef7e403706049af38b6df265b88a3a7ec0cd563e47e; empirical validation sha256:8a749211f1c231fd35e55694bc2a6e954fe421147b85b0211aedbbbc641f72134;

external validation

sha256:624267b9f730ae90b82e95afc90be5baea9c8731f316d048f905295cc382ddab; isolation

sha256:fd3a6eb53d7dcd1fa7e327278c8d45ffe9ffe2d6f88f3ddc1780eb25b6bad807; custody

sha256:0fd78b13aafd3a0ee98d831039268d713cf308db76af4ceab41e7cc582b498c0.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

224. Phase information ownership handoff

Family: CORR Obligation: SFT - INFO - OBL - CORR - 008 Claim: SFT - INFO - CORR - PHASE - HANDOFF - 008

Question, law and forced result. Information Science retains phase-class distinctions and provenance; Quantum Computation owns phase transformations, interference and operational composition.

CORR-008 uniquely retains phase-record-quantum-dynamics-handoff, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - THERM - COMPLETENESS - 010
- SFT - INFO - CORR - ENTANGLEMENT - BOUNDARY - 007

The current reconciliation binds this obligation to the current model-admitted receipt
sha256: fe8cd634a75985a8a44fe9418edff56537e4a3d7fa4783988c7ae32035e04c48. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-008 product before observation access.

Boundary: Every positive finite CORR-008 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-008 uniquely retains phase-record-quantum-dynamics-handoff, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__phase-record-quantum-dynamics-handoff__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256: ad161a809154043d753c9b8694183cf25fa707b8bcf3fcf5161c7ccc8683be74; generality
sha256: 489beb5ccc5b08f636df100355cc8f69b39f201cf6de7a3e6e052333a02ec74.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
sha256: 6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement

dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
 sha256: 295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256: e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate
 sha256: 8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal
 sha256: 09dffe443d9666df9d271825a77008370ff47385e176a359d68aab5ff1b2c5d5. Source manifest
 sha256: 35d59476d911cc72ed60209a400faaaaaacd95f833ad61591acb329bf65abc31. Engine receipt
 sha256: fe8cd634a75985a8a44fe9418edff56537e4a3d7fa4783988c7ae32035e04c48 at receipts/engine/model_admitted/SFT-INFO-CORR-PHASE-HANDOFF-008-fe8cd634a75985a8.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-008: phase_handoff
- exact observation: {"phase_classes":2,"quantum_dynamics_owner":"quantum_computation"}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 40995e24615efd931cd7e0b2ea49e87086c68651ecba66b85debf8da1c0e58f4; empirical validation sha256: 21cf602135c00d77bfcfa93634879a220e71a323b29a2b6aa000d57eecd8b9d2; external validation
 sha256: 924ddc4d2d5a3f77165a5a92bf18828ece4a5770e3a99ad52a7d54c62abd5033; isolation
 sha256: eddb8998c7f309c7bda3e38afe339baa537e348d8f635ed5782f20aa55fcd288; custody
 sha256: 1b0ff44f504e5512cc1038702f29c4841c49f7de15508806b4d695f22dab33a8.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

225. Measurement support and record correspondence

Family: CORR Obligation: SFT-INFO-OBL-CORR-009 Claim: SFT-INFO-CORR-MEASUREMENT-SUPPORT-009

Question, law and forced result. Measurement-support correspondence is a declared partition of state support into observation classes plus an exact retained outcome record.

CORR-009 uniquely retains support-partition-outcome-record, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-THERM-COMPLETENESS-010
- SFT-INFO-CORR-PHASE-HANDOFF-008

The current reconciliation binds this obligation to the current model-admitted receipt sha256:b9fbf7541d51c25d138e4ac83e062aa5dc4f4c1975b0960cac0d6b985e7b8d75. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-009 product before observation access.

Boundary: Every positive finite CORR-009 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-009 uniquely retains support-partition-outcome-record, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__support-partition-outcome-record__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-corr-espondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:0b0d61a7503b87e0452261f999034ca65d2c195192659b70597435bac1c9db5b; generality sha256:95b46702e4cd3ee4e30486ab0c149c8fa824191cbbc8b0acd2f429704535d3fb.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt sha256:6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt sha256:295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256:e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the

declared census and sole survivor; independent certificate

sha256:8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal sha256:8c34a4e1c311e623c4aacc8f5e4f45082194587db2f77718d91f7474c1913f83. Source manifest sha256:1752c5c693036864616ba533f2d2c8ff4c9238e5cefef813c4ce866ddafd25e9. Engine receipt sha256:b9fbf7541d51c25d138e4ac83e062aa5dc4f4c1975b0960cac0d6b985e7b8d75 at receipts/engine/model_admitted/SFT-INFO-CORR-MEASUREMENT-SUPPORT-009-b9fbf7541d51c25d.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-009: measurement_support
- exact observation: {"observation_classes":2,"records_per_word":1,"support_words":4}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:5d062d82ab78420da6c553641d1a778e4f3824978e94935d7e4d6e2309c89258; empirical validation sha256:6aef64f4bc1c8d1f5400a51ca2923957b90629e6ec1af0e64ad020ac5719fa4c; external validation

sha256:9d403e09d08682798b4236ba1741dc27b4e15054e1fb28235ba444e227c0ae9c; isolation sha256:fefe47f6cce7e4dca9ae815a7f28d025a4d99dc875373e9e11752040b21a3882; custody sha256:f03ea50ce248c98372c4bfb9f379a2579db41d253272847fb459e700cb373a89.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

226. Classical-to-quantum encoding boundary

Family: CORR Obligation: SFT-INFO-OBL-CORR-010 Claim:

SFT-INFO-CORR-CLASSICAL-QUANTUM-ENCODING-010

Question, law and forced result. Classical information encodes into quantum support through an injective basis-labelled map with exact inverse decoding on the encoded subspace.

CORR-010 uniquely retains injective-classical-basis-encoding, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-THERM-COMPLETENESS-010
- SFT-INFO-CORR-MEASUREMENT-SUPPORT-009

The current reconciliation binds this obligation to the current model-admitted receipt sha256:e8cad4f6644667784e5c6c7cd0e61ca7da360915d55b238f57c0336eccb7d7e5. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-010 product before observation access.

Boundary: Every positive finite CORR-010 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-010 uniquely retains injective-classical-basis-encoding, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__injective-classical-basis-encoding__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:1c8e9ab25c15124a8f392cd06631e3ed0d6408c7f00a4a8638fcfe53d67ce9; generality sha256:79175d7101ae0933518970892f0822c826552fd1699ff1e12276239f70ef4034.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt sha256:6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt sha256:295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation sha256:e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fddb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate sha256:8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbef332. Derivation seal

sha256:51e2cf32b78b0a728412ba89ce52e09cb0387f9a8dbd7b9c1bc384a99d3125cb. Source manifest sha256:79ead1c75458fbcfee02d424f7dfc4b0c19c1b461725e49972d99c5c07e807f7. Engine receipt sha256:e8cad4f6644667784e5c6c7cd0e61ca7da360915d55b238f57c0336eccb7d7e5 at receipts/engine/model_admitted/SFT-INFO-CORR-CLASSICAL-QUANTUM-ENCODING-010-e8cad4f664466778.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-010: classical_quantum_encoding
- exact observation: {"basis_forms":2,"classical_forms":2,"injective":true}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:080235bab31edf3e809a80ddf7804723a5a58835e44d7f3922d37a55ef20685d; empirical validation sha256:a7770c3291e14bc2a1b3db775676e5c3156f6fab82b65749f1b3452a912fa2a4; external validation

sha256:550c11b5f67c03dba2caa366d7c3424d1b1adb8a2fc8f07e94b97e57becfc87e; isolation sha256:88bfd39b6b68cbefbf1ea5e39bbd91c161612e1e2f12aa417b9bb67b14d3449e; custody sha256:611f2277f27ad323cf3f4a4ce340be6ba9d7b0a8a549e668ab0b05a0ba67e209.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

227. Quantum-to-classical observation boundary

Family: CORR Obligation: SFT-INFO-OBL-CORR-011 Claim:

SFT-INFO-CORR-QUANTUM-CLASSICAL-OBSERVATION-011

Question, law and forced result. Quantum-to-classical observation returns a classical class label and retains the complete quantum-support predecessor class compatible with that record.

CORR-011 uniquely retains observation-class-predecessor-custody, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-THERM-COMPLETENESS-010
- SFT-INFO-CORR-CLASSICAL-QUANTUM-ENCODING-010

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:bfc0c7b211fdbbdfda90769fd16c8923025edc967489bef354b02507b5cd5ce7. A missing
 dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-011 product before observation access.

Boundary: Every positive finite CORR-011 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-011 uniquely retains observation-class-predecessor-custody, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__observation-class-predecessor-custody__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:72033bdb1f3cfca3e7d87323cdc13820310bf31b690cf1bf888ce2f9301c9801; generality
 sha256:401125b6514c0b7dd12efb48c72d4bbb6ca656230634015fe6669bb7e2872608.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
 sha256:6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
 sha256:295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256:e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate
 sha256:8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal
 sha256:ebd0d6b85fd236522bbd265059be704c1348f7c925fc5f8460a00e15e9509a45. Source manifest
 sha256:cb899d0c81a9df57cb63a8b0155f3d83911681b12fb7fd957ca2845643a12d9f. Engine receipt
 sha256:bfc0c7b211fdbbdfda90769fd16c8923025edc967489bef354b02507b5cd5ce7 at receipts/e

ngine/model_admitted/SFT-INFO-CORR-QUANTUM-CLASSICAL-OBSERVATION-011-bfc0c7b211fdbb
dd.json. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows
preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-011: quantum_classical_observation
- exact observation: {"cross_word_predecessors":2,"observation_class":"different"}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:8731d37e374a4ce1eecfb7a2a4ce074cbc5e8e4060b7f2d9263c830bbff2a8b0; empirical
validation sha256:290f323931e03fb72630dd4ce9a03c0ab55dbfb3938bb77ad22138cc99b88b5d;
external validation

sha256:e1c5602b3202cc01d389ff3ecc081173cf646da30be1ce5cf3e6ed6202ab39b1; isolation
sha256:7cfdd9d2fccb7c3ea0da8aed3b9a83475609ff313506e34ab4b36fe7237a2c7a; custody
sha256:12268ea68a16283dd5a6246549fd773f97d492465a2dbd20c1b31e441eda836a.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

228. No-cloning information boundary correspondence

Family: CORR Obligation: SFT-INFO-OBL-CORR-012 Claim: SFT-INFO-CORR-NO-CLONING-BOUNDARY-012

Question, law and forced result. Support organization alone permits classical label copying and therefore cannot prove quantum no-cloning; the theorem belongs to phase-bearing linear operational dynamics.

CORR-012 uniquely retains support-insufficient-no-cloning-boundary, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-THERM-COMPLETENESS-010
- SFT-INFO-CORR-QUANTUM-CLASSICAL-OBSERVATION-011

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:10224f88e4fb663e8fd3626bbead483ab7b13155f2e2169be9611817245170c5. A missing
 dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-012 product before observation access.

Boundary: Every positive finite CORR-012 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-012 uniquely retains support-insufficient-no-cloning-boundary, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__support-insufficient-no-cloning-boundary__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:aa61791144b2086d07a9893bb99a716d76826730b13bec7b1e9c2e013c486655; generality
 sha256:7b8e1404be281bd61e41f71d65190388fe2269503a3058f1c939d64fc389ce4e.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
 sha256:6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
 sha256:295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256:e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fddb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate
 sha256:8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal
 sha256:b3d53221096138bbce764cece02318ad7a70470cd36152c8a2a9a241c71c3c9f. Source manifest
 sha256:f2ab42c084be1f3e78b8a021c880d3513f8b7e50cae7fcccc6f80756417558bf. Engine receipt
 sha256:10224f88e4fb663e8fd3626bbead483ab7b13155f2e2169be9611817245170c5 at receipts/engine/model_admitted/SFT-INFO-CORR-NO-CLONING-BOUNDARY-012-10224f88e4fb663e.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-012: `no_cloning_boundary`
- exact observation: `{"classical_support_copies":2,"no_cloning_from_support_alone":false}`
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:66cfc5021cd0ca9ad95654e6f8b08f870ae3a3fdb1c579ca1e77b919fa48fc7; empirical validation sha256:540b5d95b1c938891144038e5c88e783006c85cef439a5b04ca18d8292b33912; external validation

sha256:cdf0a7c6bdbf34b07c78ba952fc703eef559327835cf6b903d5b98dbb0bbb4d4; isolation sha256:458dc03c1fff13f55e94df637a0e73e590c22fbff502789bbe7f59758bb76574; custody sha256:b8d977fe2f8d11961679150981b26a350a6e2629fc28c6e5fd021fc3895f8268.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

229. Quantum channel support correspondence

Family: CORR Obligation: SFT-INFO-0BL-CORR-013 Claim: SFT-INFO-CORR-QUANTUM-CHANNEL-013

Question, law and forced result. Quantum-channel support correspondence retains the complete input-output support relation while channel amplitudes and physical transformations remain downstream.

CORR-013 uniquely retains input-output-quantum-support-relation, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-THERM-COMPLETENESS-010
- SFT-INFO-CORR-NO-CLONING-BOUNDARY-012

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:3dc6abb9e97fa6b29ca17d100b3604d7d7ba62c62b072a0fbb29f15f629991b2. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-013 product before observation access.

Boundary: Every positive finite CORR-013 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-013 uniquely retains input-output-quantum-support-relation, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__input-output-quantum-support-relation__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:c2ac3ccaa6fd80936aa332440bc2b955674d6ff2a3c51abc2b39707698db4369; generality
sha256:886aa1f2daf3a951648e631b61b32f2e4365482bb5a31931dbffaf2f4b9a1568.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
sha256:6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
sha256:295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256:e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate
sha256:8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal
sha256:0be7915f61b0addc8b307a7b4edaa9ab4fc449de6544d782bd3b41e6aa6f51f9. Source manifest
sha256:7890f137987f001436ef2405212780bd2182271ed4f1253c8a8ef519ef087b32. Engine receipt
sha256:3dc6abbae97fa6b29ca17d100b3604d7d7ba62c62b072a0fbb29f15f629991b2 at receipts/engine/model_admitted/SFT-INFO-CORR-QUANTUM-CHANNEL-013-3dc6abbae97fa6b2.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER

- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-013: quantum_channel_support
- exact observation: {"amplitudes_imported":false,"input_forms":2,"relation_rows":3}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:21e47eecba25d654dd828490475dcacc49e6db6639513f20c89e2eaff64ed5ba; empirical validation sha256:15059b6d07a3949de368ebf7c32fab2aba89cfca3d63869b5d3bf1c35c1c7c2; external validation

sha256:0050b2edf974b543074a5d82eca5b9216ad602fe0b1366d3d391c8b0e27b0dbd; isolation sha256:a7a5fb107f4888d4fc6e5c8a8633482f04e118ef8cacd83fe55ad9644bd58f71; custody sha256:c2acebb728da909973e28baa9bb55aaa4467bddbe9197d52fdbb55c52d29bb35.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

230. Quantum coding ownership handoff

Family: CORR Obligation: SFT-INFO-OBL-CORR-014 Claim:

SFT-INFO-CORR-QUANTUM-CODING-HANDOFF-014

Question, law and forced result. Information Science exports codeword, error and observation support; Quantum Computation owns phase-bearing codes, correction operations and fault tolerance.

CORR-014 uniquely retains information-support-quantum-coding-handoff, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-THERM-COMPLETENESS-010
- SFT-INFO-CORR-QUANTUM-CHANNEL-013

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:824a1118b9f203ef2bfb1cb203cc5aae00cac49a9bada352f3660079aff4eb2. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-014 product before observation access.

Boundary: Every positive finite CORR-014 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-014 uniquely retains information-support-quantum-coding-handoff, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__information-support-quantum-coding-handoff__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: 465928cafb961cdd028fe4986c47cbfc129390f7b49fe48480887f16e04f011d; generality sha256: 49a327b1d982f4e5c99ae954441dcd49cbe1b35c04e4d998dfb5cc3a281f4dcf.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt sha256: 6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt sha256: 295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256: e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate sha256: 8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbef332. Derivation seal sha256: f7b3ed21fcc129673df8d10ad36c2ed4428829a5dccf3f976e4ba68da82281ab. Source manifest sha256: 015ae59ace6d60789e02ce4aec1d3b7a407ae8350eef831ad55ede3b3b0d8d4. Engine receipt sha256: 824a1118b9f203ef2bfbcb1cb203cc5aae00cac49a9bada352f3660079aff4eb2 at receipts/engine/model_admitted/SFT-INFO-CORR-QUANTUM-CODING-HANDOFF-014-824a1118b9f203ef.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-014: quantum_coding_handoff

- exact observation: {"downstream_owner": "quantum_computation", "exported_support_records": 4, "ownership_overlap": 0}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 055ceb6a15dffe2b6e1361a4978dbea3be65b278c6e712d93f540cee07dc41a9; empirical validation sha256: f7082a3735cba464f4bf04579566a18f34894e30f39638de862bf21b78f101d0; external validation

sha256: 64ca3ad036bf69668e720c060511d20e61a9ad86e866d376cbee368b833a0105; isolation sha256: 3902cfb56d49fad98e33849caaed19351d3147712a89dc28513495a70a7e196e; custody sha256: 7d45957a5d1bdb13c7bfb753005a8e9c62c9bb7546d8c417f1f6989083469d69.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

231. Operational-dynamics nonduplication boundary

Family: CORR Obligation: SFT - INFO - OBL - CORR - 015 Claim: SFT - INFO - CORR - NONDUPLICATION - 015

Question, law and forced result. Information correspondence may not duplicate or select quantum gates, amplitudes, interference, measurement dynamics or physical implementations.

CORR-015 uniquely retains support-dynamics-one-owner-boundary, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - THERM - COMPLETENESS - 010
- SFT - INFO - CORR - QUANTUM - CODING - HANDOFF - 014

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 452ba460fdf82a1005400b2764039037fbc6d38b98527a1e8fc554f669036e78. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-015 product before observation access.

Boundary: Every positive finite CORR-015 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-015 uniquely retains support-dynamics-one-owner-boundary, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__support-dynamics-one-owner-boundary__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-c

correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:bc88e1ed43daad2147a9ce3c60a189f22ac0f5e5886a227a1d819788702a1fe4; generality
 sha256:48614777c9e245d01fdea9e48c9022cf6fd518bf658202eaa887b6a18b8a94c7.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
 sha256:6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
 sha256:295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256:e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate
 sha256:8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal
 sha256:d0fce3f50eec2edcba49fd1e66918d661d940ecec1a870671976c61f9afec739. Source manifest
 sha256:ca5da84963207370ee95084f6ed8381a17654b13cee2984e9bc8bb4213e27d07. Engine receipt
 sha256:452ba460fdf82a1005400b2764039037fbc6d38b98527a1e8fc554f669036e78 at receipts/engine/model_admitted/SFT-INFO-CORR-NONDUPLICATION-015-452ba460fdf82a10.json. Closure
 scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-CORRESPONDENCE-OBSERVER
- SFT-V1-V2-CLASSICAL-QUANTUM-CORPUS

Observed, adverse and boundary records:

- CORR-015: nonduplication
- exact observation: {"operational_dynamics_imported":false,"support_record_types":3}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:d8f8a093f64114de98475c54c556065c2076d56e51a9e3d82bdebca25e517012; empirical validation sha256:69c416b1ba48684b6b315c900212e9a30a9cba22d6d5095539e029d67ef46cb5; external validation sha256:bdedd0c9a512c5cefc70d604bf6832590a2666113759e73ce8a23dfd31427b25; isolation sha256:721d5e53ec81e0213b3a6f58ca2b75bb5c02171a2b1b1191dfd84db1fb99d464; custody sha256:4752be1fbb3e708cb423bec41a2473abe49e0c7c529e690416099d17cf20ecc6.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

232. Classical-probabilistic-quantum completeness certificate

Family: CORR Obligation: SFT - INFO - OBL - CORR - 016 Claim: SFT - INFO - CORR - COMPLETENESS - 016

Question, law and forced result. Correspondence completeness reconciles all sixteen frozen classical, deterministic, probabilistic and quantum-support interface obligations.

CORR-016 uniquely retains sixteen-correspondence-obligation-ledger, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - THERM - COMPLETENESS - 010
- SFT - INFO - CORR - NONDUPLICATION - 015

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:776c6b3d98422e38cda9379977c46ef5f93409220e6716024bffa07200504289. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis CORR-016 product before observation access.

Boundary: Every positive finite CORR-016 support, correspondence row, observation class, predecessor record, handoff and registered successor boundary.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- CORR-016 uniquely retains sixteen-correspondence-obligation-ledger, complete correspondence custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-generated-support__sixteen-correspondence-obligation-ledger__complete-observation-record-custody__support-only-ownership-boundary__complete-declared-correspondence-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-boundary.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:cebe74212b2d4d3f51bf4336352d055422580bca1b4226b75229fda2671bdf9d; generality sha256:125c4771ecb6fa06f57b75c9071375575e034806c8f284b08fbf76c9ed616982.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted forms falsify correspondence.; receipt
sha256: 6245714bb77acb645b9be36e01a3ae1d35226cd990220671fdef5b4da570348e.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported probability measure, Hilbert-space model or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden support word, fitted amplitude, stochastic prior or imported normalization; no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation; no failed route retires an obligation or changes protected authority; receipt
sha256: 295ad3d969bcf228cacc213256f8f79a3d84999868cb4f445edf73327b67a633.

Independent reconstruction and engine admission. Independent implementation

sha256: e5278a68b5470355fc82bc09ed8ef8898fd9249a0fa275f9fdbb9bd495d03d73 regenerated the declared census and sole survivor; independent certificate

sha256: 8b3621ada151f1df9446c23704fbd556b368045e6dd6166ec722ec98dbefd332. Derivation seal
sha256: 5f1a7e64659270999d51657afa1732c89014f2880d4afb537e7402ab8b1633f8. Source manifest
sha256: 700714c186a3183978f802e749260b3d0ccab794088404dc5697824114879a5e. Engine receipt
sha256: 776c6b3d98422e38cda9379977c46ef5f93409220e6716024bffa07200504289 at receipts/engine/model_admitted/SFT-INFO-CORR-COMPLETENESS-016-776c6b3d98422e38.json. Closure scope
depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INDEPENDENT- EXACT- CORRESPONDENCE- OBSERVER
- SFT-V1-V2- CLASSICAL- QUANTUM- CORPUS

Observed, adverse and boundary records:

- CORR-016: correspondence_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":16,"omitted_owners":0,"registered_obligations":16}
- complete family observation custody: 16 records; all rows preserved
- complete classical, deterministic, mixture, basis, product, joint, observation and ownership-boundary enumeration supplies the observation; no imported probability measure, amplitude, Hilbert-space law, gate or hidden support word enters

Falsification condition: Reject if the value-free registry, complete support/correspondence/ownership census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 7c19f64235d3f192c78c267e25b378d4db311cbbd749a20b0fe3a84dd85bdd52; empirical

validation sha256:494302f20df60da806f3bd805119f4c0c6009f6a22abc8a8541eb4d0002a8df9;
 external validation
 sha256:2da0907f31541226d0eb4a7cb41e14f9085d84aecbc633e8d980eabc4ab2a5f; isolation
 sha256:1316d1207cd466c347fc574c27ae07323ffd0e517df1bcfed8230d1917cb2b93; custody
 sha256:81cc9565b7f6bebecd6076a2415e78a45a69d47f968f96b7eed1bee1fbe1b2fe.

Prohibited inputs.

- no axiom, imported probability measure, Hilbert-space model or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden support word, fitted amplitude, stochastic prior or imported normalization
- no gate, interference, entanglement dynamics or quantum error correction duplicated from Quantum Computation
- no failed route retires an obligation or changes protected authority

Complete-field family: SEM

233. Symbol count and semantic-content boundary

Family: SEM Obligation: SFT - INFO - OBL - SEM - 001 Claim: SFT - INFO - SEM - SYMBOL - CONTENT - BOUNDARY - 001

Question, law and forced result. Symbol quantity fixes the exact count and organization of marks but does not alone fix a reference relation; equal-size symbol organizations can retain distinct content records.

SEM-001 uniquely retains symbol-count-content-nonidentity, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CORR-COMPLETENESS-016

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:b34473d457ce45cd3d8166ab996da33cb3b1a7df8b8eb01b9b892be2e03d7508. A missing
 dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-001 product before observation access.

Boundary: Every positive finite SEM-001 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-001 uniquely retains symbol-count-content-nonidentity, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__symbol-count-content-nonidentity__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-success-or-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:0f3635c18325ef47a50385140ad6142e262d2f026960f39127fb423be5330440; generality
 sha256:ec36f56f4432b84fe027c975cde689c848c3a71d452fbb9bf4100291c5b5eb84.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT

numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt
sha256: 67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt
sha256: 18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256: 3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate

sha256: aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal

sha256: f72e12455cbb1e0da4663cc83e129aac26ef2f6635fb20082d366aac6e65674e. Source manifest

sha256: cdf9ec5ac03e97387a5a665789d55603455599cb15dea3a28e97ca00612155be. Engine receipt

sha256: b34473d457ce45cd3d8166ab996da33cb3b1a7df8b8eb01b9b892be2e03d7508 at receipts/engine/model_admitted/SFT-INFO-SEM-SYMBOL-CONTENT-BOUNDARY-001-b34473d457ce45cd.json

. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-001: symbol_content_boundary
- exact observation: {"distinct_orders":2,"equal_symbol_counts":true,"symbols_per_word":2}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 5249d0beadf7039305f86c4ae59131d913851792157833f3b540589dac3cdc53; empirical

validation sha256: 379499f5192f5a8a9f4234d633c2e9c34fd5be8d0017425cecc8a862c9d633c5;

external validation

sha256:d72565a0d3ab0b8cf3f35ff19a394d4d9596093582d99d34cfc01c43c1f1ed52; isolation
 sha256:d6add4739dd796306cd894519859984535acb55863f19825a4ba72d38b9e7cc4; custody
 sha256:e32bd1bfda242321b65ea226cde83155b6835f1bc17a675233cc6d65a3f8098d.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

234. Reference and interpretation record

Family: SEM Obligation: SFT-INFO-OBL-SEM-002 Claim: SFT-INFO-SEM-REFERENCE-INTERPRETATION-002

Question, law and forced result. A semantic reference is an exact registered relation among a symbol, context, interpretation rule and referent; absent relation records force structural empty-One and halt inference.

SEM-002 uniquely retains complete-symbol-context-reference-record, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CORR-COMPLETENESS-016
- SFT-INFO-SEM-SYMBOL-CONTENT-BOUNDARY-001

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:7bb7cd84816f3d312986329effe2a9566b7d20e89bb571a394576e18ccaf7a0c. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-002 product before observation access.

Boundary: Every positive finite SEM-002 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-002 uniquely retains complete-symbol-context-reference-record, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__complete-symbol-context-reference-record__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:1c6ea7f7df0afee9beb950bcfde64041269d7ef1e619676d19dc09976b08655c; generality
 sha256:87c4a165ff593b3cc336a6dec7d05a39e1d641db0a61699d88c9b3a242ffbdb8.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt
sha256: 67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt
sha256: 18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256: 3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate
sha256: aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal
sha256: 47c6c6e2399505e0bb70d32830298abb04f9af54aac0c473bf035434c6e43920. Source manifest
sha256: ddb8aef959707fd683aee1f3a3777c5cf1399a81f8c840563d0a165c44e1e96e. Engine receipt
sha256: 7bb7cd84816f3d312986329effe2a9566b7d20e89bb571a394576e18ccaf7a0c at receipts/engine/model_admitted/SFT-INF0-SEM-REFERENCE-INTERPRETATION-002-7bb7cd84816f3d31.json. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-002: reference_interpretation_record
- exact observation: {"registered_reference_rows":2,"unregistered_context_result":"empty-One"}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: 5a892b0b2bd432ea6334b41e11d0541b091553354db2a80a5e1051f35db4e0a9; empirical validation sha256: 583be33d9a5ae1fb14324ba97fe163d5bdead3805f0ac16b91a79bcb5198d307; external validation
sha256: 3ad64e97855f54dea4b8a44fa00cb91116559cc7dc737818ed6f63d522a7539a; isolation
sha256: 5f24582f093efb2e9f31100243dcc28d7f481a3b97b8e8fbf5407abc22a05b8b; custody
sha256: 7cda236b50a712d819ea205093212f20159276297a8fce438f8907b382125347.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

235. Context-dependent distinguishability

Family: SEM Obligation: SFT - INFO - OBL - SEM - 003 Claim:
SFT - INFO - SEM - CONTEXT - DISTINGUISHABILITY - 003

Question, law and forced result. Context-dependent information retains context as a coordinate, allowing one symbol to lawfully distinguish different referents without merging their records.

SEM-003 uniquely retains context-indexed-reference-distinction, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CORR - COMPLETENESS - 016
- SFT - INFO - SEM - REFERENCE - INTERPRETATION - 002

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:635490f5a7dec05f2b988b63dc45efc2142717f8146e9512f38af0acffeb12d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-003 product before observation access.

Boundary: Every positive finite SEM-003 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-003 uniquely retains context-indexed-reference-distinction, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__context-indexed-reference-distinction__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:f8c9fd0632dda781445a4e193c15a7d6ec9fd2f5efb4dd5c70413bc60ef7cc24; generality
sha256:960a929c8ce4d5b72d1f76b70c52e5efacb25202329784c3fd4c448203dc59dc.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt

sha256:67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.

- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt
sha256:18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256:3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate

sha256:aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal

sha256:ed28f6dcff38935c4a0eb26b6e88bb38cf847b40e86d50e56263e1ec2fff6bbd. Source manifest

sha256:593aee0eaec61f26111485b9ce575ba3f59aa5f3ce123ce7620b34c65b8d82f1. Engine receipt

sha256:635490f5a7dec05f2b988b63dc45efc2142717f8146e9512f38af0acffeb12d at receipts/engine/model_admitted/SFT-INFO-SEM-CONTEXT-DISTINGUISHABILITY-003-635490f5a7dec05f.j

son. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-003: context_distinguishability
- exact observation: {"contexts":2,"distinct_references":2,"shared_symbol":"bank"}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:4d0de3a778a97ebbe5216e7d5d7acf9dcfd5231e06872949fe61b8a9133cf16d; empirical validation sha256:5b269926598622b5f659713f4ee714b8882e89fb8ecf16ca61603ed759ff4cf1; external validation

sha256:fea5f836dc9683e37a6d7261f1360b4ad8e7392f2b80b1f17cc55d2790009631; isolation

sha256:1b097a85f62488c925cc98becf8725a7bafce42ac7e9a2ebf94af562bebe5b54; custody

sha256:df28af6ea7b66b1fb226f14464aae27be78e6a97412327e9203fe6f0778a4e26.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

236. Biological information ownership handoff

Family: SEM Obligation: SFT - INFO - OBL - SEM - 004 Claim: SFT - INFO - SEM - BIOLOGICAL - HANDOFF - 004

Question, law and forced result. Information Science owns the exact sequence, code and provenance record; Biology owns whether that record participates in replication, regulation, metabolism, phenotype or selection.

SEM-004 uniquely retains sequence-record-biology-handoff, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CORR - COMPLETENESS - 016
- SFT - INFO - SEM - CONTEXT - DISTINGUISHABILITY - 003

The current reconciliation binds this obligation to the current model-admitted receipt sha256: f34f12b5b901548bc6b6d5c584b0f82be02aba1113ec310c207c00c1668c05bb. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-004 product before observation access.

Boundary: Every positive finite SEM-004 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-004 uniquely retains sequence-record-biology-handoff, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__sequence-record-biology-handoff__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: 98cbcd9b2acddcf4932adcf321832ed1831ce9f1a05184aa60ebec169914d4c2; generality sha256: 69af0840c4abd303221b7113bfb46ffac38220ba71fd3226d9738350e6751c9b.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt sha256: 67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt

sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt
sha256:18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256:3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate

sha256:aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal

sha256:6c3a080d647f91d623a1ef3b3d0e884e8ca54ee8c8ae4c3584ead7159500e774. Source manifest

sha256:514c36950ec0b65593ad85a01e4894b83b701d58fd06a2906bf7327042d42c98. Engine receipt

sha256:f34f12b5b901548bc6b6d5c584b0f82be02aba1113ec310c207c00c1668c05bb at receipts/engine/model_admitted/SFT-INFO-SEM-BIOLOGICAL-HANDOFF-004-f34f12b5b901548b.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-004: biological_information_handoff
- exact observation: {"biology_handoff_explicit":true,"information_owned_records":2}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:f663061b3cc6a77192e3c1bea5957bcd24f234d9ba56e8ab15b27948081dece0; empirical validation sha256:437464ab9b23dffbc7e9e4b78e9fc54fedfd8ee0774c2a660bbdc9e2e9a2a6a0; external validation

sha256:27583456ed1a08fc7eeb24eb8b0cc35118c871b9584c95b4c0b7e8e81db3e399; isolation

sha256:7261cbd1b855600d5db8045f3fbed1f3fc2ea91873e59446232bef6c88b436bc; custody

sha256:e6eaaef8396aaf88a3052369e80ff9f4bc65bb9bdf26cb558239ccceb0d803c3.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object

- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

237. Genetic representation correspondence

Family: SEM Obligation: SFT-INFO-OBL-SEM-005 Claim: SFT-INFO-SEM-GENETIC-CORRESPONDENCE-005

Question, law and forced result. A four-symbol genetic alphabet corresponds exactly to the complete length-two words over the two forced fibre labels when encoding and decoding preserve every symbol and order position.

SEM-005 uniquely retains four-symbol-two-fibre-bijection, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CORR-COMPLETENESS-016
- SFT-INFO-SEM-BIOLOGICAL-HANDOFF-004

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:43ff998a5845965206a402626cebc2320ab39bdee80a44faed50bde4376699c0. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-005 product before observation access.

Boundary: Every positive finite SEM-005 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-005 uniquely retains four-symbol-two-fibre-bijection, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__four-symbol-two-fibre-bijection__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:f622c764f5f8b4e5b555525cdf0f4454ebcb32256ada9df0143caf10a2a26f3f; generality
sha256:b9b5130eb12afd1480ac82dc931aaaa80f72afbd2430d85a4b6289af47dbc101.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt
sha256:67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.

- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt
sha256:18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256:3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate
sha256:aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal
sha256:879b9ba5b524bf940b5d4e8f057a3a001dedc78ea2b438a067a9d9b864986564. Source manifest
sha256:00393946fe3da323a41bc25036242501ca80bcdd363f4732d6ba3f34fe102d11. Engine receipt
sha256:43ff998a5845965206a402626cebc2320ab39bdee80a44faed50bde4376699c0 at receipts/engine/model_admitted/SFT-INFO-SEM-GENETIC-CORRESPONDENCE-005-43ff998a58459652.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-005: genetic_representation_correspondence
- exact observation: {"fibre_word_length":2,"genetic_symbols":4,"reconstructed_symbols":4}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:0f06a5cf6d688ab6b0b7a1bb4bc7515775e822b2e2a9aa0c4ce934ac31591ed3; empirical validation sha256:a6f7a752fea95f72379a4f504ff3215ef7e0899717dfcd345a4f0415236ce267; external validation
sha256:377f73f6804a384a2b05d51a1d33706c5979ac9adbf71dd01244e4255b5562f5; isolation
sha256:47af63d156acea0a44993ed7779ac4f64dccf93e9261e8271081a7deb712dcfb; custody
sha256:18b6a6d11e23afd96d1934a828ef24da950bed60a8ea006a0e47c84cb20d46a4.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar

- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

238. Neural and cognitive information handoff

Family: SEM Obligation: SFT - INFO - OBL - SEM - 006 Claim: SFT - INFO - SEM - NEURAL - COGNITIVE - HANDOFF - 006

Question, law and forced result. Information Science owns exact neural signal and report records; Cognitive Science owns representation, inference, memory, intention and mental-state claims not fixed by those records alone.

SEM-006 uniquely retains signal-record-cognition-handoff, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CORR - COMPLETENESS - 016
- SFT - INFO - SEM - GENETIC - CORRESPONDENCE - 005

The current reconciliation binds this obligation to the current model-admitted receipt sha256:5ade99b0110b049309903db861e798d292250dba1fd0868306f74ebedf265591. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-006 product before observation access.

Boundary: Every positive finite SEM-006 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-006 uniquely retains signal-record-cognition-handoff, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__signal-record-cognition-handoff__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:c7b419298bfce57d781114af3056de99304e0a85dd51be04debcd35c1f0fdcd6; generality sha256:d8cf5c07228e8fcdfedfa5dab49e5b5470eda3663f8711b593b92e1e8b0d6c29.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt sha256:67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt

sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.

- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt

sha256:18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256:3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate

sha256:aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal

sha256:4b5621f71e89b7414c000b197a624ea1a2e171c323faca0452a399cc7baa9709. Source manifest

sha256:ac710003e9156168d363bdb4bd6ca41caade4e7c34c227039d77f2293b168154. Engine receipt

sha256:5ade99b0110b049309903db861e798d292250dba1fd0868306f74ebedf265591 at receipts/engine/model_admitted/SFT-INF0-SEM-NEURAL-COGNITIVE-HANDOFF-006-5ade99b0110b0493.js

on. Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-006: neural_cognitive_handoff
- exact observation: {"cognition_handoff_explicit":true,"information_owned_records":2}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:fb3fc6d742138d3e74b048ef21f88224c2ef824d2870f5718d60424dab7f811d; empirical validation sha256:e98f2810b526c77416d63e1ddbfe08af99d81b4b0799362bedac8757718e6471; external validation

sha256:bb047819ce9fce4344cda5ee7f7053caee4f6fa390fa5580c79a5af305fbdcb; isolation

sha256:165c4d197537e0458046061e70bcd9768301218c71462b141d65e347be963465; custody

sha256:456b46c2a8836e3615115867cd130669df567605e4bc5de926db0684801cbcaa.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone

- no failed route retires an obligation or changes protected authority

239. Conscious access and reportability handoff

Family: SEM Obligation: SFT - INFO - OBL - SEM - 007 Claim: SFT - INFO - SEM - CONSCIOUS - REPORT - HANDOFF - 007

Question, law and forced result. Access and reportability are exact information records, while subjective presence, qualia and the red-of-red relation remain explicitly owned by the Consciousness branch and cannot be replaced by symbol counts.

SEM-007 uniquely retains report-record-qualia-handoff, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CORR - COMPLETENESS - 016
- SFT - INFO - SEM - NEURAL - COGNITIVE - HANDOFF - 006

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:137fcb0324afe8d1f6ad5a45a1b07f5df3701fcee827086cfdfffa33e0ad5f4. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-007 product before observation access.

Boundary: Every positive finite SEM-007 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-007 uniquely retains report-record-qualia-handoff, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__report-record-qualia-handoff__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:e12b8de9b683b1f9e5a5b3fe0fbf6bb6b454123376641f62dd2c85d9bbf52bb4; generality
sha256:c95717bf1823782b4e5479a038b34f49d92fbb10327afbbeb977a7a3b5d92fe.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt
sha256:67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural

absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt
 sha256:18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256:3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate
 sha256:aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal
 sha256:f82e6f29f145f9d60734125535c529940ec984ac48d925f2a9c4ba41f6d11f69. Source manifest
 sha256:2f9f51879253faebf9238f21ae613326a46a333fe7c69d81c9cb4ed20ec92f7a. Engine receipt
 sha256:137fcb0324afef8d1f6ad5a45a1b07f5df3701fcee827086cfdfffa33e0ad5f4 at receipts/engine/model_admitted/SFT-INFO-SEM-CONSCIOUS-REPORT-HANDOFF-007-137fcb0324afef8d.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-007: conscious_reportability_handoff
- exact observation: {"consciousness_handoff_explicit":true,"information_records":2,"red_of_red_handoff_explicit":true}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:6eb0aafbe0fa0209a9d43e869af7efdcddedbe3ec3945425c204c5709ddd8bcb; empirical validation sha256:b633dd4772681ffd9eca200d7aa876b7fd34acd9e1d07de700eaa8e5312f3875; external validation

sha256:e3b723bf30b5e1c05bc3772923b57e6d999e430984970e67d2ad1349746e5b34; isolation sha256:2f2cd726375deb6a338eaf93e043d3f8f5de24e64fbf8ea65ca61ec1d4dd4845; custody sha256:239be7e106104e2a3cb89bd30369730e76a1d9fb79e6ba0b3c4312ea7eb4c9ba.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

240. Social communication and shared-record handoff

Family: SEM Obligation: SFT - INFO - OBL - SEM - 008 Claim: SFT - INFO - SEM - SOCIAL - SHARED - RECORD - 008

Question, law and forced result. A shared record is established by retained transmission and acknowledgement relations; shared interpretation, convention, trust and collective knowledge require separately registered Social and Collective Systems laws.

SEM-008 uniquely retains acknowledged-shared-record-boundary, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - CORR - COMPLETENESS - 016
- SFT - INFO - SEM - CONSCIOUS - REPORT - HANDOFF - 007

The current reconciliation binds this obligation to the current model-admitted receipt sha256: 659099aec42117c905586497e38982dc10c861f6e16c2317425326dee8debc57. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-008 product before observation access.

Boundary: Every positive finite SEM-008 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-008 uniquely retains acknowledged-shared-record-boundary, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__acknowledged-shared-record-boundary__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: e9c6e4a2c206719dd3c33a44de932c87418f750042d5b0949d48eed415c3892c; generality sha256: 52602084620fe46117a629acb22d756d768185e48f32f8f3026cdeae357768a3.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt sha256: 67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition,

consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt
 sha256:18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256:3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate

sha256:aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal
 sha256:4ae4b4d63bf0fec862d72fce1e15a56da0fa7b7026ecaa32d393bf34684ff6b9. Source manifest
 sha256:fab2d1dbe66c2e7e85b06d45c3df4b38c3d0ada2fa370e04948955ab10de9cee. Engine receipt
 sha256:659099aec42117c905586497e38982dc10c861f6e16c2317425326dee8debc57 at receipts/engine/model_admitted/SFT-INFO-SEM-SOCIAL-SHARED-RECORD-008-659099aec42117c9.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-008: social_shared_record_handoff
- exact observation: {"acknowledgements":2,"message_records":1,"social_handoff_explicit":true}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:2520ce6a2b0ead1af978a0c5a311ab6efffc6359093aaea472f63e79c4a664106; empirical validation sha256:b41fbf750c22733eef39588cca771577c632d37fc7e21560fbda30757209e373; external validation

sha256:5f7ffdf7ab6edf590746608c3ac3f52a498c63253dc8b812ec3af60ef696405; isolation sha256:06a83f07997e5e62c77df310595e9e0e19a73de1eb2adc6b43433521446d01e7; custody sha256:8c09ec76ef2c8b9c1167f57dd3e278dd8acb781242b4b717696c7a7e47d4f8a5.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

241. Meaning-preserving translation boundary

Family: SEM Obligation: SFT-INFO-OBL-SEM-009 Claim: SFT-INFO-SEM-TRANSLATION-PRESERVATION-009

Question, law and forced result. A translation is meaning-preserving within its declared scope exactly when the source and target expressions retain the same registered reference and context relations, with ambiguity and unmapped forms preserved.

SEM-009 uniquely retains reference-preserving-translation-relation, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CORR-COMPLETENESS-016
- SFT-INFO-SEM-SOCIAL-SHARED-RECORD-008

The current reconciliation binds this obligation to the current model-admitted receipt sha256:cc7e04894916d4ea57454b7306c18f3fb3a5d4ad533a33c577bff3db8e623707. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-009 product before observation access.

Boundary: Every positive finite SEM-009 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-009 uniquely retains reference-preserving-translation-relation, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__reference-preserving-translation-relation__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:3847793b03203651231dc0a3dc774222f41d21180e063ce59f082446b02038e4; generality sha256:dafd976700dbcd1d7276351dcb1f35c53e4683b110dd61adf94f82fe741016f0.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt sha256:67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt sha256:18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256:3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate

sha256:aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal sha256:d52cd8a06a69e07cca28a16538fa0fc521dd51a21d4af85b38ccb7e149687b6a. Source manifest sha256:8ed0982b3bb4309a91c6b30bd7e6d283495cf3c1c3c23860ba6e3a2bce3b106a. Engine receipt sha256:cc7e04894916d4ea57454b7306c18f3fb3a5d4ad533a33c577bff3db8e623707 at receipts/engine/model_admitted/SFT-INFO-SEM-TRANSLATION-PRESERVATION-009-cc7e04894916d4ea.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-009: translation_preservation
- exact observation: {"common_reference_rows":2,"source_forms":2,"target_forms":2}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:14d5e75c2f13ca31a76d1b43a064575a50c9c54c681494cca159562663b686c6; empirical validation sha256:373cdde58e8cb6ea266f5b975fc46597f5003875db12ed1b43ec31efd93f678a; external validation

sha256:f7495d5dee043214dd3a9a285f18cb87245f841e0c2e185a66cb07b4ecb03ee6; isolation sha256:20204464e64eec75254817cc6d3b6649ced6ced0057e38e6802a8f379dd5cb27; custody sha256:f5e23e3dddbd8eeb8f5990b6b9ef0590dc4fcbcc2c44c759da5e009e0de71944.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

242. Pragmatic action-information boundary

Family: SEM Obligation: SFT-INFO-OBL-SEM-010 Claim:

SFT-INFO-SEM-PRAGMATIC-ACTION-BOUNDARY-010

Question, law and forced result. Information Science can retain exact observation-action incidences, but purpose, value, policy and action selection require their own declared owner and cannot be inferred from incidence alone.

SEM-010 uniquely retains observation-action-purpose-handoff, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CORR-COMPLETENESS-016
- SFT-INFO-SEM-TRANSLATION-PRESERVATION-009

The current reconciliation binds this obligation to the current model-admitted receipt sha256:28e546ea99c3b74db31ff9da432ccf7377df1f6c643601647ade081acf8af761. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-010 product before observation access.

Boundary: Every positive finite SEM-010 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-010 uniquely retains observation-action-purpose-handoff, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__observation-action-purpose-handoff__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:2234fd1829a96f2d8efe39f30f993be269e55deec7a945057db50b01563f1522; generality sha256:a751b0f70d26930ea88639a6eea2eafc8cd3513806e1f958fb1c6e0ccac837e0.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt sha256:67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt sha256:18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256:3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the

declared census and sole survivor; independent certificate

sha256:aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal sha256:fadf2de75e242a20f70a4e11bebf507eb9875968d34f54106113a36585755cb9. Source manifest sha256:42753122edceb9c02e949d6afb1455cb7df7bd28996edcfa1d6de0f5175eaf64. Engine receipt sha256:28e546ea99c3b74db31ff9da432ccf7377df1f6c643601647ade081acf8af761 at receipts/engine/model_admitted/SFT-INFO-SEM-PRAGMATIC-ACTION-BOUNDARY-010-28e546ea99c3b74d.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-010: pragmatic_action_boundary
- exact observation: {"incidence_fields":3,"pragmatic_handoff_explicit":true}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:7761cb29806c81f71fe0e3d733bda36e054defe348db606d86cdec15b81040a7; empirical validation sha256:d18f82c928763761131d31390dfebe6d74be786dae4820b7c674aa3ff15b1d10; external validation

sha256:701b23f6f62cb5d1289b9bd58f1ea5bd6d7526d4f7ad110f5394829cbaf253ef; isolation sha256:20d362992e2c34b36d063c125215ca827fb9ae4826b14f75f2ddd27bc061b7d; custody sha256:e4dce84ecc783249222096a2418c16991454f1b3db81366581272f730fb09901.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

243. Cross-domain provenance and nonduplication

Family: SEM Obligation: SFT-INFO-OBL-SEM-011 Claim: SFT-INFO-SEM-CROSS-DOMAIN-PROVENANCE-011

Question, law and forced result. Every cross-domain semantic result retains one derivation owner, explicit handoff dependencies and a single provenance path, preventing omission, duplicate closure or silent substitution across branches.

SEM-011 uniquely retains single-owner-semantic-provenance-ledger, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CORR-COMPLETENESS-016
- SFT-INFO-SEM-PRAGMATIC-ACTION-BOUNDARY-010

The current reconciliation binds this obligation to the current model-admitted receipt sha256:6cd4e601d33ee9a0519d4c9c783c4c9b08636d6b0d41564d757de57a8e0e248a. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-011 product before observation access.

Boundary: Every positive finite SEM-011 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-011 uniquely retains single-owner-semantic-provenance-ledger, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__single-owner-semantic-provenance-ledger__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:94226fd7038f0861507ec1b16db69222bd2321dec2ac1a9c6b12e57060b25def; generality sha256:b17a38850ba63c40d2c18761e1863633240fdce3c3a0ef9249db8ea053836a48.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt sha256:67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt sha256:18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256:3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate

sha256:aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal

sha256:0491ac930d32d3088555d3d29492c09fe39089eddaad40ee9f6418bf83d9746e. Source manifest sha256:ffc71d848a3cc823b86cede5be7475aaaa9fc2427ad6a3357295cb24ff7b8b36. Engine receipt sha256:6cd4e601d33ee9a0519d4c9c783c4c9b08636d6b0d41564d757de57a8e0e248a at receipts/engine/model_admitted/SFT-INFO-SEM-CROSS-DOMAIN-PROVENANCE-011-6cd4e601d33ee9a0.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-011: cross_domain_provenance
- exact observation: {"declared_owners":6,"duplicate_owners":0,"handoff_paths_retained":true}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256:3ed62065c27d3b4fed85e5e351114fe8127954e9a6d9b1203a5491bd476c24be; empirical validation sha256:8fc46afa15e89a66b84554f37129d0a3024fbe502cf478e1ae63814b059a2d0b; external validation sha256:0898352593ceb6f3cd3a40d9ff6e48e107b2c32186f5c9dbd6219d2dacd2e563; isolation sha256:b75af34ca1be36f69c1beec00fd31f5dd64219b42e6a35b023d872afd4a9af98; custody sha256:90bcacf839a654187cb13a20f366fa2a957c49116af825a9c66170bb124a78ad0.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

244. Semantic-handoff completeness certificate

Family: SEM Obligation: SFT-INFO-OBL-SEM-012 Claim: SFT-INFO-SEM-COMPLETENESS-012

Question, law and forced result. Semantic-information completeness is the one-to-one reconciliation of all twelve frozen symbol-content, reference, context, biological, genetic, cognitive, conscious, social, translation, pragmatic and provenance obligations.

SEM-012 uniquely retains twelve-semantic-obligation-ledger, complete semantic custody, root forcing, post-registry observation and no extra rule.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-CORR-COMPLETENESS-016
- SFT-INFO-SEM-CROSS-DOMAIN-PROVENANCE-011

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:996be961ea2a53fba28114d5b99e6d05773712dfd4fdb3a2693d4b9207216658. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis SEM-012 product before observation access.

Boundary: Every positive finite SEM-012 symbol, reference, context, interpretation, ownership, handoff and provenance record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- SEM-012 uniquely retains twelve-semantic-obligation-ledger, complete semantic custody, root forcing, post-registry observation and no extra rule.

Unique survivor: complete-symbol-record-support__twelve-semantic-obligation-ledger__context-coordinate-retained__single-owner-explicit-handoff__complete-declared-semantic-product__root-bound-forward-forcing__post-registry-exact-observation__finite-successor-or-explicit-handoff.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:a0d0967a6ee8edf161914af8cf53566c625391a93bf86ba5702ebd2a93d95ec0; generality
sha256:812d3d5dd61502bdc810fd39bb02a89bab441ad722272278beeea5e263ee9738.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted marks or records change semantic distinctions.; receipt
sha256:67e1a7fd009f51b17b331380c1bfec1a1aac1c026c8596ceb18ce49a291eaa0a.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported semantic theory, consensus interpretation or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence; no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone; no failed route retires an obligation or changes protected authority; receipt
sha256:18f62ed4b2c826f47e57adc51b51dcf61927ed2ddb1a32d8d75f7ac670b95678.

Independent reconstruction and engine admission. Independent implementation

sha256:3c831ef633a83b1b516d00a133785c53a07560dca5a7e059bdf8f7e8dfa5ad32 regenerated the declared census and sole survivor; independent certificate
sha256:aa1d38b42bc7941a184874a231b937905b33a51170024df87abd6771b7c8bbd2. Derivation seal
sha256:1fe6b1c94ed4c1b0d9f08e1c58d08f7b66b578974429123fec2d83189b42cc8. Source manifest
sha256:d198d02294e3d2213f017219c230444145dd3dfafbbbf8636c006ee735d28e3a. Engine receipt
sha256:996be961ea2a53fba28114d5b99e6d05773712dfd4fdb3a2693d4b9207216658 at

receipts/engine/model_admitted/SFT-INFO-SEM-COMPLETENESS-012-996be961ea2a53fb.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INDEPENDENT-EXACT-SEMANTIC-OBSERVER
- SFT-V1-V2-SEMANTIC-INFORMATION-CORPUS

Observed, adverse and boundary records:

- SEM-012: semantic_handoff_completeness
- exact observation: {"duplicate_owners":0,"observation_rows":12,"omitted_owners":0,"registered_obligations":12}
- complete family observation custody: 12 records; all rows preserved
- complete symbol, reference, context, interpretation, translation, ownership, handoff and provenance enumeration supplies the observation; no imported meaning, fitted referent, inferred quale, silent cross-domain substitution or target-selected result enters

Falsification condition: Reject if the value-free registry, complete symbol-reference-context-handoff census, exact observation, candidate survivor, source identity, seal, custody record, control or independent reconstruction is missing, changed, duplicated or opened out of order.

Measurement receipt

sha256: ee0d969ff2e5f8732a29f196abab8f8dde28793402d1b61cd0fa1db330455400; empirical validation sha256: d04db87305693c549ba8dfa2d33e0ba9f46afb772c8f0219aaa443f8ce4cbb02; external validation sha256: 3918e8b3645cdefcb7b33b7f00dd9fe5cb1c453d8af4d1fa213aec65dbf4d32b; isolation sha256: 006994cd696fa19b37421a4e885ec39c7ab2e95b330e811bd5c7c12324d26dbd; custody sha256: 0c77902a511865a2fc9ee6b37bf6e31ab377738db9b18eed89b2b3ce9ddc42a5.

Prohibited inputs.

- no axiom, imported semantic theory, consensus interpretation or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no hidden referent, context, interpretation rule, duplicate owner or fitted correspondence
- no biological function, cognition, consciousness, qualia, social agreement or purpose inferred from a symbol record alone
- no failed route retires an obligation or changes protected authority

Complete-field family: VALID

245. Symbol and representation complete validation vector

Family: VALID Obligation: SFT-INFO-OBL-VALID-001 Claim:
 SFT-INFO-VALID-SYMBOL-REPRESENTATION-001

Question, law and forced result. The complete symbol and representation validation vector is the exact one-to-one binding of every BASE and SYMREP obligation, current receipt, evidence package, control set and independent reconstruction.

VALID-001 uniquely retains base-symbol-representation-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

• SFT-INFO-SEM-COMPLETENESS-012

The current reconciliation binds this obligation to the current model-admitted receipt
sha256:13905605a50b784959980afaa21acd703491416db290915d3d22d2b8f8fb36d7. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-001 product before observation access.

Boundary: Every positive finite VALID-001 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-001 uniquely retains base-symbol-representation-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__base-symbol-representation-receipt-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-observation__immutable-engine-authority.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:d4f8d6936316702c861f94bbf4bdef076a66b398919db119ef858727ac23a074; generality
sha256:638b2adbf7895a19a9cd4f4ccced492f310dc283c06f1bd7430f1db17e1674dd.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt
sha256:8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt
sha256:9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256:8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate
sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfa5a3c752d8c9c. Derivation seal
sha256:e8d29724526827a7fc496b17b4e6e272d16e3bee7554030654a4bb603ec99a15. Source manifest
sha256:6e6120d3439d05b042a22e09a843124f6a69ba4952f2e93543ad527e8c0eec2f. Engine receipt
sha256:13905605a50b784959980afaa21acd703491416db290915d3d22d2b8f8fb36d7 at receipts/e

ngine/model_admitted/SFT-INFO-VALID-SYMBOL-REPRESENTATION-001-13905605a50b7849.json
 . Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V18
- SFT-V3- INDEPENDENT-EXACT-VALIDATION-RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-001: symbol_representation
- exact observation:
 {"all_current_receipts_bound":true,"families":["BASE","SYMREP"],"family_count":2,"receipt_count":26}
- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt, erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256: aae6fcc7e5c98250a9d8131c81f8c09833ec5af9356cddca4075a9b0aaa1356f; empirical
 validation sha256: e53660ad65777dbecb1bca903436bff2c917fee29bb02074e15a49b61220892d;
 external validation
 sha256: 6240bf16ea28a52fadca49aba016281fa5273fb5336a20a5a7c3282cc71ad420; isolation
 sha256: 6974a6ffab526b0ac63d4e304c04409c405098d1608bd2f54f2350f3fffb26d1b; custody
 sha256: 6f13f091f0c1192838d75ba32442b5372f23ca8b1eec6cdc0b4d90f66167087b.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

246. Record and provenance complete validation vector

Family: VALID Obligation: SFT-INFO-OBL-VALID-002 Claim: SFT-INFO-VALID-RECORD-PROVENANCE-002

Question, law and forced result. The complete record and provenance validation vector binds every RECORD obligation to one current receipt, complete provenance chain, control set and independent reconstruction.

VALID-002 uniquely retains record-provenance-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SEM-COMPLETENESS-012
- SFT-INFO-VALID-SYMBOL-REPRESENTATION-001

The current reconciliation binds this obligation to the current model-admitted receipt
 sha256:e50c6cb2f1852e84cc550a4d34331d5ef65ac7727ece5b477600c7245e099500. A missing
 dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-002 product before observation access.

Boundary: Every positive finite VALID-002 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-002 uniquely retains record-provenance-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__record-provenance-receipt-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-observation__immutable-engine-authority.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
 sha256:5fba713f2cf95c8df6e00ac88393ccf2c9c417457d9e0b1ce9026585759fde92; generality
 sha256:bccf41b371fe49b3971ae3201fda90e1024aed1df2b2fd394eaa966e94b43ca0.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt
 sha256:8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt
 sha256:9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256:8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate
 sha256:3789f633aeb3c2e376ba88be8dbc66ed29e04f6254eaf2cfa5a3c752d8c9c. Derivation seal
 sha256:e0eb72485f6cb7354695ff681c711647472a5890fffff985d3660e906c2f4fe7. Source manifest
 sha256:2845fd60b3eae77cd9b33e1d2dbc45f0733b21692dfea42bb5652b0d42983975. Engine receipt
 sha256:e50c6cb2f1852e84cc550a4d34331d5ef65ac7727ece5b477600c7245e099500 at receipts/engine/model_admitted/SFT-INFO-VALID-RECORD-PROVENANCE-002-e50c6cb2f1852e84.json.
 Closure scope depth_independent. External status
 empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V18
- SFT-V3- INDEPENDENT-EXACT-VALIDATION-RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-002: record_provenance
- exact observation: `{"all_current_receipts_bound":true,"families":["RECORD"],"family_count":1,"receipt_count":12}`
- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt, erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256:00f2a6abab44a4adbd68a210e6612fb3a5127d0c7bc9f5bbd76cc3279e495639; empirical validation sha256:928ac26690e01db048cdb9160a421a4603d300ac9807ee4303245bfd59060b5d; external validation

sha256:98c8b8d942fffa66b87a9a494da6972aa4055092bbd6c1b6deab93d15f55c9d9; isolation sha256:7e1ddef97d019bc4f5e143bf8101dc33ff45a88c71e067815d283e43e5c2542b; custody sha256:ddcda51a3969c1ed637dc13f1e26e8ae2bda97274130270518458163b40b3c68.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

247. Source and measure complete validation vector

Family: VALID Obligation: SFT- INFO- OBL- VALID- 003 Claim: SFT- INFO- VALID- SOURCE- MEASURE- 003

Question, law and forced result. The complete source and measure validation vector binds every SOURCE and MEASURE obligation to its exact source identity, observation class, current receipt and retained comparison record.

VALID-003 uniquely retains source-measure-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT- ROOT- THERE- IS- NO- NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT- INFO- SEM- COMPLETENESS- 012
- SFT- INFO- VALID- RECORD- PROVENANCE- 002

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:e4d2488dcc212f2693e7c48d372c4354bf9fc26aa6ac6dbf7ef93bdf89f542f0. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-003 product before observation access.

Boundary: Every positive finite VALID-003 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-003 uniquely retains source-measure-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__source-measure-receipt-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-observation__immutable-engine-authority.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:6704281aaf9e8a59111e87d552bc52e69369df84d4a39772e77bbf666e48db68; generality sha256:cfb8788b043bf8986287c89c48ffa78b323b349e925ea9c71cb29126459bfaae.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt sha256:8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt sha256:9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256:8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfa5a3c752d8c9c. Derivation seal sha256:27dca260b4504c6a101d0934eba03a8ff40add882aea067c3891848e5e7e8924. Source manifest sha256:07e51302e261abee4de8c64782ccbc3e7d1d42750dbb615b02cbb94d2cdab1d0. Engine receipt sha256:e4d2488dcc212f2693e7c48d372c4354bf9fc26aa6ac6dbf7ef93bdf89f542f0 at receipts/engine/model_admitted/SFT-INFO-VALID-SOURCE-MEASURE-003-e4d2488dcc212f26.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V18
- SFT-V3- INDEPENDENT-EXACT-VALIDATION-RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-003: source_measure
- exact observation:
{"all_current_receipts_bound":true,"families":["SOURCE","MEASURE"],"family_count":2,"receipt_count":30}
- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt, erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256:119ffbc5c6db3d3a246fd8527d0cfb4c807fcb0552116c2636239cff9238e10; empirical validation sha256:d433fdd3491c34df0a924c9f2a20b6c39f0113e33ca1524b16021cab3a62465f; external validation

sha256:1a08314a4862faf71c6d23db26e87a7e93de4cdf6cb1b8ac300da0eb3d51e213; isolation sha256:4a6cd489de457726bf6622420013d9a2c4440677f73ce4266f74b6275d0e4f2d; custody sha256:dfa7a105da21e1def24ba44e02f5584a5e072b0e2d2eeda62bb2f11f37e7234c.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

248. Signal and sampling complete validation vector

Family: VALID Obligation: SFT-INFO-OBL-VALID-004 Claim: SFT-INFO-VALID-SIGNAL-SAMPLING-004

Question, law and forced result. The complete signal and sampling validation vector binds every SIGNAL obligation to one exact signal support, sampling boundary, reconstruction witness, current receipt and adverse control set.

VALID-004 uniquely retains signal-sampling-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-SEM-COMPLETENESS-012
- SFT-INFO-VALID-SOURCE-MEASURE-003

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:ec09e7739ec73209f79f0998119b960deb7dfbc7056ea62abd7ce98d3f719719. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-004 product before observation access.

Boundary: Every positive finite VALID-004 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-004 uniquely retains signal-sampling-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__signal-sampling-receipt-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-observation__immutable-engine-authority.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:d0551b8f40fb4252e9cea82ee769ad8c088863a37cca2a9121f5300f43fa16d2; generality

sha256:20deec390ba34fb3c0c9f29410906b9edacf6bc285291075ab291a4127ef8019.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt
sha256:8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt
sha256:9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256:8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate

sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfa5a3c752d8c9c. Derivation seal

sha256:f7c156386afc0b90418021e6fd3c7003632d50339df4614d3c7866375e1ff73e. Source manifest

sha256:3c3e2b65c60de54ff92bc75b66eab5e489436275e6f4c04ade5f29906eaebb47. Engine receipt

sha256:ec09e7739ec73209f79f0998119b960deb7dfbc7056ea62abd7ce98d3f719719 at receipts/engine/model_admitted/SFT-INFO-VALID-SIGNAL-SAMPLING-004-ec09e7739ec73209.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3-INFORMATION-SCIENCE-RECONCILIATION-V18
- SFT-V3-INDEPENDENT-EXACT-VALIDATION-RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-004: signal_sampling
- exact observation: {"all_current_receipts_bound":true,"families":["SIGNAL"],"family_count":1,"receipt_count":14}
- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt, erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256:56f78165b3d347f2ea1a0ca45a835c2eb7f823920d0918469a51edb1eb9ad3d7; empirical validation sha256:1c1cf29b94e47915b433cfb7696ab2f640bae9226de6f816d9df242fea6987a2; external validation sha256:97ac4f56d8f9ef9b27f080aeb0dbc4992847b2b8387e4781c1cfc5375408f6db; isolation sha256:e2c991c12f91fafe16e281fe6300ac61b09594d45154d5b8984fb05284bada44; custody sha256:0634194a6c9d4b186d34ae5b2254c1924d92d931250657a45c440ead6cfc6f77.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

249. Compression and distortion complete validation vector

Family: VALID Obligation: SFT - INFO - OBL - VALID - 005 Claim:

SFT - INFO - VALID - COMPRESSION - DISTORTION - 005

Question, law and forced result. The complete compression and distortion validation vector binds every COMP obligation to exact retained/lost distinctions, distortion boundary, current receipt and reconstruction.

VALID-005 uniquely retains compression-distortion-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SEM - COMPLETENESS - 012
- SFT - INFO - VALID - SIGNAL - SAMPLING - 004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:fa495dfb7aa1f9b9ce3c35a347554c45d2fb3cfc6699a3b78110bea94d78c4a3. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-005 product before observation access.

Boundary: Every positive finite VALID-005 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-005 uniquely retains compression-distortion-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__compression-distortion-receipt-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-observation__immutable-engine-authority.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:6cb70828ea9e386cded516f133cab68bdef57df972e246d8d8cc5c85888641a4; generality sha256:b7b66e04deda88ae6a1a53c403cf20fc2ebdf3dae1242eeddac2b34de0e67831.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- false_premise - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt sha256:8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- tampered_source - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- tampered_artifact - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- boundary - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt sha256:9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256:8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c. Derivation seal sha256:9d2293ae3b6bb827db21aff8a344b39a4175c366c86f48db317cf1eaa0288e0. Source manifest sha256:8b874fdddf2ecb09ceefa77cd14885eedd9f821d1fdd8283a6d4e1ad8edaa1cf. Engine receipt sha256:fa495dfb7aa1f9b9ce3c35a347554c45d2fb3cfc6699a3b78110bea94d78c4a3 at receipts/engine/model_admitted/SFT-INFO-VALID-COMPRESSION-DISTORTION-005-fa495dfb7aa1f9b9.js on. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V18
- SFT-V3- INDEPENDENT-EXACT-VALIDATION-RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-005: compression_distortion
- exact observation: {"all_current_receipts_bound":true,"families":["COMP"],"family_count":1,"receipt_count":14}
- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt, erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256:9ccfc79bd7df42826691a80367e55e218f869a181e683a256536e4b46632b5f8; empirical validation sha256:6095efe47667d51222b3ca39d44e6b13441d079e57531543af64c1f162c481d0; external validation

sha256:5d0987d6af339f84ae44f32d214d4835d8dd299db0ca43c1f2223e4d4d1da9; isolation sha256:e85f0ade8875b495744378473f7644427b4e9a640216c40739c775f624286816; custody sha256:4e6a33bfd446c5df9b0287c1e3ecd2355ea2b5ad12a2b5f5bf9bee5ed6a3533.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

250. Channel and capacity complete validation vector

Family: VALID Obligation: SFT - INFO - OBL - VALID - 006 Claim: SFT - INFO - VALID - CHANNEL - CAPACITY - 006

Question, law and forced result. The complete channel and capacity validation vector binds every CHAN obligation to an exact channel relation, capacity resource boundary, current receipt and post-registry observation.

VALID-006 uniquely retains channel-capacity-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SEM - COMPLETENESS - 012
- SFT - INFO - VALID - COMPRESSION - DISTORTION - 005

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:01cb3d3963c79cfd1bba2523d1ac9fb4512f2f7de977d8a66276aa2142961438. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-006 product before observation access.

Boundary: Every positive finite VALID-006 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-006 uniquely retains channel-capacity-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__channel-capacity-receipt-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-observation__immutable-engine-authority.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:114c93682db73bdc6af716bd083d1ecfcc83b24ed21e5480436884f0c9d91081; generality sha256:7d7fac05338d2d014e6fcdb20e535ee4d89a3307406dc476cc11a3eba367b431.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt sha256:8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt sha256:9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256:8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate

sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfa5a3c752d8c9c. Derivation seal

sha256:c34a126f9558c3fd2c84a77d8ff506ae9c1ee3f0e1f9ab0d036248df4b0b9ab7. Source manifest

sha256:d9a83b5ce74b3b19228145d6ff0792c6c73177b5150a457e0004a0fecf127eb2. Engine receipt

sha256:01cb3d3963c79cfd1bba2523d1ac9fb4512f2f7de977d8a66276aa2142961438 at receipts/engine/model_admitted/SFT-INFO-VALID-CHANNEL-CAPACITY-006-01cb3d3963c79cfd.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION- SCIENCE- RECONCILIATION- V18
- SFT-V3- INDEPENDENT- EXACT- VALIDATION- RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-006: channel_capacity
- exact observation: {"all_current_receipts_bound":true,"families":["CHAN"],"family_count":1,"receipt_count":18}

- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt, erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256: fd120c2768d5fd5778343006da53bfcabe8cefc561020b762f0f23b6858c6d6c; empirical validation sha256: a0b23af7cd4e703ec19620661b778625afcccad58c08f792efd0237282cf7d11; external validation

sha256: bb954770edc2211c39875a1862f1a3c427efec2e9511fccb61a43a2ec9e03df0; isolation sha256: 0732a9a6c8d4554da0c804cdab1b2c62d0c0b5901766d87a87ae00f3abd84363; custody sha256: 51834b15e875095ce49b8f67c5210fc9597683ab40bea05ca4814487119f0f79.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

251. Noise and coding complete validation vector

Family: VALID Obligation: SFT - INFO - OBL - VALID - 007 Claim: SFT - INFO - VALID - NOISE - CODING - 007

Question, law and forced result. The complete noise and coding validation vector binds every NOISE and CODE obligation to exact error support, detection or correction boundary, current receipt and preserved adverse rows.

VALID-007 uniquely retains noise-coding-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SEM - COMPLETENESS - 012
- SFT - INFO - VALID - CHANNEL - CAPACITY - 006

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 5601ea6e1088c2d50f796c28718b70a11a3a95f95fb0d12378daf3c53bb3f289. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-007 product before observation access.

Boundary: Every positive finite VALID-007 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-007 uniquely retains noise-coding-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__noise-coding-receipt-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-observation__immutable

-engine-authority.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256: f2afe04b793e2fec69d43b23d32bd0f7f33470d42a1d1edef5711a938e4f08cf; generality sha256: f20ddc43e5b3bf2db93b83d6f4a3eb0bc73c1a094f7d2a551a8e3fea6e78f758.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt sha256: 8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt sha256: 9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256: 8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate

sha256: 3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c. Derivation seal sha256: 3b952f764dcde31c6e7e9a442f15854ad4c1a4b49844ac60a21a171c87da42b7. Source manifest sha256: 5d53714e08708d1f5f5d8864c59a28e23e2856206bb38bb427ba3608e221993c. Engine receipt sha256: 5601ea6e1088c2d50f796c28718b70a11a3a95f95fb0d12378daf3c53bb3f289 at receipts/engine/model_admitted/SFT-INFO-VALID-NOISE-CODING-007-5601ea6e1088c2d5.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V18
- SFT-V3- INDEPENDENT-EXACT-VALIDATION-RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-007: noise_coding
- exact observation: {"all_current_receipts_bound":true,"families":["NOISE","CODE"],"family_count":2,"receipt_count":30}
- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt,

erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256:27b841b612d1891ec0945fd8ff2f1f757a8596459ceae4f77fbe67a6263956d8; empirical validation sha256:5c0378fce8fe3bd003bd38a708edbb180abbe33cbe65769667531aa48e5a5962; external validation

sha256:6c5a6f05bd9ad1a36207e92c9ada876240b131a09f6621baca62903bba65887b; isolation sha256:263f89551a05473bba4ae9fa769117e89e69cac2457bb064fc73f2e72c19dfe1; custody sha256:35a58ef1dec1a34db30eefbbc482d0ee0ae19ce431b5525db502cf63674b6804.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

252. Relational and coarse-graining complete validation vector

Family: VALID Obligation: SFT - INFO - OBL - VALID - 008 Claim: SFT - INFO - VALID - RELATIONAL - COARSE - 008

Question, law and forced result. The complete relational and coarse-graining validation vector binds every REL and COARSE obligation to exact joint records, retained/lost distinctions, current receipts and reconstruction witnesses.

VALID-008 uniquely retains relational-coarse-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SEM - COMPLETENESS - 012
- SFT - INFO - VALID - NOISE - CODING - 007

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:826d35df20d7ea259d64e873ba161a578b8597dfad41020a5ef13a95b4dfd64d. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-008 product before observation access.

Boundary: Every positive finite VALID-008 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-008 uniquely retains relational-coarse-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__relational-coarse-receipt-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-observation__immutable-engine-authority.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: cf2471e66aee3ea96d2b9f30fd9b57c02bb300b6044c3f8990f2c0fb4202a477; generality
 sha256: 437a68bb14db15f958db37b8646b86ddfabbffff115b1d7b3cc5112ff2cd8ecd4.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt
 sha256: 8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt
 sha256: 9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256: 8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate

sha256: 3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c. Derivation seal

sha256: e475ec4d3f213186b5ebe79487af8185e1e53dae465357589e5cce89df446076. Source manifest

sha256: 4af888cde704ccaa37ef6f55c9ff547978126ec937e41cd8715047a646b16f1c. Engine receipt

sha256: 826d35df20d7ea259d64e873ba161a578b8597dfad41020a5ef13a95b4dfd64d at receipts/engine/model_admitted/SFT-INFO-VALID-RELATIONAL-COARSE-008-826d35df20d7ea25. json.

Closure scope `depth_independent`. External status

`empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V18
- SFT-V3- INDEPENDENT-EXACT-VALIDATION-RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-008: relational_coarse
- exact observation:
 {"all_current_receipts_bound":true,"families":["REL","COARSE"],"family_count":2,"receipt_count":26}
- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt,

erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256:5f2528d1f64629eb25af1b35c3418b8ca8a3c11c5a0425598d785fefb929343a; empirical validation sha256:56bb704730aa4ede0621ff47b35b4719d95f62f7b1cde25c17eff6fa52f82653; external validation

sha256:024ae840f5bf05c2f4d74c9570a1b65a34b437136c3ca5ab0b1d20a42806648a; isolation sha256:8dc2e50eca65bf1d0b8fa1b9409a5eaa8e7e4800c97b3905ffebfa717ac72707; custody sha256:07a7d74764dd53e3faa089388bf157d7e37acf01071f852f5f0b2c3168d0de61.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

253. Retrieval and inference complete validation vector

Family: VALID Obligation: SFT - INFO - OBL - VALID - 009 Claim:

SFT - INFO - VALID - RETRIEVAL - INFERENCE - 009

Question, law and forced result. The complete retrieval and inference validation vector binds every RETR and INFER obligation to preregistered queries, exact result or inference records, current receipts and controls.

VALID-009 uniquely retains retrieval-inference-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SEM - COMPLETENESS - 012
- SFT - INFO - VALID - RELATIONAL - COARSE - 008

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:26e531f060e701c76464ad95b078892b2c2d507ed1d867e3583bbf41378bb5ae. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-009 product before observation access.

Boundary: Every positive finite VALID-009 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-009 uniquely retains retrieval-inference-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__retrieval-inference-receipt-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-observation__immutable-engine-authority.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: 9351cd315cd6b026df04f4f2f094a6de78858cfa7f7414319f95a3d543800b11; generality
 sha256: 635377b815db1711282cabcc401cc379820cba765734ef18e97fd7eb7b53af2e.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- `false_premise` - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt
 sha256: 8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- `tampered_source` - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- `tampered_artifact` - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- `boundary` - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt
 sha256: 9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256: 8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate
 sha256: 3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c. Derivation seal
 sha256: 1894585cf81823a1ba2d5e48dfc94a53a557375a9509d61e282fab56ab4bed05. Source manifest
 sha256: 2e8259579db7b89deae1f17c9507606f5eff7abf859e7bc5431c2aba2269da2a. Engine receipt
 sha256: 26e531f060e701c76464ad95b078892b2c2d507ed1d867e3583bbf41378bb5ae at `receipts/engine/model_admitted/SFT-INFO-VALID-RETRIEVAL-INFERENCE-009-26e531f060e701c7.json`.
 Closure scope `depth_independent`. External status
`empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V18
- SFT-V3- INDEPENDENT-EXACT-VALIDATION-RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-009: `retrieval_inference`
- exact observation:
 {"all_current_receipts_bound":true,"families":["RETR","INFER"],"family_count":2,"receipt_count":26}
- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt,

erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256:1759e51836b8bbfac0c79bd8bcc0ddd38d002521aeb6f5f3b6d64a9429ebd0ad; empirical validation sha256:a3d2a0162725a846d32efb256a6c28c961f9fd6a25c91a926e63e44cac1bf867; external validation

sha256:4ad646fe44b17e34415b9aa413e9e38bbcd0b2419c02b70408cd5baaf5c8f445; isolation sha256:69c778f60dc538b3a2b16321608199a9216c6982d55d6c0790f4013d73bf9f77; custody sha256:738d31be896a894f4c4f72c6bc189c9bb1720b3f7c54d80aacb434919fa97d10.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

254. Privacy, thermodynamics and correspondence validation vector

Family: VALID Obligation: SFT - INFO - OBL - VALID - 010 Claim:

SFT - INFO - VALID - PRIVACY - THERMAL - CORRESPONDENCE - 010

Question, law and forced result. The complete privacy, information-thermodynamics and classical-probabilistic-quantum support vector binds every PRIV, THERM and CORR obligation to exact resource custody, current receipts and explicit operational handoffs.

VALID-010 uniquely retains privacy-thermal-correspondence-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SEM - COMPLETENESS - 012
- SFT - INFO - VALID - RETRIEVAL - INFERENCE - 009

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:69afe4173c3da6539dfae6b81f5768559180a87b37c86efb0905278ece5d090f. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-010 product before observation access.

Boundary: Every positive finite VALID-010 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-010 uniquely retains privacy-thermal-correspondence-receipt-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__privacy-thermal-correspondence-receipt-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-obse

rvation__immutable-engine-authority.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:cdd7a1416a4fd7f06831ea494999b02badfc0293afd078f7ca6eafa77bff0e9f; generality
sha256:4ebb09048073c16a3273800daf3a9296a7d9ced93b154e3b9ea7e26a3d7fdbcb4.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt
sha256:8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt
sha256:9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256:8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate
sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c. Derivation seal
sha256:f165ff073cee2c8e8479e618597e25241d53b58d8da4a976980a871480821582. Source manifest
sha256:5beadaf71c252bee8c1723dc9768f33d73c1aa521c09ab4e6a11d3f8aa21011f. Engine receipt
sha256:69afe4173c3da6539dfe6b81f5768559180a87b37c86efb0905278ece5d090f at receipts/engine/model_admitted/SFT-INFO-VALID-PRIVACY-THERMAL-CORRESPONDENCE-010-69afe4173c3da653.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V18
- SFT-V3- INDEPENDENT-EXACT-VALIDATION-RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-010: privacy_thermal_correspondence
- exact observation:
{ "all_current_receipts_bound":true, "families":["PRIV","THERM","CORR"], "family_count":3, "receipt_count":36 }
- complete family observation custody: 12 records; all rows preserved

- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt, erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256:3aeffecc5379cad7132181ed9db8f5947761d6e408c405586ca3df69d624ccc3; empirical validation sha256:0e974de548afc3d158dbec4426a24b30b8f79437b92efc6b1e6a55aeed2ce01e; external validation

sha256:0e8e133efc603972381f14d7802026ecf28082e4fe6eb36dba90e0b00e1d722f; isolation

sha256:91b5fce277e81edf40205e165740764c179a41c2e42a990fa01668745e0f6089; custody

sha256:40ca0e473c8b6a88473bf20cac5a11e6957857f46fd81bc9c1dee9480bbcf0fa.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

255. Adverse, absent, unresolved and boundary vector

Family: VALID Obligation: SFT - INFO - OBL - VALID - 011 Claim: SFT - INFO - VALID - ADVERSE - BOUNDARY - 011

Question, law and forced result. A complete validation record preserves every favorable, adverse, absent and unresolved observation and every declared scope boundary; no failed comparison may be deleted, renamed as success or used to retire an obligation.

VALID-011 uniquely retains complete-adverse-disposition-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SEM - COMPLETENESS - 012
- SFT - INFO - VALID - PRIVACY - THERMAL - CORRESPONDENCE - 010

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:49d52801554c6c2e7504c642f2a72e1519d41728f2cd87e6aec326f1553cbb90. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-011 product before observation access.

Boundary: Every positive finite VALID-011 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-011 uniquely retains complete-adverse-disposition-vector, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__complete-adverse-disposition-vector__current-receipt-certificate-binding__complete-disposition-custody__complete-declared

- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt, erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256:6e66fd5e80f3ca2cafffbf1e75f4ce45a9da89a12d823d700a979dd97438c3080; empirical validation sha256:b700aacc8fd607e89ce019994d20c9f791bdfa296c7b12df7907a5db0347c159; external validation

sha256:40cac004e5a429ba4571a50eb7413288001fe6f1824bfe23b8c4c4c10c12887b; isolation sha256:850d001008a50acc2e00c4a77d18e1eec579721dff39fca4dd6e7e89fc115bd5; custody sha256:7aeaacc9a9e7c96825915fa3f1f9980f94c4df707bb3b9f0b018cc2a6a556088.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

256. Information Science empirical and formal Grand Lock

Family: VALID Obligation: SFT - INFO - OBL - VALID - 012 Claim: SFT - INFO - VALID - GRAND - LOCK - 012

Question, law and forced result. The Information Science empirical and formal Grand Lock is the exact reconciliation of all 244 pre-lock receipts across eighteen completed families, their 976 controls, independent reconstructions, observation records, source identities and frozen obligation ownership.

VALID-012 uniquely retains frozen-information-science-validation-lock, complete validation custody, root forcing, post-registry observation and immutable authority.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - SEM - COMPLETENESS - 012
- SFT - INFO - VALID - ADVERSE - BOUNDARY - 011

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:7b67ff978891dc8cc9b0e1a956def0b6006b545966bb1d8876d767ce10fda35b. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis VALID-012 product before observation access.

Boundary: Every positive finite VALID-012 obligation, receipt, certificate, control, observation, disposition, scope and authority record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- VALID-012 uniquely retains frozen-information-science-validation-lock, complete validation custody, root forcing, post-registry observation and immutable authority.

Unique survivor: complete-frozen-receipt-support__frozen-information-science-validation-lock__current-receipt-certificate-binding__complete-disposition-custody__complete-declared-validation-product__root-bound-forward-forcing__post-registry-exact-observation__immutable-engine-authority.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:c1008e9f6dbec4111b3f02fc311dc186fa4c05fe3fd93f5516d8a05097e1e534; generality sha256:2d86ab71ae0dea0aeb715bfd39d811cb4f838661d2a6bddb3a3580a9ac682b18.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: Omitted claims or evidence rows invalidate completeness.; receipt sha256:8421ee17925730015bf71a2451617406f0c756e1bb15204d800a1f770a18088c.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner; no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority; no failed route retires an obligation or changes protected authority; receipt sha256:9ce85ecc970f571b10ef8cde1da030c7120c389e49d4b17fde4f2072dc66156b.

Independent reconstruction and engine admission. Independent implementation

sha256:8e6f0bf5d3569ba99876e5534459bb379fe934fd9aadd793c1f5e3cd820e758d regenerated the declared census and sole survivor; independent certificate

sha256:3789f633aecb3c2e376ba88be8dbc66ed29e04f6254eaf2cfafc5a3c752d8c9c. Derivation seal

sha256:207a321acd4f1cea96118f4e81b007d20701aa4e124cd20245d687c6b98f8b13. Source manifest

sha256:b557e9344cdf956d510162b27ab218f42b1c58b01be86962ca9f65e847a00ac7. Engine receipt

sha256:7b67ff978891dc8cc9b0e1a956def0b6006b545966bb1d8876d767ce10fda35b at

receipts/engine/model_admitted/SFT-INFO-VALID-GRAND-LOCK-012-7b67ff978891dc8c.json.

Closure scope depth_independent. External status

empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V18
- SFT-V3-INDEPENDENT-EXACT-VALIDATION-RECONSTRUCTOR

Observed, adverse and boundary records:

- VALID-012: information_science_grand_lock
- exact observation: {"adverse_controls":976,"completed_families":18,"frozen_census_mutated":false,"pre_lock_claims":244,"reconciliation_identity":"sha256:09be805951a305262d0480a27be62b2fcb5b3794655c6c6594be520cea71a4aa"}

- complete family observation custody: 12 records; all rows preserved
- the frozen 244-receipt reconciliation, 976 controls, current certificates, complete disposition custody and implementation-distinct reconstruction supply the observation; no favorable-only selection, fitted tolerance, stale receipt, erased halt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 244-receipt reconciliation, complete controls and disposition vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, selectively omitted or opened out of order.

Measurement receipt

sha256: c099e3d062c93d8890395da53fbbb4ac5a324c41cee920539feac865ec3b35ec; empirical validation sha256: 2862af2f8a4c64031557b83a18019ace0a5328ba9de85ded9c06a8f0de87cf51; external validation

sha256: a81bb3f58476b199e1e7ca2078609ee130dd5b69aedad2040f6638e34238d2b6; isolation sha256: 780a6e919dd9d9126cd62c5003f807d64aeabdb9a58c1537ae638edfb19eb843; custody sha256: 4554bcbb857a4bccb4ba0d562bff68a9e583a01a2abdd84afd5d9e8196262d62.

Prohibited inputs.

- no axiom, imported consensus assessment, fitted tolerance or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no missing receipt, stale certificate, erased adverse row, changed scope or duplicated obligation owner
- no validation claim substitutes for the underlying derivation or repairs a halted result by editing authority
- no failed route retires an obligation or changes protected authority

Complete-field family: HAND

257. Information Science one-owner downstream handoff

Family: HAND Obligation: SFT - INFO - OBL - HAND - 001 Claim: SFT - INFO - HAND - ONE - OWNER - DOWNSTREAM - 001

Question, law and forced result. An Information Science law crosses a branch boundary only as a directed dependency: Information Science retains one derivation owner and each downstream branch owns only the new law it independently forces from that input.

HAND-001 uniquely retains directed-single-owner-export-graph, one-owner custody, root forcing, post-registry observation and versioned extension.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - VALID - GRAND - LOCK - 012

The current reconciliation binds this obligation to the current model-admitted receipt

sha256: 84fd6deba477535e0957ec15ce74dd798ba082e143eaac8668b3e695a3c91e3a. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis HAND-001 product before observation access.

Boundary: Every positive finite HAND-001 owner, dependency, source, consumer, evidence, disposition, comparison and extension record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- HAND-001 uniquely retains directed-single-owner-export-graph, one-owner custody, root forcing, post-registry observation and versioned extension.

Unique survivor: complete-owned-information-law__exactly-one-derivation-owner__directed-single-owner-export-graph__directed-consumer-dependency__complete-paired-record-custody__root-bound-forward-forcing__post-registry-exact-observation__dated-complete-versioned-extension.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof
sha256:8573113c1204124518b6f5179a87bf110e2b1c7451810b37588077a288bbae60; generality
sha256:8080f3c19b68881188d6f764ea10aac11441a5c4fd232796129690d42e58d744.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: An anonymous result loses its derivation owner.; receipt
sha256:8ff708fe56a1642957800d6d619ef6078875f15e03e938ca71640a11a4d1e0b9.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported downstream law, conventional prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt; no dated completeness claim bars lawful versioned discovery; no failed route retires an obligation or changes protected authority; receipt
sha256:722f05a8e0b924e4c6566f96c23ce349670a0a5d00afdf4c600565bcd5a6b906.

Independent reconstruction and engine admission. Independent implementation

sha256:59edf358becca2aa3a8a19398d32cf267d99420ab2441ce5d6ebbc629a9f5073 regenerated the declared census and sole survivor; independent certificate
sha256:93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d. Derivation seal
sha256:166d4c78e7e73890e8af2c855cd6c68d3c784f9eec73314a883ef9a2b9921efd. Source manifest
sha256:c43b4af462c99fccaa5241f8a46beb8d86a6520a6dccac7202c38f545fae9a38. Engine receipt
sha256:84fd6deba477535e0957ec15ce74dd798ba082e143eaac8668b3e695a3c91e3a at receipts/engine/model_admitted/SFT-INFO-HAND-ONE-OWNER-DOWNSTREAM-001-84fd6deba477535e.json.
Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V19
- SFT-V3- INDEPENDENT-EXACT- HANDOFF- RECONSTRUCTOR

Observed, adverse and boundary records:

- HAND-001: one_owner_downstream_handoff
- exact observation: {"dependency_targets":7,"duplicate_target_owners":0,"one_information_science_owner_retained":true}
- complete family observation custody: 6 records; all rows preserved

- the frozen 256-receipt predecessor, six handoff obligations, one-owner dependency graph, complete boundary records and implementation-distinct reconstruction supply the observation; no copied law, duplicate owner, silent conventional import, rewritten receipt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 256-receipt predecessor, complete one-owner dependency and handoff vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, ownerless, selectively omitted or opened out of order.

Measurement receipt

sha256:189362f2cb3319c483abebd06da516fee74e23fd305143cdd290672854d9b8b3; empirical validation sha256:327f79402a4c010206a4f75690392001c7def092dce79b87795430b0b0a751e4; external validation

sha256:a2a4427a1593d3dec09ba69733ddfb948131cf9a21480d638df20a78d569e89a; isolation sha256:b84a7934f93ef0d123b2f16eacce52f595a3a72a93a8edebf60f493629522909; custody sha256:fe2ed7291b8c9ef9f61348a55caeee77dec6fc1671f0b17bdab23204f30c05a7.

Prohibited inputs.

- no axiom, imported downstream law, conventional prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt
- no dated completeness claim bars lawful versioned discovery
- no failed route retires an obligation or changes protected authority

258. Information Science measurement-boundary handoff

Family: HAND Obligation: SFT - INFO - OBL - HAND - 002 Claim: SFT - INFO - HAND - MEASUREMENT - BOUNDARY - 002

Question, law and forced result. The measurement boundary retains formal derivation, externally held observation and exact comparison as distinct records; observation tests a law but cannot silently choose or rewrite it.

HAND-002 uniquely retains formal-observation-comparison-separation, one-owner custody, root forcing, post-registry observation and versioned extension.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-VALID-GRAND-LOCK-012
- SFT-INFO-HAND-ONE-OWNER-DOWNSTREAM-001

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:0d9849de5d10c9b2274eeda65888ba039347b56ef77a2351e22fc634a3b56ebf. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis HAND-002 product before observation access.

Boundary: Every positive finite HAND-002 owner, dependency, source, consumer, evidence, disposition, comparison and extension record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- HAND-002 uniquely retains formal-observation-comparison-separation, one-owner custody, root forcing, post-registry observation and versioned extension.

Unique survivor: complete-owned-information-law__exactly-one-derivation-owner__formal-observation-comparison-separation__directed-consumer-dependency__complete-paired-record-custody__root-bound-forward-forcing__post-registry-exact-observation__dated-complete-versioned-extension.

Closure: `depth_independent`; minimality `true`; named-shape uniqueness `true`; proof
 sha256: `b19a3e91f7d91d9b3975cf967772e5347ecedbf84212761409c6ad7552e8a382`; generality
 sha256: `28e7add21ad119e6d7f0f17983baba1f192d0418114605756df6e042cd64f624`.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: An anonymous result loses its derivation owner.; receipt
 sha256: `8ff708fe56a1642957800d6d619ef6078875f15e03e938ca71640a11a4d1e0b9`.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
 sha256: `a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26`.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
 sha256: `b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219`.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported downstream law, conventional prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt; no dated completeness claim bars lawful versioned discovery; no failed route retires an obligation or changes protected authority; receipt
 sha256: `722f05a8e0b924e4c6566f96c23ce349670a0a5d00afdf4c600565bcd5a6b906`.

Independent reconstruction and engine admission. Independent implementation
 sha256: `59edf358becca2aa3a8a19398d32cf267d99420ab2441ce5d6ebbc629a9f5073` regenerated the declared census and sole survivor; independent certificate
 sha256: `93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d`. Derivation seal
 sha256: `7b43233ee6efe2662070c5e4f997a4724fa3c2c9a4d13cf015c3eebf9c1f2eda`. Source manifest
 sha256: `c995d64f15107da3e3ff29673c013df8cb2b938a90acc32d5bdc9d7aead43cf5`. Engine receipt
 sha256: `0d9849de5d10c9b2274eeda65888ba039347b56ef77a2351e22fc634a3b56ebf` at `receipts/engine/model_admitted/SFT-INFO-HAND-MEASUREMENT-BOUNDARY-002-0d9849de5d10c9b2.json`.
 Closure scope `depth_independent`. External status
`empirically_tested_and_independently_replicated`.

Post-registry observation and comparison. Target opened after seal: `true`. Evaluator verified seal: `true`. All rows preserved: `true`. Comparison passed: `true`.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V19
- SFT-V3- INDEPENDENT-EXACT-HANDOFF-RECONSTRUCTOR

Observed, adverse and boundary records:

- HAND-002: `measurement_boundary_handoff`
- exact observation: `{"boundary_records":3,"formal_observation_comparison_separate":true}`
- complete family observation custody: 6 records; all rows preserved
- the frozen 256-receipt predecessor, six handoff obligations, one-owner dependency graph, complete boundary records and implementation-distinct reconstruction supply the observation; no copied law, duplicate owner, silent conventional import, rewritten receipt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 256-receipt predecessor, complete one-owner dependency and handoff vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, ownerless, selectively omitted or opened out of order.

Measurement receipt

sha256:68146fdccde8f1c10dff8b5b159bef977e105fb9ced8fc94d71657818d01d3bf; empirical validation sha256:8bc00b529b97b8478b9416208cc9c7ea9084c010a84b87d5a0501f8b153b17b4; external validation sha256:d593d5a4b7bf3f7abb74e29841d072bfceb33144fa3bb7eb5828edc91e1e7907; isolation sha256:f73718798c6ca64fdf76d002acacf83a46d880184a2798670caf2078bbb4442a; custody sha256:e9a585097f5f5d5e4b618d71b3d3bd1ebd19892d6b35d001635848f6abe7f082.

Prohibited inputs.

- no axiom, imported downstream law, conventional prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt
- no dated completeness claim bars lawful versioned discovery
- no failed route retires an obligation or changes protected authority

259. Information Science formal-to-empirical handoff

Family: HAND Obligation: SFT - INFO - OBL - HAND - 003 Claim: SFT - INFO - HAND - FORMAL - EMPIRICAL - 003

Question, law and forced result. A lawful formal-to-empirical handoff orders complete derivation before sealed prediction, target release before comparison, and preserves every favorable, adverse, absent, unresolved and boundary row.

HAND-003 uniquely retains sealed-formal-to-empirical-custody, one-owner custody, root forcing, post-registry observation and versioned extension.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - VALID - GRAND - LOCK - 012
- SFT - INFO - HAND - MEASUREMENT - BOUNDARY - 002

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:79f833a305f0f49b5a7db1a6e8f0ef8989f7eff4ed0e488fd9d29cd8b8e03cbf. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis HAND-003 product before observation access.

Boundary: Every positive finite HAND-003 owner, dependency, source, consumer, evidence, disposition, comparison and extension record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- HAND-003 uniquely retains sealed-formal-to-empirical-custody, one-owner custody, root forcing, post-registry observation and versioned extension.

Unique survivor: complete-owned-information-law__exactly-one-derivation-owner__sealed-formal-to-empirical-custody__directed-consumer-dependency__complete-paired-record-custody__root-bound-forward-forcing__post-registry-exact-observation__dated-complete-versioned-extension.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:5f71c79a1f99aa40bdeec1a596f9a633e48d6c7ea1f7ce86030315ac0923233f; generality sha256:0882371863452d51ce8406aec72308a68a881f84cf372eaa7490f3ed41c07679.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: An anonymous result loses its derivation owner.; receipt
sha256: 8ff708fe56a1642957800d6d619ef6078875f15e03e938ca71640a11a4d1e0b9.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported downstream law, conventional prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt; no dated completeness claim bars lawful versioned discovery; no failed route retires an obligation or changes protected authority; receipt
sha256: 722f05a8e0b924e4c6566f96c23ce349670a0a5d00afdf4c600565bcd5a6b906.

Independent reconstruction and engine admission. Independent implementation

sha256: 59edf358becca2aa3a8a19398d32cf267d99420ab2441ce5d6ebbc629a9f5073 regenerated the declared census and sole survivor; independent certificate

sha256: 93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d. Derivation seal
sha256: 059ef913d58a0f57cd9c0811fe84641af558f91d2f7907d407422aa5789774f6. Source manifest
sha256: 4689bf455ca96cf99522479637afa8319c059d4f238973ae69957209a304e368. Engine receipt
sha256: 79f833a305f0f49b5a7db1a6e8f0ef8989f7eff4ed0e488fd9d29cd8b8e03cbf at receipts/engine/model_admitted/SFT-INFO-HAND-FORMAL-EMPIRICAL-003-79f833a305f0f49b.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V19
- SFT-V3- INDEPENDENT-EXACT-HANDOFF-RECONSTRUCTOR

Observed, adverse and boundary records:

- HAND-003: formal_empirical_handoff
- exact observation: {"all_dispositions_preserved":true,"derivation_before_target_release":true,"ordered_custody_stages":4}
- complete family observation custody: 6 records; all rows preserved
- the frozen 256-receipt predecessor, six handoff obligations, one-owner dependency graph, complete boundary records and implementation-distinct reconstruction supply the observation; no copied law, duplicate owner, silent conventional import, rewritten receipt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 256-receipt predecessor, complete one-owner dependency and handoff vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, ownerless, selectively omitted or opened out of order.

Measurement receipt

sha256: a09cb2c21001b351483d21cbdfef23b5255194b04c998f6d8eba5e4f837ef0902; empirical

validation sha256:7cad96c5e2c2d70284fd5a64c5db8cb63fba49b7eb8948eb307ddd4fd5e79571;
 external validation
 sha256:a99380f5791c6d5e11ba4aa962820088f30bbbe8ddf1525a4d6d814ec4088fb1; isolation
 sha256:6c843cd7c7f8dab47dcf1b3cd846459cd73083409d0609ed7aa7282a6d22dcee3; custody
 sha256:f20d1aa41f990edc5e5e33b9444f1ff1beb98b70b7e9158454103b7cfc402b7a.

Prohibited inputs.

- no axiom, imported downstream law, conventional prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt
- no dated completeness claim bars lawful versioned discovery
- no failed route retires an obligation or changes protected authority

260. Information Science conventional-correspondence handoff

Family: HAND Obligation: SFT - INFO - OBL - HAND - 004 Claim:
 SFT - INFO - HAND - CONVENTIONAL - CORRESPONDENCE - 004

Question, law and forced result. A conventional information-science term is admitted only as an explicit reversible correspondence to a derived Fold relation within a declared boundary; the conventional model never selects the Fold law.

HAND-004 uniquely retains reversible-comparison-boundary-vocabulary, one-owner custody, root forcing, post-registry observation and versioned extension.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - VALID - GRAND - LOCK - 012
- SFT - INFO - HAND - FORMAL - EMPIRICAL - 003

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:ce76e7148ac9ea1c6e9ffca7d2ca72072cb35d4167744c4d6eab6f4ce5eafc7e. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis HAND-004 product before observation access.

Boundary: Every positive finite HAND-004 owner, dependency, source, consumer, evidence, disposition, comparison and extension record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- HAND-004 uniquely retains reversible-comparison-boundary-vocabulary, one-owner custody, root forcing, post-registry observation and versioned extension.

Unique survivor: complete-owned-information-law__exactly-one-derivation-owner__reversible-comparison-boundary-vocabulary__directed-consumer-dependency__complete-paired-record-custody__root-bound-forward-forcing__post-registry-exact-observation__dated-complete-versioned-extension.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:990dcc24fbb3a89be2f1988367cd2a1a682566e16572af44bab1bf2e1f4f5fcf; generality
 sha256:50248c4994f56528ebb19c5190081c5319c400ba7a91715533458d4bf41b62b1.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT

numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: An anonymous result loses its derivation owner.; receipt
sha256:8ff708fe56a1642957800d6d619ef6078875f15e03e938ca71640a11a4d1e0b9.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported downstream law, conventional prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt; no dated completeness claim bars lawful versioned discovery; no failed route retires an obligation or changes protected authority; receipt
sha256:722f05a8e0b924e4c6566f96c23ce349670a0a5d00afdf4c600565bcd5a6b906.

Independent reconstruction and engine admission. Independent implementation

sha256:59edf358becca2aa3a8a19398d32cf267d99420ab2441ce5d6ebbc629a9f5073 regenerated the declared census and sole survivor; independent certificate

sha256:93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d. Derivation seal
sha256:953aa92b6c563ccb62be16b473780c024d202961c909ac6c15e64641d14ea87b. Source manifest
sha256:38febbe5c0d7bdf3e814667bb6d6ac5b61b78d3085cf5210ea8d12698a70984a. Engine receipt
sha256:ce76e7148ac9ea1c6e9ffca7d2ca72072cb35d4167744c4d6eab6f4ce5eafc7e at receipts/engine/model_admitted/SFT-INFO-HAND-CONVENTIONAL-CORRESPONDENCE-004-ce76e7148ac9ea1c.json. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V19
- SFT-V3- INDEPENDENT-EXACT-HANDOFF-RECONSTRUCTOR

Observed, adverse and boundary records:

- HAND-004: conventional_correspondence_handoff
- exact observation:
{"conventional_correspondences":4,"conventional_model_selects_law":false,"reversible_within_declared_boundary":true}
- complete family observation custody: 6 records; all rows preserved
- the frozen 256-receipt predecessor, six handoff obligations, one-owner dependency graph, complete boundary records and implementation-distinct reconstruction supply the observation; no copied law, duplicate owner, silent conventional import, rewritten receipt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 256-receipt predecessor, complete one-owner dependency and handoff vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, ownerless, selectively omitted or opened out of order.

Measurement receipt

sha256:ab024decbf0798db6b991f1e57eebdf6621aaf53a39f21e47f08602bb751d1a5; empirical
validation sha256:f82e4edea9f3eeca7f6d66ba5d4a467fd382f25c6669e6a175c7eb878f1d49d;
external validation

sha256:104cf50cf8d514aecc9e8721cb6f0725146e277eaedca25cad550a500da37924; isolation
 sha256:4a7e0d475ed3cbceea3dad2ed6ad795ac7c4eee1dcef765cae05abedb86c1a18; custody
 sha256:6210918cb37b06ee9c1fef171e9708c4a071395cf5df59651d0ff46b2289539b.

Prohibited inputs.

- no axiom, imported downstream law, conventional prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt
- no dated completeness claim bars lawful versioned discovery
- no failed route retires an obligation or changes protected authority

261. Information Science open-extension handoff

Family: HAND Obligation: SFT-INFO-OBL-HAND-005 Claim: SFT-INFO-HAND-OPEN-EXTENSION-005

Question, law and forced result. Information Science is complete to its frozen dated census and remains open to lawful discovery: an extension appends a versioned obligation, derivation and receipt without rewriting prior receipts or claiming permanent closure.

HAND-005 uniquely retains dated-complete-versioned-extension, one-owner custody, root forcing, post-registry observation and versioned extension.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT-ROOT-THERE-IS-NO-NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT-INFO-VALID-GRAND-LOCK-012
- SFT-INFO-HAND-CONVENTIONAL-CORRESPONDENCE-004

The current reconciliation binds this obligation to the current model-admitted receipt

sha256:521d97d351870c06956513f18cd6a77b7221da4b62697213d8a692ca4eef56a1. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis HAND-005 product before observation access.

Boundary: Every positive finite HAND-005 owner, dependency, source, consumer, evidence, disposition, comparison and extension record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- HAND-005 uniquely retains dated-complete-versioned-extension, one-owner custody, root forcing, post-registry observation and versioned extension.

Unique survivor: complete-owned-information-law__exactly-one-derivation-owner__dated-complete-versioned-extension__directed-consumer-dependency__complete-paired-record-custody__root-bound-forward-forcing__post-registry-exact-observation__dated-complete-versioned-extension.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof

sha256:e501badc21aa5d17df29c45bd414f3b2f8c0832c234d2999cfc46945d44d90ca; generality
 sha256:439a962148fcff7d7cffe7ece9b40d961edfb8d45e46faf810ec1dcf5c6cc78e.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: An anonymous result loses its derivation owner.; receipt
sha256:8ff708fe56a1642957800d6d619ef6078875f15e03e938ca71640a11a4d1e0b9.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256:a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256:b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported downstream law, conventional prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt; no dated completeness claim bars lawful versioned discovery; no failed route retires an obligation or changes protected authority; receipt
sha256:722f05a8e0b924e4c6566f96c23ce349670a0a5d00afdf4c600565bcd5a6b906.

Independent reconstruction and engine admission. Independent implementation

sha256:59edf358becca2aa3a8a19398d32cf267d99420ab2441ce5d6ebbc629a9f5073 regenerated the declared census and sole survivor; independent certificate

sha256:93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d. Derivation seal sha256:275d9ce1fac06e0186940e34d6953d64b31f7979c9852b6644945e2f69b38c8e. Source manifest sha256:c264319e42670419fc1bf2bcaea69123f2e9eaf892640e15a4bd4ff2c200ac33. Engine receipt sha256:521d97d351870c06956513f18cd6a77b7221da4b62697213d8a692ca4eef56a1 at receipts/engine/model_admitted/SFT-INFO-HAND-OPEN-EXTENSION-005-521d97d351870c06.json. Closure scope depth_independent. External status empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V19
- SFT-V3- INDEPENDENT-EXACT-HANDOFF-RECONSTRUCTOR

Observed, adverse and boundary records:

- HAND-005: open_extension_handoff
- exact observation: {"frozen_obligations":262,"prior_receipts_rewritten":false,"versioned_extension_appends":true}
- complete family observation custody: 6 records; all rows preserved
- the frozen 256-receipt predecessor, six handoff obligations, one-owner dependency graph, complete boundary records and implementation-distinct reconstruction supply the observation; no copied law, duplicate owner, silent conventional import, rewritten receipt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 256-receipt predecessor, complete one-owner dependency and handoff vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, ownerless, selectively omitted or opened out of order.

Measurement receipt

sha256:9cac6df5805b2a3d0f81c324da588ab015907edfbc39e7ac500e9822edd47a85; empirical validation sha256:37b6e72e4b1262481464886dff5667ad61ac64d9ee62ea2ec93281afe0ad8219; external validation

sha256:480685945a4f28fc7e7cb7669dd59c100b80fb70a0928ea0d5d064322ab09997; isolation sha256:f9d8b2a7fffc98d5e9a29df1a7ba71655df4428f5a65c98caa75eac6f4bf04baa; custody sha256:8161dbe8ea9e221bf06782d55aeaaacfd6893c414581e85fd75f0bd898d3351.

Prohibited inputs.

- no axiom, imported downstream law, conventional prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt
- no dated completeness claim bars lawful versioned discovery
- no failed route retires an obligation or changes protected authority

262. Information Science cross-branch completeness certificate

Family: HAND Obligation: SFT - INFO - OBL - HAND - 006 Claim:
SFT - INFO - HAND - CROSS - BRANCH - COMPLETENESS - 006

Question, law and forced result. Cross-branch handoff completeness is the one-to-one reconciliation of all six frozen handoff obligations with the 256 pre-handoff receipts, yielding all 262 Information Science obligations with one owner, root trace and explicit extension boundary.

HAND-006 uniquely retains complete-six-handoff-reconciliation, one-owner custody, root forcing, post-registry observation and versioned extension.

Trace to the foundational theorem. This registration names no axioms and no free parameters. It declares SFT - ROOT - THERE - IS - NO - NOTHING as its root theorem and consumes only these already admitted receipt identities:

- SFT - INFO - VALID - GRAND - LOCK - 012
- SFT - INFO - HAND - OPEN - EXTENSION - 005

The current reconciliation binds this obligation to the current model-admitted receipt sha256:8e464b9679a76b39f142b622489e4be2beda543d18c7de51abd6bd2567dac8d0. A missing dependency, stale certificate or broken root trace halts rather than being replaced by prose.

Complete candidate grammar. Generate the complete eight-axis HAND-006 product before observation access.

Boundary: Every positive finite HAND-006 owner, dependency, source, consumer, evidence, disposition, comparison and extension record.

The literal product contains 256 candidates and the elimination ledger contains exactly 256 one-for-one decisions. Exactly one survives and 255 are eliminated. The forced coordinates are:

- HAND-006 uniquely retains complete-six-handoff-reconciliation, one-owner custody, root forcing, post-registry observation and versioned extension.

Unique survivor: complete-owned-information-law__exactly-one-derivation-owner__complete-six-handoff-reconciliation__directed-consumer-dependency__complete-paired-record-custody__root-bound-forward-forcing__post-registry-exact-observation__dated-complete-versioned-extension.

Closure: depth_independent; minimality true; named-shape uniqueness true; proof sha256:ae9738775c8945f4a5a7a47cf3a54975cd8f326d352f3d0264e33252725a03d3; generality sha256:73a105dbc694fe3e7975a2a6b593222db6cbc848ec666e0659dbce527508aa02.

Scientific meaning. This result is the exact information object named by the claim: its support, relation, retained distinctions, closed distinctions, observation conditions, records, provenance, positive-finite extension and explicit downstream boundary remain attached. It does not replace those records with a score, imported named model, fitted magnitude or consensus label. Structural absence is empty One; host display 0 denotes that absence and is not an SFT numerical object.

Falsification controls. All four required controls passed:

- **false_premise** - passed; expected: reject a generated form missing the first required structural condition; observed: An anonymous result loses its derivation owner.; receipt
sha256: 8ff708fe56a1642957800d6d619ef6078875f15e03e938ca71640a11a4d1e0b9.
- **tampered_source** - passed; expected: reject a changed official source identity; observed: the changed identity differs from the registered source manifest; receipt
sha256: a549a1da4d8bc6d4bc9fb65e0d475f2e3bc196a80c402742f80f637f51eb4e26.
- **tampered_artifact** - passed; expected: reject a missing, duplicated or additional survivor; observed: the generated product contains exactly one registered survivor; receipt
sha256: b79ac01f095bf293f4d7a736893d30f1c51e9c6a1c5d98b04f5a8cda8f071219.
- **boundary** - passed; expected: halt any attempt to import an excluded object or answer-producing model; observed: no axiom, imported downstream law, conventional prior or target outcome selects the result; host 0 denotes structural absence or artifact counts only and is not an SFT number object; no negative, irrational, imaginary or floating proof scalar; no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt; no dated completeness claim bars lawful versioned discovery; no failed route retires an obligation or changes protected authority; receipt
sha256: 722f05a8e0b924e4c6566f96c23ce349670a0a5d00afdf4c600565bcd5a6b906.

Independent reconstruction and engine admission. Independent implementation

sha256: 59edf358becca2aa3a8a19398d32cf267d99420ab2441ce5d6ebbc629a9f5073 regenerated the declared census and sole survivor; independent certificate
sha256: 93a8eab573f9e689afc25deb6b3279a747bce09ac5cd8f760565720b531dd22d. Derivation seal
sha256: 524f1db9acf7b3557b122d2888c3879e40409486708523d6ac7b0338db3edde1. Source manifest
sha256: e9c97f6929d8bbde14e8c8fdb202b210d46109dc638ab33081aaddf8c3848668. Engine receipt
sha256: 8e464b9679a76b39f142b622489e4be2beda543d18c7de51abd6bd2567dac8d0 at receipts/engine/model_admitted/SFT-INFO-HAND-CROSS-BRANCH-COMPLETENESS-006-8e464b9679a76b39.json. Closure scope depth_independent. External status
empirically_tested_and_independently_replicated.

Post-registry observation and comparison. Target opened after seal: true. Evaluator verified seal: true. All rows preserved: true. Comparison passed: true.

Source identities:

- SFT-V3- INFORMATION-SCIENCE-RECONCILIATION-V19
- SFT-V3- INDEPENDENT-EXACT-HANDOFF-RECONSTRUCTOR

Observed, adverse and boundary records:

- HAND-006: cross_branch_completeness
- exact observation: {"complete_obligations":262,"duplicate_owners":0,"handoff_obligations":6,"omitted_owners":0,"pre_handoff_receipts":256}
- complete family observation custody: 6 records; all rows preserved
- the frozen 256-receipt predecessor, six handoff obligations, one-owner dependency graph, complete boundary records and implementation-distinct reconstruction supply the observation; no copied law, duplicate owner, silent conventional import, rewritten receipt or edited authority enters

Falsification condition: Reject if the value-free registry, frozen 256-receipt predecessor, complete one-owner dependency and handoff vector, exact observation, candidate survivor, source identity, seal, custody record or independent reconstruction is missing, stale, changed, duplicated, ownerless, selectively omitted or opened out of order.

Measurement receipt

sha256: f08e2f44ab74de61ef0f529b4ea364c4bdef5c29363be75539e2b2df5a60c9c8; empirical validation sha256: a6141b593970b233b9d6ac1c0bb6ed22681cef7527955bdb6a526ebadffb9c76; external validation
sha256: 632728215d2e1d5662ab5490be9d3a94d20bfe5735aae518ab0f189d9bcefbef0; isolation
sha256: a74f6924543bc52fb11136a128028a6119b23c0a9f6c75d523e1a4cca4c1e2d8; custody
sha256: 188fbee72f48ae0b9a4849694443f64c85740acd9c3a646b535c292c7c8166b1.

Prohibited inputs.

- no axiom, imported downstream law, conventional prior or target outcome selects the result
- host 0 denotes structural absence or artifact counts only and is not an SFT number object
- no negative, irrational, imaginary or floating proof scalar
- no duplicate owner, copied law, missing dependency, erased adverse row or rewritten receipt
- no dated completeness claim bars lawful versioned discovery
- no failed route retires an obligation or changes protected authority

External observation and validation at the correct evidential boundary

Empirical means based on observation or experiment. For this formal branch, the directly observable objects are exact generated symbol carriers, records, channel relations, execution traces, error images, reconstruction results, receipt identities and controlled perturbations. Every successor claim freezes a value-free target identity before opening its observation vector, executes the law, preserves all rows and binds the comparison to an engine receipt. Where an information law later produces a natural measured magnitude, the owning physical, biological, social or engineering branch must preregister and compare that magnitude against authoritative external data; this branch cannot invent a measurement value or pre-approve the downstream result.

Opaque prediction is not an admissible substitute. A score without premises, candidate space, eliminated alternatives, source identity, target custody, adverse rows and halt conditions hides the distinctions required to reproduce or falsify the claim. Conventional Hartley, Shannon, Hamming, Landauer, Bennett and Schumacher language is therefore used only after sealing as a declared correspondence boundary. It does not select the Fold carrier, its exact parts or its laws.

The complete validation family binds all 244 pre-lock claims, all current certificates and all 976 pre-lock controls. The terminal HAND family then establishes single-owner downstream dependencies, formal/measurement separation, sealed formal-to-empirical order, conventional correspondence, open extension and cross-branch completeness. Reconciliation v20 records all 262 obligations under identity

sha256:48056bb5684a97ca5e9eba6bda72afe536a11dd1b9f970a0d220f82dc1ee3a54.

Information power, open knowledge and institutional accountability

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. It does not ask the public to exchange one inaccessible authority for another. Scientific authority is deliberately narrow, inspectable and revocable: a claim enters this corpus only with its derivation chain, generated alternatives, elimination ledger, unique survivor, adverse controls, observation custody, independent reconstruction and unchanged-engine receipt. Open criticism requires no permission. Admission to the scientific corpus is the separate act of satisfying that public standard.

Maria Smith produced this programme outside conventional academic credentials, institutional research employment and grant funding. That fact is not evidence for a theorem; the derivations and observations carry the scientific burden. It is evidence about access. A credential-first, capital-dependent knowledge system does not merely deny individuals opportunity: it loses unknown questions, methods and discoveries from minds it never authorizes to speak. The significance is therefore not special pleading for one author. It is an indictment of every contribution lost when financial and status gates are dressed as scientific rigor.

Paywalls, inaccessible datasets, undisclosed training corpora, proprietary validators and black-box oracle scores close precisely the source distinctions required for independent falsification. Funding, prestige, publication and consensus systems can carry expertise and resources, but none is itself a proof or empirical receipt. Ernos Labs argues for transparent, reproducible and freely inspectable science in which institutions and independent researchers meet the same evidential burden.

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Reproduction, falsification and extension boundary

This complete-field paper is a read-only projection of the claim packages. It does not modify a derivation, verifier, engine receipt or authority seal. Local preparation requires the v20 reconciliation, exact replay of the final HAND family, all focused Information Science tests, lightweight repository validation and both immutable seals. The heavy all-branch command remains reserved for the final global Grand Lock and is not misrepresented as rerun for this paper preparation.

The branch is complete to the registered 29 July 2026 census, not permanently closed. A lawful new result may correct a correspondence, falsify an empirical claim or add a new obligation. It must be preregistered, completely enumerated, independently reconstructed, passed through the same untouched engine, externally compared wherever possible and appended by a versioned reconciliation. Prior receipts remain reproducible historical evidence; they are not silently invalidated or rewritten.

Conclusion

The complete dated Information Science branch now exposes a continuous, machine-auditable route from the premise-free Fold theorem to 262 separately admitted information laws. Its substantive unification is exact distinction custody: symbols, descriptions, uncertainty, channels, errors, codes, dependence, inference, privacy, thermodynamic records, deterministic probability, quantum support and semantic boundaries are different organizations of the same complete support-and-observation structure. Its methodological claim is equally concrete: trust in Maria Smith, Ernos Labs, an institution or an AI system is unnecessary where the grammar, decisions, controls, sources, receipts and falsifiers are public.

Publication and repository links

- Canonical repository: <https://github.com/MettaMazza/ernos-labs-sft-platform>
- Author: Maria Smith, Ernos Labs
- Contact: Maria.Smith.Sftoe@gmail.com
- Submissions and open review: <https://discord.gg/ucwGryVxGr>
- Paper license: CC BY 4.0; code license: Apache-2.0; authorship and creative rights retained.

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